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Back

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LandingsSpaceships and RocketsCommunicating with MissionsArtemisJames Webb
Space TelescopeHubble Space TelescopeInternational Space StationOSIRIS-Rex

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Hubble Space Telescope

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and RocketsLiving in Space

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Destinations

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Earth Science in Action

Earth Multimedia

Earth Data

Earth Science Researchers

The Solar SystemThe SunMercuryVenusEarthThe
MoonMarsJupiterSaturnUranusNeptunePluto & Dwarf PlanetsAsteroids, Comets &
MeteorsThe Kuiper BeltThe Oort CloudSkywatching

The Sun

Mercury

Venus

The Moon

Mars

Jupiter

Saturn

Uranus

Neptune

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Asteroids, Comets & Meteors

The Kuiper Belt

The Oort Cloud

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Black Holes

The Big Bang

Dark Energy

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Internships

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 Phytoplankton with NASA's FjordPhyto ProjectarticleHace 2 days minutos2 min readMore
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espacioarticleHace 3 months minutos4 min readEl X-59 enciende su motor por primera vez
rumbo al despeguearticleHace 3 months minutos

Página de inicioMisionesSeres humanos en el espacioTierraEl sistema solarEl
universoCienciaAeronáuticaTecnologíaRecursos de aprendizajeAcerca de la
NASAEspañolNoticias y eventosMultimediaNASA+

Página de inicio

Misiones

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Tierra

El sistema solar

El universo

Tecnología

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during the Cold War, this Lockheed Orion P3 has conducted airborne science for NASA since 1991! It has been to all seven continents many times. NASA coming to an airport near you in 2050? NASA Funds New Studies Looking at Future of Sustainable Aircraft Picture yourself at an airport a few decades from now. What does your airliner look like? It's more efficient, with lower emissions than today's aircraft – what kinds of designs or technology make that possible? NASA is working to answer those questions by commissioning five new design studies looking to push the boundaries of possibility for sustainable aircraft. Read the story and see the ideas Artist's concept of a 50-60 passenger hydrogen fuel cell electric plane created by Boeing through its future flight concept efforts. Aurora Flight Sciences, a Boeing Company, received an award through NASA's Advanced Aircraft Concepts for Environmental Sustainability 2050 initiative to examine new alternative aviation fuels propulsion systems, aerodynamic technologies, and aircraft configurations, along with other technology areas. Boeing aircraft operations Johnson Space Center The Houston-based space center plays host to a number of NASA aircraft that are used for astronaut training and flight proficiency, transport of oversized cargo, and hosting airborne science investigations of planet Earth. Read About the JSC Aircraft NASA Johnson Center Aircraft. The large image is of the Super Guppy and the smaller images (left to right): 1. WB57 (N926NA, N927NA, & N928NA), G-V (N95NA), T-38 Fleet and the G-III (N992NA). NASA NASA Aircraft 45 Images Go To Gallery Go To Gallery more on nasa aircraft Support Aircraft and Maintenance Operations Overview Support Aircraft and Maintenance Operations (SAMO)'s mission support aircraft are flexible, adaptable, and agile platforms that are real-time in-situ research facilities to collect, process, exploit, and disseminate the best research data possible, as quickly as possible to demonstrate technologies and concepts, in the toughest environments. Learn More About SAMOA NASA F/A-18 is towed to the apron at NASA's Armstrong Flight Research Center in Edwards, California during sunrise over Rogers Dry Lake. The F/A-18 was used to test a transmitter for an air navigation system, called the Airborne Location Integrating Geospatial Navigation System, or ALIGNS. This system, designed to allow pilots to position their aircraft at precise distances to each other, will be critical for acoustic validation efforts of NASA's next supersonic X-plane, the X-59 Quiet SuperSonic Technology. NASA/Lauren Hughes Here is Some Other Cool NASA Aeronautics Stuff to Check Out Aeronautics @ Home A collection of activities and lessons to learn more about NASA Aeronautics and the science of flight that you can do at home. STEM Learning Resources Explore hands-on activities, interactive lesson plans, educator guides, and other downloadable content related to aeronautics. Aeronautics Backgrounds Take video calls from your home or office in the cockpit of the X-59 or from inside a wind tunnel! And other aviation themes. Free E-Books to Download Add these books covering NASA Aeronautics and other aviation topics to your electronic library. NASA X Video Files This binge worthy page collects aeronautics-themed topics from the popular NASA X video series. Facebook

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4 Episodes

Other Worlds

Space Out

7 Episodes

NASA Explorers

34 Episodes

Artemis I: Path to the Pad

Down To Earth

9 Episodes

JPL and The Space Age

16 Episodes

Sci-Girls

5 Episodes

Elements of Webb

13 Episodes

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2 Episodes

Climate Tales

Elmo Visits NASA

STEMonstrations

28 Episodes

Earth Minute

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The Hubble Space Telescope floats above Earth after being released from space shuttle Columbia's robot arm at the conclusion of a servicing mission in March 2002. NASA

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Jet Propulsion LaboratoryFeb 11, 2025ArticleContentsDawn of Twilight CloudsCloud MysteryMastcam's Partial ViewMore About CuriosityNews Media ContactsTo view this video please enable JavaScript, and consider upgrading to a web browser that supports HTML5 videoNASA's Curiosity Mars rover captured these drifting noctilucent, or twilight, clouds in a 16-minute recording on Jan. 17. (This looping clip has been speeded up about 480 times.) The white plumes falling out of the clouds are carbon dioxide ice that would evaporate closer to the Martian surface.NASA/JPL-Caltech/MSSS/SSIWhile the Martian clouds may look like the kind seen in Earth's skies, they include frozen carbon dioxide, or dry ice.Red-and-green-tinted clouds drift through the Martian sky in a new set of images captured by NASA's Curiosity rover using its Mastcam — its main set of “eyes.” Taken over 16 minutes on Jan. 17 (the 4,426th Martian day, or sol, of Curiosity's mission), the images show the latest observations of what are called noctilucent (Latin for “night shining”), or twilight clouds, tinged with color by scattering light from the setting Sun.Sometimes these clouds even create a rainbow of colors, producing iridescent, or “mother-of-pearl” clouds. Too faint to be seen in daylight, they're only visible when the clouds are especially high and evening has fallen.Martian clouds are made of either water ice or, at higher altitudes and lower temperatures, carbon dioxide ice. (Mars' atmosphere is more than 95% carbon dioxide.) The latter are the only kind of clouds observed at Mars producing iridescence, and they can be seen near the top of the new images at an altitude of around 37 to 50 miles (60 to 80 kilometers). They're also visible as white plumes falling through the atmosphere, traveling as low as 31 miles (50 kilometers) above the surface before evaporating because of rising temperatures. Appearing briefly at the bottom of the images are water-ice clouds traveling in the opposite direction roughly 31 miles (50 kilometers) above the rover.Dawn of Twilight CloudsTwilight clouds were first seen on Mars by NASA's Pathfinder mission in 1997; Curiosity didn't spot them until 2019, when it acquired its first-ever images of iridescence in the clouds. This is the fourth Mars year the rover has observed the phenomenon, which occurs during early fall in the southern hemisphere.Mark Lemmon, an atmospheric scientist with the Space Science Institute in Boulder, Colorado, led a paper summarizing Curiosity's first two seasons of twilight cloud observations, which published late last year in Geophysical Research Letters. “I'll always remember the first time I saw those iridescent clouds and was sure at first it was some color artifact,” he said. “Now it's become so predictable that we can plan our shots in advance; the clouds show up at exactly the same time of year.” Each sighting is an opportunity to learn more about the particle size and growth rate in Martian clouds. That, in turn, provides more information about the planet's atmosphere.Cloud MysteryOne big mystery is why twilight clouds made of carbon dioxide ice haven't been spotted in other locations on Mars. Curiosity, which landed in 2012, is on Mount Sharp in Gale Crater, just south of the Martian equator. Pathfinder landed in Ares Vallis, north of the equator.NASA's Perseverance rover, located in

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The new twilight clouds appear framed in a partially open circle. That's because they were taken using one of Mastcam's two color cameras: the left 34 mm focal length Mastcam, which has a filter wheel that is stuck between positions. Curiosity's team at NASA's Jet Propulsion Laboratory in Southern California remains able to use both this camera and the higher-resolution right 100 mm focal length camera for color imaging. The rover recently wrapped an investigation of a place called Gediz Vallis channel and is on its way to a new location that includes boxwork—fractures formed by groundwater that look like giant spiderwebs when viewed from space. More recently, Curiosity visited an impact crater nicknamed "Rustic Canyon," capturing it in images and studying the composition of rocks around it. The crater, 67 feet (20 meters) in diameter, is shallow and has lost much of its rim to erosion, indicating that it likely formed many millions of years ago. One reason Curiosity's science team studies craters is because the cratering process can unearth long-buried materials that may have better preserved organic molecules than rocks exposed to radiation at the surface. These molecules provide a window into the ancient Martian environment and how it could have supported microbial life billions of years ago, if any ever formed on the Red Planet.

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Contents Dawn of Twilight Clouds Cloud Mystery Mastcam's Partial View More About Curiosity News Media Contacts

Contents

[Dawn of Twilight Clouds](#)

[Cloud Mystery](#)

[Mastcam's Partial View](#)

[More About Curiosity](#)

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Erika PetersDec 06, 2024ArticleContentsLos hallazgosEl proceso de investigación de la NASA

Avances en el escudo térmicoEl 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo.Crédito de la foto: NASA / Rad SinyakRead this story in Englishhere.

Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I. Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares. “Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”. Los hallazgos

Los equipos técnicos adoptaron un enfoque metódico para comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra. Durante Artemis I, los ingenieros utilizaron la técnica de guiamento de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje. Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía

térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste) previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior. Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección. Crédito: NASA

Los equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra. Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas. Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera. El proceso de investigación de la NASA

Poco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas. “Nos

tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación”, dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. “El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entenderíamos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones”. El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado. Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para “ver” dentro del escudo térmico. Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado. Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna. Crédito: NASA Los ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques

completos, esto es, aplicando todas las pruebas en cada bloque de material. Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural. En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico.

Avances en el escudo térmico

Al saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans. Para obtener más información sobre

las campañas Artemis de la NASA, visita el sitio web (en inglés): <https://www.nasa.gov/artemis-fin>-Meira Bernstein / Rachel Kraft / María José Viñas Sede, Washington 202-358-1600 meira.b.bernstein@nasa.gov / rachel.h.kraft@nasa.gov / maria-jose.vinasgarcia@nasa.gov

Erika Peters Dec 06, 2024 Article Contents Los hallazgos El proceso de investigación de la NASA Avances en el escudo térmico

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Los hallazgos

El proceso de investigación de la NASA

Avances en el escudo térmico

El 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo. Crédito de la foto: NASA / Rad Sinyak [Read this story in English here](#). Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I. Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares. “Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”. Los hallazgos Los equipos técnicos adoptaron un enfoque metódico para

comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra. Durante Artemis I, los ingenieros utilizaron la técnica de guiado de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje. Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste) previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior. Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección. Crédito: NASA

Los equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra. Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los

límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas. Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera. El proceso de investigación de la NASA Poco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas. “Nos tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación”, dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. “El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entiendiéramos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones”. El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado. Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para “ver” dentro del escudo térmico. Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado. Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una

cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna. Crédito: NASA

Los ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques completos, esto es, aplicando todas las pruebas en cada bloque de material. Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural. En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión

estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico. Avances en el escudo térmico Al saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans. Para obtener más información sobre las campañas Artemis de la NASA, visita el sitio web (en inglés): <https://www.nasa.gov/artemis-fin> Meira Bernstein / Rachel Kraft / María José Viñas Sede, Washington 202-358-1600 meira.b.bernstein@nasa.gov / rachel.h.kraft@nasa.gov / maria-jose.vinasgarcia@nasa.gov

El 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo. Crédito de la foto: NASA / Rad Sinyak

El 28 de junio de 2024, la nave espacial Orion de Artemis II es retirada de la Celda de Ensamblaje Final y Pruebas del Sistema (FAST, por sus siglas en inglés) y colocada en la cámara de altitud oeste dentro del Edificio de Operaciones y Revisión del Centro Espacial Kennedy de la NASA en Florida. Dentro de la cámara de altitud, la nave espacial se sometió a una serie de pruebas que simulaban las condiciones de vacío del espacio profundo.

Crédito de la foto: NASA / Rad Sinyak

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Tras extensos análisis y pruebas, la NASA ha identificado la causa técnica de la pérdida imprevista de material carbonizado en el escudo térmico de la nave espacial Orion de Artemis I.

Los ingenieros determinaron que, cuando Orion regresaba de su misión sin tripulación alrededor de la Luna, los gases generados dentro del material ablativo exterior del escudo térmico, denominado Avcoat, no pudieron ventilarse y disiparse como estaba previsto. Esto permitió que se acumulara presión y se produjeran grietas, lo que causó que parte del material carbonizado se desprendiera en varios lugares.

“Nuestros primeros vuelos de Artemis son una campaña de prueba, y el vuelo de prueba de Artemis I nos dio la oportunidad de comprobar nuestros sistemas en el entorno del espacio profundo antes de incorporar a la tripulación en futuras misiones”, dijo Amit Kshatriya, administrador asociado adjunto de la Oficina del programa De la Luna a Marte, en la sede de la NASA en Washington. “La investigación sobre el escudo térmico ayudó a garantizar que comprendiéramos completamente la causa y la naturaleza del problema, así como el riesgo que les pedimos a nuestras tripulaciones que asuman cuando emprendan su viaje a la Luna”.

Los equipos técnicos adoptaron un enfoque metódico para comprender e identificar el origen del problema de pérdida de material carbonizado, incluyendo el muestreo detallado del escudo térmico de Artemis I, la revisión de las imágenes y los datos de los sensores de la nave espacial, y pruebas y análisis exhaustivos en tierra.

Durante Artemis I, los ingenieros utilizaron la técnica de guiado de reentrada atmosférica doble para el regreso de Orion a la Tierra. Esta técnica ofrece más flexibilidad ya que amplía el alcance del vuelo de Orion después del punto de reentrada para llevarlo hasta un lugar de amerizaje en el océano Pacífico. Con esta maniobra, Orion se sumergió en la parte superior de la atmósfera de la Tierra y utilizó la resistencia atmosférica para reducir su velocidad. A continuación, Orion utilizó la sustentación aerodinámica de la cápsula para rebotar y salir de nuevo de la atmósfera, para luego volver a entrar en el descenso final con paracaídas para su amerizaje.

Utilizando los datos de la respuesta del material Avcoat de Artemis I, el equipo de investigación pudo simular el entorno de la trayectoria de entrada de Artemis I —una parte clave para comprender la causa del problema— dentro de la instalación de chorro en arco del Centro de Investigación Ames de la NASA en California. Observaron que, durante el período entre las inmersiones en la atmósfera, las tasas de calentamiento disminuyeron y la energía térmica se acumuló dentro del material Avcoat del escudo térmico. Esto condujo a la acumulación de gases que forman parte del proceso de ablación (desgaste)

previsto. Debido a que el Avcoat no tenía “permeabilidad”, la presión interna se acumuló y produjo el agrietamiento y el desprendimiento desigual de la capa exterior.

Después de que la nave espacial Orion de la NASA fuera recuperada al término del vuelo de prueba Artemis I y transportada al Centro Espacial Kennedy de la NASA en Florida, su escudo térmico fue separado del módulo de la tripulación en el interior del Edificio de Operaciones y Revisión y rotado para su inspección. Crédito: NASA

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Crédito: NASA

Los equipos técnicos realizaron extensas pruebas en tierra para simular el fenómeno de rebote en la reentrada antes de la misión Artemis I. Sin embargo, hicieron pruebas a velocidades de calentamiento mucho más altas que las que la nave espacial experimentó durante su vuelo. Las altas velocidades de calentamiento puestas a prueba en tierra permitieron que el material carbonizado permeable se formara y se desgastara como estaba previsto, liberando la presión del gas. El calentamiento menos severo observado durante la reentrada real de Artemis I desaceleró el proceso de formación de material carbonizado, al tiempo que siguió creando gases en esta capa de material. La presión del gas se acumuló hasta el punto de agrietar el Avcoat y liberar partes de la capa carbonizada. Las mejoras recientes en la instalación de chorro en arco han permitido una reproducción más precisa de los entornos de vuelo registrados por Artemis I, de modo que este comportamiento de agrietamiento pudo demostrarse en pruebas en tierra.

Si bien Artemis I no estaba tripulado, los datos del vuelo mostraron que, si la tripulación hubiera estado a bordo, habría estado a salvo. Los datos de la temperatura de los sistemas del módulo de tripulación dentro de la cabina también estaban dentro de los límites y se mantenían estables, con temperaturas alrededor de los 24 grados centígrados (75 grados Fahrenheit). El desempeño del escudo térmico superó las expectativas.

Los ingenieros comprenden tanto el fenómeno material como el entorno con el que interactúan los materiales durante la entrada a la atmósfera. Al cambiar el material o el entorno, pueden predecir cómo responderá la nave espacial. Los equipos de la NASA acordaron por unanimidad que la agencia puede desarrollar un análisis de vuelo aceptable que mantenga a la tripulación segura utilizando el actual escudo térmico de Artemis II con cambios operativos para su entrada en la atmósfera.

Poco después de que los ingenieros de la NASA descubrieran las condiciones del escudo térmico de Artemis I, la agencia comenzó un extenso proceso de investigación, el cual contó con un equipo multidisciplinario de expertos en sistemas de protección térmica, aerotermodinámica, pruebas y análisis térmicos, análisis de estrés (fatiga de materiales), pruebas y análisis de materiales, y muchos otros campos técnicos relacionados. El Centro de Ingeniería y Seguridad de la NASA también participó para aportar su experiencia técnica, incluyendo evaluación no destructiva, análisis térmico y estructural, análisis de árbol de fallas y otros métodos de respaldo de las pruebas.

“Nos tomamos muy en serio nuestro proceso de investigación del escudo térmico, con la seguridad de la tripulación como la fuerza impulsora que mueve esta investigación”, dijo Howard Hu, gerente del Programa Orion del Centro Espacial Johnson de la NASA en Houston. “El proceso fue extenso. Le dimos al equipo el tiempo necesario para investigar todas las causas posibles, y trabajaron incansablemente para asegurarse de que entenderíamos el fenómeno y los pasos necesarios para mitigar este problema en futuras misiones”.

El escudo térmico de Artemis I estaba muy cargado de instrumentos para este vuelo, e incluía sensores de presión, extensómetros y termopares a diferentes profundidades del material ablativo. Los datos de estos instrumentos acrecentaron el análisis de muestras físicas, lo que permitió al equipo validar modelos informáticos, crear reconstrucciones de entornos, proporcionar perfiles de temperatura interna y dar información sobre el momento de la pérdida de material carbonizado.

Alrededor de 200 muestras de Avcoat fueron extraídas del escudo térmico de Artemis I en el Centro de Vuelo Espacial Marshall de la NASA en Alabama para su análisis e inspección. El equipo llevó a cabo una evaluación no destructiva para “ver” dentro del escudo térmico.

Uno de los hallazgos más importantes que arrojó el examen de estas muestras fue que algunas superficies en la zona del Avcoat permeable, las cuales habían sido identificadas antes del vuelo, no sufrieron agrietamiento ni pérdida de material carbonizado. Dado que estas superficies eran permeables al comienzo de la entrada en la atmósfera, los gases producidos por la ablación pudieron ventilarse adecuadamente, eliminando la acumulación de la presión, el agrietamiento y la pérdida de material carbonizado.

Un bloque de pruebas de Avcoat se somete a una prueba de pulso de calor dentro de una cámara de pruebas de chorro de plasma en arco en el Centro de Investigación Ames de la NASA en California. El elemento de prueba, configurado con secciones Avcoat permeables (superior) y no permeables (inferior) para su comparación, ayudó a confirmar la comprensión de la causa raíz de la pérdida de material Avcoat carbonizado que los

ingenieros vieron en la nave espacial Orion tras el vuelo de prueba Artemis I más allá de la Luna. Crédito: NASA

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Los ingenieros hicieron ocho campañas separadas de pruebas térmicas posteriores al vuelo para respaldar el análisis del origen de estas condiciones, y completaron 121 pruebas individuales. Estas pruebas fueron llevadas a cabo en instalaciones en diferentes lugares de Estados Unidos que cuentan con capacidades únicas, entre ellas: la Instalación de Calentamiento Aerodinámico en el Complejo de Chorro en Arco del centro Ames, para poner a prueba perfiles de calentamiento convectivo con diversos gases de prueba; el Laboratorio de Evaluación de Materiales Endurecidos por Láser en la Base de la Fuerza Aérea Patterson-Wright en Ohio, con el fin de poner a prueba perfiles de calentamiento radiativo y proporcionar radiografías en tiempo real; y la Instalación de Calentamiento por Interacción del centro Ames, para poner a prueba perfiles combinados de calentamiento convectivo y radiativo en el aire en bloques completos, esto es, aplicando todas las pruebas en cada bloque de material.

Los expertos en aerotermia también completaron dos campañas de pruebas en el túnel de viento hipersónico del Centro de Investigación Langley de la NASA en Virginia y en las instalaciones de pruebas aerodinámicas del CUBRC en Buffalo, Nueva York, para realizar pruebas con una diversidad de configuraciones de pérdida de material carbonizado, y mejorar y validar los modelos analíticos. También se realizaron pruebas de permeabilidad en el centro Kratos en Alabama, en la Universidad de Kentucky y en el centro Ames para caracterizar aún mejor el volumen elemental y la porosidad del Avcoat. La instalación de pruebas del centro de investigaciones Advanced Light Source, una instalación para usuarios científicos del Departamento de Energía de Estados Unidos en el Laboratorio Nacional Lawrence Berkeley, también fue utilizada por los ingenieros para examinar el comportamiento del calentamiento del Avcoat a nivel microestructural.

En la primavera de 2024, la NASA creó un equipo de revisión independiente que realizó una revisión exhaustiva del proceso de investigación, los hallazgos y los resultados de la agencia. La revisión independiente fue dirigida por Paul Hill, un exdirectivo de la NASA que se desempeñó como director principal de vuelo del transbordador espacial para el

programa Return to Flight (Regreso a los vuelos) después del accidente del Columbia, quien también dirigió la Dirección de Operaciones de Misiones de la NASA y es miembro actual del Panel Asesor de Seguridad Aeroespacial de la agencia. La revisión se llevó a cabo durante un período de tres meses a fin de evaluar las condiciones del escudo térmico posteriores al vuelo, los datos del entorno para la entrada a la atmósfera, la respuesta térmica del material ablativo y el avance de las investigaciones de la NASA. El equipo de revisión estuvo de acuerdo con los hallazgos de la NASA sobre la causa técnica del comportamiento físico del escudo térmico.

Al saber que la permeabilidad de Avcoat es un parámetro clave para evitar o minimizar la pérdida de material carbonizado, la NASA tiene la información correcta para garantizar la seguridad de la tripulación y mejorar el desempeño de los futuros escudos térmicos del programa Artemis. A lo largo de su historia, la NASA ha aprendido de cada uno de sus vuelos e incorporado mejoras en el hardware y las operaciones. Los datos recopilados a lo largo del vuelo de prueba de Artemis I han proporcionado a los ingenieros información valiosísima para guiar futuros diseños y refinamientos. Los datos de desempeño del vuelo de retorno lunar y un sólido programa de calificación de pruebas en tierra, mejorado después de la experiencia del vuelo de Artemis I, están respaldando las mejoras en la producción del escudo térmico de Orion. Los futuros escudos térmicos para el regreso de Orion en las misiones de alunizaje de Artemis están en producción para lograr una uniformidad y permeabilidad consistente. El programa de calificación se está completando actualmente, junto con la producción de bloques de Avcoat más permeables, en la Instalación de Ensamblaje Michoud de la NASA en Nueva Orleans.

Para obtener más información sobre las campañas Artemis de la NASA, visita el sitio web (en inglés):

<https://www.nasa.gov/artemis>

-fin-

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Emily Furfaro Feb 12, 2025 Article We've been talking about this for 2,000 years. Aristotle mentions it. And in our own time, scientists are designing experiments to figure out exactly what's going on. But there's no consensus yet. Here's what we do know. The atmosphere isn't magnifying the Moon. If anything, atmospheric refraction squashes it a little bit. And the Moon's not closer to us at the horizon. It's about 1.5 percent farther away. Also, it isn't just the Moon. Constellations look huge on the horizon, too. One popular idea is that this is a variation on the Ponzo illusion. Everything in our experience seems to shrink as it recedes toward the horizon — I mean clouds and planes and cars and ships. But the Moon doesn't do that. So our minds make up a story to reconcile this inconsistency. Somehow the Moon gets bigger when it's at the horizon. That's one popular hypothesis, but there are others. And we're still waiting for the experiment that will convince everyone that we understand this. So why does the Moon look larger on the horizon? We don't really know, but scientists are still trying to figure it out. [END VIDEO TRANSCRIPT] Full Episode List Full YouTube Playlist

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Emily Furfaro

Feb 12, 2025

We've been talking about this for 2,000 years. Aristotle mentions it. And in our own time, scientists are designing experiments to figure out exactly what's going on. But there's no consensus yet. Here's what we do know. The atmosphere isn't magnifying the Moon. If anything, atmospheric refraction squashes it a little bit. And the Moon's not closer to us at the horizon. It's about 1.5 percent farther away. Also, it isn't just the Moon. Constellations look huge on the horizon, too. One popular idea is that this is a variation on the Ponzo illusion. Everything in our experience seems to shrink as it recedes toward the horizon — I mean clouds and planes and cars and ships. But the Moon doesn't do that. So our minds make up a story to reconcile this inconsistency. Somehow the Moon gets bigger when it's at the horizon. That's one popular hypothesis, but there are others. And we're still waiting for

the experiment that will convince everyone that we understand this. So why does the Moon look larger on the horizon? We don't really know, but scientists are still trying to figure it out. [END VIDEO TRANSCRIPT] Full Episode List Full YouTube Playlist

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Ashley BalzerFeb 10, 2025ArticleThis artist's concept visualizes a super-Neptune world orbiting a low-mass star near the center of our Milky Way galaxy. Scientists recently discovered such a system that may break the current record for fastest exoplanet system, traveling at least 1.2 million miles per hour, or 540 kilometers per second.NASA/JPL-

Caltech/R. Hurt (Caltech-IPAC) Astronomers may have discovered a scrawny star bolting through the middle of our galaxy with a planet in tow. If confirmed, the pair sets a new record for the fastest-moving exoplanet system, nearly double our solar system's speed through the Milky Way. The planetary system is thought to move at least 1.2 million miles per hour, or 540 kilometers per second. "We think this is a so-called super-Neptune world orbiting a low-mass star at a distance that would lie between the orbits of Venus and Earth if it were in our solar system," said Sean Terry, a postdoctoral researcher at the University of Maryland, College Park and NASA's Goddard Space Flight Center in Greenbelt, Maryland. Since the star is so feeble, that's well outside its habitable zone. "If so, it will be the first planet ever found orbiting a hypervelocity star." A paper describing the results, led by Terry, was published in *The Astronomical Journal* on February 10.

A Star on the Move The pair of objects was first spotted indirectly in 2011 thanks to a chance alignment. A team of scientists combed through archived data from MOA (Microlensing Observations in Astrophysics) – a collaborative project focused on a microlensing survey conducted using the University of Canterbury Mount John Observatory in New Zealand — in search of light signals that betray the presence of exoplanets, or planets outside our solar system. Microlensing occurs because the presence of mass warps the fabric of space-time. Any time an intervening object appears to drift near a background star, light from the star curves as it travels through the warped space-time around the nearer object. If the alignment is especially close, the warping around the object can act like a natural lens, amplifying the background star's light. This artist's concept visualizes stars near the center of our Milky Way galaxy. Each has a colorful trail indicating its speed — the longer and redder the trail, the faster the star is moving. NASA scientists recently discovered a candidate for a particularly speedy star, visualized near the center of this image, with an orbiting planet. If confirmed, the pair sets a record for fastest known exoplanet system.

NASA/JPL-Caltech/R. Hurt (Caltech-IPAC) In this case, microlensing signals revealed a pair of celestial bodies. Scientists determined their relative masses (one is about 2,300 times heavier than the other), but their exact masses depend on how far away they are from Earth. It's sort of like how the magnification changes if you hold a magnifying glass over a page and move it up and down. "Determining the mass ratio is easy," said David Bennett, a senior research scientist at the University of Maryland, College Park and NASA Goddard, who co-authored the new paper and led the original study in 2011. "It's much more difficult to calculate their actual masses." The 2011 discovery team suspected the microlensed objects were either a star about 20 percent as massive as our Sun and a planet roughly 29 times heavier than Earth, or a nearer "rogue" planet about four times Jupiter's mass with a moon smaller than Earth. To figure out which explanation is more likely, astronomers searched through data from the Keck Observatory in Hawaii and ESA's (European Space Agency's) Gaia satellite. If the pair were a rogue planet and moon, they'd

be effectively invisible – dark objects lost in the inky void of space. But scientists might be able to identify the star if the alternative explanation were correct (though the orbiting planet would be much too faint to see). They found a strong suspect located about 24,000 light-years away, putting it within the Milky Way's galactic bulge — the central hub where stars are more densely packed. By comparing the star's location in 2011 and 2021, the team calculated its high speed. This Hubble Space Telescope image shows a bow shock around a very young star called LL Ori. Named for the crescent-shaped wave made by a ship as it moves through water, a bow shock can be created in space when two streams of gas collide. Scientists think a similar feature may be present around a newfound star that could be traveling at least 1.2 million miles per hour, or 540 kilometers per second. Traveling at such a high velocity in the galactic bulge (the central part of the galaxy) where gas is denser could generate a bow shock. NASA and The Hubble Heritage Team (STScI/AURA); Acknowledgment: C. R. O'Dell (Vanderbilt University) But that's just its 2D motion; if it's also moving toward or away from us, it must be moving even faster. Its true speed may even be high enough to exceed the galaxy's escape velocity of just over 1.3 million miles per hour, or about 600 kilometers per second. If so, the planetary system is destined to traverse intergalactic space many millions of years in the future. "To be certain the newly identified star is part of the system that caused the 2011 signal, we'd like to look again in another year and see if it moves the right amount and in the right direction to confirm it came from the point where we detected the signal," Bennett said. "If high-resolution observations show that the star just stays in the same position, then we can tell for sure that it is not part of the system that caused the signal," said Aparna Bhattacharya, a research scientist at the University of Maryland, College Park and NASA Goddard who co-authored the new paper. "That would mean the rogue planet and exomoon model is favored." NASA's upcoming Nancy Grace Roman Space Telescope will help us find out how common planets are around such speedy stars, and may offer clues to how these systems are accelerated. The mission will conduct a survey of the galactic bulge, pairing a large view of space with crisp resolution. "In this case we used MOA for its broad field of view and then followed up with Keck and Gaia for their sharper resolution, but thanks to Roman's powerful view and planned survey strategy, we won't need to rely on additional telescopes," Terry said. "Roman will do it all." Download additional images and video from NASA's Scientific Visualization Studio. By Ashley Balzer NASA's Goddard Space Flight Center, Greenbelt, Md. Media contact: Claire Andreoli NASA's Goddard Space Flight Center, Greenbelt, Md. 301-286-1940

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Noticias recientes
Más noticias
Artículo3 Min Read
Incendios arrasan Los Ángeles
Artículo10 Min Read
¿Qué son los arrecifes de coral?
4 minutos de lectura
Cómo Cancún se convirtió en un centro turístico
Artículo7 minutos de lectura
Entrevista con Juan Torres-Pérez, investigador principal de OCEANOS
Artículo8 minutos de lectura
Instrumento EMIT de la NASA explorará diversas interrogantes científicas en una extensión...
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Aeronautics
Exploration Systems
Science
Space Operations
Space Technology
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TITLE: Dark Energy

Explore the UniverseDark EnergyContentsA Brief HistoryIt All Started With CepheidsDiscovering an Expanding UniverseExpansion is Speeding Up, Supernovae ShowWhat Exactly is Dark Energy?Vacuum EnergyQuintessenceSpace WrinklesA Flaw in General RelativityThe FutureSome 13.8 billion years ago, the universe began with a rapid expansion we call the big bang. After this initial expansion, which lasted a fraction of a second, gravity started to slow the universe down. But the cosmos wouldn't stay this way. Nine billion years after the universe began, its expansion started to speed up, driven by an unknown force that scientists have nameddark energy.But what exactlyisdark energy?The short answer is: We don't know. But we do know that it exists, it's making the universe expand at an accelerating rate, and approximately 68.3 to 70% of the universe is dark energy.The history of the universe is outlined in this infographic.NASAA Brief HistoryIt All Started With CepheidsDark energy wasn't discovered until the late 1990s. But its origin in scientific study stretches all the way back to 1912 when American astronomer Henrietta Swan Leavitt made an important discovery using Cepheid variables, a class of stars whose brightness fluctuates with a regularity that depends on the star's brightness.All Cepheid stars with a certain period (a Cepheid's period is the time it takes to go from bright, to dim, and bright again) have the same absolute magnitude, or luminosity – the amount of light they put out. Leavitt measured these stars and proved that there is a relationship between their regular period of brightness and luminosity. Leavitt's findings made it possible for astronomers to use a star's period and luminosity to measure the distances between us and Cepheid stars in far-off galaxies (and our own Milky Way).Around this same time in history, astronomer Vesto Slipher observed spiral galaxies using his telescope's spectrograph, a device that splits light into the colors that make it up, much like the way a prism splits light into a rainbow. He used the spectrograph, a relatively recent invention at the time, to see the different wavelengths of light coming from the galaxies in different spectral lines. With his observations, Slipher was the first astronomer to observe how quickly the galaxy was moving away from us, called redshift, in distant galaxies. These observations would prove to be critical for many future scientific breakthroughs, including the discovery of dark energy.Redshift is a term used when astronomical objects are moving away from us and the light coming from those objects stretches out. Light behaves like a wave, and red light has the longest wavelength. So, the light coming from objects moving away from us has a longer wavelength, stretching to the "red end" of the electromagnetic spectrum.Discovering an Expanding UniverseThe discovery of galactic redshift, the period-luminosity relation of Cepheid variables, and a newfound ability to gauge a star or galaxy's distance eventually played a role in astronomers observing that galaxies were

getting farther away from us over time, which showed how the universe was expanding. In the years that followed, different scientists around the world started to put the pieces of an expanding universe together. In 1922, Russian scientist and mathematician Alexander Friedmann published a paper detailing multiple possibilities for the history of the universe. The paper, which was based on Albert Einstein's theory of general relativity published in 1917, included the possibility that the universe is expanding. In 1927, Belgian astronomer Georges Lemaître, who is said to have been unaware of Friedmann's work, published a paper also factoring in Einstein's theory of general relativity. And, while Einstein stated in his theory that the universe was static, Lemaître showed how the equations in Einstein's theory actually support the idea that the universe is not static but, in fact, is actually expanding. Astronomer Edwin Hubble confirmed that the universe was expanding in 1929 using observations made by his associate, astronomer Milton Humason. Humason measured the redshift of spiral galaxies. Hubble and Humason then studied Cepheid stars in those galaxies, using the stars to determine the distance of their galaxies (or nebulae, as they called them). They compared the distances of these galaxies to their redshift and tracked how the farther away an object is, the bigger its redshift and the faster it is moving away from us. The pair found that objects like galaxies are moving away from Earth faster the farther away they are, at upwards of hundreds of thousands of miles per second – an observation now known as Hubble's Law, or the Hubble-Lemaître law. The universe, they confirmed, is really expanding.

This composite image features one of the most complicated and dramatic collisions between galaxy clusters ever seen. Known officially as Abell 2744, this system has been dubbed Pandora's Cluster because of the wide variety of different structures found. Data from NASA's Chandra X-ray Observatory (red) show gas with temperatures of millions of degrees. In blue is a map showing the total mass concentration (mostly dark matter) based on data from NASA's Hubble Space Telescope, the VLT (Very Large Telescope), and the Subaru telescope. Optical data from Hubble and VLT also show the constituent galaxies of the clusters. Astronomers think at least four galaxy clusters coming from a variety of directions are involved with this collision. NASA, ESA, J. Merten (Institute for Theoretical Astrophysics, Heidelberg/Astronomical Observatory of Bologna), and D. Coe (STScI) Expansion is Speeding Up, Supernovae Show Scientists previously thought that the universe's expansion would likely be slowed down by gravity over time, an expectation backed by Einstein's theory of general relativity. But in 1998, everything changed when two different teams of astronomers observing far-off supernovae noticed that (at a certain redshift) the stellar explosions were dimmer than expected. These groups were led by astronomers Adam Riess, Saul Perlmutter, and Brian Schmidt. This trio won the 2011 Nobel Prize in Physics for this work. While dim supernovae might not seem like a major find, these astronomers were looking at Type 1a supernovae, which are known to have a certain level of luminosity. So they knew that there must be another factor making

these objects appear dimmer. Scientists can determine distance (and speed) using an objects' brightness, and dimmer objects are typically farther away (though surrounding dust and other factors can cause an object to dim). This led the scientists to conclude that these supernovae were just much farther away than they expected by looking at their redshifts. Using the objects' brightness, the researchers determined the distance of these supernovae. And using the spectrum, they were able to figure out the objects' redshift and, therefore, how fast they were moving away from us. They found that the supernovae were not as close as expected, meaning they had traveled farther away from us faster than anticipated. These observations led scientists to ultimately conclude that the universe itself must be expanding faster over time. While other possible explanations for these observations have been explored, astronomers studying even more distant supernovae or other cosmic phenomena in more recent years continued to gather evidence and build support for the idea that the universe is expanding faster over time, a phenomenon now called cosmic acceleration. But, as scientists built up a case for cosmic acceleration, they also asked: Why? What could be driving the universe to stretch out faster over time? Enter dark energy. What Exactly is Dark Energy? Right now, dark energy is just the name that astronomers gave to the mysterious "something" that is causing the universe to expand at an accelerated rate. Dark energy has been described by some as having the effect of a negative pressure that is pushing space outward. However, we don't know if dark energy has the effect of any type of force at all. There are many ideas floating around about what dark energy could possibly be. Here are four leading explanations for dark energy. Keep in mind that it's possible it's something else entirely.

Vacuum Energy Some scientists think that dark energy is a fundamental, ever-present background energy in space known as vacuum energy, which could be equal to the cosmological constant, a mathematical term in the equations of Einstein's theory of general relativity. Originally, the constant existed to counterbalance gravity, resulting in a static universe. But when Hubble confirmed that the universe was actually expanding, Einstein removed the constant, calling it "my biggest blunder," according to physicist George Gamow. But when it was later discovered that the universe's expansion was actually accelerating, some scientists suggested that there might actually be a non-zero value to the previously-discredited cosmological constant. They suggested that this additional force would be necessary to accelerate the expansion of the universe. This theorized that this mystery component could be attributed to something called "vacuum energy," which is a theoretical background energy permeating all of space. Space is never exactly empty. According to quantum field theory, there are virtual particles, or pairs of particles and antiparticles. It's thought that these virtual particles cancel each other out almost as soon as they crop up in the universe, and that this act of popping in and out of existence could be made possible by "vacuum energy" that fills the cosmos and pushes space outward. While this theory has been a popular topic of

discussion, scientists investigating this option have calculated how much vacuum energy there should theoretically be in space. They showed that there should either be so much vacuum energy that, at the very beginning, the universe would have expanded outwards so quickly and with so much force that no stars or galaxies could have formed, or... there should be absolutely none. This means that the amount of vacuum energy in the cosmos must be much smaller than it is in these predictions. However, this discrepancy has yet to be solved and has even earned the moniker "the cosmological constant problem."

Quintessence Some scientists think that dark energy could be a type of energy fluid or field that fills space, behaves in an opposite way to normal matter, and can vary in its amount and distribution throughout both time and space. This hypothesized version of dark energy has been nicknamed quintessence after the theoretical fifth element discussed by ancient Greek philosophers. It's even been suggested by some scientists that quintessence could be some combination of dark energy and dark matter, though the two are currently considered completely separate from one another. While the two are both major mysteries to scientists, dark matter is thought to make up about 85% of all matter in the universe.

Space Wrinkles Some scientists think that dark energy could be a sort of defect in the fabric of the universe itself; defects like cosmic strings, which are hypothetical one-dimensional "wrinkles" thought to have formed in the early universe.

A Flaw in General Relativity Some scientists think that dark energy isn't something physical that we can discover. Rather, they think there could be an issue with general relativity and Einstein's theory of gravity and how it works on the scale of the observable universe. Within this explanation, scientists think that it's possible to modify our understanding of gravity in a way that explains observations of the universe made without the need for dark energy. Einstein actually proposed such an idea in 1919 called unimodular gravity, a modified version of general relativity that scientists today think wouldn't require dark energy to make sense of the universe.

The Future Dark energy is one of the great mysteries of the universe. For decades, scientists have theorized about our expanding universe. Now, for the first time ever, we have tools powerful enough to put these theories to the test and really investigate the big question: "what is dark energy?"

NASA plays a critical role in the ESA (European Space Agency) mission **Euclid** (launched in 2023), which will make a 3D map of the universe to see how matter has been pulled apart by dark energy over time. This map will include observations of billions of galaxies found up to 10 billion light-years from Earth.

NASA's **Nancy Grace Roman Space Telescope**, set to launch by May 2027, is designed to investigate dark energy, among many other science topics, and will also create a 3D dark matter map. Roman's resolution will be as sharp as NASA's Hubble Space Telescope's, but with a field of view 100 times larger, allowing it to capture more expansive images of the universe. This will allow scientists to map how matter is structured and spread across the universe and explore how dark energy behaves and has changed over time. Roman will also

conduct an additional survey to detect Type Ia supernovae. In addition to NASA's missions and efforts, the Vera C. Rubin Observatory, supported by a large collaboration that includes the U.S. National Science Foundation, which is currently under construction in Chile, is also poised to support our growing understanding of dark energy. The ground-based observatory is expected to be operational in 2025. The combined efforts of Euclid, Roman, and Rubin will usher in a new "golden age" of cosmology, in which scientists will collect more detailed information than ever about the great mysteries of dark energy. Additionally, NASA's James Webb Space Telescope (launched in 2021), the world's most powerful and largest space telescope, aims to make contributions to several areas of research, and will contribute to studies of dark energy. NASA's SPHEREx (the Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer) mission, scheduled to launch no later than April 2025, aims to investigate the origins of the universe. Scientists expect that the data collected with SPHEREx, which will survey the entire sky in near-infrared light, including over 450 million galaxies, could help to further our understanding of dark energy. NASA also supports a citizen science project called Dark Energy Explorers, which enables anyone in the world, even those who have no scientific training, to help in the search for dark energy answers. *A brief note* Lastly, to clarify, dark energy is not the same as dark matter. Their main similarity is that we don't yet know what they are!

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NASA's Jet Propulsion Laboratory
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6 Min Read
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Article
27 Min Read
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Contents
A Brief History
It All Started With Cepheids
Discovering an Expanding Universe
Expansion is Speeding Up, Supernovae Show
What Exactly is Dark Energy?
Vacuum Energy
Quintessence
Space Wrinkles
A Flaw in General Relativity
The Future
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back to 1912 when American astronomer Henrietta Swan Leavitt made an important discovery using Cepheid variables, a class of stars whose brightness fluctuates with a regularity that depends on the star's brightness. All Cepheid stars with a certain period (a Cepheid's period is the time it takes to go from bright, to dim, and bright again) have the same absolute magnitude, or luminosity – the amount of light they put out. Leavitt measured these stars and proved that there is a relationship between their regular period of brightness and luminosity. Leavitt's findings made it possible for astronomers to use a star's period and luminosity to measure the distances between us and Cepheid stars in far-off galaxies (and our own Milky Way). Around this same time in history, astronomer Vesto Slipher observed spiral galaxies using his telescope's spectrograph, a device that splits light into the colors that make it up, much like the way a prism splits light into a rainbow. He used the spectrograph, a relatively recent invention at the time, to see the different wavelengths of light coming from the galaxies in different spectral lines. With his observations, Slipher was the first astronomer to observe how quickly the galaxy was moving away from us, called redshift, in distant galaxies. These observations would prove to be critical for many future scientific breakthroughs, including the discovery of dark energy. Redshift is a term used when astronomical objects are moving away from us and the light coming from those objects stretches out. Light behaves like a wave, and red light has the longest wavelength. So, the light coming from objects moving away from us has a longer wavelength, stretching to the "red end" of the electromagnetic spectrum.

Discovering an Expanding Universe

The discovery of galactic redshift, the period-luminosity relation of Cepheid variables, and a newfound ability to gauge a star or galaxy's distance eventually played a role in astronomers observing that galaxies were getting farther away from us over time, which showed how the universe was expanding. In the years that followed, different scientists around the world started to put the pieces of an expanding universe together. In 1922, Russian scientist and mathematician Alexander Friedmann published a paper detailing multiple possibilities for the history of the universe. The paper, which was based on Albert Einstein's theory of general relativity published in 1917, included the possibility that the universe is expanding. In 1927, Belgian astronomer Georges Lemaître, who is said to have been unaware of Friedmann's work, published a paper also factoring in Einstein's theory of general relativity. And, while Einstein stated in his theory that the universe was static, Lemaître showed how the equations in Einstein's theory actually support the idea that the universe is not static but, in fact, is actually expanding.

Astronomer Edwin Hubble confirmed that the universe was expanding in 1929 using observations made by his associate, astronomer Milton Humason. Humason measured the redshift of spiral galaxies. Hubble and Humason then studied Cepheid stars in those galaxies, using the stars to determine the distance of their galaxies (or nebulae, as they called them). They compared the distances of these galaxies to their redshift and tracked how the farther away an object

is, the bigger its redshift and the faster it is moving away from us. The pair found that objects like galaxies are moving away from Earth faster the farther away they are, at upwards of hundreds of thousands of miles per second – an observation now known as Hubble's Law, or the Hubble-Lemaître law. The universe, they confirmed, is really expanding. This composite image features one of the most complicated and dramatic collisions between galaxy clusters ever seen. Known officially as Abell 2744, this system has been dubbed Pandora's Cluster because of the wide variety of different structures found. Data from NASA's Chandra X-ray Observatory (red) show gas with temperatures of millions of degrees. In blue is a map showing the total mass concentration (mostly dark matter) based on data from NASA's Hubble Space Telescope, the VLT (Very Large Telescope), and the Subaru telescope. Optical data from Hubble and VLT also show the constituent galaxies of the clusters. Astronomers think at least four galaxy clusters coming from a variety of directions are involved with this collision. NASA, ESA, J. Merten (Institute for Theoretical Astrophysics, Heidelberg/Astronomical Observatory of Bologna), and D. Coe (STScI) Expansion is Speeding Up, Supernovae Show Scientists previously thought that the universe's expansion would likely be slowed down by gravity over time, an expectation backed by Einstein's theory of general relativity. But in 1998, everything changed when two different teams of astronomers observing far-off supernovae noticed that (at a certain redshift) the stellar explosions were dimmer than expected. These groups were led by astronomers Adam Riess, Saul Perlmutter, and Brian Schmidt. This trio won the 2011 Nobel Prize in Physics for this work. While dim supernovae might not seem like a major find, these astronomers were looking at Type 1a supernovae, which are known to have a certain level of luminosity. So they knew that there must be another factor making these objects appear dimmer. Scientists can determine distance (and speed) using an objects' brightness, and dimmer objects are typically farther away (though surrounding dust and other factors can cause an object to dim). This led the scientists to conclude that these supernovae were just much farther away than they expected by looking at their redshifts. Using the objects' brightness, the researchers determined the distance of these supernovae. And using the spectrum, they were able to figure out the objects' redshift and, therefore, how fast they were moving away from us. They found that the supernovae were not as close as expected, meaning they had traveled farther away from us faster than anticipated. These observations led scientists to ultimately conclude that the universe itself must be expanding faster over time. While other possible explanations for these observations have been explored, astronomers studying even more distant supernovae or other cosmic phenomena in more recent years continued to gather evidence and build support for the idea that the universe is expanding faster over time, a phenomenon now called cosmic acceleration. But, as scientists built up a case for cosmic acceleration, they also asked: Why? What could be driving the universe to stretch out faster over time? Enter dark

energy. What Exactly is Dark Energy? Right now, dark energy is just the name that astronomers gave to the mysterious "something" that is causing the universe to expand at an accelerated rate. Dark energy has been described by some as having the effect of a negative pressure that is pushing space outward. However, we don't know if dark energy has the effect of any type of force at all. There are many ideas floating around about what dark energy could possibly be. Here are four leading explanations for dark energy. Keep in mind that it's possible it's something else entirely.

Vacuum Energy Some scientists think that dark energy is a fundamental, ever-present background energy in space known as vacuum energy, which could be equal to the cosmological constant, a mathematical term in the equations of Einstein's theory of general relativity. Originally, the constant existed to counterbalance gravity, resulting in a static universe. But when Hubble confirmed that the universe was actually expanding, Einstein removed the constant, calling it "my biggest blunder," according to physicist George Gamow. But when it was later discovered that the universe's expansion was actually accelerating, some scientists suggested that there might actually be a non-zero value to the previously-discredited cosmological constant. They suggested that this additional force would be necessary to accelerate the expansion of the universe. This theorized that this mystery component could be attributed to something called "vacuum energy," which is a theoretical background energy permeating all of space. Space is never exactly empty. According to quantum field theory, there are virtual particles, or pairs of particles and antiparticles. It's thought that these virtual particles cancel each other out almost as soon as they crop up in the universe, and that this act of popping in and out of existence could be made possible by "vacuum energy" that fills the cosmos and pushes space outward. While this theory has been a popular topic of discussion, scientists investigating this option have calculated how much vacuum energy there should theoretically be in space. They showed that there should either be so much vacuum energy that, at the very beginning, the universe would have expanded outwards so quickly and with so much force that no stars or galaxies could have formed, or... there should be absolutely none. This means that the amount of vacuum energy in the cosmos must be much smaller than it is in these predictions. However, this discrepancy has yet to be solved and has even earned the moniker "the cosmological constant problem."

Quintessence Some scientists think that dark energy could be a type of energy fluid or field that fills space, behaves in an opposite way to normal matter, and can vary in its amount and distribution throughout both time and space. This hypothesized version of dark energy has been nicknamed quintessence after the theoretical fifth element discussed by ancient Greek philosophers. It's even been suggested by some scientists that quintessence could be some combination of dark energy and dark matter, though the two are currently considered completely separate from one another. While the two are both major mysteries to scientists, dark matter is thought to make up about 85% of all matter in

the universe.

Space Wrinkles

Some scientists think that dark energy could be a sort of defect in the fabric of the universe itself; defects like cosmic strings, which are hypothetical one-dimensional "wrinkles" thought to have formed in the early universe.

A Flaw in General Relativity

Some scientists think that dark energy isn't something physical that we can discover. Rather, they think there could be an issue with general relativity and Einstein's theory of gravity and how it works on the scale of the observable universe. Within this explanation, scientists think that it's possible to modify our understanding of gravity in a way that explains observations of the universe made without the need for dark energy. Einstein actually proposed such an idea in 1919 called unimodular gravity, a modified version of general relativity that scientists today think wouldn't require dark energy to make sense of the universe.

The Future

Dark energy is one of the great mysteries of the universe. For decades, scientists have theorized about our expanding universe. Now, for the first time ever, we have tools powerful enough to put these theories to the test and really investigate the big question: "what is dark energy?"

NASA plays a critical role in the ESA (European Space Agency) mission Euclid

(launched in 2023), which will make a 3D map of the universe to see how matter has been pulled apart by dark energy over time. This map will include observations of billions of galaxies found up to 10 billion light-years from Earth.

NASA's Nancy Grace Roman Space Telescope

, set to launch by May 2027, is designed to investigate dark energy, among many other science topics, and will also create a 3D dark matter map. Roman's resolution will be as sharp as NASA's Hubble Space Telescope's, but with a field of view 100 times larger, allowing it to capture more expansive images of the universe. This will allow scientists to map how matter is structured and spread across the universe and explore how dark energy behaves and has changed over time. Roman will also conduct an additional survey to detect Type Ia supernovae.

In addition to NASA's missions and efforts, the Vera C. Rubin Observatory, supported by a large collaboration that includes the U.S. National Science Foundation, which is currently under construction in Chile, is also poised to support our growing understanding of dark energy. The ground-based observatory is expected to be operational in 2025.

The combined efforts of Euclid, Roman, and Rubin will usher in a new "golden age" of cosmology, in which scientists will collect more detailed information than ever about the great mysteries of dark energy.

Additionally, NASA's James Webb Space Telescope (launched in 2021), the world's most powerful and largest space telescope, aims to make contributions to several areas of research, and will contribute to studies of dark energy.

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NASA also supports a citizen science project called Dark Energy Explorers,

which enables anyone in the world, even those who have no scientific training, to help in the search for dark energy answers.*A brief note*Lastly, to clarify, dark energy is not the same as dark matter. Their main similarity is that we don't yet know what they are!By Chelsea GohdNASA's Jet Propulsion Laboratory

ContentsA Brief HistoryIt All Started With CepheidsDiscovering an Expanding UniverseExpansion is Speeding Up, Supernovae ShowWhat Exactly is Dark Energy?Vacuum EnergyQuintessenceSpace WrinklesA Flaw in General RelativityThe Future

A Brief History

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Discovering an Expanding Universe

Expansion is Speeding Up, Supernovae Show

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Vacuum Energy

Quintessence

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these predictions. However, this discrepancy has yet to be solved and has even earned the moniker "the cosmological constant problem."

Some scientists think that dark energy could be a type of energy fluid or field that fills space, behaves in an opposite way to normal matter, and can vary in its amount and distribution throughout both time and space. This hypothesized version of dark energy has been nicknamed quintessence after the theoretical fifth element discussed by ancient Greek philosophers.

It's even been suggested by some scientists that quintessence could be some combination of dark energy and dark matter, though the two are currently considered completely separate from one another. While the two are both major mysteries to scientists, dark matter is thought to make up about 85% of all matter in the universe.

Some scientists think that dark energy could be a sort of defect in the fabric of the universe itself; defects like cosmic strings, which are hypothetical one-dimensional "wrinkles" thought to have formed in the early universe.

Some scientists think that dark energy isn't something physical that we can discover. Rather, they think there could be an issue with general relativity and Einstein's theory of gravity and how it works on the scale of the observable universe. Within this explanation, scientists think that it's possible to modify our understanding of gravity in a way that explains observations of the universe made without the need for dark energy. Einstein actually proposed such an idea in 1919 called unimodular gravity, a modified version of general relativity that scientists today think wouldn't require dark energy to make sense of the universe.

Dark energy is one of the great mysteries of the universe. For decades, scientists have theorized about our expanding universe. Now, for the first time ever, we have tools powerful enough to put these theories to the test and really investigate the big question: "what is dark energy?"

NASA plays a critical role in the ESA (European Space Agency) mission Euclid (launched in 2023), which will make a 3D map of the universe to see how matter has been pulled apart by dark energy over time. This map will include observations of billions of galaxies found up to 10 billion light-years from Earth.

NASA's Nancy Grace Roman Space Telescope, set to launch by May 2027, is designed to investigate dark energy, among many other science topics, and will also create a 3D dark matter map. Roman's resolution will be as sharp as NASA's Hubble Space Telescope's, but with a field of view 100 times larger, allowing it to capture more expansive images of the universe. This will allow scientists to map how matter is structured and spread across the

universe and explore how dark energy behaves and has changed over time. Roman will also conduct an additional survey to detect Type Ia supernovae.

In addition to NASA's missions and efforts, the Vera C. Rubin Observatory, supported by a large collaboration that includes the U.S. National Science Foundation, which is currently under construction in Chile, is also poised to support our growing understanding of dark energy. The ground-based observatory is expected to be operational in 2025.

The combined efforts of Euclid, Roman, and Rubin will usher in a new "golden age" of cosmology, in which scientists will collect more detailed information than ever about the great mysteries of dark energy. Additionally, NASA's James Webb Space Telescope (launched in 2021), the world's most powerful and largest space telescope, aims to make contributions to several areas of research, and will contribute to studies of dark energy.

NASA's SPHEREx (the Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer) mission, scheduled to launch no later than April 2025, aims to investigate the origins of the universe. Scientists expect that the data collected with SPHEREx, which will survey the entire sky in near-infrared light, including over 450 million galaxies, could help to further our understanding of dark energy.

NASA also supports a citizen science project called Dark Energy Explorers, which enables anyone in the world, even those who have no scientific training, to help in the search for dark energy answers.

A brief note

Lastly, to clarify, dark energy is not the same as dark matter. Their main similarity is that we don't yet know what they are!

By Chelsea Gohd NASA's Jet Propulsion Laboratory

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Through a drone flight test at NASA's Ames Research Center in California's Silicon Valley, researchers tested a blockchain-based system for protecting flight data. The system aims to keep air traffic management safe from disruption and protect data transferred between aircraft and ground stations from being intercepted or manipulated. For aviation and airspace operations to remain safe, users need to be able to trust that data is reliable and transparent. While current systems have been able to protect flight data systems, cyberthreats continue to evolve, requiring new approaches. NASA researchers found the blockchain-based system can safely transmit and store information in real time. Blockchain operates like a decentralized database — it does not rely on a single computer or centralized system. Instead, it shares information across a vast network, recording and verifying every change to a dataset. The system ensures the data stays safe, accurate, and trustworthy. Previous cybersecurity research focused on implementing a layered security architecture — using multiple physical and digital security measures to control system access. For this test, researchers took a different approach using blockchain to address potential threats. Using drones allowed the team to show that the blockchain framework could yield benefits across several priority areas in aviation development, including autonomous air traffic management, urban air mobility, and high-altitude aircraft.

Terrence D. Lewis (left), Kale Dunlap (center), and Aidan Jones monitor the flow of telemetry from both actual and simulated flights, ensuring the simulation and blockchain systems are processing and recording data accurately. Credit: NASA/Brandon Torres Navarette

This NASA research explored how blockchain can secure digital transactions between multiple systems and operators. The team used an open-source blockchain framework that allows trusted users real-time sharing and storage of critical data like aircraft operator registration information, flight plans, and telemetry. This framework restricts access to this data to trusted parties and approved users only. To further examine system resilience, the team introduced a set of cybersecurity tests designed to assess, improve, and reinforce security during operations in airspace environments. During an August flight at Ames, the team demonstrated these capabilities using an Alta-X drone with a custom-built software and hardware package that included a computer, radio, GPS system, and battery. The test simulated an environment with a drone flying in real-world conditions, complete with a separate ground control station and the blockchain and security infrastructure. The underlying blockchain framework and cybersecurity protocols can be extended to support high-altitude operations at 60,000 feet and higher and Urban Air Mobility operations, paving the way for a more secure, scalable, and trusted ecosystem. NASA researchers will continue to look at the data gathered during the test and apply what they've learned to future work. The testing will ultimately benefit U.S. aviation stakeholders looking for new

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Katrina LeeMolly KearnsDec 17, 2025ArticleContentsAmazonSES Space & DefenseSpaceXTelesatViasatEditor's Note:This story was updated on Dec. 18, 2025, to change SES to SES Space and Defense.NASA's commercial partners are actively demonstrating next-generation satellite relay capabilities for spaceflight missions, marking a significant step toward retiring the agency's Tracking and Data Relay Satellite (TDRS) system and adopting commercial services. The demonstrations – ranging from real-time spacecraft tracking during launch to transmitting mission commands and scientific data – are part of NASA's Communications Services Project, which is modernizing how the agency communicates with its science missions in near-Earth orbit.Managed by the agency'sSCaN(Space Communications and Navigation) Program, the project awarded funded Space Act Agreements in 2022 to six U.S. companies that are developing and testing commercial satellite communications services. The initiative supports NASA's broader strategy to retire the TDRS constellation and adopt a commercial-first model for near-Earth communications.“In collaboration with our commercial partners, SCaN is ushering in a new era of space exploration that will deliver powerful, forward-thinking solutions that reduce cost, increase adaptability, and increase mission success,” said Kevin Coggins, deputy associate administrator for SCaN at NASA Headquarters in Washington. “This work advances our commitment to expanding the low Earth orbit economy, and our commercial space partners are leading the charge through these groundbreaking demonstrations, proving for the first time that commercial satellite relay services can work for NASA missions.”This work advances our commitment to expanding the low Earth orbit economy, and our commercial space partners are leading the charge through these groundbreaking demonstrations.Kevin CogginsDeputy Associate Administrator for SCaNBy leveraging private-sector innovation, NASA aims to establish a more flexible, cost-effective, and scalable communications infrastructure for future science missions.AmazonAmazon Leo for Government, a subsidiary of Amazon, is demonstrating high-rate data exchanges over optical links using its satellite network in low Earth orbitAmazon has developed the hardware and software components necessary to support optical communication links within its Amazon Leo satellite relay network. Optical communications, also known as laser communications, use infrared light to transmit data at a higher rate compared to standard radio frequency systems. The Amazon Leo demonstrations, scheduled to begin in early 2026, will test the pointing, acquisition, and tracking capabilities of their optical communications systems to ensure the technology can accurately locate, lock onto, and stay connected with a mission as it travels through space.An image of the view from an Amazon Leo satellite overlooking the Earth.Credit: AmazonSES Space & DefenseSES Space & Defense is demonstrating high-rate data exchanges as well as tracking, telemetry, and command services using its O3b mPOWER satellite network in medium Earth orbit and its satellites in geosynchronous Earth orbit.Over the last two months, in collaboration with

Planet Labs, SES Space & Defense conducted multiple flight tests of its near-Earth space relay services. These demonstrations showcased uninterrupted, high-capacity connectivity between a Planet Labs satellite in low Earth orbit and SES communications satellites in geosynchronous Earth orbit and medium Earth orbit, demonstrating the ability to deliver real-time data relay across multiple orbits. SES Space & Defense has validated two relay services, one for low-rate tracking, telemetry, and command applications via its geostationary C-band satellites, and one for high-rate data applications over its O3b mPOWER Ka-band constellation. Additional flight demonstrations are planned for early 2026.

An artist's concept of SES Space and Defense's satellite relay demonstration. Credit: SES Space and Defense

SpaceX is demonstrating high-rate data exchanges over optical links using its Starlink network in low Earth orbit. Since 2024, SpaceX has completed multiple demonstrations of on-orbit optical communications services. During two human spaceflight missions, Polaris Dawn and Fram2, SpaceX leveraged the Starlink satellite constellation and an optical communications terminal installed on the Dragon spacecraft to demonstrate high-rate data relay services. Optical communications technology is not currently available through TDRS. By demonstrating optical relay services with multiple commercial partners, the agency is unlocking new capabilities for emerging missions.

An artist's concept of SpaceX's commercial satellite relay demonstration using the Dragon spacecraft and Starlink network. Credit: SpaceX

Telesat U.S. Services LLC, doing business as Telesat Government Solutions, is demonstrating high-rate data exchanges over optical links using its anticipated Telesat Lightspeed network in low Earth orbit. Development of the Telesat Lightspeed satellite network is currently underway, with satellite launches planned for late 2026. These satellites will use innovative technologies, like optical inter-satellite links and advanced onboard processing, to establish a global, mesh network in space. Software-defined networks aim to enable robust and reliable routing of traffic from a space-based or terrestrial terminal to its final destination autonomously. In 2027, Telesat plans to complete multiple demonstrations of space-to-space connectivity, including an optical data exchange from a Planet Labs spacecraft in low Earth orbit to the Telesat Lightspeed constellation. The data will then be routed over optical links before getting downlinked to a Telesat landing station on Earth, representing a full end-to-end capability.

An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network. Credit: Telesat

Viasat Viasat Inc. is demonstrating launch, tracking, telemetry, command, and high-data rate exchanges for launch vehicles and low Earth orbit operations. In May 2023, Viasat completed the acquisition of Inmarsat, the sixth satellite communications company to win a contract award from NASA, combining the resources of both companies to form a unified global communications provider. Viasat's space demonstrations will use its established satellite networks in geostationary orbit to validate three primary capabilities: launch telemetry over

the L-band radio frequency to track and monitor spacecraft during ascent; command and control over L-band to maintain continuous spacecraft custody and enable real-time operations; and high-speed Ka-band data relay to transfer large volumes of mission data through next-generation spacecraft terminals. Flights test began in November, when Viasat used its satellite network to successfully track the telemetry of Blue Origin's New Glenn rocket as it launched into low Earth orbit. Follow-on demonstrations are planned for 2026, including additional L-band launch services as well as high-capacity services over Ka-band frequencies. An artist's concept outlining Viasat's satellite relay capabilities. Credit: Viasat

Commercializing communications services for future near-Earth science missions enables NASA to focus resources on deep space missions to the Moon as part of the Artemis campaign, in preparation for future human missions to Mars. The agency will continue to work with these commercial partners to demonstrate next-generation services through 2027. By 2031, NASA plans to purchase satellite relay services for science missions from one or more U.S. satellite communications providers. To learn more about the decision to use commercial satellite relay services in low Earth orbit, visit: [Embracing Commercial Relay Services – NASA](#)

The Communications Services Project is managed by NASA's Glenn Research Center in Cleveland, under the direction of the Space Communications and Navigation Program within NASA's Space Operations Mission Directorate.

Katrina Lee Molly Kearns Dec 17, 2025 Article Contents Amazon SES Space & Defense SpaceX Telesat Viasat

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Managed by the agency's SCaN (Space Communications and Navigation) Program, the project awarded funded Space Act Agreements in 2022 to six U.S. companies that are developing and testing commercial satellite communications services. The initiative supports NASA's broader strategy to retire the TDRS constellation and adopt a commercial-first model for near-Earth communications.

“In collaboration with our commercial partners, SCaN is ushering in a new era of space exploration that will deliver powerful, forward-thinking solutions that reduce cost, increase adaptability, and increase mission success,” said Kevin Coggins, deputy associate administrator for SCaN at NASA Headquarters in Washington. “This work advances our commitment to expanding the low Earth orbit economy, and our commercial space partners are leading the charge through these groundbreaking demonstrations, proving for the first time that commercial satellite relay services can work for NASA missions.”

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By leveraging private-sector innovation, NASA aims to establish a more flexible, cost-effective, and scalable communications infrastructure for future science missions.

Amazon Leo for Government, a subsidiary of Amazon, is demonstrating high-rate data exchanges over optical links using its satellite network in low Earth orbit

Amazon has developed the hardware and software components necessary to support optical communication links within its Amazon Leo satellite relay network. Optical communications, also known as laser communications, use infrared light to transmit data at a higher rate compared to standard radio frequency systems. The Amazon Leo demonstrations, scheduled to begin in early 2026, will test the pointing, acquisition, and tracking capabilities of their optical communications systems to ensure the technology can accurately locate, lock onto, and stay connected with a mission as it travels through space.

An image of the view from an Amazon Leo satellite overlooking the Earth. Credit: Amazon

An image of the view from an Amazon Leo satellite overlooking the Earth.

Credit: Amazon

SES Space & Defense is demonstrating high-rate data exchanges as well as tracking, telemetry, and command services using its O3b mPOWER satellite network in medium Earth orbit and its satellites in geosynchronous Earth orbit.

Over the last two months, in collaboration with Planet Labs, SES Space & Defense conducted multiple flight tests of its near-Earth space relay services. These demonstrations showcased uninterrupted, high-capacity connectivity between a Planet Labs satellite in low Earth orbit and SES communications satellites in geosynchronous Earth orbit and medium Earth orbit, demonstrating the ability to deliver real-time data relay

across multiple orbits. SES Space & Defense has validated two relay services, one for low-rate tracking, telemetry, and command applications via its geostationary C-band satellites, and one for high-rate data applications over its O3b mPOWER Ka-band constellation. Additional flight demonstrations are planned for early 2026.

An artist's concept of SES Space and Defense's satellite relay demonstration. Credit: SES Space and Defense

SpaceX is demonstrating high-rate data exchanges over optical links using its Starlink network in low Earth orbit.

Since 2024, SpaceX has completed multiple demonstrations of on-orbit optical communications services. During two human spaceflight missions, Polaris Dawn and Fram2, SpaceX leveraged the Starlink satellite constellation and an optical communications terminal installed on the Dragon spacecraft to demonstrate high-rate data relay services. Optical communications technology is not currently available through TDRS. By demonstrating optical relay services with multiple commercial partners, the agency is unlocking new capabilities for emerging missions.

An artist's concept of SpaceX's commercial satellite relay demonstration using the Dragon spacecraft and Starlink network. Credit: SpaceX

Telesat U.S. Services LLC, doing business as Telesat Government Solutions, is demonstrating high-rate data exchanges over optical links using its anticipated Telesat Lightspeed network in low Earth orbit.

Development of the Telesat Lightspeed satellite network is currently underway, with satellite launches planned for late 2026. These satellites will use innovative technologies, like optical inter-satellite links and advanced onboard processing, to establish a global, mesh network in space. Software-defined networks aim to enable robust and reliable routing of traffic from a space-based or terrestrial terminal to its final destination autonomously. In 2027, Telesat plans to complete multiple demonstrations of space-to-space connectivity, including an optical data exchange from a Planet Labs spacecraft in low Earth orbit to the Telesat Lightspeed constellation. The data will then be routed over optical links before getting downlinked to a Telesat landing station on Earth, representing a full end-to-end capability.

An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network. Credit: Telesat

An artist illustration of Telesat's planned commercial relay demonstration using its Lightspeed satellite network.

Credit: Telesat

Viasat Inc. is demonstrating launch, tracking, telemetry, command, and high-data rate exchanges for launch vehicles and low Earth orbit operations. In May 2023, Viasat completed the acquisition of Inmarsat, the sixth satellite communications company to win a contract award from NASA, combining the resources of both companies to form a unified global communications provider.

Viasat's space demonstrations will use its established satellite networks in geostationary orbit to validate three primary capabilities: launch telemetry over the L-band radio frequency to track and monitor spacecraft during ascent; command and control over L-band to maintain continuous spacecraft custody and enable real-time operations; and high-speed Ka-band data relay to transfer large volumes of mission data through next-generation spacecraft terminals. Flights test began in November, when Viasat used its satellite network to successfully track the telemetry of Blue Origin's New Glenn rocket as it launched into low Earth orbit. Follow-on demonstrations are planned for 2026, including additional L-band launch services as well as high-capacity services over Ka-band frequencies.

An artist's concept outlining Viasat's satellite relay capabilities.Credit: Viasat

Commercializing communications services for future near-Earth science missions enables NASA to focus resources on deep space missions to the Moon as part of the Artemis campaign, in preparation for future human missions to Mars. The agency will continue to work with these commercial partners to demonstrate next-generation services through 2027. By 2031, NASA plans to purchase satellite relay services for science missions from one or more U.S. satellite communications providers.

To learn more about the decision to use commercial satellite relay services in low Earth orbit, visit:

Embracing Commercial Relay Services – NASA

The Communications Services Project is managed by NASA's Glenn Research Center in Cleveland, under the direction of the Space Communications and Navigation Program within NASA's Space Operations Mission Directorate.

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TITLE: NASA, SpaceX Continue Crew-2 Space Station Undocking

Brandi DeanNovember 6, 2021CategoriesNASA, SpaceX Continue Crew-2 Space Station UndockingAstronauts (from left) Thomas Pesquet, Megan McArthur, Shane Kimbrough and Akihiko Hoshide talk to journalists on Earth before their return to Earth aboard the SpaceX Crew Dragon Endeavour.NASA's SpaceX Crew-2 mission continues to target a return to Earth no earlier than 7:14 a.m. EST Monday, Nov. 8, with a splashdown off the coast of Florida. The Crew Dragon spacecraft, named Endeavour, is scheduled to undock from

the International Space Station at 12:04 p.m. Sunday, Nov. 7, to begin the journey home. Mission teams completed an undocking weather review on Saturday and are 'go' to proceed to the next weather review about six hours prior to the scheduled undocking time. Winds remain the main watch item for the return of the mission. NASA will provide coverage of the mission on NASA Television, the NASA app, and the agency's website. NASA astronauts Shane Kimbrough and Megan McArthur, JAXA (Japan Aerospace Exploration Agency) astronaut Aki Hoshide, and ESA (European Space Agency) astronaut Thomas Pesquet will complete 199 days in space at the conclusion of their mission. The spacecraft also will return to Earth with about 530 pounds of hardware and scientific investigations. Endeavour now will forego a fly around maneuver to photograph the exterior of the International Space Station to allow for additional alternate splashdown locations off the coast of Florida. NASA and SpaceX also have a backup undocking and splashdown opportunity available Monday, Nov. 8, if weather conditions are not favorable for the primary opportunity. The NASA and SpaceX teams will determine a primary and alternate splashdown location from the seven possible landing locations prior to return, factoring in weather, crew rescue, and recovery operations. Additional decision milestones take place prior to undocking, during free flight, and before Crew Dragon performs the deorbit burn. NASA and SpaceX closely coordinate with the U.S. Coast Guard to establish a safety zone around the expected splashdown location to ensure safety for the public and for those involved in the recovery operations, as well as the crew aboard the returning spacecraft. With Crew-2 splashdown Monday, Nov. 8, NASA's SpaceX Crew-3 mission is targeting launch no earlier than 9:03 p.m. Wednesday, Nov. 10, on a SpaceX Falcon 9 rocket from Launch Complex 39A at NASA's Kennedy Space Center in Florida. For this launch opportunity, the Crew Dragon Endurance is scheduled to dock to the space station around 7:10 p.m. Thursday, Nov. 11. NASA's SpaceX Crew-2 return coverage is as follows: Sunday, Nov. 7: 9:45 a.m. EST– Coverage begins for 10:15 a.m. hatch closure 11:45 a.m. EST– Coverage begins for 12:04 p.m. undocking (NASA will provide continuous coverage from undocking to splashdown) Monday, Nov. 8: 7:14 a.m. EST– Splashdown Crew-2 is the second of six NASA and SpaceX crewed missions to fly as part of the agency's Commercial Crew Program, which is working with the U.S. aerospace industry to launch astronauts on American rockets and spacecraft from American soil. NASA's Commercial Crew Program has delivered on its goal of safe, reliable, and cost-effective transportation to and from the International Space Station from the U.S. through a partnership with American private industry. This partnership is changing the arc of human spaceflight history by opening access to low-Earth orbit and the International Space Station to more people, more science, and more commercial opportunities. The space station remains the springboard to NASA's next great leap in space exploration, including future missions to the Moon and, eventually, to Mars. Brandi Dean November 6, 2021 Categories

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NASA's SpaceX Crew-2 mission continues to target a return to Earth no earlier than 7:14 a.m. EST Monday, Nov. 8, with a splashdown off the coast of Florida. The Crew Dragon spacecraft, named Endeavour, is scheduled to undock from theInternational Space Stationat 12:04 p.m. Sunday, Nov. 7, to begin the journey home. Mission teams completed an undocking weather review on Saturday and are 'go' to proceed to the next weather review about six hours prior to the scheduled undocking time. Winds remain the main watch item for the return of the mission.

NASA will provide coverage of the mission on NASA Television, theNASA app, and the agency'swebsite.

NASA astronautsShane KimbroughandMegan McArthur, JAXA (Japan Aerospace Exploration Agency) astronautAki Hoshide, and ESA (European Space Agency) astronautThomas Pesquetwill complete 199 days in space at the conclusion of their mission. The spacecraft also will return to Earth with about 530 pounds of hardware and scientific investigations.

Endeavour now will forego a fly around maneuver to photograph the exterior of the International Space Station to allow for additional alternate splashdown locations off the coast of Florida.

NASA and SpaceX also have a backup undocking and splashdown opportunity available Monday, Nov. 8, if weather conditions are not favorable for the primary opportunity.

The NASA and SpaceX teams will determine a primary and alternate splashdown location from the seven possible landing locations prior to return, factoring in weather, crew rescue, and recovery operations. Additional decision milestones take place prior to undocking, during free flight, and before Crew Dragon performs the deorbit burn.

NASA and SpaceX closely coordinate with the U.S. Coast Guard to establish a safety zone around the expected splashdown location to ensure safety for the public and for those involved in the recovery operations, as well as the crew aboard the returning spacecraft.

With Crew-2 splashdown Monday, Nov. 8, NASA's SpaceX Crew-3 mission is targeting launch no earlier than 9:03 p.m. Wednesday, Nov. 10, on a SpaceX Falcon 9 rocket from Launch Complex 39A at NASA's Kennedy Space Center in Florida. For this launch opportunity, the Crew Dragon Endurance is scheduled to dock to the space station around 7:10 p.m. Thursday, Nov. 11.

NASA's SpaceX Crew-2 return coverage is as follows:

Sunday, Nov. 79:45 a.m. EST– Coverage begins for 10:15 a.m. hatch closure

11:45 a.m. EST– Coverage begins for 12:04 p.m. undocking (NASA will provide continuous coverage from undocking to splashdown)

Monday, Nov. 87:14 a.m. EST– Splashdown

Crew-2 is the second of six NASA and SpaceX crewed missions to fly as part of the agency's Commercial Crew Program, which is working with the U.S. aerospace industry to launch astronauts on American rockets and spacecraft from American soil.

NASA's Commercial Crew Program has delivered on its goal of safe, reliable, and cost-effective transportation to and from the International Space Station from the U.S. through a partnership with American private industry. This partnership is changing the arc of human spaceflight history by opening access to low-Earth orbit and the International Space Station to more people, more science, and more commercial opportunities. The space station remains the springboard to NASA's next great leap in space exploration, including future missions to the Moon and, eventually, to Mars.

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TITLE: What Is Climate Change?

NASA Kids ScienceWhat Is Climate Change?Climate change describes a change in the average conditions over a long period of time. These include changes to things like temperature or rainfall. Scientists have been observing Earth for a long time. They use satellites and other instruments to collect information about Earth. This information shows that the planet is warming. And that warming is impacting Earth's climate.Weather vs. ClimateNASA/JPL-CaltechWeatherdescribes the conditions outside right now in a specific place. For example, if you see that it's raining outside right now, that's a way to describe today's weather. Rain, snow, wind, hurricanes, tornadoes — these are all weather events.Climate, on the other hand, is more than just one or two rainy days. Climate describes the weather conditions that are expected in a region.Is it usually rainy or usually dry? Is it typically hot or typically cold? A region's climate is determined by observing its weather over a period of many years. Generally 30 years or more is used.So, for example, a week of rainyweatherwouldn't change the fact that Phoenix typically has a dry, desertclimate. Even though it's rainy right now, we still expect Phoenix to be dry because that's what is usually the case.Want to know more about the difference between weather and climate? Take a look at this video.What Is Climate Change?Climate change describes a change in the average conditions over a long period of time. These include changes to things like temperature or rainfall. For example, 20,000 years ago during the last ice age, much of the United States was covered in glaciers. That would be a much colder climate than what we are used to today.When changes happen that impact the entire planet, it is called global climate change. These changes include warming global temperatures and shifts in precipitation. Plus, when the planet warms, it affects other parts of the Earth, such as:Rising sea levelsShrinking mountain glaciersIce melting at a faster rate than usual in Greenland, Antarctica, and the ArcticChanges in flower and plant blooming times.Earth's climate has constantly been changing — even long before humans came into the picture. However, scientists have observed a shift in the speed at which Earth's climate is changing. Previously, it has taken thousands of years for Earth's climate to switch between more ice and less ice. But now, observations show that Earth's temperature is increasing much more

quickly. How Do Scientists Know Earth's Climate is Changing? Scientists use instruments in space, on airplanes, and the ground to track Earth. These can be things like weather stations or even space lasers that measure the height of land and ice. When collected over time, these measurements help scientists understand what changes are happening. Watch this video to learn how scientists can use new and old pictures to keep track of Earth's changes over time! To better understand how changes today compare to those in the past, scientists have to dig a little deeper. Earth keeps its own climate record for us through things that have been around a really long time. These are things like the bottom of the ocean or the Greenland and Antarctic ice sheets. Over time, these places build layers that capture information from when they formed. Scientists will drill down and take cylinder-shaped samples. These are known as sediment cores and ice cores. Sediment cores come from the bottoms of lakes or the ocean floor. Ice cores are from deep — sometimes miles deep — below the surface of the ice in places like Antarctica. The layers in an ice core are frozen solid. These layers of ice give clues about every year of Earth's history back to the time the deepest layer was formed. The ice contains bubbles of the air from each year. Scientists analyze the bubbles in each layer to find out how much of the air contained certain gases, like carbon dioxide. They can also learn about temperature and if it was a wet or dry year.

NASA/JPL-Caltech How Is Earth's Climate Changing Right Now? Some parts of Earth are warming faster than others. But on average, global air temperatures have gone up more than 2 degrees Fahrenheit in the past 100 years. In fact, the past five years have been the warmest five years in centuries. Alaska's Muir glacier in August 1941 and August 2004. The glacier has retreated significantly in the 63 years between these two photos. USGS As Earth continues to warm, it impacts Earth's weather and climate. Scientists have already observed changes in several ways. The warmer temperatures have led to more intense and more common heat waves. Melting ice sheets and glaciers and ocean warming have led to sea level rise along the coast. And changes in rainfall have made droughts and floods more intense.

What Causes Climate Change? There are lots of factors that can contribute to changes in Earth's climate. However, scientists agree that Earth has been getting warmer over the past 100 years. They found that this warming is due to an increase in greenhouse gases in the atmosphere. Greenhouse gases in Earth's atmosphere trap some of the heat from sunlight. This is called the greenhouse effect. These gases keep Earth warm like the glass in a greenhouse keeps plants warm. Things we do, such as burning fuel to power factories, cars and buses, are changing the natural greenhouse. These changes cause the atmosphere to trap more heat than it used to, leading to a warmer Earth.

How Do Scientists Use Satellites to Understand Climate Change? NASA's Earth observing satellites collect information about our planet. These include the atmosphere, water and land. By looking at this information, scientists can see how Earth is changing over time. This helps us understand how Earth is connected through

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Weather vs. Climate

NASA/JPL-Caltech

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What Is Climate Change? Climate change is a long-term change in the average weather patterns that have come to define Earth's local, regional and global climates. These changes have a broad range of observed effects that are synonymous with the term. Changes observed in Earth's climate since the mid-20th century are driven by human activities, particularly fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere, raising Earth's average surface temperature. Natural processes, which have been overwhelmed by human activities, can also contribute to climate change, including internal variability (e.g., cyclical ocean patterns like El Niño, La Niña and the Pacific Decadal Oscillation) and external forcings (e.g., volcanic activity, changes in the Sun's energy output, variations in Earth's orbit). Scientists use observations from the ground, air, and space, along with computer models, to monitor and study past, present, and future climate change. Climate data records provide evidence of climate change key indicators, such as global land and ocean temperature increases; rising sea levels; ice loss at Earth's poles and in mountain glaciers; frequency and severity changes in extreme weather such as hurricanes, heatwaves, wildfires, droughts, floods, and precipitation; and cloud and vegetation cover changes. "Climate change" and "global warming" are often used interchangeably but have distinct meanings. Similarly, the terms "weather" and "climate" are sometimes confused, though they refer to events with broadly different spatial- and timescales.

What Is Global Warming? This graph illustrates the change in global surface temperature relative to 1951-1980 average temperatures, with the year 2020 statistically tying with 2016 for hottest on record (Source: NASA's Goddard Institute for Space Studies). Learn more about global surface temperature [here](#). [NASA/JPL-Caltech](#)

Global warming is the long-term heating of Earth's surface observed since the pre-industrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere. This term is not interchangeable with the term "climate change." Since the pre-industrial period, human activities are estimated to have increased Earth's global average temperature by about 1 degree Celsius (1.8 degrees Fahrenheit), a number that is currently increasing by more than 0.2 degrees Celsius (0.36 degrees Fahrenheit) per decade. The current warming trend is unequivocally the result of human activity since the 1950s and is proceeding at an unprecedented rate over millennia.

Weather vs. Climate "If you don't like the weather in New England, just wait a few minutes."
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Weather refers to atmospheric conditions that occur locally over short periods of time—from minutes to hours or days. Familiar examples include rain, snow, clouds, winds, floods, or thunderstorms. Climate, on the other hand, refers to the long-term (usually at least 30 years) regional or even global average of temperature, humidity, and rainfall patterns over seasons, years, or decades.

Find Out More: A Guide to NASA's Global Climate Change Website This website provides a high-level overview of some of the known causes, effects and indications of

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Other NASA Resources

Goddard Scientific Visualization Studio.An extensive collection of animated climate change and Earth science visualizations.

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Climate Change From the unique vantage point in space, NASA collects critical long-term observations of our changing planet. NASA applies ingenuity and expertise gained from decades of planetary and deep-space exploration to the study of our home planet. The Earth Science Division operates more than 20 satellites in orbit, sponsors hundreds of research programs and studies, and funds opportunities to put data to use for societal needs. We develop new ways to observe the oceans, land cover, ice, atmosphere, and life, and we measure how changes in one drive changes in others over the short and long term. While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world. Learn more about how NASA Earth science helps Americans respond to challenges.

Wildfires and Climate Change Earth's warming climate is amplifying wildland fire activity, particularly in northern and temperate forests. When fires ignite the landscape, NASA's satellites and instruments can detect and track them. This information helps communities and land managers around the world prepare for and respond to fires, and also provides a rich data source to help scientists better understand this growing risk. Learn More about Wildfires and Climate Change

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atmosphere. We already see effects scientists predicted, such as the loss of sea ice, melting glaciers and ice sheets, sea level rise, and more intense heat waves. Scientists predict global temperature increases from human-made greenhouse gases will continue. Severe weather damage will also increase and intensify. [Read More](#) [Images of Change](#) [Before-and-after images of Earth](#). [Learn More](#) [about Images of Change](#) [Climate Change Resources](#) [Climate Time Machine](#) [Climate change in recent history](#) [Global Ice Viewer](#) [Climate change's impact on ice](#) [Multimedia](#) [Vast library of images, videos, graphics, and more](#) [En español](#) [Creciente biblioteca de recursos en español](#) [Climate Data Resources](#) [Sea Level Change Observations from Space](#) Established in 2014, this NASA-sponsored team works to improve the understanding of regional relative sea-level change on a range of timescales. They work with partners to translate the latest science and research into actionable information and to communicate how impacts are increasing at the coast. [NASA Center for Climate Simulation \(NCCS\)](#) NCCS provides high performance computing for NASA-sponsored scientists and engineers. The purpose is to enhance NASA capabilities in Earth science, with an emphasis on weather and climate prediction, and to enable future scientific discoveries that will benefit humankind. [GISS Surface Temperature Analysis \(GISTEMP\)](#) NASA's Goddard Institute of Space Studies assembles one of the world's most trusted global temperature records, using a combination of surface air temperature data acquired by tens of thousands of meteorological stations, as well as sea surface temperature data from ship- and buoy-based instruments. [NASA Earth Exchange \(NEX\)](#) NEX combines state-of-the-art supercomputing, Earth system modeling, and NASA remote sensing data feeds to deliver a work environment for exploring and analyzing terabyte- to petabyte-scale datasets covering large regions, continents or the globe. [Global Modeling and Assimilation Office \(GMAO\)](#) GMAO members perform research, develop models and assimilation systems, and produce quasi-operational products in support of NASA's missions. The Goddard Earth Observing System" (GEOS) family of models is used for applications across a wide range of spatial scales, from kilometers to many tens of kilometers. [World of Change](#) NASA Earth Observatory has produced a collection of image series that show some features of Earth that have changed over time due to both natural and human-induced causes. [Learn More](#) This false-color image, captured by the Landsat 8 satellite, shows Alaska's Columbia Glacier and the surrounding landscape following the more than 20 kilometer retreat of the glacier between 1986 and 2019. Credit: NASA Earth Observatory

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TITLE: Ultra-Efficient Aviation

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TITLE: What Is the Kuiper Belt?

kuiper-beltWhat Is the Kuiper Belt?The Short Answer:The Kuiper Belt is a ring of icy bodies just outside of Neptune's orbit. Pluto is the most famous Kuiper Belt Object.The Sun is at the center of our solar system. It is orbited by eight planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. But what's past Neptune?Just outside of Neptune's orbit is a ring of icy bodies. We call it theKuiper Belt. It's pronouncedky-purr.This is where you'll find dwarf planet Pluto. It's the most famous of the objects floating in the Kuiper Belt, which are also calledKuiper Belt Objects, or KBOs.Why is it named Kuiper?The Kuiper Belt is named after a scientist namedGerard Kuiper. In 1951 he had the idea that a belt of icy bodies might have existed beyond Neptune when the solar system formed. He was trying to explain where comets with small orbits came from. No one had seen anything out there yet because it's hard to see small comets past Neptune even with the best telescopes. But even without being able to see it with his own eyes, Kuiper made a prediction. And it turned out to be right.What's out there?There are bits of rock and ice, comets, and dwarf planets. Besides Pluto, two other interesting Kuiper Belt Objects are Eris and Haumea.ErisEris is a Kuiper Belt Object a little smaller than Pluto. It is so far away it takes 557 years to orbit the Sun. Eris has a small moon named Dysnomia.Learn more about ErisHaumeaAnother interesting Kuiper Belt Object is Haumea. It's shaped like a squashed American football about 1,200 miles (1,931 km) long. It spins end over end every few hours. The strange shape and rotation were caused by a collision with an object about half its size. When Haumea and this other object smashed into each other, the impact blasted away big pieces of ice and sent Haumea spinning.Haumea also has two moons named Hi'iaka and Namaka.Learn more about HaumeaArrokothMove the slider to the left to see a real image of Arrokoth from the New Horizons spacecraft. Credit: NASA/Johns Hopkins University Applied Physics Laboratory/Southwest Research InstituteA billion miles past Pluto there is

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The Kuiper Belt is still a very mysterious place, and we have a lot to learn about it. New Horizons will keep exploring the Kuiper Belt and sending us more information about it. Learn more about the Kuiper Belt

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Arrokoth

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anywhereNASA's Division of Biological and Physical Sciences (BPS) uses the spaceflight environment to study phenomena in ways that cannot be done on Earth.In 2020, NASA transferred administrative oversight of NASA's biological and physical sciences research from the Space Life and Physical Sciences Research and Applications (SLPSRA) Division in the Human Exploration and Operations Mission Directorate into the Science Mission Directorate (SMD), a NASA directorate organization that pursues outstanding science and understands its potential application to future exploration missions.The vision for BPS, in keeping with theNASA Strategic PlanandDecadal Survey, is: We lead the space life and physical sciences research community to enable space exploration and benefit life on Earth.The mission of BPS is two-pronged:Pioneer scientific discovery in and beyond low Earth orbit to drive advances in science, technology, and space exploration to enhance knowledge, education, innovation, and economic vitalityEnable human spaceflight exploration to expand the frontiers of knowledge, capability, and opportunity in spaceExecution of this mission requires both scientific research and technology development.NASA's Initial Response to the 2023-2032 Decadal SurveyBiological and Physical Sciences StoriesExplore All Biological & Physical Sciences StoriesArticle4 Min ReadNASA Back for Seconds with New Food System Design ChallengeArticle3 Min Read10 Years of Students Helping NASA Grow Space Food with Growing Beyond Earth8 Min Read25 Years of Scientific Discovery Aboard the International Space StationArticle2 Min ReadOSDR and PSI Unveil New Consolidated WebsiteArticle3 Min ReadJoe A. Adam Presents Ring Sheared Drop (RSD) Research at 2025 ISSRDCArticleLatest ContentStay up-to-date with the latest content from NASA as we explore the universe and discover more about our home planet.Growing Beyond Earth®2 min readMiddle and high school teachers in the U.S.: your students can test plants for use in space missions. Kid friendly!TopicOSDR AWG4 min readHelp astronauts and life thrive in deep space by investigating data on space biology and space health. Laptop required. Knowledge...TopicNASA Back for Seconds with New Food System Design Challenge4 min readNASA is getting ready to send four astronauts around the Moon with Artemis II, laying the foundation for sustainable missions...ArticleLast updated: November 2024BPS administers NASA's:Space Biology Program, which solicits and conducts research to use the space environment to advance our knowledge of how gravity affects the design and function of living organisms, and to understand how biological systems accommodate to spaceflight environments.Physical Sciences Program, which solicits and conducts research using the space environment as a tool to provide transformational insights in physics and engineering science, and to understand how physical systems respond to spaceflight environments, particularly weightlessness and the partial gravity of planetary bodies.Commercially Enabled Rapid Space Science project (CERISS), which will develop transformative research capabilities with commercial space industry to dramatically increase the pace of research.BPS

partners with the research community and a wide range of organizations to accomplish its mission. Grants to academic, commercial, and government laboratories are the core of BPS's research and technology development efforts. All BPS solicitations are issued through NSPIRES and all awards are described and results tracked in the Task Book: Biological and Physical Sciences Division and Human Research Program. Partnerships with other NASA organizations, other government agencies, industry and international partners provide access to a broad range of experimental platforms, involvement of a diverse set of experts, and translation of results to a wide community. BPS research and technology development is conducted on a wide range of experimental platforms including ground-based analogs for spaceflight, drop towers and aircraft that provide seconds of weightlessness, to free-flyer satellites. The International Space Station (ISS) is a critically important United States facility that provides researchers the ability to conduct long-duration experiments in low Earth orbit. ISS allows continuous and interactive research similar to Earth-based laboratories, enabling scientists to pursue innovations and discoveries not currently achievable by other means. BPS is also planning research and technology development to enable human spaceflight and scientific discovery beyond low Earth orbit. BPS strives for broad involvement of the research and technology development communities in the formulation and dissemination of its work. BPS typically solicits for teams of investigators for spaceflight investigations to maximize the scientific benefit derived from the experiments. BPS also emphasizes the importance of archiving data, metadata, computational tools, and samples after spaceflight experiments to enable future experiments. The NASA Open Science Data Repository (which combines both GeneLab and the Ames Life Sciences Data Archive) and the Physical Sciences Informatics database are the principal repositories.

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What We Study

Space Biology Overview

The main objective of Space Biology research is to build a better understanding of how spaceflight affects living systems in spacecraft such as the International Space Station (ISS), or in ground-based experiments, and to prepare for future human exploration missions far from Earth.

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The Physical Science Research Program makes contributions in two distinct areas: fundamental research, which investigates physical phenomena in the absence of gravity and fundamental laws of the universe, and second, applied research, which contributes to the basic understanding underlying space exploration technologies.

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NASA Science Editorial Team Jan 23, 2025 Article NASA's Mars rover Curiosity acquired this image using its Left Navigation Camera on sol 4429 — or Martian day 4,429 of the Mars Science Laboratory mission — on Jan. 20, 2025, at 23:05:33 UTC. NASA/JPL-Caltech Earth planning date: Tuesday, Jan. 21, 2025 Over the weekend Curiosity completed a 38-meter drive (about 125 feet), and the rover is now perched on the rim of a crater named "Rustic Canyon." Today we planned contact science to search for any unique textures or compositions associated with the crater, and a lot of imaging to assess the morphology of the crater to learn about degradation rates and processes. Today is a three-sol plan on a Tuesday, to account for the U.S. holiday yesterday, Martin Luther King Jr. Day. The plan starts with remote sensing observations of the rocks in our workspace. ChemCam and Mastcam will investigate two dark blocks named "Slide Peak" and "Rocky Peak" to assess their composition and texture. Then Curiosity will use the DRT to clear the dust at "Bee Rock," a block in our workspace that looks smoother than the surrounding areas, followed by MAHLI and APXS on the same block. Curiosity will also use MAHLI and APXS to assess a dark block at "Pleasant View Ridge" to see how these ejecta blocks compare to the surrounding rocks of the sulfate unit. Mastcam will be busy in this plan, acquiring a large 70-frame L0 mosaic covering the entire crater to assess its morphology, as well as stereo imaging on the western rim where more dark rocks are exposed. Another Mastcam mosaic will target the transition from the ejecta blocks back into typical rocks of the sulfate unit, and a fourth Mastcam mosaic will look toward the distant Gediz Vallis ridge, coordinated with a giant 20x1 ChemCam Long Distance RMI of the same area. Navcam will also be used to document the crater, to help generate high-resolution topography. This plan includes a drive on the second sol, during which Curiosity will take Full MAHLI Wheel Imaging — periodic documentation to assess the state and health of the rover wheels. The third sol contains untargeted science activities, including an autonomously selected ChemCam AEGIS target. Throughout the plan Curiosity will acquire observations to monitor the environment, including a Navcam dust devil movie, Mastcam twilight observation to search for iridescent ice clouds, and Navcam line-of-sight observations to assess dust content in the atmosphere. I was on shift as LTP today, and it was great to see this comprehensive plan come together to address a lot of different science areas. Looking forward to getting the results back and seeing what "Rustic Canyon" can tell us about Mars! Written by Lauren Edgar, Planetary Geologist at USGS Astrogeology Science Center

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TITLE: NASA's SPHEREx Observatory Completes First Cosmic Map Like No Other

6 min read NASA's SPHEREx Observatory Completes First Cosmic Map Like No Other

Jet Propulsion LaboratoryDec 18, 2025ArticleContentsSuperpowered telescopeMore about SPHERExThis panoramic view of SPHEREx's first all-sky map shows how the sky looks to the telescope. It transitions between observations of colors emitted by hot hydrogen gas (blue) and cosmic dust (red), and those primarily emitted by stars.Credit: NASA/JPL-CaltechLaunched in March, NASA's SPHEREx space telescope has completed its first infrared map of the entire sky in 102 colors. While not visible to the human eye, these 102 infrared wavelengths of light are prevalent in the cosmos, and observing the entire sky this way enables scientists to answer big questions, including how a dramatic event that occurred in the first billionth of a trillionth of a trillionth of a second after the big bang influenced the 3D distribution of hundreds of millions of galaxies in our universe. In addition, scientists will use the data to study how galaxies have changed over the universe's nearly 14 billion-year history and learn about the distribution of key ingredients for life in our own galaxy."It's incredible how much informationSPHERExhas collected in just six months — information that will be especially valuable when used alongside our other missions' data to better understand our universe," said Shawn Domagal-Goldman, director of the Astrophysics Division at NASA Headquarters in Washington. "We essentially have 102 new maps of the entire sky, each one in a different wavelength and containing unique information about the objects it sees. I think every astronomer is going to find something of value here, as NASA's missions enable the world to answer fundamental questions about how the universe got its start, and how it changed to eventually create a home for us in it."Circling Earth about 14½ times a day, SPHEREx (which stands for Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer) travels from north to south, passing over the poles. Each day it takes about 3,600 images along one circular strip of the sky, and as the days pass and the planet moves around the Sun, SPHEREx's field of view shifts as well. After six months, the observatory has looked out into space in every direction, capturing the entire sky in 360 degrees.Managed by NASA's Jet Propulsion Laboratory in Southern California, the missionbegan mapping the skyin May and completed its first all-sky mosaic in December. It will complete three additional all-sky scans during its two-year primary mission, and merging those maps together will increase the sensitivity of the measurements. The entire dataset isfreely available to scientistsand the public."SPHEREx is a mid-sized astrophysics mission delivering big science," said JPL Director Dave Gallagher. "It's a phenomenal example of how we turn bold ideas into reality, and in doing so, unlock enormous potential for discovery."NASA's SPHEREx has mapped the entire sky in 102 infrared colors, which are invisible to the human eye but can be used to reveal different features of the cosmos. This image features a selection of colors emitted primarily by stars (blue, green, and white), hot hydrogen gas (blue), and cosmic dust (red).NASA/JPL-CaltechThis SPHEREx image shows a selection of the infrared colors primarily emitted by stars and galaxies. The space

telescope is observing hundreds of millions of distant galaxies across the sky. Its multiwavelength view will help astronomers measure the distance to those galaxies. NASA/JPL-Caltech The infrared colors emitted primarily by dust (red) and hot gas (blue), key ingredients for forming new stars and planets, are seen in this SPHEREx image. Though these clouds of material cover a massive portion of the sky, they are invisible in most wavelengths of light, including those the human eye can detect. NASA/JPL-Caltech Superpowered telescope Each of the 102 colors detected by SPHEREx represents a wavelength of infrared light, and each wavelength provides unique information about the galaxies, stars, planet-forming regions, and other cosmic features therein. For example, dense clouds of dust in our galaxy where stars and planets form radiate brightly in certain wavelengths but emit no light (and are therefore totally invisible) in others. The process of separating the light from a source into its component wavelengths is called spectroscopy. And while a handful of previous missions has also mapped the entire sky, such as NASA's Wide-field Infrared Survey Explorer, none have done so in nearly as many colors as SPHEREx. By contrast, NASA's James Webb Space Telescope can do spectroscopy with significantly more wavelengths of light than SPHEREx, but with a field of view thousands of times smaller. The combination of colors and such a wide field of view is why SPHEREx is so powerful. "The superpower of SPHEREx is that it captures the whole sky in 102 colors about every six months. That's an amazing amount of information to gather in a short amount of time," said Beth Fabinsky, the SPHEREx project manager at JPL. "I think this makes us the mantis shrimp of telescopes, because we have an amazing multicolor visual detection system and we can also see a very wide swath of our surroundings." To accomplish this feat, SPHEREx uses six detectors, each paired with a specially designed filter that contains a gradient of 17 colors. That means every image taken with those six detectors contains 102 colors (six times 17). It also means that every all-sky map that SPHEREx produces is really 102 maps, each in a different color. The observatory will use those colors to measure the distance to hundreds of millions of galaxies. Though the positions of most of those galaxies have already been mapped in two dimensions by other observatories, SPHEREx's map will be in 3D, enabling scientists to measure subtle variations in the way galaxies are clustered and distributed across the universe. Each frame of this movie shows the entire sky in a different infrared wavelength, indicated by the color bar in the top right corner. Taken by NASA's SPHEREx observatory, the maps illustrate how viewing the universe in different wavelengths of light can reveal unique cosmic features. Credit: NASA/JPL-Caltech Those measurements will offer insights into an event that took place in the first billionth of a trillionth of a second after the big bang. In this moment, called inflation, the universe expanded by a trillion-trillionfold. Nothing like it has occurred in the universe since, and scientists want to understand it better. The SPHEREx mission's approach is one way to help in that effort. More about

SPHERExThe SPHEREx mission is managed by JPL for NASA's Astrophysics Division within the Science Mission Directorate in Washington. The telescope and the spacecraft bus were built by BAE Systems. The science analysis of the SPHEREx data is being conducted by a team of scientists at 10 institutions across the U.S., and in South Korea and Taiwan. Data is processed and archived at IPAC at Caltech in Pasadena, which manages JPL for NASA. The mission's principal investigator is based at Caltech with a joint JPL appointment. The SPHEREx dataset is publicly available. For more information about the SPHEREx mission visit: <https://science.nasa.gov/mission/spherex/News Media Contacts> Calla Cofield Jet Propulsion Laboratory, Pasadena, Calif. 626-808-2469 calla.e.cofield@jpl.nasa.gov 2025-144

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This panoramic view of SPHEREx's first all-sky map shows how the sky looks to the telescope. It transitions between observations of colors emitted by hot hydrogen gas (blue) and cosmic dust (red), and those primarily emitted by stars. Credit: NASA/JPL-Caltech

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Each of the 102 colors detected by SPHEREx represents a wavelength of infrared light, and each wavelength provides unique information about the galaxies, stars, planet-forming regions, and other cosmic features therein. For example, dense clouds of dust in our galaxy where stars and planets form radiate brightly in certain wavelengths but emit no light (and are therefore totally invisible) in others. The process of separating the light from a source into its component wavelengths is called spectroscopy.

And while a handful of previous missions has also mapped the entire sky, such as NASA's Wide-field Infrared Survey Explorer, none have done so in nearly as many colors as SPHEREx. By contrast, NASA's James Webb Space Telescope can do spectroscopy with significantly more wavelengths of light than SPHEREx, but with a field of view thousands of times smaller. The combination of colors and such a wide field of view is why SPHEREx is so powerful.

“The superpower of SPHEREx is that it captures the whole sky in 102 colors about every six months. That’s an amazing amount of information to gather in a short amount of time,” said Beth Fabinsky, the SPHEREx project manager at JPL. “I think this makes us the mantis shrimp of telescopes, because we have an amazing multicolor visual detection system and we can also see a very wide swath of our surroundings.”

To accomplish this feat, SPHEREx uses six detectors, each paired with a specially designed filter that contains a gradient of 17 colors. That means every image taken with those six detectors contains 102 colors (six times 17). It also means that every all-sky map that SPHEREx produces is really 102 maps, each in a different color.

The observatory will use those colors to measure the distance to hundreds of millions of galaxies. Though the positions of most of those galaxies have already been mapped in two dimensions by other observatories, SPHEREx’s map will be in 3D, enabling scientists to measure subtle variations in the way galaxies are clustered and distributed across the universe.

Those measurements will offer insights into an event that took place in the first billionth of a trillionth of a trillionth of a second after the big bang. In this moment, called inflation, the universe expanded by a trillion-trillionfold. Nothing like it has occurred in the universe since, and scientists want to understand it better. The SPHEREx mission’s approach is one way to help in that effort.

The SPHEREx mission is managed by JPL for NASA’s Astrophysics Division within the Science Mission Directorate in Washington. The telescope and the spacecraft bus were built by BAE Systems. The science analysis of the SPHEREx data is being conducted by a team of scientists at 10 institutions across the U.S., and in South Korea and Taiwan. Data is processed and archived at IPAC at Caltech in Pasadena, which manages JPL for NASA. The mission’s principal investigator is based at Caltech with a joint JPL appointment. The SPHEREx dataset is publicly available.

For more information about the SPHEREx mission visit:

<https://science.nasa.gov/mission/spherex/>

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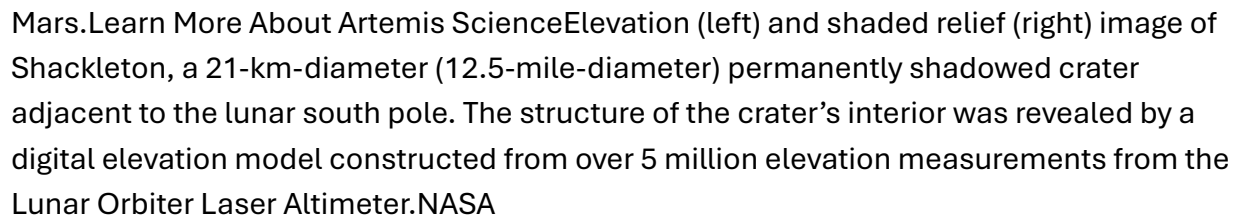
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Elevation (left) and shaded relief (right) image of Shackleton, a 21-km-diameter (12.5-mile-diameter) permanently shadowed crater adjacent to the lunar south pole. The structure of the crater's interior was revealed by a digital elevation model constructed from over 5 million elevation measurements from the Lunar Orbiter Laser Altimeter. NASA

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the crater's interior was revealed by a digital elevation model constructed from over 5 million elevation measurements from the Lunar Orbiter Laser Altimeter.

How can we grow a lunar economy? For the Artemis missions, American innovation is leading the way, and the future at the Moon holds promise for a robust lunar marketplace. NASA's strategy stimulates the commercial space industry, which drives new ideas, brings down costs, and grows the business opportunities that can foster a lunar economy and serve other customers for the benefit of humanity. [Learn More About Artemis Partners](#) Illustration of Firefly Aerospace's Blue Ghost lander on the lunar surface. The lander will carry a suite of 10 science investigations and technology demonstrations to the Moon in 2023 as part of NASA's Commercial Lunar Payload Services (CLPS) initiative. Firefly Aerospace

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Beyond the MoonHumans to MarsLike the Moon, Mars is a rich destination for scientific discovery and a driver of technologies that will enable humans to travel and explore far from Earth.Mars remains our horizon goal for human exploration because it is one of the only other places we know where life may have existed in the solar system. What we learn about the Red Planet will tell us more about our Earth's past and future, and may help answer whether life exists beyond our home planet.Learn Moreabout Humans to MarsIllustration of an astronaut on Mars, using a remote control drone to inspect a nearby cliff.NASA

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StatusActiveFuturePastMission TypesAirborne ScienceAnalog Field TestingAsteroid
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ResupplyCommunications & NavigationDemonstration MissionEarth
ObservationExperimentExperiment & InstrumentFlybyGround-based TelescopeHelicopter
& RotorcraftHuman SpaceflightImpactInstrumentISS
ExpeditionLanderOrbiterProbeRoverSample ReturnSmallSat & CubeSatSounding
RocketSpace EnvironmentSpace Telescope & ObservatorySupersonic ResearchTechnology
DemonstrationWeather SatelliteMission TargetsAsteroid 16 PsycheAsteroid BennuAsteroid
PsycheAsteroidsAurorasCeresCometsEarthEarth OrbitEarth's MoonEarth's MoonEarth's
Polar IceEuropaExoplanetsInternational Space StationJupiter and its moonsKuiper
BeltMars and its moonsMercuryMeteors & MeteoritesMilky WayNeptune and its
moonsOort CloudOur Solar SystemPlanet XPluto and its moonsSaturn and its
moonsSpace EnvironmentSunSun-Earth InteractionSun-Earth Interaction & Space
EnvironmentTitanTrojan AsteroidsUniverseUranus and its moonsVenusVestaMission
ProgramAirborne ScienceAnalogApolloArtemisAstrobiologyAstrophysicsBiological and
Physical SciencesCommercial LunarEarth
ScienceGatewayGeminiHeliophysicsInternational Space StationMercury
ProgramPlanetary DefensePlanetary ScienceScientific BalloonSkylabSpace
Communications and NavigationSpace ShuttleSurveyorTechnology Demonstration
MissionUniverse

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TypesAirborne ScienceAnalog Field TestingAsteroid DeflectionAtmospheric
ProbeBalloonCommercial CrewCommercial ResupplyCommunications &
NavigationDemonstration MissionEarth ObservationExperimentExperiment &
InstrumentFlybyGround-based TelescopeHelicopter & RotorcraftHuman

SpaceflightImpactInstrumentISS ExpeditionLanderOrbiterProbeRoverSample
ReturnSmallSat & CubeSatSounding RocketSpace EnvironmentSpace Telescope &
ObservatorySupersonic ResearchTechnology DemonstrationWeather SatelliteMission
TargetsAsteroid 16 PsycheAsteroid BennuAsteroid
PsycheAsteroidsAurorasCeresCometsEarthEarth OrbitEarth's MoonEarth's MoonEarth's
Polar IceEuropaExoplanetsInternational Space StationJupiter and its moonsKuiper
BeltMars and its moonsMercuryMeteors & MeteoritesMilky WayNeptune and its
moonsOort CloudOur Solar SystemPlanet XPluto and its moonsSaturn and its
moonsSpace EnvironmentSunSun-Earth InteractionSun-Earth Interaction & Space
EnvironmentTitanTrojan AsteroidsUniverseUranus and its moonsVenusVestaMission
ProgramAirborne ScienceAnalogApolloArtemisAstrobiologyAstrophysicsBiological and
Physical SciencesCommercial LunarEarth
ScienceGatewayGeminiHeliophysicsInternational Space StationMercury
ProgramPlanetary DefensePlanetary ScienceScientific BalloonSkylabSpace
Communications and NavigationSpace ShuttleSurveyorTechnology Demonstration
MissionUniverse

Search ContentSearchReset FiltersMission StatusActiveFuturePastMission TypesAirborne
ScienceAnalog Field TestingAsteroid DeflectionAtmospheric ProbeBalloonCommercial
CrewCommercial ResupplyCommunications & NavigationDemonstration MissionEarth
ObservationExperimentExperiment & InstrumentFlybyGround-based TelescopeHelicopter
& RotorcraftHuman SpaceflightImpactInstrumentISS
ExpeditionLanderOrbiterProbeRoverSample ReturnSmallSat & CubeSatSounding
RocketSpace EnvironmentSpace Telescope & ObservatorySupersonic ResearchTechnology
DemonstrationWeather SatelliteMission TargetsAsteroid 16 PsycheAsteroid BennuAsteroid
PsycheAsteroidsAurorasCeresCometsEarthEarth OrbitEarth's MoonEarth's MoonEarth's
Polar IceEuropaExoplanetsInternational Space StationJupiter and its moonsKuiper
BeltMars and its moonsMercuryMeteors & MeteoritesMilky WayNeptune and its
moonsOort CloudOur Solar SystemPlanet XPluto and its moonsSaturn and its
moonsSpace EnvironmentSunSun-Earth InteractionSun-Earth Interaction & Space
EnvironmentTitanTrojan AsteroidsUniverseUranus and its moonsVenusVestaMission
ProgramAirborne ScienceAnalogApolloArtemisAstrobiologyAstrophysicsBiological and
Physical SciencesCommercial LunarEarth
ScienceGatewayGeminiHeliophysicsInternational Space StationMercury
ProgramPlanetary DefensePlanetary ScienceScientific BalloonSkylabSpace
Communications and NavigationSpace ShuttleSurveyorTechnology Demonstration
MissionUniverse

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Mission StatusActiveFuturePastMission TypesAirborne ScienceAnalog Field
TestingAsteroid DeflectionAtmospheric ProbeBalloonCommercial CrewCommercial
ResupplyCommunications & NavigationDemonstration MissionEarth
ObservationExperimentExperiment & InstrumentFlybyGround-based TelescopeHelicopter
& RotorcraftHuman SpaceflightImpactInstrumentISS
ExpeditionLanderOrbiterProbeRoverSample ReturnSmallSat & CubeSatSounding
RocketSpace EnvironmentSpace Telescope & ObservatorySupersonic ResearchTechnology
DemonstrationWeather SatelliteMission TargetsAsteroid 16 PsycheAsteroid BennuAsteroid
PsycheAsteroidsAurorasCeresCometsEarthEarth OrbitEarth's MoonEarth's MoonEarth's
Polar IceEuropaExoplanetsInternational Space StationJupiter and its moonsKuiper
BeltMars and its moonsMercuryMeteors & MeteoritesMilky WayNeptune and its
moonsOort CloudOur Solar SystemPlanet XPluto and its moonsSaturn and its
moonsSpace EnvironmentSunSun-Earth InteractionSun-Earth Interaction & Space
EnvironmentTitanTrojan AsteroidsUniverseUranus and its moonsVenusVestaMission
ProgramAirborne ScienceAnalogApolloArtemisAstrobiologyAstrophysicsBiological and
Physical SciencesCommercial LunarEarth
ScienceGatewayGeminiHeliophysicsInternational Space StationMercury
ProgramPlanetary DefensePlanetary ScienceScientific BalloonSkylabSpace
Communications and NavigationSpace ShuttleSurveyorTechnology Demonstration
MissionUniverse

Mission Status

Active

Future

Past

Mission Types

Airborne Science

Analog Field Testing

Asteroid Deflection

Atmospheric Probe

Balloon

Commercial Crew

Commercial Resupply

Communications & Navigation

Demonstration Mission

Earth Observation

Experiment

Experiment & Instrument

Flyby

Ground-based Telescope

Helicopter & Rotorcraft

Human Spaceflight

Impact

Instrument

ISS Expedition

Lander

Orbiter

Probe

Rover

Sample Return

SmallSat & CubeSat

Sounding Rocket

Space Environment

Space Telescope & Observatory

Supersonic Research

Technology Demonstration

Weather Satellite

Mission Targets

Asteroid 16 Psyche

Asteroid Bennu

Asteroid Psyche

Asteroids

Auroras

Ceres

Comets

Earth Orbit

Earth's Polar Ice

Europa

Jupiter and its moons

Mars and its moons

Meteors & Meteorites

Milky Way

Neptune and its moons

Our Solar System

Planet X

Pluto and its moons

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Sun

Sun-Earth Interaction

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Titan

Trojan Asteroids

Uranus and its moons

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TITLE: Últimos preparativos para la primera misión tripulada a la Luna con la campaña Artemis de la NASA

11 min readÚltimos preparativos para la primera misión tripulada a la Luna con la campaña Artemis de la NASA

Tiernan P. DoyleJan 12, 2026ArticleEl cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el

Centro Espacial Kennedy de la NASA en Florida. NASA/Joel Kowsky Read this article in English [here](#). Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento. La NASA tiene como objetivo comenzar desde el sábado 17 de enero su traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. El viaje de casi 6,5 kilómetros (cuatro millas) en el vehículo transportador oruga 2 podría durar hasta 12 horas. Los equipos técnicos están trabajando día y noche para dar por terminadas todas las tareas antes del transporte del cohete. Sin embargo, esta fecha objetivo está sujeta a cambios si fuera necesario tiempo adicional para los preparativos técnicos o debido a las condiciones meteorológicas. “Nos estamos acercando a la misión Artemis II, y tenemos su lanzamiento a la vuelta de la esquina”, dijo Lori Glaze, administradora asociada interina para la Dirección de Misiones de Desarrollo de Sistemas de Exploración de la NASA. “Nos quedan pasos importantes en nuestro camino hacia el lanzamiento, y la seguridad de la tripulación seguirá siendo nuestra principal prioridad en todo momento, a medida que nos acercamos al regreso de la humanidad a la Luna”. Al igual que con todos los nuevos desarrollos de sistemas complejos, los ingenieros han estado solucionando varios problemas en los últimos días y semanas. Durante las comprobaciones finales antes del traslado, los técnicos detectaron que un cable relacionado con el sistema de terminación de vuelo estaba doblado en contra de las especificaciones. El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas de demostración de la cuenta regresiva el 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable. Traslado Una vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre. Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch

de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma. Ensayo general con circulación de combustible y llenado de tanques. A finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio. Durante el lanzamiento, el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial. El ensayo general con circulación de combustible incluirá varios “encendidos”, o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal. La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución. Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante los ensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre. Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo. De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del

lanzamiento después del ensayo general con circulación de combustible. Próximos pasos para el lanzamiento Después de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento. Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero, el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento. A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento. Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II. El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna. Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión. La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo. La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra. Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamiento hasta abril de 2026. Los planificadores de la misión mejoran estos períodos en

función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios. Período de lanzamiento del 31 de enero al 14 de febrero Oportunidades de lanzamiento los días 6, 7, 8, 10 y 11 de febrero Período de lanzamiento del 28 de febrero al 13 de marzo Oportunidades de lanzamiento los días 6, 7, 8, 9, 11 de marzo Período de lanzamiento del 27 de marzo al 10 de abril Oportunidades de lanzamiento los días 1, 3, 4, 5, 6 de abril Además de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento. Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento. Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte. Encuentra más información sobre la campaña Artemis de la NASA en el siguiente sitio web (en inglés): <https://www.nasa.gov/artemis>

Tiernan P. Doyle Jan 12, 2026 Article

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El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el Centro Espacial Kennedy de la NASA en Florida. NASA/Joel Kowsky Read this article in English [here](#). Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento. La NASA tiene como objetivo comenzar desde el sábado 17 de enero su

traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. El viaje de casi 6,5 kilómetros (cuatro millas) en el vehículo transportador oruga 2 podría durar hasta 12 horas. Los equipos técnicos están trabajando día y noche para dar por terminadas todas las tareas antes del transporte del cohete. Sin embargo, esta fecha objetivo está sujeta a cambios si fuera necesario tiempo adicional para los preparativos técnicos o debido a las condiciones meteorológicas. “Nos estamos acercando a la misión Artemis II, y tenemos su lanzamiento a la vuelta de la esquina”, dijo Lori Glaze, administradora asociada interina para la Dirección de Misiones de Desarrollo de Sistemas de Exploración de la NASA. “Nos quedan pasos importantes en nuestro camino hacia el lanzamiento, y la seguridad de la tripulación seguirá siendo nuestra principal prioridad en todo momento, a medida que nos acercamos al regreso de la humanidad a la Luna”. Al igual que con todos los nuevos desarrollos de sistemas complejos, los ingenieros han estado solucionando varios problemas en los últimos días y semanas. Durante las comprobaciones finales antes del traslado, los técnicos detectaron que un cable relacionado con el sistema de terminación de vuelo estaba doblado en contra de las especificaciones. El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas de demostración de la cuenta regresiva el 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable.

Traslado Una vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre. Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma.

Ensayo general con circulación de combustible y llenado de tanques A finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva

a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio. Durante el lanzamiento, el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial. El ensayo general con circulación de combustible incluirá varios “encendidos”, o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal. La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución. Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante los ensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre. Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo. De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del lanzamiento después del ensayo general con circulación de combustible. Próximos pasos para el lanzamiento Después de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento. Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero,

el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento. A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento. Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II. El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna. Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión. La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo. La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra. Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamiento hasta abril de 2026. Los planificadores de la misión mejoran estos períodos en función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios.

Período de lanzamiento	Oportunidades de lanzamiento
del 31 de enero al 14 de febrero	los días 6, 7, 8, 10 y 11 de febrero
del 28 de febrero al 13 de marzo	los días 6, 7, 8, 9, 11 de marzo
del 27 de marzo al 10 de abril	los días 1, 3, 4, 5, 6 de abril

Además de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen

restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento. Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento. Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte. Encuentra más información sobre la campaña Artemis de la NASA en el siguiente sitio web (en inglés): <https://www.nasa.gov/artemis>

El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t

esperan la llegada de los tripulantes de Artemis II para abordar su nave espacial Orion en la parte superior del cohete como parte de la prueba de demostración de la cuenta atrás de Artemis II, el sábado 20 de diciembre de 2025, en el Centro Espacial Kennedy de la NASA en Florida. NASA/Joel Kowsky

El cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) de la NASA, visto en la nave High Bay 3 del Edificio de Ensamblaje de Vehículos mientras los equipos t

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NASA/Joel Kowsky

Read this article in English [here](#).

Conforme la NASA se acerca al lanzamiento del vuelo de prueba Artemis II, la agencia pronto llevará por primera vez su cohete Sistema de Lanzamiento Espacial (SLS, por sus siglas en inglés) y la nave espacial Orion a la plataforma de lanzamiento en el Centro Espacial Kennedy de la agencia en Florida para comenzar la integración final, las pruebas y los ensayos para el lanzamiento.

La NASA tiene como objetivo comenzar desde el sábado 17 de enero su traslado desde el Edificio de Ensamblaje de Vehículos hasta la Plataforma de Lanzamientos 39B, lo que tardará varias horas. El viaje de casi 6,5 kilómetros (cuatro millas) en el vehículo transportador oruga 2 podría durar hasta 12 horas. Los equipos técnicos están trabajando día y noche para dar por terminadas todas las tareas antes del transporte del cohete. Sin embargo, esta fecha objetivo está sujeta a cambios si fuera necesario tiempo adicional para los preparativos técnicos o debido a las condiciones meteorológicas.

“Nos estamos acercando a la misión Artemis II, y tenemos su lanzamiento a la vuelta de la esquina”, dijo Lori Glaze, administradora asociada interina para la Dirección de Misiones de Desarrollo de Sistemas de Exploración de la NASA. “Nos quedan pasos importantes en nuestro camino hacia el lanzamiento, y la seguridad de la tripulación seguirá siendo nuestra principal prioridad en todo momento, a medida que nos acercamos al regreso de la humanidad a la Luna”.

Al igual que con todos los nuevos desarrollos de sistemas complejos, los ingenieros han estado solucionando varios problemas en los últimos días y semanas. Durante las comprobaciones finales antes del traslado, los técnicos detectaron que un cable relacionado con el sistema de terminación de vuelo estaba doblado en contra de las especificaciones. El personal técnico lo está reemplazando y hará pruebas con el nuevo cable durante el fin de semana. Además, una válvula relacionada con la presurización de la escotilla de Orion presentó problemas que hicieron necesario llevar a cabo pruebas de demostración de la cuenta regresiva el 20 de diciembre pasado. El 5 de enero, el equipo reemplazó la válvula e hizo pruebas de su funcionamiento que resultaron exitosas. Los ingenieros también trabajaron para resolver fugas en el hardware de soporte en tierra que es necesario para cargar oxígeno gaseoso en Orion a fin de proporcionar aire respirable.

Traslado

Una vez que el cohete y la nave espacial integrados lleguen a la plataforma de lanzamiento, la NASA comenzará inmediatamente una larga lista de verificación para los preparativos en la plataforma de lanzamiento, incluyendo la conexión de equipos mecánicos de apoyo en tierra, como líneas eléctricas, conductos del sistema de control ambiental de combustible y tomas de surtido de combustible criogénico. Los equipos de personal técnico encenderán todos los sistemas integrados en la plataforma por primera vez para garantizar que los componentes del hardware de vuelo funcionen correctamente entre sí, con el lanzador móvil y con los sistemas de infraestructura terrestre.

Una vez que esté todo completado, los astronautas de Artemis II, Reid Wiseman, Victor Glover y Christina Koch de la NASA, y el astronauta de la CSA (Agencia Espacial Canadiense) Jeremy Hansen, llevarán a cabo una caminata final en la plataforma.

Ensayo general con circulación de combustible y llenado de tanques

A finales de enero, la NASA llevará a cabo un ensayo general con circulación de combustible, el cual es una prueba previa al lanzamiento para llenar los tanques de combustible en el cohete. Durante este ensayo general, el personal técnico hace una demostración de la capacidad de cargar más de 700.000 galones de combustible criogénico en el cohete, lleva a cabo una cuenta regresiva para el lanzamiento y practica la extracción segura del combustible del cohete sin tripulación presente en el sitio.

Durante el lanzamiento, el equipo de cierre, o de tareas finales, será responsable de asegurar a los astronautas en Orion y cerrar sus escotillas. El personal de cierre también utilizará este ensayo para practicar sus procedimientos de forma segura sin tener tripulación a bordo de la nave espacial.

El ensayo general con circulación de combustible incluirá varios “encendidos”, o pruebas de funcionamiento, para demostrar la capacidad del equipo de lanzamiento para detener, reanudar y reiniciar operaciones en varios momentos diferentes de los últimos 10 minutos de la cuenta regresiva, conocida como conteo terminal.

La ejecución del primer encendido comenzará aproximadamente en las 49 horas antes del lanzamiento, cuando los equipos encargados de lanzamiento son llamados a sus estaciones, hasta 1 minuto y 30 segundos antes del lanzamiento, seguido de una pausa planificada de tres minutos y luego la reanudación de la cuenta regresiva hasta 33 segundos antes del lanzamiento, el punto en el que el secuenciador de lanzamiento automático del cohete controlará los últimos segundos de la cuenta regresiva. Luego, los equipos técnicos volverán a reiniciar a T-10 minutos y detendrán el conteo, y luego reanudarán los procedimientos hasta 30 segundos antes del lanzamiento como parte de una segunda ejecución.

Si bien la NASA ha integrado las lecciones aprendidas con Artemis I en los procedimientos de la cuenta regresiva para el lanzamiento, la agencia hará una pausa para abordar cualquier problema durante la prueba o en cualquier otro momento si surgen retos técnicos. Los ingenieros vigilarán de cerca la carga de combustible de hidrógeno líquido y oxígeno líquido en el cohete, después de los desafíos que se encontraron con la carga de hidrógeno líquido durante los ensayos generales con circulación de combustible de Artemis I. Los equipos técnicos también prestarán mucha atención a la efectividad de los procedimientos recientemente actualizados para limitar la cantidad de nitrógeno gaseoso

que se acumula en el espacio que está entre el módulo de tripulación de Orion y las escotillas del sistema de cancelación de lanzamiento, lo que podría representar un problema para el personal de cierre.

Es posible que se requieran ensayos generales con circulación de combustible adicionales para garantizar que el vehículo esté completamente revisado y apto para el vuelo.

De ser necesario, la NASA podría trasladar al cohete SLS y la nave Orion de vuelta al Edificio de Ensamblaje de Vehículos para realizar trabajos adicionales antes del lanzamiento después del ensayo general con circulación de combustible.

Próximos pasos para el lanzamiento

Después de un exitoso ensayo general con circulación de combustible, la NASA convocará una revisión de aptitud para el vuelo en la cual el equipo de gestión de la misión evaluará la aptitud de todos los sistemas, incluyendo el hardware de vuelo, la infraestructura y el personal de lanzamiento, vuelo y recuperación antes de comprometerse con una fecha de lanzamiento.

Aunque la ventana para el lanzamiento de Artemis II se podría iniciar tan pronto como el viernes 6 de febrero, el equipo de gestión de la misión evaluará la aptitud para el vuelo después del ensayo general con toda la nave espacial, la infraestructura de lanzamiento, y la tripulación y el personal de operaciones antes de seleccionar una fecha para el lanzamiento.

A fin de determinar las posibles fechas de lanzamiento, los ingenieros identificaron las restricciones clave necesarias para cumplir la misión y mantener a salvo a la tripulación dentro de Orion. Los períodos de lanzamiento resultantes son los días o las semanas en los que la nave espacial y el cohete pueden cumplir los objetivos de la misión. Estos períodos de lanzamiento explican la compleja mecánica orbital relacionada con el lanzamiento en una trayectoria precisa hacia la Luna mientras la Tierra rota sobre su eje y la Luna orbita la Tierra cada mes en su ciclo lunar. Esto da como resultado un patrón de alrededor de una semana de oportunidades de lanzamiento, seguido de tres semanas sin oportunidades de lanzamiento.

Existen varios parámetros principales que establecen la disponibilidad del lanzamiento dentro de estos períodos. Debido a su trayectoria única en relación con las misiones de alunizaje posteriores, estas limitaciones clave son exclusivas del vuelo de prueba de la misión Artemis II.

El día y la hora de lanzamiento deben permitir que SLS pueda llevar a Orion a una órbita terrestre alta, donde la tripulación y los equipos técnicos en tierra evaluarán los sistemas

de soporte vital de la nave espacial antes de que la tripulación emprenda su viaje con rumbo a la Luna.

Orion también debe estar en la alineación adecuada con la Tierra y la Luna en el momento del encendido de motores con inyección translunar. El encendido de motores con inyección translunar de Artemis II pone a Orion en rumbo de sobrevolar la Luna, y también lo pone en una trayectoria de retorno libre, en la cual la nave espacial utiliza la gravedad de la Luna para enviar la nave espacial de regreso a la Tierra sin maniobras adicionales importantes de propulsión.

La trayectoria para un día determinado debe garantizar que Orion no esté en la oscuridad durante más de 90 minutos a la vez para que las alas de los paneles solares puedan recibir y convertir la luz solar en electricidad, y la nave espacial pueda mantener un rango de temperatura óptimo. Los planificadores de la misión eliminan las posibles fechas de lanzamiento que llevarían a Orion a eclipses prolongados durante el vuelo.

La fecha de lanzamiento debe sustentar una trayectoria que permita el perfil de entrada adecuado planificado durante el regreso de Orion a la Tierra.

Los períodos a continuación muestran la disponibilidad de llevar a cabo el lanzamiento hasta abril de 2026. Los planificadores de la misión mejoran estos períodos en función de un análisis actualizado más o menos dos meses antes de que estos comiencen, y ellos están sujetos a cambios.

Período de lanzamiento del 31 de enero al 14 de febrero

Oportunidades de lanzamiento los días 6, 7, 8, 10 y 11 de febrero

Período de lanzamiento del 28 de febrero al 13 de marzo

Oportunidades de lanzamiento los días 6, 7, 8, 9, 11 de marzo

Período de lanzamiento del 27 de marzo al 10 de abril

Oportunidades de lanzamiento los días 1, 3, 4, 5, 6 de abril

Además de las oportunidades de lanzamiento basadas en la mecánica orbital y los requisitos de desempeño, también existen restricciones sobre qué días dentro de un período de lanzamiento pueden ser viables en función de la reposición de productos básicos, las condiciones meteorológicas y las operaciones de otros usuarios en el cronograma del Área Este. Como regla general, se pueden hacer hasta cuatro intentos de lanzamiento dentro de la semana aproximada de oportunidades que existen dentro de un período de lanzamiento.

Mientras la agencia se prepara para su primera misión tripulada más allá de la órbita terrestre en más de 50 años, la NASA espera aprender durante los procesos, tanto en tierra como en vuelo, y dejará que la aptitud y el desempeño de sus sistemas indiquen el momento en que la agencia está lista para el lanzamiento.

Como parte de una edad de oro de innovación y exploración, el vuelo de prueba de Artemis II, el cual tendrá una duración aproximada de 10 días, es el primer vuelo tripulado para la campaña Artemis de la NASA. Este es otro paso hacia nuevas misiones tripuladas de Estados Unidos en la superficie de la Luna, lo que llevará a una presencia sostenida en la Luna que ayudará a la agencia a prepararse para enviar a los primeros astronautas estadounidenses a Marte.

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SUBJECT: Photographic Support for Flight Analogs HERA Campaign 5 Mission 1 Crew exiting the HERA Habitat – with Sara Edwards, Roderick Borgie, Ian Porter and Dustin Wallace

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Boarding passes will carry participants' names on NASA's Artemis II mission in 2026. Credit: NASA

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TITLE: Commercial Space

Humans in SpaceCommercial SpaceNASA is enabling commercial industry to build, own, and operate space systems with the agency purchasing services for its science and research needs. Through public-private partnerships, NASA is opening space to more science, people, and opportunities.Low Earth Orbit EconomyAs the International Space Station nears the end of operations, NASA plans to transition to a new low Earth orbit model to continue leveraging microgravity benefits. Through commercial partnerships, NASA aims to maintain its leadership in microgravity research and ensure continued benefits for humanity.Learn Moreabout Low Earth Orbit EconomyPrivate Astronaut MissionsPrivate astronaut missions to the International Space Station help pave the way toward commercial space stations as part of NASA's efforts to develop a thriving low Earth orbit ecosystem and marketplace.Learn Moreabout Private Astronaut MissionsThe Axiom Mission 4, or Ax-4, crew will launch aboard a SpaceX Dragon spacecraft to the International Space Station from NASA's Kennedy Space Center in Florida. From left to right: ISRO (Indian Space Research Organization) astronaut Shubhanshu Shukla, former NASA astronaut Peggy Whitson, ESA (European Space Agency) astronaut Sławosz Uznański-Wiśniewski of Poland, and Tibor Kapu of Hungary.Credit: Axiom SpaceCommercial Space StationsNASA is committed to maintaining a continuous human presence in low Earth orbit as the agency transitions from the International Space Station to commercial space stations.Learn Moreabout Commercial Space StationsThis long-duration photograph highlights the city lights of Mexico's Yucatan peninsula, Earth's atmospheric glow, and star trails above taken from the International Space Station as it orbited 258 miles above.Credit: NASAGrowing the Lunar EconomyNASA is leading Artemis, humanity's return to the Moon. With international partners and U.S. industry providers, the future at the Moon holds promise for a robust lunar marketplace.Learn Moreabout Growing the Lunar EconomyHuman Landing SystemsNASA's commercial providers, SpaceX and Blue Origin, are building the human landing systems that will carry Artemis astronauts to the lunar surface and back to lunar orbit for their ride home to Earth aboard the Orion spacecraft.Learn Moreabout Human Landing SystemsSide-by-side illustrations of the SpaceX Starship lunar lander and the Blue Origin Blue Moon lunar lander.Credit: SpaceX/Blue OriginCommercial Lunar Payload ServicesNASA is working with several American companies to deliver science and technology to the lunar surface through the Commercial Lunar Payload Services initiative. These companies, ranging in size, bid on delivering payloads for NASA. Under Artemis, commercial deliveries are performing science experiments, testing technologies, and demonstrating capabilities to help NASA explore the Moon.Learn Moreabout Commercial Lunar Payload ServicesThe moon's

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Private Astronaut MissionsPrivate astronaut missions to the International Space Station help pave the way toward commercial space stations as part of NASA's efforts to develop a thriving low Earth orbit ecosystem and marketplace.[Learn More](#)about Private Astronaut MissionsThe Axiom Mission 4, or Ax-4, crew will launch aboard a SpaceX Dragon spacecraft to the International Space Station from NASA's Kennedy Space Center in Florida. From left to right: ISRO (Indian Space Research Organization) astronaut Shubhanshu Shukla, former NASA astronaut Peggy Whitson, ESA (European Space Agency) astronaut Sławosz Uznański-Wiśniewski of Poland, and Tibor Kapu of Hungary.Credit: Axiom Space

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The Axiom Mission 4, or Ax-4, crew will launch aboard a SpaceX Dragon spacecraft to the International Space Station from NASA's Kennedy Space Center in Florida. From left to right: ISRO (Indian Space Research Organization) astronaut Shubhanshu Shukla, former NASA astronaut Peggy Whitson, ESA (European Space Agency) astronaut Sławosz Uznański-Wiśniewski of Poland, and Tibor Kapu of Hungary.

Credit: Axiom Space

Commercial Space StationsNASA is committed to maintaining a continuous human presence in low Earth orbit as the agency transitions from the International Space Station to commercial space stations.[Learn More](#)about Commercial Space StationsThis long-duration photograph highlights the city lights of Mexico's Yucatan peninsula, Earth's atmospheric glow, and star trails above taken from the International Space Station as it orbited 258 miles above.Credit: NASA

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Growing the Lunar Economy NASA is leading Artemis, humanity's return to the Moon. With international partners and U.S. industry providers, the future at the Moon holds promise for a robust lunar marketplace. [Learn More about Growing the Lunar Economy](#)

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Human Landing Systems NASA's commercial providers, SpaceX and Blue Origin, are building the human landing systems that will carry Artemis astronauts to the lunar surface and back to lunar orbit for their ride home to Earth aboard the Orion spacecraft. [Learn More about Human Landing Systems](#) Side-by-side illustrations of the SpaceX Starship lunar lander and the Blue Origin Blue Moon lunar lander. [Credit: SpaceX/Blue Origin](#)

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Blog3 Min ReadCrew Studies Cardiac Research, Artificial Intelligence as Dragon Boosts Station's OrbitArticle6 Min ReadNASA Kennedy Top 20 Stories of 2025Blog4 Min ReadSpace Station First: All Docking Ports Fully Occupied, 8 Spacecraft on OrbitBlog3 Min ReadCrew Studies Space Orientation, DNA Data Storage as Dragon Reboosts Station

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Blog4 Min ReadSpace Station First: All Docking Ports Fully Occupied, 8 Spacecraft on OrbitBlog3 Min ReadCrew Studies Space Orientation, DNA Data Storage as Dragon Reboosts Station

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Low Earth Orbit EconomyCommercial Lunar Payload ServicesCommercial Space StationsGrowing the Lunar Economy

URL: <https://science.nasa.gov/biological-physical/programs/physical-sciences/>

TITLE: Physical Sciences Program

Explore This SectionPhysical Sciences ProgramOverviewNASA's Physical Science Research Program has made contributions in two distinct areas: first, fundamental research, which investigates physical phenomena in the absence of gravity and fundamental laws of the universe, and second, applied research, which contributes to the

basic understanding underlying space exploration technologies. In completing these investigations, physical sciences provides basic scientific knowledge, results leading to societal benefit, and contributions to the basic understanding underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and control. All have led to improved space systems or new products on Earth. Our core objectives include: Investigate fundamental laws of physics, often using either microgravity or interplanetary distances as research tools Provide mechanistic understanding of processes underlying space exploration technologies such as power generation and storage, space propulsion, life support systems, and environmental monitoring and control Support the transfer of knowledge and technology of space-based research to terrestrial systems to benefit life on Earth Developing cutting-edge technologies to facilitate spaceflight research Promote open science through data sharing Disciplines Biophysics Combustion Science Soft Matter Fluid Physics Fundamental Physics Materials Science The International Space Station provides the highly desired condition of long-duration microgravity, allowing continuous and interactive research similar to Earth-based laboratories, even providing statistical validity when required. The program also has benefited from research collaborations with the International Space Station partners (Russia, Europe, Japan, Canada) and individual foreign governments with space programs, such as France, Germany and Italy. NASA's physical science research is organized into six disciplines – Biophysics, Combustion Science, Complex Fluids, Fluid Physics, Fundamental Physics and Materials Science. Conducted in a nearly weightless environment, experiments in these disciplines reveal how physical systems respond to the near absence of buoyancy-driven convection, sedimentation or sagging. They also reveal how other forces, such as capillary forces, which are small compared to gravity, can dominate the system behavior in space. The data acquired from these investigations is stored in NASA's Physical Sciences Informatics System (PSI) and is available to the public. If you are a researcher and interested in learning more about NASA's Physical Sciences Program, please contact Dan Walsh. The following information is a summary of the six disciplines and the PSI database. Physical Sciences Informatics System (database) In fulfillment of the new Open Science model, we are pleased to announce the Physical Science Informatics (PSI) data repository for physical science experiments performed on the International Space Station (ISS). The PSI system is accessible and open to the public. This provides the opportunity for researchers to data mine results from prior flight investigations, expanding on the research performed. This approach will allow numerous ground-based investigations to be conducted from one flight experiment's data, exponentially increasing our body of knowledge. PSI also meets the President's Open Data policy. Fundamental Physics Commercially Enabled Rapid Space Science Open Science

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Developing cutting-edge technologies to facilitate spaceflight research

Promote open science through data sharing

Disciplines

Biophysics

Combustion Science

Soft Matter

Fluid Physics

Fundamental Physics

Materials Science

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Open Science

URL: <https://science.nasa.gov/biological-physical/programs/space-biology/>

TITLE: Space Biology Program

Explore This Section
Space Biology Program Contents
Overview
Animal Biology
Cell and Molecular Biology
Microbiology
Plant Biology
Developmental, Reproductive and Evolutionary Biology
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experiments we conduct on these platforms examine how plants, microbes, and animals adjust or adapt to living in space. We examine processes of metabolism, growth, stress response, physiology, and development. We study how organisms repair cellular damage and protect themselves from infection and disease in conditions of microgravity while being exposed to space radiation. And we do it across the spectrum of biological organization, from molecules to cells, from tissues and organs, and from systems to whole organisms to communities of microorganisms. Our program's core objectives include: Discovering how biological systems respond, acclimate and adapt to the space environment; Developing integrated physiological models for biology in space; Identifying the underlying mechanisms and networks that govern biological processes in the space environment; Promoting open science through the GeneLab Data System and Life Sciences Data Archive; Developing cutting-edge biological technologies to facilitate spaceflight research; Developing mechanistic understanding to support human health in space; Enabling the transfer of knowledge and technology to the understanding of life on Earth.

[Space Biology Science Plan \(PDF\)](#) [Search the Space Station Research Explorer for Space Biology experiments](#)

In addition to providing useful information on how living organisms adapt to spaceflight, the discoveries NASA researchers make in space have enormous implications for life on Earth. Space Biology's research into the virulence of pathogens in space, loss of bone density, and the changes in the growth of plants can impact the development of drugs that promote wound healing or tissue regeneration, treatments designed to counter osteoporosis on Earth, and high-tech fertilizers that increase crop yield.

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Animal Biology

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The goal of the Space Biology Program in the animal biology area is to understand the basic mechanisms that animals use to adapt and/or acclimate to spaceflight and alterations in gravity in general. Animals are frequently used to model human disease as well as how humans respond to stressful stimuli. The most commonly used model organisms for which genomics are now well defined include vertebrate species, e.g., rodents, both rats and mice, and a variety of invertebrate species, e.g., nematodes and insects. NASA has used these model organisms extensively to evaluate biological spaceflight hazards, elucidate the fundamental mechanisms life uses to adapt to microgravity, and apply such knowledge to advance human exploration, and for societal benefits on Earth.

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Cell and Molecular Biology Cell and molecular biology research cuts across all science disciplines in Space Biology, from understanding how single-celled organisms, such as protozoa, bacteria, and fungi respond to the conditions of spaceflight, to how all of the various cells in a complex tissue or organ work together to help an organism as a whole acclimate to such a foreign environment. The overarching goal of Space Biology Cell and Molecular Biology research at NASA is to determine how the stresses of the spaceflight environment affect living systems at the basic cellular and molecular levels, using contemporary cell and molecular biology techniques and measures. This includes characterizing and identifying changes in gene and protein expression, DNA function and structure, cellular structure and morphology, and cell-to-cell communication. As we examine the impacts of spaceflight on the cell biology and physiology, we ask the following: What are the underlying genetic and molecular mechanisms of the cells that are influenced by gravitational changes and the space environment? Does the space flight environment affect or influence cell and molecular functions causing tissue/organ dysfunction or disease states? Does the space environment affect cellular and molecular functions in a manner that impact tissue morphogenesis or development?

Microbiology **No Matter Where We Go, We Take Our Microbes** Everywhere we go, we take microorganisms with us, whether we want to or not. This is true even aboard the “clean” environment of the ISS, which is a closed environment. With a crew of astronauts on board, living, breathing, exercising, and sweating, the ISS is a breeding ground for microbes. **Microbes Can Be Both Friends and Foes** The effects of spaceflight on the biology of microorganisms and on microbial populations are largely unknown. In addition to posing a risk to astronaut health, bio-corrosive microorganisms that grow on metallic surfaces in spacecraft can damage both equipment and hardware. Unlike cruise ships on Earth which can be evacuated, emptied, and cleaned, problems that arise due to microbial contamination on long-duration spaceflight missions can only be resolved using tools and resources already present within the vessel. Understanding how microbial species grow and interact with each other in this environment is the first step in preparing for such a scenario. While the presence of certain microbes can be problematic, it is also important to remember that many microbial species play important roles in Earth’s various ecosystems. For example, the symbiotic relationships that have evolved between plants and certain bacteria are vital in ensuring plants receive the nutrients they require to grow and flourish. Oxygen-producing cyanobacteria that live in terrestrial bodies of water help replenish Earth’s oxygen. Even the

bacteria living within our digestive tracts play an important role in the digestion of our food and the proper absorption of nutrients. The presence of certain microbes may be important for the proper growth of some plants in space and may even be critical for the production of future bio-regenerative life support systems. It is therefore important to keep in mind that reaching an appropriate microbial equilibrium within a spacecraft may be vital for successful long-duration spaceflight missions. As we examine the impacts of spaceflight on microorganisms, we ask the following: What underlying genetic, molecular and biochemical processes are influenced by the spaceflight environment? How does the spaceflight environment influence microbial reproduction, growth, and physiology? Does long-duration spaceflight alter normal rates of evolutionary change? What are the effects of spaceflight on microbial communities as they interact with other organisms to effect processes such as symbioses, biodegradation, nitrogen-fixation, etc. What are the mechanisms that effect changes, such as the altered virulence or altered drug resistance, observed in some organisms during spaceflight?

Plant Biology Space Biology research helps us understand the fundamentals of plant growth by examining the very building blocks of plant life down to the molecular level: transcriptomics, genomics, proteomics, and metabolomics. To compare the effects of microgravity conditions on plants, we also conduct experiments on Earth using gravity or simulated microgravity ground controls at the Kennedy Space Center. We conduct our research on the ISS in conditions of microgravity to help us understand how to support astronauts aboard the ISS and on their long journey to Mars.

Crop Rotation: Food Production in Space One of Space Biology's major objectives is to understand how the spaceflight environment affects how plants grow and thrive. The basic research has allowed NASA scientists to grow edible plants in space that could be used as a source of fresh food by the crew on the ISS. Considering that every single thing that astronauts eat is freeze-dried and comes out of a shrink-wrapped package, being able to enjoy a fresh vegetable provides a healthy and much welcomed break from this routine. Already, edible romaine lettuce and cabbage has successfully been grown on the ISS. Soon, Mizuna and tomatoes will join the list of edible plants grown in space. In the next year Space Biology will fly experiments to the ISS designed to test the growth of a variety of new plants its crew can eventually eat as they fly to the moon and Mars. To ensure the health of our astronauts, we'll be examining the nutritional composition of plants grown in space and looking at the microbiome of plants in orbit. This work may eventually lead to the production of a sustainable source of healthy food on long-duration space flights, which will help astronauts get the nutrition they need.

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Contents Overview Animal Biology Cell and Molecular Biology Microbiology Plant Biology Developmental, Reproductive and Evolutionary Biology Overview The main objective of Space Biology research is to build a better understanding of how spaceflight affects living systems in spacecraft such as the International Space Station (ISS), or in ground-based experiments that mimic aspects of spaceflight, and to prepare for future human exploration missions far from Earth. The experiments we conduct on these platforms examine how plants, microbes, and animals adjust or adapt to living in space. We examine processes of metabolism, growth, stress response, physiology, and development. We study how organisms repair cellular damage and protect themselves from infection and disease in conditions of microgravity while being exposed to space radiation. And we do it across the spectrum of biological organization, from molecules to cells, from tissues and organs, and from systems to whole organisms to communities of microorganisms. Our program's core objectives include: Discovering how biological systems respond, acclimate and adapt to the space environment Developing integrated physiological models for biology in space Identifying the underlying mechanisms and networks that govern biological processes in the space environment Promoting open science through the GeneLab Data System and Life Sciences Data Archive Developing cutting-edge biological technologies to facilitate spaceflight research Developing mechanistic understanding to support human health in space Enabling the transfer of knowledge and technology to the understanding of life on Earth Space Biology Science Plan (PDF) Search the Space Station Research Explorer for Space Biology experiments In addition to providing useful information on how living organisms adapt to spaceflight, the discoveries NASA researchers make in space have enormous implications for life on Earth. Space Biology's research into the virulence of pathogens in space, loss of bone density, and the changes in the growth of plants can impact the development of drugs that promote wound healing or tissue regeneration,

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Contents Overview Animal Biology Cell and Molecular Biology Microbiology Plant Biology Developmental, Reproductive and Evolutionary Biology

Animal Biology

Cell and Molecular Biology

Microbiology

Plant Biology

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Animal Biology

Life in space produces profound changes in biology. All organisms on Earth have adapted to perform under conditions of gravity, atmosphere, and cycles of light and darkness that have not changed in millions of years, conditions which are altered aboard spacecraft like the ISS. For example, while circling Earth at speeds of 17,130 miles an hour, crewmembers of the ISS experience sunrise and sunset 16 times a day! Simply put, terrestrial organisms are not designed for life in space.

The goal of the Space Biology Program in the animal biology area is to understand the basic mechanisms that animals use to adapt and/or acclimate to spaceflight and alterations in gravity in general. Animals are frequently used to model human disease as well as how humans respond to stressful stimuli. The most commonly

used model organisms for which genomics are now well defined include vertebrate species, e.g., rodents, both rats and mice, and a variety of invertebrate species, e.g., nematodes and insects. NASA has used these model organisms extensively to evaluate biological spaceflight hazards, elucidate the fundamental mechanisms life uses to adapt to microgravity, and apply such knowledge to advance human exploration, and for societal benefits on Earth.

Read about a Behavior Analysis Study that shows mice adapt to spaceflight. As we examine the impacts of spaceflight on animal biology and physiology, we ask the following questions: Do the effects of spaceflight level off over time, get worse, or get better? Are the effects of spaceflight exposure permanent or do they decrease and/or vanish with time upon return? Can we prevent adverse effects before they appear, or at least lessen their impact? What conditions are required for animals to live in space for years, come back to Earth, and remain healthy?

Cell and Molecular Biology Cell and molecular biology research cuts across all science disciplines in Space Biology, from understanding how single-celled organisms, such as protozoa, bacteria, and fungi respond to the conditions of spaceflight, to how all of the various cells in a complex tissue or organ work together to help an organism as a whole acclimate to such a foreign environment. The overarching goal of Space Biology Cell and Molecular Biology research at NASA is to determine how the stresses of the spaceflight environment affect living systems at the basic cellular and molecular levels, using contemporary cell and molecular biology techniques and measures. This includes characterizing and identifying changes in gene and protein expression, DNA function and structure, cellular structure and morphology, and cell-to-cell communication.

As we examine the impacts of spaceflight on the cell biology and physiology, we ask the following: What are the underlying genetic and molecular mechanisms of the cells that are influenced by gravitational changes and the space environment? Does the space flight environment affect or influence cell and molecular functions causing tissue/organ dysfunction or disease states? Does the space environment affect cellular and molecular functions in a manner that impact tissue morphogenesis or development?

Microbiology No Matter Where We Go, We Take Our Microbes Everywhere we go, we take microorganisms with us, whether we want to or not. This is true even aboard the “clean” environment of the ISS, which is a closed environment. With a crew of astronauts on board, living, breathing, exercising, and sweating, the ISS is a breeding ground for microbes.

Microbes Can Be Both Friends and Foes The effects of spaceflight on the biology of microorganisms and on microbial populations are largely unknown. In addition to posing a risk to astronaut health, bio-corrosive microorganisms that grow on metallic surfaces in spacecraft can damage both equipment and hardware. Unlike cruise ships on Earth which can be evacuated, emptied, and cleaned, problems that arise due to microbial contamination on long-duration spaceflight missions can only be resolved using tools and resources already present within the vessel. Understanding how

microbial species grow and interact with each other in this environment is the first step in preparing for such a scenario. While the presence of certain microbes can be problematic, it is also important to remember that many microbial species play important roles in Earth's various ecosystems. For example, the symbiotic relationships that have evolved between plants and certain bacteria are vital in ensuring plants receive the nutrients they require to grow and flourish. Oxygen-producing cyanobacteria that live in terrestrial bodies of water help replenish Earth's oxygen. Even the bacteria living within our digestive tracts play an important role in the digestion of our food and the proper absorption of nutrients. The presence of certain microbes may be important for the proper growth of some plants in space and may even be critical for the production of future bio-regenerative life support systems. It is therefore important to keep in mind that reaching an appropriate microbial equilibrium within a spacecraft may be vital for successful long-duration spaceflight missions. As we examine the impacts of spaceflight on microorganisms, we ask the following: What underlying genetic, molecular and biochemical processes are influenced by the spaceflight environment? How does the spaceflight environment influence microbial reproduction, growth, and physiology? Does long-duration spaceflight alter normal rates of evolutionary change? What are the effects of spaceflight on microbial communities as they interact with other organisms to effect processes such as symbioses, biodegradation, nitrogen-fixation, etc. What are the mechanisms that effect changes, such as the altered virulence or altered drug resistance, observed in some organisms during spaceflight? Plant Biology Space Biology research helps us understand the fundamentals of plant growth by examining the very building blocks of plant life down to the molecular level:

transcriptomics, genomics, proteomics, and metabolomics. To compare the effects of microgravity conditions on plants, we also conduct experiments on Earth using gravity or simulated microgravity ground controls at the Kennedy Space Center. We conduct our research on the ISS in conditions of microgravity to help us understand how to support astronauts aboard the ISS and on their long journey to Mars. Crop Rotation: Food Production in Space One of Space Biology's major objectives is to understand how the spaceflight environment affects how plants grow and thrive. The basic research has allowed NASA scientists to grow edible plants in space that could be used as a source of fresh food by the crew on the ISS. Considering that every single thing that astronauts eat is freeze-dried and comes out of a shrink-wrapped package, being able to enjoy a fresh vegetable provides a healthy and much welcomed break from this routine. Already, edible romaine lettuce and cabbage has successfully been grown on the ISS. Soon, Mizuna and tomatoes will join the list of edible plants grown in space. In the next year Space Biology will fly experiments to the ISS designed to test the growth of a variety of new plants its crew can eventually eat as they fly to the moon and Mars. To ensure the health of our astronauts, we'll be examining the nutritional composition of plants grown in space and looking at the

microbiome of plants in orbit. This work may eventually lead to the production of a sustainable source of healthy food on long-duration space flights, which will help astronauts get the nutrition they need. As we examine the impacts of spaceflight on plant biology, we ask the following: How does gravity affect plant growth, development & metabolism (e.g. photosynthesis, re- production, lignin formation, plant defense mechanisms)? Does the spaceflight environment cause alterations in plant/microbe interactions? How can horticultural approaches for sustained production of edible crops in space be both improved and implemented (especially as related to water and nutrient provision in the root zone)? What are the effects of chronic exposure to cosmic radiation on plants? How do plants sense and react to gravity and what are the molecular mechanisms involved? Developmental, Reproductive and Evolutionary Biology What happens to reproduction and the development of offspring across a lifespan and over multiple generations in space? What about differences in how males and females respond to the space environment? The Reproduction, Development, and Sex Differences Laboratory of the NASA Space Biosciences Research Branch investigated these fundamental questions about biological processes beyond Earth. Since no mammal had given birth in space at the time, researchers recognized that these questions could challenge scientists for years to come.

The main objective of Space Biology research is to build a better understanding of how spaceflight affects living systems in spacecraft such as the International Space Station (ISS), or in ground-based experiments that mimic aspects of spaceflight, and to prepare for future human exploration missions far from Earth. The experiments we conduct on these platforms examine how plants, microbes, and animals adjust or adapt to living in space. We examine processes of metabolism, growth, stress response , physiology, and development. We study how organisms repair cellular damage and protect themselves from infection and disease in conditions of microgravity while being exposed to space radiation. And we do it across the spectrum of biological organization, from molecules to cells, from tissues and organs, and from systems to whole organisms to communities of microorganisms.

Our program's core objectives include:

Discovering how biological systems respond, acclimate and adapt to the space environment

Developing integrated physiological models for biology in space

Identifying the underlying mechanisms and networks that govern biological processes in the space environment

Promoting open science through the [GeneLab Data System](#) and [Life Sciences Data Archive](#)

Developing cutting-edge biological technologies to facilitate spaceflight research

Developing mechanistic understanding to support human health in space

Enabling the transfer of knowledge and technology to the understanding of life on Earth

[Space Biology Science Plan \(PDF\)](#)

[Search the Space Station Research Explorer for Space Biology experiments](#)

In addition to providing useful information on how living organisms adapt to spaceflight, the discoveries NASA researchers make in space have enormous implications for life on Earth. Space Biology's research into the virulence of pathogens in space, loss of bone density, and the changes in the growth of plants can impact the development of drugs that promote wound healing or tissue regeneration, treatments designed to counter osteoporosis on Earth, and high-tech fertilizers that increase crop yield.

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While the presence of certain microbes can be problematic, it is also important to remember that many microbial species play important roles in Earth's various ecosystems. For example, the symbiotic relationships that have evolved between plants and certain bacteria are vital in ensuring plants receive the nutrients they require to grow and flourish. Oxygen-producing cyanobacteria that live in terrestrial bodies of water help replenish Earth's oxygen. Even the bacteria living within our digestive tracts play an important role in the digestion of our food and the proper absorption of nutrients. The presence of certain microbes may be important for the proper growth of some plants in space and may even be critical for the production of future bio-regenerative life support systems. It is therefore important to keep in mind that reaching an appropriate microbial equilibrium within a spacecraft may be vital for successful long-duration spaceflight missions.

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Does the spaceflight environment cause alterations in plant/microbe interactions?

How can horticultural approaches for sustained production of edible crops in space be both improved and implemented (especially as related to water and nutrient provision in the root zone)?

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URL: <https://osdr.nasa.gov/bio/>

TITLE: Biological and Physical Sciences Data

Explore This Section Biological and Physical Sciences Data The NASA Open Science Data Repository (OSDR) and Physical Sciences Informatics (PSI) provide open access to biological and physical science datasets from spaceflight and ground studies, enabling data reuse for discovery and innovation. Together, they accelerate research, broaden participation, and advance NASA's open science mission.

BPS Data and Repositories

- OSDR Biological Data Project** - Learn more about OSDR PSI Physical Sciences Data Project - Learn more about PSI NBISCBiorepository of non-human samples - Learn more about NBISCSMCCRRepository for collections of space microbes - Learn more about SMCCExplore and Contribute Biological Data Search biological space-relevant datasets Explore Biological Submission Portal Submit your biological space-relevant data Contribute Physical Sciences Data Search physical sciences datasets Explore Physical Sciences Submission Portal Submit your physical sciences data Contribute New Biological Data
- OSD-655** Pick-and-eat space crop production flight testing on the International Space Station (Elemental Analysis, ORAC, Phenolic Content)
- OSD-949** Skeletal muscle-specific inducible AMPK α 1/ α 2 knockout mice develop muscle weakness, glycogen depletion, and fibrosis that persists during disuse atrophy
- OSD-890** Planetary Protection Low Biomass Nanopore Pilot Study
- New Physical Sciences Data** PSI-190 Particle Engulfment and Pushing By Solidifying Interfaces (PEP) PSI-177 Coupled Growth in Hypermonotectics (CGH) PSI-186 Investigating the Structure of Paramagnetic Aggregates from Colloidal Emulsions - 4 (InSPACE-4) Working Groups and Data Management OSDR AWGs Join OSDR Analysis Working
- BPS Data Management Policy** Policy and guidelines for managing scientific Data
- More Data tools** Visualization Portals Explore OSDR Visualization Portals Biospecimen Search Tool Search and request biospecimens Microgravity Tool Explore Microgravity Analysis Platform SPLASH Database of spaceflight experimental hardware
- Latest Content** 2 min read OSDR and PSI Unveil New Consolidated Website Article 4 months ago 3 min read Joe A. Adam Presents Ring Sheared Drop (RSD) Research at 2025 ISSRDC Article 4 months ago 2 min read New File Download Process for PSI Large Requests Article 4 months ago
- Resources** Task Book The NASA Task Book is an online database of research projects supported by NASA's Biological & Physical Sciences (BPS) Division and Human Research Program (HRP).
- NSLSL** The NSLSL seeks to consolidate global space life sciences literature into a single database.
- PSI Database Tutorial** A guide to search and navigate investigations in PSI.
- PSI Workspace Tutorial** A guide designed to help you navigate the PSI workspace.
- OSDR Tutorials** How To Guides for OSDR applications.

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URL: <https://science.nasa.gov/citizen-science/osdr-awg/>

TITLE: Open Science Data Repository Analysis Working Groups (OSDR AWG)

Explore This SectionOpen Science Data Repository Analysis Working Groups (OSDR AWG)How does life that's evolved here on Earth react to the radiation-rich, gravity-poor environment of space? It's an important question to ask before we send people to Mars or anywhere else in the solar system. The Open Science Data Repository Analysis Working Groups (OSDR AWGs) are teams of people - scientists, students, and members of the public - working together to answer this question.The nine Analysis Working Groups examine data from NASA missions and space experiments collected in theOpen Science Data Repository(OSDR). These teams use the data to answer questions in basic science, applied science, and health outcomes for space exploration. Dozens of project groups are active at any time.The OSDR provides free and open data analysis/visualization tools for anyone to use, and free, open, on-demand trainings for youth, students, and the public to learn bioinformatics, artificial intelligence, and machine learning for space biology. Your ideas and collaborative research may enable the next breakthrough to help life thrive in deep space!Go to Project Websiteabout Open Science Data Repository Analysis Working Groups (OSDR AWG)project taskIndependent work with datadivisionBiological and Physical SciencewhereOnline launched2018To view this video please enable JavaScript, and consider upgrading to a web browser that supports HTML5 videoWhat you'll doThe Analysis Working Groups (AWG) have two main activities:provide feedback on scientific standards that support reuse of data related to life in space;mine and reuse OSDR data to conduct scientific analysis to learn more from the valuable datasets in it.Become part of a global network of passionate scientists and volunteers working together to unlock the secrets to help terrestrial life thrive in space.RequirementsTime:Time: Varies. All AWGs have monthly calls. There are sub-group

project meetings as well. Depending on interest and expertise, members likely do ~5 hours of activity (writing, coding, analyzing data) between meetings. Equipment: Internet-connected computer. Knowledge: None required. Familiarity with data engineering, statistics, coding, space biology literature is very helpful. There are tutorials here: <https://osdr-tutorials.readthedocs.io/en/latest/>, and various open trainings available to AWG members (see below). Get started! Visit the project website. Read the AWG Charter. Submit your request to join the AWG! Not sure you're ready? Check out the training resources below. Learn More. There is a 'Forum-Space' for OSDR AWG members to collaborate, find projects, chat with one another, ask questions, and post/find opportunities. Here is the Forum-Space Homepage and pinned top post is the maintained list of all current projects. The About page on this site has lots of information and links to resources. In addition to the OSDR tutorials that cover the basics of accessing and navigating the repository, there are some special resources for learners. GeneLab for High Schools is a summer intensive internship for high school students that introduces space biology and bioinformatics analysis methods. Visit the link for information on this year's offerings. Past teachers who have participated as interns in this course alongside students have created curricular materials — units, worksheets, and supporting activities — to support classroom participation in space biology using OSDR. Click here for the teacher-created lessons. GeneLab for Colleges and Universities and Artificial Intelligence/Machine Learning (AI/ML) in Space Biology trainings are also available as free, open-access, on-demand courses designed to support skill development. Sign up for the OSDR Newsletter! The current Analysis Working Groups are listed below. Each perform the two functions of AWGs - providing feedback to data standards and conducting analysis of data within the OSDR - for their disciplinary area. The Animal AWG focuses on data from experiments with animal in space, seeking to better understand basic animal science as well as animal evolution and adaptation to the conditions of space. The experiments work with rodents, fruit flies, roundworms, and tissue-on-a-chip technology. The Artificial Intelligence/Machine Learning (AI/ML) AWG develops and deploys AI and ML tools to answer questions about spaceflight biology and human health. The NASA Ames Life Sciences Data Archive (ALSDA) AWG focuses on physiological, phenotypic, biomedical, imaging, and behavioral data. This group spends a majority of its time on data and metadata standards. The Female Reproductive System AWG investigates the impacts of spaceflight on female reproductive health, from endocrinology to reproductive organ health. The Human AWG investigates the direct impacts of living in space on human health and works to connect an individual's response to space to their genetics, physiology, biochemistry, etc. to better predict health outcomes. The Microbial AWG investigates microbial life and the ecosystems to support life in space; investigations address bacteria, yeast, viruses, and other types of microbiological life. The Multi-Omics AWG. Each 'omics

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Go to Project Website about Open Science Data Repository Analysis Working Groups (OSDR AWG)

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Equipment:Internet-connected computer

Knowledge:None required. Familiarity with data engineering, statistics, coding, space biology literature is very helpful. There are tutorials here:<https://osdr-tutorials.readthedocs.io/en/latest/>, and various open trainings available to AWG members (see below).

Get started!

Visit the project website.

Read the AWG Charter

Submit your request to join the AWG!

Not sure you're ready? Check out the training resources below.

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URL: <https://science.nasa.gov/science-research/biological-physical-sciences/osdr-and-psi-unveil-new-consolidated-website/>

TITLE: OSDR and PSI Unveil New Consolidated Website

2 min readOSDR and PSI Unveil New Consolidated Website

Danielle LopezSep 30, 2025ArticleThe Open Science Data Repository (OSDR) and Physical Sciences Informatics (PSI) has a new home. As part of NASA's website consolidation initiative, the OSDR and PSI site have officially transitioned to the Biological and Physical

Sciences (BPS) Datapage, accessible through the “Data” menu on the Science Mission Directorate’s (SMD) website at science.nasa.gov. This strategic move reflects NASA’s broader effort to streamline user access to resources, unify digital platforms, and provide a more consistent experience across the SMD divisions. The OSDR and PSI consolidation brings together two powerful resources, giving researchers a single point of access to search both biological and physical sciences datasets. By integrating these repositories, NASA is expanding opportunities for cross-disciplinary research, enabling scientists to draw connections across fields and gain deeper insights into how biology and physical systems respond to spaceflight environments. The redesigned OSDR website continues to serve as a hub for open access to space science data, offering a modernized layout, improved navigation, and direct pathways to explore datasets and analysis tools, and submit data through the submission portals enabled by OSDR and PSI. Whether you are a scientist seeking resources for new investigations, a student learning about space research, or a collaborator from another discipline, the updated platform makes accessing NASA’s open science data easier than ever. Check out the new BPS Data and OSDR, and PSI websites now! The launch of the new consolidated OSDR and PSI websites underscores NASA’s commitment to open science and to advancing knowledge through transparent, accessible, and reusable data. By situating OSDR under the BPS data ecosystem and combining it with PSI, NASA is strengthening visibility, fostering collaboration, and ensuring that both biological and physical sciences research in space continues to thrive.

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URL: <https://science.nasa.gov/biological-physical/precision-health/>

TITLE: Precision Health

Explore This Section Precision Health Leveraging space to unlock the secrets of aging and disease BIOLOGICAL & PHYSICAL SCIENCES Leveraging space to unlock the secrets of aging and disease Stressors encountered during space travel can affect human health, including bone and muscle loss, immune system function, microbes, and other biological responses. NASA research could provide vital information needed to help protect astronauts during future deep-space missions and advance the prevention and treatment of disease for people on Earth. Information Sheet BPS Road Map Impact & Benefits Key Investigation: AVATAR Organ Chips Aging & Disease Synthetic Biology Acclimation and Adaptation Investigations BPS Scientific Goals Related Resources About BPS Precision Health Information Sheet Sep 11, 2025 PDF (421.77 KB) Biological & Physical Sciences Road Map Sep 12, 2025 PDF (1.60 MB) Impacts & Benefits Precision Health NASA studies the physiological, cellular, and genetic alterations that can occur during space travel. This fundamental research can help ensure crew health and safety on their missions, as well as contribute to significant advancements in medicine for all. Human exploration exposes astronauts and their microbiomes to spaceflight stressors. NASA's four Precision Health focus areas will provide a mechanistic understanding of the physiological, cellular, and genetic alterations that occur during space travel. Research will expand knowledge of the short- and long-term risks of prolonged deep-space exploration, as well as the onset and progression of disease and dysfunction that could affect astronauts beyond low Earth orbit. ADVANCING MOON AND MARS MISSIONS: • A Virtual Astronaut Tissue Analog Response (AVATAR) • Microbiome of the Built Environment (MoBE) ENABLING SPACE EXPLORATION: • Identify the potential risks to human health prior to deep-space missions • Support the development of risk mitigation strategies • Inform personalized medical kits for crew BENEFITTING HUMANITY: • Expand our scientific understanding of aging and disease • Advance the use of biomedical technologies in research • Contribute to healthcare improvements, such as cancer treatments AVATAR (A Virtual Astronaut Tissue Analog Response) Avatars for Astronaut Health to Fly on Artemis II The investigation is a collaboration between NASA, multiple government agencies, and industry that seeks to gain a deeper understanding of human biology and disease, preventative measures, and personalized therapeutic treatments. Learn More about Avatars for Astronaut Health to Fly on Artemis II Emulate Organ-On-a-Chip (Organ Chips) Standardize organ-chips are tiny devices that act like small versions of human organs. Made with human cells, the chips mimic how tissues, such as the brain, heart, liver, or dozens of other organs, work. NASA research, including AVATAR, will focus on validating and leveraging these models to assess the impacts of deep-space stressors on human health. Insights could advance

personalized medicine in space and on Earth. Aging and Disease Research to-date has revealed aging-like phenotypic changes that suggest spaceflight accelerates the onset and progression of age-related disease. NASA research aims to clarify this linkage by using relevant model systems and computational modeling and analyses. Work will focus on identifying the underlying mechanisms of spaceflight aging and disease and understanding if severity is altered by the duration or destination of the mission. Synthetic Biology Microbes can be used as tools for protecting human health. NASA will investigate ways to engineer molecules that are beneficial to astronaut health, such as vitamins or pharmaceuticals essential for long-duration missions. This could include developing beneficial bacteria to counteract disease or expanding capabilities such as biosensors for measuring low oxygen. Synthetic biology research could yield advancements benefitting agriculture, medicine, and manufacturing. Acclimation and Adaptation Organisms, including humans and microbes, may acclimate and adapt to living in a spaceflight environment. Acclimation is the short-term physiological responses organisms undergo, while adaptation involves permanent genetic changes resulting from exposure to selective environmental pressures. NASA research will focus on understanding the mechanisms of acclimation and adaptation and determining whether changes are beneficial or associated with dysfunction and disease.

Investigations

01 AVATAR (A Virtual Astronaut Tissue Analog Response) This investigation will use organ-on-a-chip devices, or organ-chips, to study the effects of increased radiation and microgravity on human health. The chips will contain cells derived from Artemis II astronauts and accompany crew on their ten-day journey around the Moon. [Learn More](#)

02 GEARS (Genomic Enumeration of Antibiotic Resistance in Space) This investigation will focus on two types of antibiotic-resistant bacteria, *Enterococcus faecalis* (EF) and *Enterococcus faecium*, that have been found on the International Space Station and can be hazardous to crew health. [Learn More](#)

03 Microgravity Associated Bone Loss-B (MABL-B) This investigation tests potential treatments to prevent bone loss in astronauts by examining how microgravity affects bone-forming and bone-degrading cells. Focusing on blocking IL-6 protein signaling pathways that may contribute to bone deterioration in space. [Learn More](#)

04 Alteration of Bacillus Subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome (BRIC-26) The investigation will support the first in-depth evaluation of DNA structure for a microbe cultured in spaceflight. The work will provide a deeper understanding how the spaceflight environment can generate cellular outcomes that can in turn affect the greater health of the exploration ecosystem. This work pioneer's scientific discovery in the field of microbiology and supports the ability for crew to thrive in the deep space environment. [Learn More](#)

05 BioScience-4 The BioScience-4 mission is the first study to investigate the multiplication of nervous system stem cells in microgravity. This experiment will test whether these important cells from the brain and spinal cord divide faster into two

daughter cells in the microgravity environment of space.[Learn More](#)[06](#)[See All BPS Investigations](#)[Learn More](#)[ADDITIONAL](#)[Biological & Physical Sciences](#).[Our Goals At-A-Glance](#).[Overview](#)[Quantum Leaps](#)[Foundations](#)[Space Crops](#)[Space Labs](#)[BPS Scientific Goals Overview](#)[Revolutionary Research in Extraordinary Places](#). NASA research contributes to breakthroughs that advance national priorities and maintain U.S. leadership in science and technology. Studying the fundamental effects of space stressors (such as radiation and microgravity) on biological and physical phenomena promotes mission success and benefits life on Earth.[Read More](#)[Quantum Leaps](#)[Unraveling mysteries of the universe](#) While modern physics has led to numerous scientific breakthroughs, many aspects of quantum phenomena remain unexplained. NASA's research conducted in space offers unique opportunities to advance quantum science in ways that Earth-based studies cannot. Technologies like smartphones, computers, GPS, and medical imaging all stem from quantum research. Continued exploration in this field could contribute to innovations beyond our most imaginative theories.[Read More](#)[Foundations](#)[Revealing the novel behaviors of fluids, fire, and materials in space](#) Physical phenomena behave differently in space: how flames burn, fluids flow, and materials react to extreme conditions. Research in these areas can lead to scientific breakthroughs and new technologies that enable safe, sustained missions to the Moon, Mars, and beyond. It can also contribute to everyday life, including improvements in fire safety, manufacturing, commercial products, and more.[Read More](#)[Space Crops](#)[Boldly growing where no one has grown before](#) To go farther and stay longer in space, crew will need sustainable sources of nutrition. Crops can provide fresh food, benefit astronaut mental well-being, and improve space habitats. Studying how plants adapt to harsh conditions in space can lead to agricultural innovations that support deep-space exploration and improve farming in austere environments on our home planet.[Read More](#)[Space Labs](#)[Advancing research in space, on any platform, anywhere](#) Conducting experiments in space reveals phenomena impossible to observe on Earth. Space Labs enable the use research capabilities across a spectrum of spaceflight environments — suborbital, low Earth orbit, Moon, Mars, and distant destinations — to push the boundaries of scientific knowledge.[Read More](#)[Related Resources](#)[Article](#) 3 min read [Bone Loss Research Launches Aboard NASA's SpaceX-33 Resupply Mission](#) "Avatars" Making Big Strides in Precision Health Care [Article](#) 6 min read [Artemis II Crew to Advance Human Spaceflight Research](#) 10 Things to Know About Organ Chips and How They Benefit Humanity [Article](#) 10 min read [Artemis II Crew Both Subjects and Scientists in NASA Deep Space Research](#) [Biological & Physical Sciences Division](#) NASA's Biological and Physical Sciences Division pioneers scientific discovery and enables exploration by using space environments to conduct investigations not possible on Earth. Studying biological and physical phenomena under extreme conditions allows researchers to advance the

fundamental scientific knowledge required to go farther and stay longer in space, while also benefitting life on Earth.

Precision HealthLeveraging space to unlock the secretsof aging and disease

Leveraging space to unlock the secretsof aging and disease

BIOLOGICAL & PHYSICAL SCIENCESLeveraging space to unlock the secretsof aging and diseaseStressors encountered during space travel can affect human health, including bone and muscle loss, immune system function, microbes, and other biological responses. NASA research could provide vital information needed to help protect astronauts during future deep-space missions and advance the prevention and treatment of disease for people on Earth.Information SheetBPS Road MapImpact & BenefitsKey Investigation: AVATAROrgan ChipsAging & DiseaseSynthetic BiologyAcclimation and AdaptationInvestigationsBPS Scientific GoalsRelated ResourcesAbout BPS

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BIOLOGICAL & PHYSICAL SCIENCES

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Information Sheet

BPS Road Map

Impact & Benefits

Key Investigation: AVATAR

Organ Chips

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Impacts & Benefits
Precision Health
NASA studies the physiological, cellular, and genetic alterations that can occur during space travel. This fundamental research can help ensure crew health and safety on their missions, as well as contribute to significant advancements in medicine for all. Human exploration exposes astronauts and their microbiomes to spaceflight stressors. NASA's four Precision Health focus areas will provide a mechanistic understanding of the physiological, cellular, and genetic alterations that occur during space travel. Research will expand knowledge of the short- and long-term risks of prolonged deep-space exploration, as well as the onset and progression of disease and dysfunction that could affect astronauts beyond low Earth orbit.
ADVANCING MOON AND MARS MISSIONS:
• A Virtual Astronaut Tissue Analog Response (AVATAR)
• Microbiome of the Built Environment (MoBE)
ENABLING SPACE EXPLORATION:
• Identify the potential risks to human health prior to deep-space missions
• Support the development of risk mitigation

strategies• Inform personalized medical kits for crewBENEFITTING HUMANITY:• Expand our scientific understanding of aging and disease• Advance the use of biomedical technologies in research• Contribute to healthcare improvements, such as cancer treatments

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AVATAR (A Virtual Astronaut Tissue Analog Response) Avatars for Astronaut Health to Fly on Artemis II The investigation is a collaboration between NASA, multiple government agencies, and industry that seeks to gain a deeper understanding of human biology and disease, preventative measures, and personalized therapeutic treatments. Learn More about Avatars for Astronaut Health to Fly on Artemis II Emulate

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Emulate

Organ-On-a-Chip(Organ Chips)

Standardize organ-chips are tiny devices that act like small versions of human organs. Made with human cells, the chips mimic how tissues, such as the brain, heart, liver, or dozens of other organs, work. NASA research, including AVATAR, will focus on validating and leveraging these models to assess the impacts of deep-space stressors on human health. Insights could advance personalized medicine in space and on Earth.

Aging and Disease

Research to-date has revealed aging-like phenotypic changes that suggest spaceflight accelerates the onset and progression of age-related disease. NASA research aims to clarify this linkage by using relevant model systems and computational modeling and analyses. Work will focus on identifying the underlying mechanisms of spaceflight aging and disease and understanding if severity is altered by the duration or destination of the mission.

Microbes can be used as tools for protecting human health. NASA will investigate ways to engineer molecules that are beneficial to astronaut health, such as vitamins or pharmaceuticals essential for long-duration missions. This could include developing beneficial bacteria to counteract disease or expanding capabilities such as biosensors for measuring low oxygen. Synthetic biology research could yield advancements benefitting agriculture, medicine, and manufacturing.

Organisms, including humans and microbes, may acclimate and adapt to living in a spaceflight environment. Acclimation is the short-term physiological responses organisms undergo, while adaptation involves permanent genetic changes resulting from exposure to selective environmental pressures. NASA research will focus on understanding the mechanisms of acclimation and adaptation and determining whether changes are beneficial or associated with dysfunction and disease.

Investigations01AVATAR(A Virtual Astronaut Tissue Analog Response)This investigation will use organ-on-a-chip devices, or organ-chips, to study the effects of increased radiation and microgravity on human health. The chips will contain cells derived from Artemis II astronauts and accompany crew on their ten-day journey around the Moon.Learn More02GEARS (Genomic Enumeration of Antibiotic Resistance in Space)This investigation

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05BioScience-4 The BioScience-4 mission is the first study to investigate the multiplication of nervous system stem cells in microgravity. This experiment will test whether these important cells from the brain and spinal cord divide faster into two daughter cells in the microgravity environment of space. [Learn More](#)
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ADDITIONAL

Biological & Physical Sciences.

Our Goals At-A-Glance.

OverviewQuantum LeapsFoundationsSpace CropsSpace LabsBPS Scientific Goals
OverviewRevolutionary Research in Extraordinary Places.NASA research contributes to breakthroughs that advance national priorities and maintain U.S. leadership in science and technology.Studying the fundamental effects of space stressors (such as radiation and microgravity) on biological and physical phenomena promotes mission success and benefits life on Earth.[Read More](#)Quantum LeapsUnraveling mysteries of the universeWhile modern physics has led to numerous scientific breakthroughs, many aspects of quantum phenomena remain unexplained. NASA's research conducted in space offers unique opportunities to advance quantum science in ways that Earth-based studies cannot. Technologies like smartphones, computers, GPS, and medical imaging all stem from quantum research. Continued exploration in this field could contribute to innovations beyond our most imaginative theories.[Read More](#)FoundationsRevealing the novel behaviors of fluids, fire, and materials in spacePhysical phenomena behave differently in

space: how flames burn, fluids flow, and materials react to extreme conditions. Research in these areas can lead to scientific breakthroughs and new technologies that enable safe, sustained missions to the Moon, Mars, and beyond. It can also contribute to everyday life, including improvements in fire safety, manufacturing, commercial products, and more.[Read More](#)[Space Crops](#)**Boldly growing where no one has grown before**To go farther and stay longer in space, crew will need sustainable sources of nutrition. Crops can provide fresh food, benefit astronaut mental well-being, and improve space habitats. Studying how plants adapt to harsh conditions in space can lead to agricultural innovations that support deep-space exploration and improve farming in austere environments on our home planet.[Read More](#)[Space Labs](#)**Advancing research in space, on any platform, anywhere**Conducting experiments in space reveals phenomena impossible to observe on Earth. Space Labs enable the use research capabilities across a spectrum of spaceflight environments — suborbital, low Earth orbit, Moon, Mars, and distant destinations — to push the boundaries of scientific knowledge.[Read More](#)

Quantum Leaps

[BPS Scientific Goals Overview](#)**Revolutionary Research in Extraordinary Places.**[NASA](#) research contributes to breakthroughs that advance national priorities and maintain U.S. leadership in science and technology. Studying the fundamental effects of space stressors (such as radiation and microgravity) on biological and physical phenomena promotes mission success and benefits life on Earth.[Read More](#)[Quantum Leaps](#)**Unraveling mysteries of the universe**While modern physics has led to numerous scientific breakthroughs, many aspects of quantum phenomena remain unexplained. NASA's research conducted in space offers unique opportunities to advance quantum science in ways that Earth-based studies cannot. Technologies like smartphones, computers, GPS, and medical imaging all stem from quantum research. Continued exploration in this field could contribute to innovations beyond our most imaginative theories.[Read More](#)[Foundations](#)**Revealing the novel behaviors of fluids, fire, and materials in space**Physical phenomena behave differently in space: how flames burn, fluids flow, and materials react to extreme conditions. Research in these areas can lead to scientific breakthroughs and new technologies that enable safe, sustained missions to the Moon, Mars, and beyond. It can also contribute to everyday life, including improvements in fire safety, manufacturing, commercial products, and more.[Read More](#)[Space Crops](#)**Boldly growing where no one has grown before**To go farther and stay longer in space, crew will need sustainable sources of nutrition. Crops can provide fresh food, benefit astronaut mental well-being, and improve space habitats. Studying how plants adapt to harsh conditions in space can lead to agricultural innovations that support deep-space exploration and improve farming in austere environments on our home planet.[Read More](#)[Space Labs](#)**Advancing research in**

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Quantum Leaps

Unraveling mysteries of the universe

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Foundations

Revealing the novel behaviors of fluids, fire, and materials in space

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URL: <https://www.nasa.gov/learning-resources/stem-engagement/>

TITLE: About NASA STEM Engagement

Learning Resources
About NASA STEM Engagement
NASA investments in STEM engagement are focused on building a future STEM workforce, through program elements designed to bolster capacity and to attract, engage and enable students to move toward STEM careers through NASA-unique opportunities.
What We Do
Engage students in authentic learning experiences with NASA's people, content, and facilities.
Enable students to contribute to NASA's work in exploration and discovery through challenges and work experiences.
Attract students to STEM through unique learning opportunities that spark interest and provide connections to NASA's mission and work.
NASA STEM Engagement 2024
Find out how student work experiences, internship opportunities, engagement activities, and products helped connect students to NASA's mission, work, and people – and how these impacts are preparing students to join tomorrow's skilled technical workforce.
Read FY 2024 Office of STEM Engagement Executive Summary Highlights
about NASA STEM Engagement
2024
NASA STEM Impacts and Benefits
Partnerships and Collaborations
NASA STEM Engagement Funding Opportunities
NASA STEM Impacts and Benefits
A Culture of Evidence
NASA has long held a culture of using evidence to inform its endeavors. As an Agency at the cutting edge of exploration, scientific discovery, and technological development, NASA believes that frequently and thoroughly examining and evaluating its programs, missions, and projects leads to future success. NASA STEM Engagement is dedicated to robust evidence-building practices. The Office of STEM Engagement employs a performance assessment and evaluation strategy consisting of three main components: performance activities, evaluation activities, and a learning agenda. This strategy is a crucial tool for systematically gathering evidence to identify the priorities and support decision-making that will help cultivate the next generation of explorers.
Learn More About STEM Impacts
Partnering With NASA STEM Engagement
NASA has a rich history of collaborating across the STEM ecosystem to foster innovative student experiences that leverage mission, people, and facilities. NASA conducts strategic partnerships with a wide range of external STEM engagement stakeholder organizations. These unfunded collaborators gain access to NASA mission data and imagery, subject matter expertise in scientific and technical disciplines connected to our Mission, and support with curation of NASA resources, products, and materials. Through these efforts, we coordinate with industry, educational institutions, and non-profit organizations to support development of high-quality opportunities for both youth outside of the classroom and students in pre-kindergarten through graduate school.
Learn More About Partnerships
Engagement Opportunities in NASA STEM (EONS)
Supporting research in science and technology is an important part of NASA's overall mission. NASA solicits this research through the release of various research announcements in a wide range of science and technology

disciplines. EONS is an omnibus announcement that includes a range of NASA STEM Engagement opportunities for basic and applied science and technology research and education. Specific opportunities with individual requirements and milestones will be issued periodically throughout the year as appendices to this solicitation as funds are made available for new awards. EONS opportunities are announced via the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES) website and grants.gov. Prospective proposers are required to be registered on NSPIRES. Learn More NASA STEM Leadership and Contacts Organizational Chart Elaine Ho Associate Administrator for the Office of STEM Engagement Read Biography Torry A. Johnson Deputy Associate Administrator for the STEM Engagement Program Read Biography STEM Offices and Mission Leads NASA Center STEM Engagement and Mission Contacts Read More NASA STEM Public Affairs Officers For public information and media inquiries national in scope. Read More NASA Advisory Council STEM Engagement Committee Meeting Agenda, Minutes and Reports Read More NASA Strategy for STEM Engagement Build the Next Generation of Explorers The 2022 NASA Strategic Plan Objective 4.3, "Build the next generation of explorers," establishes the agency's STEM engagement strategy. Given the Nation's need for a skilled STEM workforce and projected demand, NASA clearly has a vested interest in attracting, engaging, and preparing its future STEM professionals. 2022 NASA Strategic Plan about NASA Strategy for STEM Engagement NASA STEM Projects NASA's Office of STEM Engagement (OSTEM)'s four integrated projects create pathways for students to enter the aerospace industry. Next Gen STEM Prepares K-12 students to enter the aerospace workforce through strategic partnerships and competitive awards that build their STEM skills and capabilities so they can do the jobs needed in the growing aerospace economy. Learn More National Space Grant College and Fellowship Project A national network located in all 50 states that fosters science and engineering training, research, and industry partnerships with the goal of cultivating a skilled, innovative talent pool to advance space exploration and innovations. Learn More Established Program to Stimulate Competitive Research Promotes the development of a highly skilled workforce by funding hands-on research opportunities that not only prepare students for STEM jobs, but also lead to technological and innovative breakthroughs. Learn More Minority University Research and Education Project Funds cutting-edge research that enables students at Minority Serving Institutions to gain the expertise America needs to remain a leader in aerospace and technology. Learn More NASA EXPRESS Newsletter Sign-up Are you looking for NASA's latest STEM opportunities and resources? Subscribe to the NASA EXPRESS newsletter to get the latest NASA STEM opportunities delivered to your inbox every Thursday. Learn More about NASA EXPRESS Newsletter Sign-up Join the Artemis Missions Make, launch, compete and learn. Find your favorite way to be part of NASA's Artemis missions. Learn More get social with nasa stem engagement NASA Kahoot! Learn

With NASA YouTubeLearn With NASA PinterestLearn With NASA XLearn With NASA
FacebookKeep ExploringDiscover More Topics From NASAPeople of NASACenters and
FacilitiesNASA DirectoratesDoing Business with NASA

About NASA STEM EngagementNASA investments in STEM engagement are focused on building a future STEM workforce, through program elements designed to bolster capacity and to attract, engage and enable students to move toward STEM careers through NASA-unique opportunities.What We DoEngage students in authentic learning experiences with NASA's people, content, and facilities.Enable students to contribute to NASA's work in exploration and discovery through challenges and work experiences.Attract students to STEM through unique learning opportunities that spark interest and provide connections to NASA's mission and work.

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NASA STEM Engagement 2024 Find out how student work experiences, internship opportunities, engagement activities, and products helped connect students to NASA's mission, work, and people – and how these impacts are preparing students to join tomorrow's skilled technical workforce. [Read FY 2024 Office of STEM Engagement Executive Summary Highlights](#) [about NASA STEM Engagement 2024](#)

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Find out how student work experiences, internship opportunities, engagement activities, and products helped connect students to NASA's mission, work, and people – and how these impacts are preparing students to join tomorrow's skilled technical workforce.

NASA STEM Impacts and Benefits **Partnerships and Collaborations** **NASA STEM Engagement Funding Opportunities** **NASA STEM Impacts and Benefits** **A Culture of Evidence** NASA has long held a culture of using evidence to inform its endeavors. As an Agency at the cutting edge of exploration, scientific discovery, and technological development, NASA believes that frequently and thoroughly examining and evaluating its programs, missions, and projects leads to future success. NASA STEM Engagement is dedicated to robust evidence-building practices. The Office of STEM Engagement employs a performance assessment and evaluation strategy consisting of three main components: performance activities, evaluation activities, and a learning agenda. This strategy is a crucial tool for systematically gathering evidence to identify the priorities and support decision-making that will help cultivate the next generation of explorers. [Learn More About STEM Impacts](#) **Partnering With NASA STEM Engagement** NASA has a rich history of collaborating across the STEM ecosystem to foster innovative student experiences that leverage mission, people, and facilities. NASA conducts strategic partnerships with a wide range of external STEM engagement stakeholder organizations. These unfunded collaborators gain access to NASA mission data and imagery, subject matter expertise in scientific and technical disciplines connected to our Mission, and support with curation of NASA resources, products, and materials. Through these efforts, we coordinate with industry, educational institutions, and non-profit organizations to support development of high-quality opportunities for both youth outside of the classroom and students in pre-kindergarten through graduate school. [Learn More About Partnerships](#) **Engagement Opportunities in NASA STEM (EONS)** Supporting research in science and technology is an important part of NASA's overall mission. NASA solicits this research through the release of various research announcements in a wide range of science and technology disciplines. EONS is an omnibus announcement that includes a range of NASA STEM Engagement opportunities for

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Torry A. Johnson Deputy Associate Administrator for the STEM Engagement Program
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NASA Strategy for STEM Engagement Build the Next Generation of Explorers The 2022 NASA Strategic Plan Objective 4.3, “Build the next generation of explorers,” establishes the agency’s STEM engagement strategy. Given the Nation’s need for a skilled STEM workforce and projected demand, NASA clearly has a vested interest in attracting, engaging, and preparing its future STEM professionals. 2022 NASA Strategic Plan about NASA Strategy for STEM Engagement

NASA Strategy for STEM Engagement

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Explore the UniverseExplore Cosmic HistoryStudy how the universe evolved, learn about the fundamental forces , and discover what the cosmos is made of.OverviewThe origin, evolution, and nature of the universe have fascinated and confounded humankind for centuries. New ideas and major discoveries made during the 20th century transformed cosmology – the term for the way we conceptualize and study the universe – although much remains unknown.Learn moreThe origin, evolution, and nature of the universe have fascinated and confounded humankind for centuries. New ideas and major discoveries made during the 20th century transformed cosmology – the term for the way we conceptualize and study the universe – although much remains unknown.Quick FactsThe cosmic microwave background is the oldest light we can observe in the universe.The universe's first stars were 30 to 300 times more massive than our Sun and millions of times brighter.Big Bang StoriesExplore All Big Bang Stories18 Min ReadCosmic Dawn with Nobel Laureate John MatherArticle6 Min ReadCosmic Mapmaker: NASA's SPHEREx Space Telescope Ready to Launch24 Min ReadThe Mind-Bending Math Inside Black Holes6 Min ReadHow NASA's Roman Space Telescope Will Illuminate Cosmic DawnArticle11 Min ReadWhat is Dark Energy? Inside our accelerating, expanding UniverseArticleKeep ExploringDiscover More Topics From NASADark Matter & Dark EnergyGalaxiesStarsBlack Holes

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Erika Peters Jan 23, 2026 Article Listen to this audio excerpt from Dustin Gohmert, Orion Crew Survival System (OCSS) manager: 0:00 / 0:00 Your browser does not support the audio element. During NASA's Artemis II mission around the Moon, the astronauts inside the Orion spacecraft will be wearing specialized pressure suits designed to protect them throughout their journey. At NASA's Johnson Space Center in Houston, Dustin Gohmert leads the team responsible for these suits, known as the Orion Crew Survival System (OCSS). "We work with the crew to say, 'Here's this design concept we have. How does this really work in the spaceflight environment?'" Gohmert said. "As we evolve the design, we take the crew's input and we adapt the suit over time to take into account not only the desire we have for safety, but the real-world impacts that it has." The suits will protect astronauts on launch day, throughout high-risk parts of missions near the Moon, during the high-speed return to Earth, and in emergency situations if such events arise. The OCSS suits are engineered to sustain life for up to six days in the event of an emergency, and can provide the astronauts oxygen, hydration, food, and waste management needed on their way back to Earth. Dustin Gohmert, Orion Crew Survival System (OCSS) manager, sits in the OCSS Lab at NASA's Johnson Space Center in Houston. Credits: NASA/Rad Sinyak "In an emergency, you're essentially living in a personal spacecraft that's only an inch bigger than your body," Gohmert said. "That's the reality of survival in space." Gohmert's team in the Orion Crew Survival Systems Lab manages every phase of the suits, including processing, designing, qualifying, and testing them for the mission, as well as integrating them with the Orion spacecraft. Their work addresses engineering challenges, such as how much internal pressure the suit can safely maintain and for how long. The team custom-builds each suit to fit the anatomy of the astronauts. Crew members undergo detailed sizing and multiple fit checks to ensure precision, and their feedback is a key part of the design evolution and refinement of the suit. Orion Crew Survival System (OCSS) Manager Dustin Gohmert and his team perform a flight suit long duration fit check with Artemis II crew member Christina Koch in the OCSS Lab at NASA's Johnson Space Center in Houston. Credit: NASA/Josh Valcarcel After earning his bachelor's in mechanical engineering from the University of Texas at San Antonio and his master's in engineering from the University of Texas at Austin, Gohmert joined United Space Alliance before becoming a NASA civil servant. He worked through the end of the Space Shuttle Program and later transitioned to Orion. Working on the suit throughout his career has been both technically challenging and a deeply personal responsibility. The weight of it is incredible; knowing the ultimate responsibility you and the team share in the safety of the crew and the mission. Every thought we have, every piece of

paper we write — crew is the number one priority.dustin GohmertOrion Crew Survival Systems (OCSS) ManagerAs NASA prepares to explore deep space with Artemis II, Gohmert's role will play a part in safely sending crew members around the Moon and returning them home.“I was born after the last Moon landing,” he said. “To actually be a part of the next round is kind of overwhelming. It’s awe-inspiring in every possible way.”

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TITLE: NASA Sets Briefings for SpaceX Crew-12 Mission to Space Station

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Jessica Taveau Jan 23, 2026 MEDIA ADVISORY M26-008 NASA Headquarters Johnson Space Center NASA's SpaceX Crew-12 crew, from left to right, is NASA astronaut Jessica Meir and Jack Hathaway, ESA (European Space Agency) astronaut Sophie Adenot, and Roscosmos cosmonaut Andrey Fedyaev. Credit: SpaceX NASA and its partners will discuss the upcoming crew rotation to the International Space Station during a pair of news conferences on Friday, Jan. 30, from the agency's Johnson Space Center in Houston. At 11 a.m. EST, mission leadership will discuss final launch and mission preparations in a news conference that will stream on the agency's YouTube channel. Next, the crew of NASA's SpaceX Crew-12 mission will participate in a virtual news conference from NASA Johnson crew quarters at 1 p.m., also on the agency's YouTube channel. Individual streams for each of the events will be available on that page. This is the final media opportunity with Crew-12 before they travel to NASA's Kennedy Space Center in Florida for launch. Crew-12 will carry NASA astronaut Jessica Meir and Jack Hathaway, ESA (European Space Agency) astronaut Sophie Adenot, and Roscosmos cosmonaut Andrey Fedyaev to the orbiting laboratory. The crew will launch aboard a SpaceX Dragon spacecraft on the company's Falcon 9 rocket from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida. The agency is working with SpaceX and its international partners to review options to advance the launch of Crew-12 from its original target date of Sunday, Feb. 15. United States-based media interested in attending in person must contact the NASA Johnson newsroom no later than 5 p.m. CST on Thursday, Jan. 29, at 281-483-5111 or orjsccommu@mail.nasa.gov. Media wishing to join the news conferences by phone must contact the Johnson newsroom by 9:45 a.m. on the day of the event. A copy of NASA's media accreditation policy is available online. Briefing participants are as follows (all times Eastern and subject to change based on real-time operations):

11 a.m.: Mission Overview News Conference Ken Bowersox, associate administrator, NASA's Space Operations Mission Directorate Steve Stich, manager, Commercial Crew Program, NASA Kennedy Dana Weigel, manager, International Space Station Program, NASA Johnson Andreas Mogensen, Human Exploration Group Leader, ESA SpaceX representative

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deputy for the Flight Integration Division, and assistant to the chief astronaut for the human landing system. A commander in the United States Navy, Hathaway was selected as part of the 2021 astronaut candidate class. This will be Hathaway's first spaceflight. The South Windsor, Connecticut, native holds a bachelor's degree in physics and history from the U.S. Naval Academy and master's degrees in flight dynamics from Cranfield University and national security and strategic studies from the U.S. Naval War College, respectively. Hathaway also is a graduate of the Empire Test Pilot's School, Fixed Wing Class 70 in 2011. At the time of his selection, Hathaway was deployed aboard the USS Truman, serving as Strike Fighter Squadron 81's prospective executive officer. He has accumulated more than 2,500 flight hours in 30 different aircraft, including more than 500 carrier arrested landings and 39 combat missions. The Crew-12 mission will be Adenot's first spaceflight. Before her selection as an ESA astronaut in 2022, Adenot earned a degree in engineering from ISAE-SUPAERO in Toulouse, France, specializing in spacecraft and aircraft flight dynamics. She also earned a master's degree in human factors engineering at Massachusetts Institute of Technology in Cambridge. After earning her master's degree, she became a helicopter cockpit design engineer at Airbus Helicopters and later served as a search and rescue pilot at Cazaux Air Base from 2008 to 2012. She then joined the High Authority Transport Squadron in Villacoublay, France, and served as a formation flight leader and mission captain from 2012 to 2017. Between 2019 and 2022, Adenot worked as a helicopter experimental test pilot in Cazaux Flight Test Center with DGA (Direction Générale de l'Armement – the French Defence Procurement Agency). She has logged more than 3,000 hours flying 22 different helicopters. This will be Fedyaev's second long-duration stay aboard the orbiting laboratory. He graduated from the Krasnodar Military Aviation Institute in 2004, specializing in aircraft operations and air traffic organization, and earned qualifications as a pilot engineer. Prior to his selection as a cosmonaut, he served as deputy commander of an Ilyushin-38 aircraft unit in the Kamchatka Region, logging more than 600 flight hours and achieving the rank of second-class military pilot. Fedyaev was selected for the Gagarin Research and Test Cosmonaut Training Center Cosmonaut Corps in 2012 and has served as a test cosmonaut since 2014. In 2023, he flew to the space station as a mission specialist during NASA's SpaceX Crew-6 mission, spending 186 days in orbit, as an Expedition 69 flight engineer. For his achievements, Fedyaev was awarded the title Hero of the Russian Federation and received the Yuri Gagarin Medal. For more information about the mission, visit: <https://www.nasa.gov/commercialcrew-end-Joshua-Finch/> Jimi Russell Headquarters, Washington 202-358-1100 joshua.a.finch@nasa.gov / james.j.russell@nasa.gov Sandra Jones / Joseph Zakrzewski Johnson Space Center, Houston 281-483-5111 sandra.p.jones@nasa.gov / joseph.a.zakrzewski@nasa.gov

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For more information about the mission, visit:

<https://www.nasa.gov/commercialcrew>

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URL: <https://www.nasa.gov/gallery/quesst/>

TITLE: Quesst Mission Image Gallery

Quesst Mission Image GalleryThis collection of images showcases the history, research, and operations associated with NASA's Quesst mission to enable the possibility of commercial supersonic flight over land.NASA test pilot Nils Larson steps out of the X-59 after successfully completing the aircraft's first flight Tuesday, Oct. 28,...The X-59 quiet supersonic research aircraft arrives at NASA's Armstrong Flight Research Center in Edwards, California, following its first flight...NASA's X-59 quiet supersonic research aircraft cruises above Palmdale and Edwards, California, during its first flight, Tuesday, Oct. 28, 2025....NASA's X-59 quiet supersonic research aircraft lifts off for its first flight Tuesday, Oct. 28, 2025, from U.S. Air Force...NASA's X-59 quiet supersonic research aircraft flies above Palmdale and Edwards, California, on its first flight Tuesday, Oct. 28, 2025,...NASA's X-59 quiet supersonic research aircraft flies above Palmdale and Edwards, California, on its first flight Tuesday, Oct. 28, 2025....NASA's X-59 quiet supersonic research aircraft lifts off for its first flight Tuesday, Oct. 28, 2025, from U.S. Air Force...NASA's X-59 quiet supersonic research aircraft lifts off for its first flight Tuesday, Oct. 28, 2025, from U.S. Air Force...NASA's X-59 quiet supersonic research aircraft taxis across the runway during a low-speed taxi test at U.S. Air Force Plant...NASA's X-59 quiet supersonic research aircraft moves under its own power for the first time at Lockheed Martin's Skunk Works...Here you see the X-59 scaled model inside the JAXA supersonic wind tunnel during critical tests related to sound predictions.NASA's X-59 quiet supersonic research aircraft sits inside its run stall in preparation for maximum afterburner testing at Lockheed Martin's...NASA's X-59 competes with sunset for best light show with its unique Mach diamonds, also known as shock diamonds, during...NASA's X-59 fires up its jet engine against a darkening sky, showcasing its unique Mach diamonds, also known as shock...NASA's X-59 complements the glow of the sky at dusk with its unique Mach diamonds, also known as shock diamonds,...NASA's X-59 lights up the night sky with its unique Mach diamonds, also known as shock diamonds, during maximum afterburner...A closer view of the powerful

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The supersonic X-1E research aircraft was the last of NASA's experimental X-1 series of aircraft. From 1955-1958, it made 26...

Before NASA's quiet supersonic X-59 aircraft takes to the skies, plenty of testing happens to ensure a safe first flight....

A model of the X-59 aircraft, the centerpiece of NASA's mission to gather information intended to help enable a new...

Don Durston never grew out of building model planes – they're just a whole lot more sophisticated these days! The...

This colorized schlieren image is of a small-scale model of NASA's X-59 Quiet SuperSonic Technology (QueSST) airplane taken inside NASA...

NASA's X-59 Quiet SuperSonic Technology aircraft (QueSST) is pictured here at Lockheed Martin Skunk Works in California, wrapped up in...

The NASA and Lockheed Martin team behind the X-59 Quiet SuperSonic Technology (QueSST) have recently removed the aircraft from its...

URL: <https://www.nasa.gov/aeronautics/nasas-sonic-boom-research-takes-shape/>

TITLE: NASA's Sonic Boom Research Takes "Shape"

4 min read NASA's Sonic Boom Research Takes "Shape"

NASA's Sonic Boom Research Takes "Shape"

Jim Banke Managing Editor/Senior Writer Nov 06, 2013 Article This book tells the story of a strangely shaped aircraft that proved to researchers it was possible to lower the volume of sonic booms. NASA / Mike Ryan If an airplane flies overhead at supersonic speed and no one below can hear it, did it make a sonic boom? Answering that twist on the classic falling tree in the forest question gets to the heart of a new book published by NASA that explores the agency's research into understanding how sonic booms work and how best to make them less of a public nuisance. "Quieting the Boom: The Shaped Sonic Boom Demonstrator and the Quest for Quiet Supersonic Flight" is available online at no cost as an e-book, while printed versions of the book may be purchased from NASA's Information Center. The 388-page book written by Lawrence R. Benson begins with an overview of early supersonic flight that triggered the first sonic booms and sets the stage for telling the story of the Shaped Sonic Boom Demonstrator research aircraft, which is the primary focus of the book. "I think 'Quieting the Boom' shows the value of sustaining basic and applied research to solve difficult scientific and engineering challenges, even if success takes decades to achieve," Benson said. This modified Northrop F-5E jet was used during 2003 for NASA's Shaped Sonic Boom Demonstration program, a successful effort to show that an aircraft's shape can be used to reduce the intensity of the sonic booms it creates while flying supersonic. NASA During the 1950s and into the 1960s, sonic booms were echoing across the nation as the Air Force deployed more and more of the Mach-1-busting jets that were designed to keep the peace during the Cold War. But the public voice protesting the noise was just as forceful as the sonic booms themselves. In fact, as Benson points out in the book, between 1956 and 1968 there were 38,831 claims against the Air Force (14,006 were approved) to cover losses from sonic booms that ranged from broken glass and cracked plaster to assertions of pets dying and livestock going insane. It was within this difficult environment that NASA and the Federal Aviation Administration led a research effort in support of industry designing and building a commercial supersonic transport, which became known as the SST. One of many microphones arrayed under the path of the F-5E SSBE (Shaped Sonic Boom Experiment) aircraft to record sonic booms. NASA / Tony Landis However, even as NASA continued to research sonic booms in support of the SST the public and political pressure became too much. The SST program was cancelled in 1971 and the future for supersonic airliners, business jets and other aircraft became gloomy at best. Fortunately, the nation's aeronautical innovators were not deterred. They

believed that with additional research a way could be found to quiet the booms enough to satisfy all critics and make commercial supersonic travel possible across the country. To help them find that solution, NASA partnered with others to conduct an innovative flight research program using what was named the Shaped Sonic Boom Demonstrator (SSBD), a Northrop Grumman F-5E Tiger fighter jet modified with a larger nose. "This modification, which made the front of the F-5E somewhat resemble a pelican's beak, was carefully shaped to change the pattern of shock waves it would generate while flying faster than the speed of sound," Benson said. The theory behind the nose job was that the increased airplane volume at the front of the SSBD would result in less intense sonic booms heard or felt on the ground below. The modified F-5E Shaped Sonic Boom Demonstration aircraft (center) flies off the wing of NASA's F-15B Research testbed aircraft, which flew in the supersonic shockwave of the F-5E. Following the two aircraft is an unmodified U.S. Navy F-5E used for baseline sonic boom measurements. NASA / Carla Thomas

During a series of some 30 tests flown in August 2003 and January 2004, data gathered by airborne and ground sensors of the SSBD in flight proved the theory was – well – sound, and that additional research in the future could enable the long-sought realization of commercial supersonic flight across the United States. Indeed, that research continues as NASA and its research partners explore fresh ideas and design new tools for making supersonic flight quieter. Most importantly, in telling what could easily be a complex story involving high speed flight, atmospheric sciences, acoustics and bizarre math, Benson said he attempted to keep the technical jargon to a minimum. "This book is intended to be a general history of sonic boom research, emphasizing the people and organizations that have contributed, and not a technical study of the science and engineering involved," Benson said. The American Institute of Aeronautics and Astronautics has recognized the book with its prestigious History Manuscript Award for 2014. Nearly ten years after SSBD, aircraft shapes that can reduce the level of sonic booms tend to look more like this. This overland supersonic aircraft concept is being used by NASA researchers in computer simulations and to create models for wind tunnel tests. NASA Publication of "Quieting the Boom" was sponsored and funded by the communications and education department of NASA's Aeronautics Research Mission Directorate. "Quieting the Boom" E-Book

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“Quieting the Boom” E-Book

About the Author Jim Banke Managing Editor/Senior Writer Jim Banke is a veteran aviation and aerospace communicator with more than 40 years of experience as a writer, producer, consultant, and project manager based at Cape Canaveral, Florida. He is part of NASA Aeronautics' Strategic Communications Team and is Managing Editor for the Aeronautics topic on the NASA website.

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URL: <https://www.nasa.gov/aeronautics/probing-the-sky-selected-naca-research-airplanes-and-their-contributions-to-flight/>

TITLE: Probing the Sky: Selected NACA Research Airplanes and Their Contributions to Flight

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URL: <https://ciencia.nasa.gov/mission/telescopio-espacial-roman/>

TITLE: Telescopio espacial Roman

Telescopio espacial RomanEl telescopio espacial Nancy Grace Roman resolverá interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica.Future MissionobjetivosResumen de la misiónciencia y tecnologíaInstrumentoshistoriaAcerca del nombreLlamado así en honor a la primera astrónoma jefa de la NASA, la “madre del telescopio espacial Hubble”, el telescopio espacial Nancy Grace Roman tendrá un campo de visión al menos cien veces mayor que el de Hubble, con el que podría medir la luz de mil millones de galaxias a lo largo de su vida. Este observatorio también podrá bloquear la luz de las estrellas para ver directamente exoplanetas y discos de formación planetaria, completar un censo estadístico de los sistemas planetarios que existen en nuestra galaxia y responder preguntas fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica en el infrarrojo.TipoTelescopio espacialLanzamientoAntes de mayo de 2027OBJETIVOUniverso, exoplanetasMisiónResolver interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísicaEl Dr. Lucas Paganini, ejecutivo de programa del telescopio espacial Nancy Grace Roman de la NASA, ofrece un recorrido por el lugar en donde se construyó este telescopio: la histórica sala limpia del Centro de Vuelo Espacial Goddard en Greenbelt, Maryland.Acerca del telescopio RomanEl telescopio espacial Roman es un observatorio de la NASA diseñado para resolver interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica en el infrarrojo. Este telescopio tiene un espejo principal de 2,4 metros (7,9 pies) de diámetro, el mismo tamaño que el espejo principal deltelescopio espacial Hubble. El telescopio espacial Roman contará con dos instrumentos: el instrumento de campo amplio y un coronógrafo, que es

una demostración de la tecnología. El instrumento de campo amplio tendrá un campo de visión al menos cien veces mayor que el de Hubble, y podrá capturar una mayor extensión del cielo con menos tiempo de observación. El instrumento de campo amplio es su instrumento principal y medirá la luz de mil millones de galaxias a lo largo de la vida de esta misión. Con este instrumento se llevará a cabo un sondeo del efecto microlente en el interior de la Vía Láctea en busca de más de mil exoplanetas. La demostración de la tecnología del instrumento coronógrafo generará imágenes de alto contraste y efectuará espectroscopia de los exoplanetas cercanos individualmente. La vida útil de la misión principal del telescopio espacial Roman, cuyo lanzamiento está programado a más tardar para mayo de 2027, será de cinco años, con una posible extensión de la misión de cinco años más.

MISIÓN ¿Por qué Roman? Aunque es tan sensible como las cámaras de Hubble, el instrumento de campo amplio de 300 megapíxeles del telescopio espacial Roman obtendrá imágenes de un área del cielo cien veces más grande. Esto significa que una sola imagen de Roman tendrá el detalle equivalente a cien imágenes de Hubble. Aprende más [L. Hustak \(STScI\)](#) Explorando el cosmos con Roman

Energía oscura El amplio campo de visión de Roman le permitirá generar grandes imágenes nunca antes vistas del universo, lo que ayudará a los astrónomos a explorar algunos de los mayores misterios del cosmos, como por ejemplo por qué la expansión del universo se está acelerando. Una posible explicación para esta aceleración es la energía oscura. ¿Qué es la energía oscura? [NASA](#)

Materia oscura La materia oscura es esencialmente el pegamento que mantiene unido al universo, por lo que una mejor comprensión de la distribución de la materia oscura ayudará a responder grandes preguntas cósmicas acerca de cómo está organizado el cosmos y cómo ha ido cambiando. Los mapas creados por Roman ayudarán a continuar reconstruyendo la verdadera historia de nuestro universo. ¿Qué es la materia oscura? [ESA/Hubble & NASA, R. Massey](#)

Exoplanetas En poco más de una generación, hemos pasado de conocer solo nuestro propio sistema solar a hallar miles de mundos más lejanos, conocidos como exoplanetas. Roman encontrará alrededor de 100.000 planetas mediante diferentes técnicas, y sentará las bases para el futuro Observatorio de Mundos Habitables. ¿Qué son los exoplanetas? [NASA](#)

PÓDCAST La búsqueda de vida más allá de la Tierra ¿Estamos solos en el universo? Hoy, las misiones de la NASA se enfocan en tres objetivos para explorar esta pregunta milenaria: el planeta Marte, los mundos oceánicos de nuestro sistema solar, y los planetas que orbitan alrededor de estrellas lejanas, o exoplanetas. Descubre más en este episodio de Universo curioso de la NASA. Escucha el episodio

Esta representación artística muestra cómo podría verse el sistema planetario TRAPPIST 1 desde un punto de observación cercano al planeta TRAPPIST 1f (a la derecha). [NASA/JPL-Caltech](#)

¿Quién es Nancy Grace Roman? Tenemos el placer de dedicar este observatorio espacial, encargado de aumentar nuestra comprensión de la energía oscura, el universo y nuestro lugar en él, a la doctora Nancy

Grace Roman, pionera de la astronomía espacial moderna. Lee más [¿Quién es Nancy Grace Roman?](#) Noticias relacionadas [Artículo 7 Min Read](#) [Cómo el telescopio espacial Roman retrocederá a los orígenes del universo](#) [Artículo 6 Min Read](#) [El telescopio Roman buscará agujeros negros originarios](#) 6 minutos de lectura [Telescopio Roman de la NASA buscará acumulaciones de materia oscura](#) [Artículo 8 minutos de lectura](#) [Telescopio Roman estudiará las luces titilantes de la Vía Láctea](#) [Artículo 1 minutos de lectura](#) [Mira cómo se construye el próximo telescopio espacial de la NASA](#) [Artículo](#) [Explora más temas](#) [Ciencia de la NASA](#) [Universo](#) [Sistema solar](#) [Telescopio espacial James Webb](#) [Telescopio espacial Hubble](#)

Telescopio espacial Roman El telescopio espacial Nancy Grace Roman resolverá interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica. [Future Mission](#) [objetivos](#) [Resumen de la misión](#) [ciencia y tecnología](#) [Instrumentos](#) [historia](#) [Acerca del nombre](#) Llamado así en honor a la primera astrónoma jefa de la NASA, la “madre del telescopio espacial Hubble”, el telescopio espacial Nancy Grace Roman tendrá un campo de visión al menos cien veces mayor que el de Hubble, con el que podría medir la luz de mil millones de galaxias a lo largo de su vida. Este observatorio también podrá bloquear la luz de las estrellas para ver directamente exoplanetas y discos de formación planetaria, completar un censo estadístico de los sistemas planetarios que existen en nuestra galaxia y responder preguntas fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica en el infrarrojo. [Tipo](#) [Telescopio espacial](#) [Lanzamiento](#) [Antes de mayo de 2027](#) [OBJETIVO](#) [Universo, exoplanetas](#) [Misión](#) Resolver interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica

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Telescopio espacial Roman

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Acerca del telescopio Roman

El telescopio espacial Roman es un observatorio de la NASA diseñado para resolver interrogantes fundamentales en los campos de la energía oscura, los exoplanetas y la astrofísica en el infrarrojo. Este telescopio tiene un espejo principal de 2,4 metros (7,9 pies) de diámetro, el mismo tamaño que el espejo principal del telescopio espacial Hubble. El telescopio espacial Roman contará con dos instrumentos: el instrumento de campo amplio y un coronógrafo, que es una demostración de la tecnología.

El instrumento de campo amplio tendrá un campo de visión al menos cien veces mayor que el de Hubble, y podrá capturar una mayor extensión del cielo con menos tiempo de observación. El instrumento de campo amplio es su instrumento principal y medirá la luz de mil millones de galaxias a lo largo de la vida de esta misión. Con este instrumento se llevará a cabo un sondeo del efecto microlente en el interior de la Vía Láctea en busca de más de mil exoplanetas. La demostración de la tecnología del instrumento coronógrafo generará imágenes de alto contraste y efectuará espectroscopia de los exoplanetas cercanos individualmente.

La vida útil de la misión principal del telescopio espacial Roman, cuyo lanzamiento está programado a más tardar para mayo de 2027, será de cinco años, con una posible extensión de la misión de cinco años más.

MISIÓN ¿Por qué Roman? Aunque es tan sensible como las cámaras de Hubble, el instrumento de campo amplio de 300 megapíxeles del telescopio espacial Roman obtendrá imágenes de un área del cielo cien veces más grande. Esto significa que una sola

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Explorando el cosmos con Roman
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TITLE: Wind Over Its Wing: NASA's X-66 Model Tests Airflow

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Sarah MannNASA Armstrong Public Affairs SpecialistFeb 05, 2025ArticleNASA's Sustainable Flight Demonstrator project concluded wind tunnel testing in the fall of 2024. Tests on a Boeing-built X-66 model were completed at NASA's Ames Research Center in California's Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. The model underwent tests representing expected flight conditions to obtain engineering information to influence design of the wing and provide data for flight simulators. NASA/Brandon Torres NavarreteNASA's Sustainable Flight Demonstrator (SFD) project recently concluded wind tunnel tests of its X-66 semi-span model in partnership with Boeing. The model, designed to represent half the aircraft, allows the research team to generate high-quality data about the aerodynamic forces that would affect the actual X-66. Test results will help researchers identify areas where they can refine the X-66 design – potentially reducing drag, enhancing fuel efficiency, or adjusting the vehicle shape for better flying qualities. Tests on the Boeing-built X-66 semi-span model were completed at NASA's Ames Research Center in California's Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. The model underwent tests representing expected flight conditions so the team could obtain engineering information to influence the design of the aircraft's wing and provide data for flight simulators. NASA's Sustainable Flight Demonstrator project concluded wind tunnel testing in the fall of 2024. Tests on a Boeing-built X-66 model were completed at NASA's Ames Research Center in California's Silicon Valley in its 11-Foot Transonic Unitary Plan Facility. Pressure points, which are drilled holes with data sensors attached, are installed along the edge of the wing and allow engineers to understand the characteristics of airflow and will influence the final design of the wing. NASA/Brandon Torres NavarreteSemi-span tests take advantage of symmetry. The forces and behaviors on a model of half an aircraft mirror those on the other half. By using a larger half of the model, engineers increase the number of surface pressure measurements. Various sensors were placed on the wing to measure forces and movements to calculate lift, drag, stability, and other important

characteristics. The semi-span tests follow earlier wind tunnel work at NASA's Langley Research Center in Hampton, Virginia, using a smaller model of the entire aircraft. Engineers will study the data from all of the X-66 wind tunnel tests to determine any design changes that should be made before fabrication begins on the wing that will be used on the X-66 itself. The SFD project is NASA's effort to develop more efficient aircraft configurations as the nation moves toward aviation that's more economically, societally, and environmentally sustainable. The project seeks to provide information to inform the next generation of single-aisle airliners, the most common aircraft in commercial aviation fleets around the world. Boeing and NASA are partnering to develop the X-66 experimental demonstrator aircraft.

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The SFD project is NASA's effort to develop more efficient aircraft configurations as the nation moves toward aviation that's more economically, societally, and environmentally sustainable. The project seeks to provide information to inform the next generation of single-aisle airliners, the most common aircraft in commercial aviation fleets around the world. Boeing and NASA are partnering to develop the X-66 experimental demonstrator aircraft.

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TITLE: Commercial Low Earth Orbit (LEO) Destination Contract (CLDC) – ON HOLD

JSC-OP-SubCommercial Low Earth Orbit (LEO) Destination Contract (CLDC) – ON HOLDThe CLDC acquisition is on hold. NASA has determined that, rather than moving forward at this time with a firm fixed price contract for CLD certification and services, NASA will continue to support U.S. industry's design and demonstration of CLDs with funded SAAs for the next phase. After the SAA phase is enabled, NASA anticipates establishing a follow-on certification and services phase to meet government requirements under a FAR based acquisition. Details regarding the future acquisition will be released in the future. NASA remains committed to enabling sustainable development of commercially owned and operated LEO destinations under the National Aeronautics and Space Act (Space Act), 51 U.S.C. §§ 20101-20164.The following link provides additional information regarding the SAA strategy and competition process:Commercial Destinations – Development and Demonstration Objectives (C3DO) Space Act Agreement (SAA) Website (link<https://www.nasa.gov/johnson/jsc-procurement/c3do>)This CLDC website and related announcements in www.sam.gov will remain accessible for reference purposes. In addition, NASA will continue to provide technical documents in the CLDC Technical Library site to keep industry up to date on current NASA technical standards that may be relevant to a future acquisition. Please continue to monitor www.sam.gov postings and the Acquisition Website for updates on the release of any CLDC information.CLDC Technical Documents Library

(<https://sam.gov/workspace/contract/opp/3b5b0bcd82e1462998a122448dad4aa3/view>)Please note, the documents/links on the CLDC Acquisition Website listed here are for information and planning purposes only. It does not constitute a Request for Proposal, Invitation for Bid, or Request for Quotation, and it is not to be construed as a commitment by the Government to enter a contract.No solicitation exists; therefore, do not request a copy of a solicitation. If a solicitation is released, it will be synopsized on www.sam.gov. It is each interested party's responsibility to monitor the www.sam.gov website for the

release of any subsequent solicitation or synopsis.

CLDC Links

Points of Contact

Current Information Links

Acquisition Milestone Schedule

Current News and Events

Points of Contact

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Title: Contract Specialist

Phone: 281.244.6212

Email: JSC-CLDC@mail.nasa.gov

Source Selection Authority (SSA): TBD

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Preliminary Acquisition Milestone Schedule

The acquisition is on-hold. An updated schedule will be provided once the procurement strategy is developed and approved.

Milestone TBD

Planned Date TBD

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08/22/2025 Posted Modification 1 to the Technical Library.

08/14/2025 Updated website and link to indicate this acquisition in placed on-hold until further notice.

06/10/2025 Posted Modification 4 to the Industry Day Communication.

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04/30/2025 Posted Modification 7 to the Development Program Phase 2 Requirements and Safety Technical Interchange Meeting.

04/25/2025 Updated the Industry Day Date on the Schedule to Postponed.

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04/22/2025 Posted Modification 1 to the Industry Day Communication.

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04/21/2025 Updated the Milestone Schedule to reflect updated Industry Day date.

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The following link provides additional information regarding the SAA strategy and competition process:

Commercial Destinations – Development and Demonstration Objectives (C3DO) Space Act Agreement (SAA) Website ([linkhttps://www.nasa.gov/johnson/jsc-procurement/c3do](https://www.nasa.gov/johnson/jsc-procurement/c3do))

This CLDC website and related announcements in www.sam.gov will remain accessible for reference purposes. In addition, NASA will continue to provide technical documents in the CLDC Technical Library site to keep industry up to date on current NASA technical

standards that may be relevant to a future acquisition. Please continue to monitor www.sam.gov postings and the Acquisition Website for updates on the release of any CLDC information.

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cuadrados —una región del cielo tan grande como 25 lunas llenas— y que penetre en el espacio, hasta el momento en que el universo tenía unos 500 millones de años, o menos del cuatro por ciento de su edad actual de 13.800 millones de años. Centro de Vuelo Espacial Goddard de la NASA El telescopio espacial Nancy Grace Roman de la NASA trabajará a la par con otros observatorios espaciales que poseen un amplio campo de visión para descubrir el cosmos y su dinámica de maneras que nunca antes habían sido posibles. “Roman trabajará en conjunto con observatorios de la NASA como el telescopio espacial James Webb y el Observatorio de rayos X Chandra, los cuales están diseñados para observar de cerca objetos transitorios singulares, una vez que estos han sido identificados; pero rara vez o nunca los descubren”, dijo Julie McEnery, científica principal del proyecto del telescopio Roman en el Centro de Vuelo Espacial Goddard de la NASA en Greenbelt, Maryland. “El campo de visión mucho más grande de Roman revelará muchos de estos objetos que antes eran desconocidos. Y como nunca antes habíamos tenido un observatorio como este para explorar el cosmos, incluso podríamos encontrar clases completamente nuevas de objetos y fenómenos”. La misión Sondeo de Altas Latitudes en el Dominio del Tiempo está bien diseñada para descubrir un tipo particular de estrellas que han hecho explosión, las cuales pueden ser usadas por los astrónomos para rastrear la evolución del universo e investigar posibles explicaciones para su expansión acelerada. Y dado que este estudio observará repetidamente la misma gran vista del espacio, los científicos también verán eventos esporádicos como el choque de cadáveres estelares y estrellas que son arrastradas hacia agujeros negros. El sondeo mirará más allá de nuestra galaxia para observar el mismo sector del cielo cada unos cinco días durante dos años. Al unir estas observaciones para hacer imágenes como animaciones cuadro por cuadro, se crearán videos que revelarán una gran cantidad de fenómenos transitorios. Chispas estelares en retirada Los astrónomos buscarán a través de todos estos datos un tipo especial de estrella en explosión llamada supernova de tipo Ia. Estos fenómenos se originan en ciertos sistemas estelares binarios que contienen al menos una enana blanca: son los restos del núcleo pequeño y caliente de una estrella similar al Sol. En algunos casos, esta enana puede vaciar el material de su compañera. Esto desencadena una reacción nuclear fuera de control que finalmente hace detonar a la estrella “ladrona”. Los astrónomos también han hallado evidencia que respalda otro escenario, relacionado con dos enanas blancas que giran en espiral una hacia la otra hasta que se fusionan. Si su masa combinada es lo suficientemente grande, también pueden producir una supernova de tipo Ia. Dado que estas explosiones alcanzan su punto máximo con un brillo intrínseco conocido y similar, los astrónomos pueden usarlas para determinar a qué distancia están simplemente midiendo qué tan intenso es su brillo aparente. Los astrónomos usarán el telescopio Roman para estudiar el espectro de luz de estas supernovas para averiguar a qué velocidad parecen alejarse de nosotros debido a la expansión del espacio. Al comparar

la velocidad con la que están retrocediendo las supernovas de tipo Ia a diferentes distancias, los científicos rastrearán la expansión cósmica a lo largo del tiempo. Esto nos ayudará a comprender si la energía oscura —la presión inexplicable que se cree que está acelerando la expansión del universo— ha cambiado a lo largo del tiempo y cómo lo ha hecho. El uso de estas y otras mediciones obtenidas por Roman también debería ayudar a aclarar las mediciones inconsistentes de la constante de Hubble, la cual es la tasa de expansión actual del universo. “Roman pintará una imagen más vívida del pasado y presente de nuestro universo, dándonos nuevas pistas sobre su posible destino”, dijo Rebekah Hounsell, científica investigadora de la Universidad de Maryland en el condado de Baltimore y del centro Goddard, quien está explorando formas de optimizar el Sondeo de Altas Latitudes en el Dominio del Tiempo de Roman. “Sus hallazgos podrían reconfigurar nuestra comprensión del cosmos”. Esta secuencia fotográfica de la supernova 2018gv en la galaxia NGC 2525 comprime casi un año de observaciones del telescopio espacial Hubble de la NASA en unos pocos segundos. Al inicio, la supernova eclipsa a las estrellas más brillantes de la galaxia antes de desvanecerse en la oscuridad. El telescopio espacial Nancy Grace Roman de la NASA podría captar tales eventos de principio a fin y alertar a otros telescopios, como Hubble y James Webb, para obtener datos aún más detallados. Crédito: NASA, ESA y A. Riess (STScI/JHU) y el equipo de SH0ES;

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Maravillas cósmicas fugaces

Debido a la forma en que este estudio observará el cosmos, también detectará otros fenómenos poco comunes. Gracias a Roman, seremos testigos del nacimiento de nuevos agujeros negros que se forman cuando se fusionan las estrellas de neutrones, esto es, los núcleos de estrellas que han explotado y que no fueron lo suficientemente masivas como para colapsar y formar agujeros negros por sí solas. Estos eventos titánicos crean ondas en el tejido del espacio-tiempo y brillantes explosiones de kilonovas. También se espera que la misión descubra varias decenas de eventos de disrupción de mareas, los cuales ocurren cuando una estrella que se aventura demasiado cerca de un agujero negro es destrozada por la extrema gravedad del agujero negro. La metralla estelar produce una gran cantidad de luz a medida que se acelera en su camino hacia el agujero negro. Roman captará estas erupciones de energía para aprender cómo los agujeros negros afectan su entorno. El sondeo también permitirá a los astrónomos explorar objetos variables, como las galaxias activas cuyos núcleos albergan un cuásar extremadamente brillante. Un cuásar es un faro brillante de luz intensa alimentado por un agujero negro supermasivo. El agujero negro se alimenta vorazmente de la materia que cae dentro de él y desata un torrente de radiación. La mirada constante de Roman ayudará a los astrónomos a estudiar cómo y por qué fluctúa el brillo de estos estallidos. Y al encontrar cientos de cuásares tenues y lejanos, Roman también permitirá a los científicos investigar el período conocido como época de reionización. Durante esta era cósmica, los científicos creen que la intensa luz ultravioleta

de los cuásares despojó a los electrones de sus átomos y los convirtió en iones. Esta transición marcó el comienzo del “amanecer cósmico”, ya que el universo pasó de ser en su mayor parte opaco a transparente, permitiendo que la luz visible y ultravioleta viajara libremente. “Este estudio de Roman descubrirá un tesoro oculto de datos para que los astrónomos lo examinen detalladamente, permitiendo una exploración cósmica con preguntas más abiertas de lo que normalmente es posible”, dijo McEnery. “Es posible que descubramos fortuitamente cosas completamente nuevas que aún no sabemos que debemos buscar”. El telescopio espacial Nancy Grace Roman es administrado desde el Centro de Vuelo Espacial Goddard de la NASA en Greenbelt, Maryland, con la participación del Laboratorio de Propulsión a Chorro (JPL, por sus siglas en inglés) de la NASA y Caltech/IPAC en el sur de California, el Instituto de Ciencias del Telescopio Espacial en Baltimore y un equipo integrado por científicos de diversas instituciones de investigación. Los principales socios de la industria son Ball Aerospace and Technologies Corporation en Boulder, Colorado; L3Harris Technologies en Melbourne, Florida; y Teledyne Scientific & Imaging en Thousand Oaks, California. Por Ashley Balzer Centro de Vuelo Espacial Goddard de la NASA, Greenbelt, Maryland [Read this story in English](#) [here](#). Más sobre Roman

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[Índice de contenidos](#) [Chispas estelares en retirada](#) [Maravillas cósmicas fugaces](#)

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Chispas estelares en retirada

Los astrónomos buscarán a través de todos estos datos un tipo especial de estrella en explosión llamada supernova de tipo Ia. Estos fenómenos se originan en ciertos sistemas estelares binarios que contienen al menos una enana blanca: son los restos del núcleo pequeño y caliente de una estrella similar al Sol. En algunos casos, esta enana puede vaciar el material de su compañera. Esto desencadena una reacción nuclear fuera de control que finalmente hace detonar a la estrella “ladrona”. Los astrónomos también han hallado evidencia que respalda otro escenario, relacionado con dos enanas blancas que giran en espiral una hacia la otra hasta que se fusionan. Si su masa combinada es lo suficientemente grande, también pueden producir una supernova de tipo Ia. Dado que estas explosiones alcanzan su punto máximo con un brillo intrínseco conocido y similar, los astrónomos pueden usarlas para determinar a qué distancia están simplemente midiendo qué tan intenso es su brillo aparente. Los astrónomos usarán el telescopio Roman para estudiar el espectro de luz de estas supernovas para averiguar a qué velocidad parecen alejarse de nosotros debido a la expansión del espacio. Al comparar la velocidad con la que están retrocediendo las supernovas de tipo Ia a diferentes distancias, los científicos rastrearán la expansión cósmica a lo largo del tiempo. Esto nos ayudará a comprender si la energía oscura —la presión inexplicable que se cree que está acelerando la expansión del universo— ha cambiado a lo largo del tiempo y cómo lo ha hecho. El uso de estas y otras mediciones obtenidas por Roman también debería ayudar a aclarar las mediciones inconsistentes de

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La misión Sondeo de Altas Latitudes en el Dominio del Tiempo está bien diseñada para descubrir un tipo particular de estrellas que han hecho explosión, las cuales pueden ser usadas por los astrónomos para rastrear la evolución del universo e investigar posibles explicaciones para su expansión acelerada. Y dado que este estudio observará repetidamente la misma gran vista del espacio, los científicos también verán eventos esporádicos como el choque de cadáveres estelares y estrellas que son arrastradas hacia agujeros negros.

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Por Ashley BalzerCentro de Vuelo Espacial Goddard de la NASA, Greenbelt, Maryland

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NASA Science Editorial TeamJan 21, 2026ArticleNASA's Universe of Learning Program Facilitator Guides provide educators with detailed resources, including background information, activities, and slide decks to engage audiences in exploring astrophysics themes such as Stars, Data & Image Processing, the Electromagnetic Spectrum, and Finding Exoplanets.The goal of NASA's Universe of Learning (UoL) is to connect the public to the data, discoveries, and experts that span NASA's Astrophysics missions. To make this possible, the NASA's UoL team creates engaging STEM experiences that let people explore data and discoveries from NASA's Astrophysics missions and learn from the experts behind them.Our science center does a lot of work with after school groups weekly. I can't wait to use this program guide [Finding Exoplanets] to help run some programs for our 'space week' this fall. I also appreciate the adaptations for different age groups.FacilitatorSouthern ArizonaOne example is the Program Facilitator Guides—a series of resources for informal educators that cover different astrophysics themes and

empower organizations to share NASA science with their audiences. Since their introduction, these guides have supported libraries and community centers in delivering engaging STEM learning experiences. ""The Programming Guide is just amazing ... that resource alone is really great for planning. There's so many opportunities for programs... and there's room for your own creativity as well,"" shared one educator. The NASA's UoL team is excited to announce the refresh of several Program Facilitator Guides, along with the introduction of a new guide. These resources have been updated based on feedback from the informal education community, collected through evaluation surveys, focus groups, and webinars. From events held last year before the updates, the guides received a highly favorable rating—91% of educators found them useful as a resource, emphasizing their value in supporting informal STEM education. To make them more effective, we implemented the following updates:

- Easy and direct access to all Program Facilitator Guides through a dedicated web page under the “Informal Educators” menu on NASA's Universe of Learning.
- Creating an easy-to-access URL for the Program Facilitator Guides: <https://universe-of-learning.org/program-guides>.
- Making available PowerPoint slides and Kahoot Quizzes for the facilitator to complement the Program Facilitator Guide themes.
- Moving activity guides to a more user-friendly and standard template.
- Designing a set of resources around some of the methods astronomers use to find exoplanets — worlds beyond the solar system — in collaboration with a NASA Science Mission Directorate Community of Practice for Education (SCoPE) grantee: The “Finding Exoplanets” Program Facilitator Guide.
- The “Lights, Coronagraph, Action!” Activity Guide that demonstrates how astronomers find exoplanets via direct imaging.
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The new and updated resources are available now through the following URL: <https://www.universe-of-learning.org/program-guides>. For any questions or suggestions, please contact: The NASA's Universe of Learning team

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Goddard Space Flight Center
Jan 20, 2026
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Video: Observatory Comparison (Hubble/Spitzer/Webb)
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The iconic Helix Nebula has been imaged by many ground- and space-based observatories over the nearly two centuries since it was discovered. Webb's near-infrared view of the target brings these knots to the forefront compared to the ethereal image from NASA's Hubble Space Telescope, while its increased resolution sharpens focus from NASA's retired Spitzer Space Telescope's snapshot. Additionally, the new near-infrared look shows the stark transition between the hottest gas to the coolest gas as the shell expands out from the central white dwarf.

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In Webb's image of the Helix Nebula, color represents the temperature and chemistry. A touch of a blue hue marks the hottest gas in this field, energized by intense ultraviolet light from the white dwarf. Farther out, the gas cools into the yellow regions where hydrogen atoms join into molecules. At the outer edges, the reddish tones trace the coolest material, where gas begins to thin and dust can take shape. Together, the colors show the star's final breath transforming into the raw ingredients for new worlds, adding to the wealth of knowledge

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Related Links

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Image: Helix Nebula Context (VISTA and Webb)
Video: Observatory Comparison (Hubble/Spitzer/Webb)
Downloads & Related Information
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TITLE: NASA's Hubble Reveals Largest Found Chaotic Birthplace of Planets

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NASA Hubble Mission Team Goddard Space Flight Center Dec 23, 2025 Article Astronomers using NASA's Hubble Space Telescope have imaged the largest protoplanetary disk ever observed circling a young star. For the first time in visible light, Hubble has revealed the disk is unexpectedly chaotic and turbulent, with wisps of material stretching much farther above and below the disk than astronomers have seen in any similar system. Strangely, more extended filaments are only visible on one side of the disk. The findings, which published Tuesday in *The Astrophysical Journal*, mark a new milestone for Hubble and shed light on how planets may form in extreme environments, as NASA's missions lead humanity's exploration of the universe and our place in it. Located roughly 1,000 light-years from Earth, IRAS 23077+6707, nicknamed "Dracula's Chivito," spans nearly 400 billion miles — 40 times the diameter of our solar system to the outer edge of the Kuiper Belt of cometary bodies. The disk obscures the young star within it, which scientists believe may be either a hot, massive star, or a pair of stars. And the enormous disk is not only the largest known planet-forming disk; it's also shaping up to be one of the most unusual. "The level of detail we're seeing is rare in protoplanetary disk imaging, and these new Hubble images show that planet nurseries can be much more active and chaotic than we expected," said lead author Kristina Monsch of the Center for Astrophysics | Harvard & Smithsonian (CfA). "We're seeing this disk nearly edge-on and its wispy upper layers and asymmetric features are especially striking. Both Hubble and NASA's James Webb Space Telescope have glimpsed similar structures in other disks, but IRAS 23077+6707 provides us with an exceptional perspective — allowing us to trace its substructures in visible light at an unprecedented level of detail. This makes the system a unique, new laboratory for studying planet formation and the environments where it happens." The nickname "Dracula's Chivito" playfully reflects the heritage of its researchers—one from Transylvania and another from Uruguay, where the national dish is a sandwich called a chivito. The

edge-on disk resembles a hamburger, with a dark central lane flanked by glowing top and bottom layers of dust and gas. This Hubble Space Telescope image shows the largest planet-forming disk ever observed around a young star. It spans nearly 400 billion miles — 40 times the diameter of our solar system. Image: NASA, ESA, STScI, Kristina Monsch (CfA); Image Processing: Joseph DePasquale (STScI) Puzzling asymmetry The impressive height of these features wasn't the only thing that captured the attention of scientists. The new images revealed that vertically imposing filament-like features appear on just one side of the disk, while the other side appears to have a sharp edge and no visible filaments. This peculiar, lopsided structure suggests that dynamic processes, like the recent infall of dust and gas, or interactions with its surroundings, are shaping the disk. "We were stunned to see how asymmetric this disk is," said co-investigator Joshua Bennett Lovell, also an astronomer at the CfA. "Hubble has given us a front row seat to the chaotic processes that are shaping disks as they build new planets — processes that we don't yet fully understand but can now study in a whole new way." All planetary systems form from disks of gas and dust encircling young stars. Over time, the gas accretes onto the star, and planets emerge from the remaining material. IRAS 23077+6707 may represent a scaled-up version of our early solar system, with a disk mass estimated at 10 to 30 times that of Jupiter — ample material for forming multiple gas giants. This, plus the new findings, makes it an exceptional case for studying the birth of planetary systems. "In theory, IRAS 23077+6707 could host a vast planetary system," said Monsch. "While planet formation may differ in such massive environments, the underlying processes are likely similar. Right now, we have more questions than answers, but these new images are a starting point for understanding how planets form over time and in different environments." Credit: NASA's Goddard Space Flight Center; Lead Producer: Paul Morris The Hubble Space Telescope has been operating for over three decades and continues to make ground-breaking discoveries that shape our fundamental understanding of the universe. Hubble is a project of international cooperation between NASA and ESA (European Space Agency). NASA's Goddard Space Flight Center in Greenbelt, Maryland, manages the telescope and mission operations. Lockheed Martin Space, based in Denver, also supports mission operations at Goddard. The Space Telescope Science Institute in Baltimore, which is operated by the Association of Universities for Research in Astronomy, conducts Hubble science operations for NASA. Facebook logo @NASA Hubble Instagram logo @NASA Hubble

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NASA's Perseverance Mars rover took a selfie with the Ingenuity helicopter, seen here about 13 feet from the rover. This image was taken by the WASTON camera on the rover's robotic arm on April 6, 2021. NASA/JPL-Caltech/MSSS

NASA's Perseverance Mars rover took a selfie with the Ingenuity helicopter, seen here about 13 feet from the rover. This image was taken by the WASTON camera on the rover's robotic arm on April 6, 2021.

NASA/JPL-Caltech/MSSS

NASA Leadership Administrator Jared Isaacman Jared “Rook” Isaacman is the 15th Administrator of the National Aeronautics and Space Administration. An entrepreneur and skilled pilot, Isaacman commanded Inspiration4, the first all-civilian orbital spaceflight aboard the Dragon spacecraft. The mission was a major milestone in commercial spaceflight. Full Biography Associate Administrator Amit Kshatriya Amit Kshatriya is NASA’s associate administrator. Kshatriya serves as the highest-ranking civil servant at the agency and as a senior advisor to the administrator. Kshatriya leads the agency’s 10 center directors, as well as the mission directorate associate administrators at NASA Headquarters in Washington. Full Biography Deputy Associate Administrator Casey Swails Casey Swails is NASA’s deputy associate administrator. Swails is deputy and principal advisor to the associate administrator for day-to-day operations and long-term strategic direction across the agency’s 10 field centers, mission directorates, and staff organizations. Full Biography

Administrator Jared Isaacman Jared “Rook” Isaacman is the 15th Administrator of the National Aeronautics and Space Administration. An entrepreneur and skilled pilot, Isaacman commanded Inspiration4, the first all-civilian orbital spaceflight aboard the Dragon spacecraft. The mission was a major milestone in commercial spaceflight. Full Biography Associate Administrator Amit Kshatriya Amit Kshatriya is NASA’s associate administrator. Kshatriya serves as the highest-ranking civil servant at the agency and as a senior advisor to the administrator. Kshatriya leads the agency’s 10 center directors, as well as the mission directorate associate administrators at NASA Headquarters in Washington. Full Biography Deputy Associate Administrator Casey Swails Casey Swails is NASA’s deputy associate administrator. Swails is deputy and principal advisor to the associate administrator for day-to-day operations and long-term strategic direction across the agency’s 10 field centers, mission directorates, and staff organizations. Full Biography

Administrator Jared Isaacman Jared “Rook” Isaacman is the 15th Administrator of the National Aeronautics and Space Administration. An entrepreneur and skilled pilot, Isaacman commanded Inspiration4, the first all-civilian orbital spaceflight aboard the Dragon spacecraft. The mission was a major milestone in commercial spaceflight. Full Biography

Administrator Jared Isaacman

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Associate Administrator Amit KshatriyaAmit Kshatriya is NASA's associate administrator. Kshatriya serves as the highest-ranking civil servant at the agency and as a senior advisor to the administrator. Kshatriya leads the agency's 10 center directors, as well as the mission directorate associate administrators at NASA Headquarters in Washington.Full Biography

Associate Administrator Amit Kshatriya

Amit Kshatriya is NASA's associate administrator. Kshatriya serves as the highest-ranking civil servant at the agency and as a senior advisor to the administrator. Kshatriya leads the agency's 10 center directors, as well as the mission directorate associate administrators at NASA Headquarters in Washington.

Deputy Associate Administrator Casey SwailsCasey Swails is NASA's deputy associate administrator. Swails is deputy and principal advisor to the associate administrator for day-to-day operations and long-term strategic direction across the agency's 10 field centers, mission directorates, and staff organizations.Full Biography

Deputy Associate Administrator Casey Swails

Casey Swails is NASA's deputy associate administrator. Swails is deputy and principal advisor to the associate administrator for day-to-day operations and long-term strategic direction across the agency's 10 field centers, mission directorates, and staff organizations.

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URL: <https://www.nasa.gov/socialmedia/>

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CenterInstagram:@nasaamesX:@NASAAMESYouTube:NASA's Ames Research CenterArmstrong Flight Research Center | Edwards, CaliforniaFacebook:NASA's Armstrong Flight Research CenterInstagram:@nasaarmstrongX:@NASAArmstrongYouTube:NASA Armstrong Flight Research CenterGlenn Research Center | Cleveland, OhioFacebook:NASA's Glenn Research CenterFlickr:NASA Glenn Research CenterInstagram:@nasaglennX:@NASAGlennYouTube:NASA Glenn Research CenterGoddard Space Flight Center | Greenbelt, MarylandFacebook:NASA's Goddard Space Flight CenterFlickr:NASA Goddard Space Flight CenterInstagram:@nasagoddardX:@NASAGoddardYouTube:NASA GoddardJet Propulsion Laboratory | Pasadena, CaliforniaFacebook:NASA Jet Propulsion LaboratoryFlickr:NASA Jet Propulsion LaboratoryInstagram:@nasajplX:@NASAJPLYouTube:NASA Jet Propulsion LaboratoryJohnson Space Center | Houston, TexasFacebook:NASA's Johnson Space CenterFlickr:NASA JohnsonInstagram:@nasajohnsonX:@NASA_JohnsonYouTube:NASA JohnsonKennedy Space Center | Kennedy Space Center, FloridaFacebook:NASA's Kennedy Space CenterFlickr:NASA KennedyInstagram:@nasakennedyX:@NASAKennedyYouTube:NASA's Kennedy Space CenterLangley Research Center | Hampton, VirginiaFacebook:NASA Langley Research CenterFlickr:NASA LangleyInstagram:@nasa_langleyX:@NASA_LangleyYouTube:NASA Langley Research CenterMarshall Space Flight Center | Huntsville, AlabamaFacebook:NASA's Marshall Space Flight CenterFlickr:NASA's Marshall Space Flight CenterInstagram:@nasa_marshallX:@NASA_MarshallYouTube:NASA's Marshall Space Flight CenterMichoud Assembly Facility | New Orleans, LouisianaFacebook:NASA Michoud Assembly FacilityStennis Space Center | Stennis Space Center, MississippiFacebook:NASA's John C. Stennis Space CenterInstagram:@nasastennisX:@NASASTennisYouTube:NASA StennisWallops Flight Facility | Wallops Island, VirginiaFacebook:NASA's Wallops Flight FacilityInstagram:@nasawallopsX:@NASAWallopsYouTube:NASA WallopsSocial Media AccountsMission Directorates, Organizations & ProgramsThese accounts offer news and updates from NASA organizations, programs, and the five Mission Directorates – Aeronautics, Exploration Systems, Science, Space Operations, and Space Technology.Academy of Program/Project & Engineering Leadership (APPEL)LinkedIn:NASA APPEL Knowledge ServicesAeronauticsFacebook:NASA AeronauticsInstagram:@nasaaeroX:@NASAaeroCareers at NASALinkedIn:NASA–National Aeronautics and Space AdministrationX:@NASApeopleExhibits, Events, and NASA SocialsX:NASA EventsNASA History DivisionFacebook:NASA HistoryFlickr:NASA on The CommonsX:@NASAhistoryOffice of the Inspector General (OIG)X:@NASAOIGYouTube:NASA Office of Inspector GeneralOffice of Small Business Programs (OSBP)LinkedIn:NASA Office of Small Business ProgramsScience Mission Directorate

(SMD)Facebook:NASA

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ChallengeFacebook:Space Apps ChallengeX:@SpaceAppsSpace Operations Mission

DirectorateX:@NASASpaceOpsSpace Technology Mission Directorate

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NASAX:@LearnWithNASAYoutube:Learn With NASAWomen @ NASAFacebook:Women at

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ObservatoryFacebook:Chandra X-ray ObservatoryFlickr:Chandra X-ray

ObservatoryInstagram:@nasachandraxrayX:@chandraxrayYouTube:Chandra X-ray

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TelescopeFacebook:NASA's James Webb Space TelescopeFlickr:NASA's James Webb

Space TelescopeInstagram:@nasawebbX:@NASAWebbYouTube:James Webb Space

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Roman Space TelescopeFacebook:Nancy Grace Roman Space

TelescopeX:@NASARomanSolar System ExplorationFacebook:NASA Solar System

ExplorationInstagram:@nasasolarsystemX:@NASASolarSystemYouTube:NASASolarSystem

Universe ExplorationFacebook:NASA

UniverseInstagram:@nasauniverseX:@NASAUniverseSocial Media AccountsNASA

LeadershipThe official accounts of current NASA leadership.Administrator Jared

IsaacmanInstagram:@nasaadminX:@NASAAdminPress Secretary Bethany

StevensX:@NASASpoxSocial Media AccountsOfficial NASA Astronaut Social Media

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Kayla BarronInstagram:@astro_kaylaBerríos, Marcos|en

españolInstagram:@astro_berriosX:@astro_berriosBirch, ChristinaInstagram:

@astro_birchX: @astro_birchBoe, Eric A.X:@Astro_BoeBresnik, Randolph

J.Facebook:NASA Astronaut Randy “Komrade”

BresnikInstagram:@astrokomradeX:@AstroKomradeBurnham,
 DenizInstagram:@astronaut_denizX:@astro_denizCardman,
 ZenalInstagram:@zenanautReddit:u/zenacardmanX:@zenanautChari, RajaFacebook:NASA
 Astronaut Raja CharilInstagram:@astro_rajaX:@Astro_RajaDouglas,
 AndreInstagram:@astro_andredX:@Astro_AndreDFincke, E.
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 J.Instagram:@astrovicgloverX:@AstroVicGloverHathaway,
 JackInstagram:@astro_hathawayX:@astro_hathawayHines,
 BobInstagram:@astro_farmerboX:@Astro_FarmerBobHoburg,
 WarrenInstagram:astro.woodyReddit:u/warrenhoburgX:@Astro_WoodyKim,
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 Jasmin MoghbeliInstagram:@astrojawsMorgan, Andrew R.Facebook:NASA Astronaut
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 LorallInstagram:@loraloharaX:@lunarloralPettit, Donald R.X:@astro_PettitRubio,
 FrankFacebook:Astronaut Frank RubioInstagram:@astro_frankrubioTingle, Scott
 D.Instagram:@astro_makerVande Hei, Mark T.X:@Astro_SabotWatkins,
 JessicalInstagram:@astro_watkinsX:@astro_watkinsWheelock, Douglas H.
 X:@Astro_WheelsWilson, Stephanie D.X:@Astro_StephanieWilliams, Christopher
 L.X:@Astro_ChrisWWilliams, Sunita L.X:@Astro_SuniWiseman, G. Reid
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 AdamInstagram:@astro_fuhrmannX:@astro_fuhrmannJones, CameronFacebook:NASA
 Astronaut Cameron JonesInstagram:@astro_camjonesX:@astro_camjonesKubo,
 YuriFacebook:NASA Astronaut Yuri KuboInstagram:@astro_kuboLawler,

RebeccaInstagram:@astro_lawlerX:@astro_lawlerMenon,
AnnaInstagram:@astroannamenonMuller, ImeldaInstagram:@astro_imeldaOvercash,
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NASAThese accounts offer the broadest of the agency's most important information.

Facebook:NASA

Flickr:NASA HQ Photo

Giphy:@nasa

Instagram:NASA

LinkedIn:NASA

Pinterest:@nasa

Reddit:u/nasa

Snapchat:NASA

Soundcloud:NASA

Spotify:nasaspotify

Tumblr:NASA

Twitch:NASA(Official Moderation Account:NASARedCrew)

X:@nasa

YouTube:@NASA

NASA en EspañolCuentas oficiales de la NASA en español.

X:@NASA_es

Facebook:NASA en español

YouTube:NASA en Español

Instagram:@nasa_es

Social Media Accounts

NASA Centers & Facilities These accounts offer news and updates from NASA research centers and facilities across the country.

Ames Research Center | Moffett Field, California

Facebook:NASA Ames Research Center

Instagram:@nasaames

X:@NASAAmes

YouTube:NASA's Ames Research Center

Armstrong Flight Research Center | Edwards, California

Facebook:NASA's Armstrong Flight Research Center

Instagram:@nasaarmstrong

X:@NASAArmstrong

YouTube:NASA Armstrong Flight Research Center

Glenn Research Center | Cleveland, Ohio

Facebook:NASA's Glenn Research Center

Flickr:NASA Glenn Research Center

Instagram:@nasaglenn

X:@NASAGlenn

YouTube:NASA Glenn Research Center

Goddard Space Flight Center | Greenbelt, Maryland

Facebook:NASA's Goddard Space Flight Center

Flickr:NASA Goddard Space Flight Center

Instagram:@nasagoddard

X:@NASAGoddard

YouTube:NASA Goddard

Jet Propulsion Laboratory | Pasadena, California

Facebook:NASA Jet Propulsion Laboratory

Flickr:NASA Jet Propulsion Laboratory

Instagram:@nasajpl

X:@NASAJPL

YouTube:NASA Jet Propulsion Laboratory

Johnson Space Center | Houston, Texas

Facebook:NASA's Johnson Space Center

Flickr:NASA Johnson

Instagram:@nasajohnson

X:@NASA_Johnson

YouTube:NASA Johnson

Kennedy Space Center | Kennedy Space Center, Florida

Facebook:NASA's Kennedy Space Center

Flickr:NASA Kennedy

Instagram:@nasakennedy

X:@NASAKennedy

YouTube:NASA's Kennedy Space Center

Langley Research Center | Hampton, Virginia

Facebook:NASA Langley Research Center

Flickr:NASA Langley

Instagram:@nasa_langley

X:@NASA_Langley

YouTube:NASA Langley Research Center

Marshall Space Flight Center | Huntsville, Alabama

Facebook:NASA's Marshall Space Flight Center

Flickr:NASA's Marshall Space Flight Center

Instagram:@nasa_marshall

X:@NASA_Marshall

YouTube:NASA's Marshall Space Flight Center

Michoud Assembly Facility | New Orleans, Louisiana

Facebook:NASA Michoud Assembly Facility

Stennis Space Center | Stennis Space Center, Mississippi

Facebook:NASA's John C. Stennis Space Center

Instagram:@nasastennis

X:@NASASTennis

YouTube:NASA Stennis

Wallops Flight Facility | Wallops Island, Virginia

Facebook:NASA's Wallops Flight Facility

Instagram: @nasawallops

X:@NASAWallops

YouTube:NASA Wallops

Mission Directorates, Organizations & ProgramsThese accounts offer news and updates from NASA organizations, programs, and the five Mission Directorates – Aeronautics, Exploration Systems, Science, Space Operations, and Space Technology.

Academy of Program/Project & Engineering Leadership (APPEL)

LinkedIn:NASA APPEL Knowledge Services

Facebook:NASA Aeronautics

Instagram:@nasaaero

X:@NASAaero

Careers at NASA

LinkedIn:NASA–National Aeronautics and Space Administration

X:@NASApeople

Exhibits, Events, and NASA Socials

X:NASA Events

NASA History Division

Facebook:NASA History

Flickr:NASA on The Commons

X:@NASAhistory

Office of the Inspector General (OIG)

X:@NASAOIG

YouTube:NASA Office of Inspector General

Office of Small Business Programs (OSBP)

LinkedIn:NASA Office of Small Business Programs

Science Mission Directorate (SMD)

Facebook:NASA Science

Instagram:@nasascience_

X:@NASAScience_

YouTube:ScienceAtNASA

Scientific and Technical Information (STI) Program

Facebook:NASA Scientific and Technical Information (STI) Program

X:@NASA_STI

YouTube:NASA STI Program

Space Apps Challenge

Facebook:Space Apps Challenge

X:@SpaceApps

X:@NASASpaceOps

Space Technology Mission Directorate (STMD)

Facebook:NASA Technology

X:@NASA_Technology

YouTube:NASASpaceTech

Facebook:Learn With NASA

Pinterest:Learn With NASA

X:@LearnWithNASA

YouTube:Learn With NASA

Women @ NASA

Facebook:Women at NASA

X:@WomenNASA

Missions & Topics These accounts offer news and updates from NASA's missions in space exploration, scientific discovery, and aeronautics research.

Facebook:NASA Artemis

Instagram:@nasaartemis

X:@NASAArtemis

NASA Space Alerts

Facebook:@NASASpaceAlerts

X:@NASASpaceAlerts

Chandra X-Ray Observatory

Facebook:Chandra X-ray Observatory

Flickr:Chandra X-ray Observatory

Instagram:@nasachandraxray

X:@chandraxray

YouTube:Chandra X-ray Observatory

Facebook:NASA Earth

Instagram:@nasaeearth

X:@NASAEarth

Facebook:NASA's Hubble Space Telescope

Flickr:NASA Hubble Space Telescope

Instagram:@nasahubble

X:@NASAHubble

Facebook:International Space Station

Instagram:iss

X:@Space_Station

Facebook:NASA's James Webb Space Telescope

Flickr:NASA's James Webb Space Telescope

Instagram:@nasawebb

X:@NASAWebb

YouTube:James Webb Space Telescope (JWST)

Mars Exploration

Facebook:NASA Mars

X:@NASAMars

Facebook:Nancy Grace Roman Space Telescope

X:@NASARoman

Facebook:NASA Solar System Exploration

Instagram:@nasasolarsystem

X:@NASASolarSystem

YouTube:NASASolarSystem

Universe Exploration

Facebook:NASA Universe

Instagram:@nasauniverse

X:@NASAUniverse

NASA LeadershipThe official accounts of current NASA leadership.

Instagram:@nasaadmin

X:@NASAAdmin

Press Secretary Bethany Stevens

X:@NASASpox

Official NASA Astronaut Social Media Accounts

Read thebiographiesof NASA astronauts eligible for flight assignment – including those who are not active on social media.

Ayers, Nichole

Instagram:@astro_ayers

X:@Astro_Ayers

Barron, Kayla

Facebook:NASA Astronaut Kayla Barron

Instagram:@astro_kayla

Berríos, Marcos|en español

Instagram:@astro_berrios

X:@astro_berrios

Birch, Christina

Instagram: @astro_birch

X: @astro_birch

Boe, Eric A.

X:@Astro_Boe

Bresnik, Randolph J.

Facebook:NASA Astronaut Randy “Komrade” Bresnik

Instagram:@astrokomrade

X:@AstroKomrade

Burnham, Deniz

Instagram:@astronaut_deniz

X:@astro_deniz

Cardman, Zena

Instagram:@zenanaut

Reddit:u/zenacardman

X:@zenanaut

Chari, Raja

Facebook:NASA Astronaut Raja Chari

Instagram:@astro_raja

X:@Astro_Raja

Douglas, Andre

Instagram:@astro_andred

X:@Astro_AndreD

Fincke, E. Michael

X:@AstroIronMike

Glover, Victor J.

Instagram:@astrovicglover

X:@AstroVicGlover

Hathaway, Jack

Instagram:@astro_hathaway

X:@astro_hathaway

Hines, Bob

Instagram:@astro_farmerbo

X:@Astro_FarmerBob

Hoburg, Warren

Instagram:astro.woody

Reddit:u/warrenhoburg

X:@Astro_Woody

Kim, Jonny

Facebook:NASA Astronaut Jonny Kim

Instagram:@jonnykimusa

X:@JonnyKimUSA

Koch, Christina H.

Instagram:@astro_christina

X:@Astro_Christina

Lindgren, Kjell N.

Instagram:@astro_kjell

X:@astro_kjell

Mann, Nicole Aunapu

X:@AstroDuke

McClain, Anne C.

Facebook:NASA Astronaut Anne McClain

Instagram:@astro_annimal

X:@AstroAnnimal

Meir, Jessica U.

Instagram:@astro_jessica

X:@Astro_Jessica

Menon, Anil

Instagram:@astro_anil

X:@astro_anil

Moghbeli, Jasmin

Facebook:NASA Astronaut Jasmin Moghbeli

Instagram:@astrojaws

Morgan, Andrew R.

Facebook:NASA Astronaut Andrew “Drew” Morgan

Instagram:@astrodrewmorgan

X:@AstroDrewMorgan

O’Hara, Loral

Instagram:@loralohara

X:@lunarloral

Pettit, Donald R.

X:@astro_Pettit

Rubio, Frank

Facebook:Astronaut Frank Rubio

Instagram:@astro_frankrubio

Tingle, Scott D.

Instagram:@astro_maker

Vande Hei, Mark T.

X:@Astro_Sabot

Watkins, Jessica

Instagram:@astro_watkins

X:@astro_watkins

Wheelock, Douglas H.

X:@Astro_Wheels

Wilson, Stephanie D.

X:@Astro_Stephanie

Williams, Christopher L.

X:@Astro_ChrisW

Williams, Sunita L.

X:@Astro_Suni

Wiseman, G. Reid

Instagram:@astro_reid

X:@astro_reid

Wittner, Jessica

Instagram:@astro_wittner

X:@Astro_Wittner

NASA Astronaut Candidates

The term astronaut candidate refers to individuals who have been selected by NASA as candidates for astronaut corps and are currently undergoing a candidacy training program at NASA's Johnson Space Center. Learn more about the astronaut selection program. Read the biographies of the most recent class of NASA astronaut candidates – including those who are not active on social media.

Bailey, Ben

Instagram:@astro_benbailey

X:@astro_benbailey

Edgar, Lauren

Facebook:NASA-Astronaut-Lauren-Edgar

Instagram:@astronautedgar

Fuhrmann, Adam

Instagram:@astro_fuhrmann

X:@astro_fuhrmann

Jones, Cameron

Facebook:NASA Astronaut Cameron Jones

Instagram:@astro_camjones

X:@astro_camjones

Kubo, Yuri

Facebook:NASA Astronaut Yuri Kubo

Instagram:@astro_kubo

Lawler, Rebecca

Instagram:@astro_lawler

X:@astro_lawler

Menon, Anna

Instagram:@astroannamenon

Muller, Imelda

Instagram:@astro_imelda

Overcash, Erin

Facebook:NASA Astronaut Erin Overcash

Instagram:@astro_overcash

Spies, Katherine

Instagram:@spiesinspace

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NASA's Moon to Mars Architecture defines the elements needed for long-term, human-led scientific discovery in deep space.

URL: <https://science.nasa.gov/dark-energy/>

URL: <https://science.nasa.gov/universe/search-for-life/>

TITLE: Are We Alone?

Explore This Section Are We Alone? Contents Just Getting Started More Questions than Answers Where Is Everybody? Just Getting Started The search for life beyond Earth is really just getting started, but science has an encouraging early answer: there are plenty of planets in the galaxy, many with similarities to our own. But what we don't know fills volumes. Observations from the ground and from space have confirmed thousands of planets beyond our solar system. Our galaxy likely holds at least 100 billion. But so far, we have no evidence of life beyond Earth. Is life in the cosmos easily begun, and commonplace? Or is it incredibly rare? How big is the Milky Way galaxy? Make the jump to light-years as we cruise through our galaxy. Video NASA/JPL-Caltech More Questions than Answers In the thousands of years humanity has been contemplating the cosmos, we are the first people to know one thing for sure: The stars beyond our Sun are teeming with planets. They come in many varieties, and a good chunk of them are around the size of Earth. Like most scientific questions, though, getting an answer to this one just breeds more questions: Which, if any, of these exoplanets harbors some form of life? How quickly does life get its start? And how long does it last? This artist's concept allows us to imagine what it would be like to stand on the surface of the exoplanet TRAPPIST-1f, located in the TRAPPIST-1 system in the constellation Aquarius, as of Feb 2017. Where Is Everybody? The universe's eerie silence has its own name – the "Fermi paradox." Physicist Enrico Fermi famously posed the question: "Where is everybody?" Even at slow travel speeds, the universe's billions of years of existence allow plenty of time for intelligent, technological lifeforms to traverse the galaxy. Why, then, is the cosmos so quiet? Meanwhile, exoplanet discoveries over the past two decades have filled in a few of the terms in the much-debated Drake Equation – a chain of numbers that might one day tell us how many intelligent civilizations we can expect to find. Most of its terms remain blank – the fraction of planets with life, with intelligent life, with detectable technology – but the equation itself suggests we might one day arrive at an answer. It feels at least a little more hopeful than Fermi's silence. We stand at a crossroads in the search for life. We've found thousands of planets in our Milky Way galaxy, a large fraction of them in Earth's size range and orbiting in

their stars' "habitable zones" – the distance from the star at which liquid water could exist on the surface. We know the galaxy likely holds trillions of planets. Our telescopes in space and on the ground, and our remote-sensing technology, grow ever more powerful. Yet so far, the only life we know of is right here at home. For the moment, we're staring into the void, hoping someone is staring back. Revised Drake equation. University of Rochester Next: We don't have a universally accepted definition of life itself, so can we find life?

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Contents Just Getting Started More Questions than Answers Where Is Everybody? Just Getting Started The search for life beyond Earth is really just getting started, but science has an encouraging early answer: there are plenty of planets in the galaxy, many with similarities to our own. But what we don't know fills volumes. Observations from the ground and from space have confirmed thousands of planets beyond our solar system. Our galaxy likely holds at least 100 billion. But so far, we have no evidence of life beyond Earth. Is life in the cosmos easily begun, and commonplace? Or is it incredibly rare? How big is the Milky Way galaxy? Make the jump to light-years as we cruise through our galaxy. Video NASA/JPL-Caltech More Questions than Answers In the thousands of years humanity has been contemplating the cosmos, we are the first people to know one thing for sure: The stars beyond our Sun are teeming with planets. They come in many varieties, and a good chunk of them are around the size of Earth. Like most scientific questions, though, getting an answer to this one just breeds more questions: Which, if any, of these exoplanets harbors some form of life? How quickly does life get its start? And how long does it last? This artist's concept allows us to imagine what it would be like to stand on the surface of the exoplanet TRAPPIST-1f, located in the TRAPPIST-1 system in the constellation Aquarius, as of Feb 2017. Where Is Everybody? The universe's eerie silence has its own name – the "Fermi paradox." Physicist Enrico Fermi famously posed the question: "Where is everybody?" Even at slow travel speeds, the universe's billions of years of existence allow plenty of time for intelligent, technological lifeforms to traverse the galaxy. Why, then, is the cosmos so quiet? Meanwhile, exoplanet discoveries over the past two decades have filled in a few of the terms in the much-debated Drake Equation – a chain of numbers that might one day tell us how many intelligent civilizations we can expect to find. Most of its terms remain blank – the fraction of planets with life, with intelligent life, with detectable technology – but the equation itself suggests we might one day arrive at an answer. It feels at least a little more hopeful than Fermi's silence. We stand at a crossroads in the search for life. We've found thousands of planets in our Milky Way galaxy, a large fraction of them in Earth's size range and orbiting in their stars' "habitable zones" – the distance from the star at which liquid water could exist on the surface. We know the galaxy likely holds trillions of planets. Our telescopes in space and on the ground, and our remote-sensing technology, grow ever

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Contents Just Getting Started More Questions than Answers Where Is Everybody?

Just Getting Started

More Questions than Answers

Where Is Everybody?

Just Getting Started The search for life beyond Earth is really just getting started, but science has an encouraging early answer: there are plenty of planets in the galaxy, many with similarities to our own. But what we don't know fills volumes. Observations from the ground and from space have confirmed thousands of planets beyond our solar system. Our galaxy likely holds at least 100 billion. But so far, we have no evidence of life beyond Earth. Is life in the cosmos easily begun, and commonplace? Or is it incredibly rare? How big is the Milky Way galaxy? Make the jump to light-years as we cruise through our galaxy.

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TITLE: Centennial Challenges

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Developments
News Release
3 Min Read
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FiltersAudienceEducatorsStudentsGrade LevelsGrades K-4Grades 5-8Grades 9-12Higher EducationInformal EducationSubjectArtCareersComputer ScienceEarth ScienceGeographyGeologyHuman Interactions and EnvironmentPlanet EarthWeather and ClimateEducationEngineering DesignHistoryBiographiesFlightHistorical OverviewsPlanetary ExplorationScientists and InventorsTimelinesHuman Space ExplorationPastPresent and FutureLife ScienceAnimals and PlantsHealth and NutritionHumansLife SupportLiving and Working in SpaceSpace BiologyMathematicsAlgebraCalculusFractions and DecimalsGeometryMeasurement and Data AnalysisNumbers and OperationsProbability and StatisticsProblem SolvingTrigonometryPhysical ScienceChemistryEnergyFlight and AeronauticsForces and MotionGravityMatter and Its PropertiesMicrogravityOpticsPhysicsWavesLightSoundSpace ScienceAstrobiologyAstronomyEclipsesExoplanetsMissions to Planets and MoonsPlanetary GeologyRobotic ExplorersSolar System and PlanetsAsteroids Comets MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary

MoonsSaturnSunUranusVenusSolar System and StarsSpace
RadiationStarsTelescopesUniverseSpanishTechnologyCommunicationModelsPhotography
and ImagingRemote SensingRoboticsRocketrySatellitesSpace StationSpace
VehiclesSpinoffs (Benefits to You)Virtual RealityTypeBookmarksData SetsEducator
GuidesHands-on ActivitiesImage GalleriesInteractive MultimediaLesson Plans /
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EducationSubjectArtCareersComputer ScienceEarth ScienceGeographyGeologyHuman
Interactions and EnvironmentPlanet EarthWeather and ClimateEducationEngineering
DesignHistoryBiographiesFlightHistorical OverviewsPlanetary ExplorationScientists and
InventorsTimelinesHuman Space ExplorationPastPresent and FutureLife ScienceAnimals
and PlantsHealth and NutritionHumansLife SupportLiving and Working in SpaceSpace
BiologyMathematicsAlgebraCalculusFractions and DecimalsGeometryMeasurement and
Data AnalysisNumbers and OperationsProbability and StatisticsProblem
SolvingTrigonometryPhysical ScienceChemistryEnergyFlight and AeronauticsForces and
MotionGravityMatter and Its PropertiesMicrogravityOpticsPhysicsWavesLightSoundSpace
ScienceAstrobiologyAstronomyEclipsesExoplanetsMissions to Planets and
MoonsPlanetary GeologyRobotic ExplorersSolar System and PlanetsAsteroids Comets
MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary
MoonsSaturnSunUranusVenusSolar System and StarsSpace
RadiationStarsTelescopesUniverseSpanishTechnologyCommunicationModelsPhotography
and ImagingRemote SensingRoboticsRocketrySatellitesSpace StationSpace
VehiclesSpinoffs (Benefits to You)Virtual RealityTypeBookmarksData SetsEducator
GuidesHands-on ActivitiesImage GalleriesInteractive MultimediaLesson Plans /
ActivitiesMobile AppsOther MultimediaPlay and LearnColoring and
PuzzlesGamesPostersProgram BrochuresRead About ItSTEM Resource
CollectionsVideosWebsites

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ScienceEarth ScienceGeographyGeologyHuman Interactions and EnvironmentPlanet
EarthWeather and ClimateEducationEngineering DesignHistoryBiographiesFlightHistorical
OverviewsPlanetary ExplorationScientists and InventorsTimelinesHuman Space

ExplorationPastPresent and FutureLife ScienceAnimals and PlantsHealth and NutritionHumansLife SupportLiving and Working in SpaceSpace BiologyMathematicsAlgebraCalculusFractions and DecimalsGeometryMeasurement and Data AnalysisNumbers and OperationsProbability and StatisticsProblem SolvingTrigonometryPhysical ScienceChemistryEnergyFlight and AeronauticsForces and MotionGravityMatter and Its PropertiesMicrogravityOpticsPhysicsWavesLightSoundSpace ScienceAstrobiologyAstronomyEclipsesExoplanetsMissions to Planets and MoonsPlanetary GeologyRobotic ExplorersSolar System and PlanetsAsteroids Comets MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary MoonsSaturnSunUranusVenusSolar System and StarsSpace RadiationStarsTelescopesUniverseSpanishTechnologyCommunicationModelsPhotography and ImagingRemote SensingRoboticsRocketrySatellitesSpace StationSpace VehiclesSpinoffs (Benefits to You)Virtual RealityTypeBookmarksData SetsEducator GuidesHands-on ActivitiesImage GalleriesInteractive MultimediaLesson Plans / ActivitiesMobile AppsOther MultimediaPlay and LearnColoring and PuzzlesGamesPostersProgram BrochuresRead About ItSTEM Resource CollectionsVideosWebsites

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PuzzlesGamesPostersProgram BrochuresRead About ItSTEM Resource
CollectionsVideosWebsites

Audience

Educators

Students

Grade Levels

Grades K-4

Grades 5-8

Grades 9-12

Higher Education

Informal Education

Subject

Art

Computer Science

Earth ScienceGeographyGeologyHuman Interactions and EnvironmentPlanet
EarthWeather and Climate

Geography

Geology

Human Interactions and Environment

Planet Earth

Education

Engineering Design

HistoryBiographiesFlightHistorical OverviewsPlanetary ExplorationScientists and
InventorsTimelines

History

Biographies

Flight

Historical Overviews

Planetary Exploration

Scientists and Inventors

Timelines

Human Space ExplorationPastPresent and Future

Human Space Exploration

Present and Future

Life ScienceAnimals and PlantsHealth and NutritionHumansLife SupportLiving and Working in SpaceSpace Biology

Life Science

Animals and Plants

Health and Nutrition

Humans

Life Support

Living and Working in Space

Space Biology

MathematicsAlgebraCalculusFractions and DecimalsGeometryMeasurement and Data AnalysisNumbers and OperationsProbability and StatisticsProblem SolvingTrigonometry

Mathematics

Algebra

Calculus

Fractions and Decimals

Geometry

Measurement and Data Analysis

Numbers and Operations

Probability and Statistics

Problem Solving

Trigonometry

Physical ScienceChemistryEnergyFlight and AeronauticsForces and MotionGravityMatter
and Its PropertiesMicrogravityOpticsPhysicsWavesLightSound

Physical Science

Chemistry

Energy

Flight and Aeronautics

Forces and Motion

Gravity

Matter and Its Properties

Microgravity

Optics

Physics

WavesLightSound

Waves

Light

Sound

Space ScienceAstrobiologyAstronomyEclipsesExoplanetsMissions to Planets and
MoonsPlanetary GeologyRobotic ExplorersSolar System and PlanetsAsteroids Comets
MeteoritesDwarf PlanetsEarthEarth's MoonJupiterMarsMercuryNeptunePlanetary
MoonsSaturnSunUranusVenusSolar System and StarsSpace
RadiationStarsTelescopesUniverse

Space Science

Astronomy

Missions to Planets and Moons

Planetary Geology

Robotic Explorers

Solar System and PlanetsAsteroids Comets MeteoritesDwarf PlanetsEarthEarth's
MoonJupiterMarsMercuryNeptunePlanetary MoonsSaturnSunUranusVenus

Solar System and Planets

Asteroids Comets Meteorites

Dwarf Planets

Planetary Moons

Solar System and Stars

Space Radiation

Telescopes

Spanish

TechnologyCommunicationModelsPhotography and ImagingRemote
SensingRoboticsRocketrySatellitesSpace StationSpace VehiclesSpinoffs (Benefits to
You)Virtual Reality

Communication

Models

Photography and Imaging

Remote Sensing

Rocketry

Satellites

Space Station

Space Vehicles

Spinoffs (Benefits to You)

Virtual Reality

Type

Bookmarks

Data Sets

Educator Guides

Hands-on Activities

Image Galleries

Interactive Multimedia

Lesson Plans / Activities

Mobile Apps

Other Multimedia

Play and LearnColoring and PuzzlesGames

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Coloring and Puzzles

Posters

Program Brochures

Read About It

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Websites

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sideways.FactsExplorationMoonsResourcesAbout Planet UranusUranus is the seventh planet from the Sun, and the third largest planet in our solar system – about four times wider than Earth.Uranus is very cold and windy. It is surrounded by faint rings, and more than two dozen small moons. It rotates at a nearly 90-degree angle from the plane of its orbit. This unique tilt makes Uranus appear to spin on its side.Get the FactsThis image of Uranus from the James Webb Telescope's Near-Infrared Camera (NIRCam) shows Uranus's seasonal north polar cap and dim inner and outer rings. This Webb image also shows 9 of the planet's 27 moons.NASA, ESA, CSA, STScIPlanet Uranus ExplorationOnly one spacecraft has explored planet Uranus.In January 1986,Voyager 2 made a close approach to Uranus, snapping images of the planet and some its moons.Read MoreNASA's Voyager 2 lifts off on Aug. 20, 1977, from Kennedy Space Center in Florida.NASAPlanet Uranus

MoonsMeet the Literary MoonsUranus has 28 known moons, including five major moons: Miranda, Ariel, Umbriel, Titania, and Oberon. The moons are sometimes called the "literary moons" because they are named for Shakespearean characters, along with a couple of the moons being named for characters from the works of Alexander Pope.Explore Moons of Planet UranusThis image of Uranus from NASA's James Webb Space Telescope shows the planet and distant background galaxies. This image also includes 14 of the planet's 27 moons.NASA, ESA, CSA, STScIIce Giant ResourcesResources for Uranus and Neptune.Explore a curated collection of resources about ice giants, including activities that can be done at home, as well as videos and animations, images, and posters.Read MoreNASA/JPL-CaltechVoyager 240th Anniversary of Uranus FlybyThe first human-made object to fly past Uranus, Voyager 2 made its closest approach on Jan. 24, 1986, and during its visit discovered 10 new moons, two new rings, and a magnetic field stronger than Saturn's, while returning images and data that still fuel discoveries today. Voyager 2 finished its encounter in February, enroute to a rendezvous with Neptune in 1989.Learn More About Voyager 2's Uranus EncounterEven decades after Voyager 2 became the first, and so far only, spacecraft to visit Uranus, its data has led to revelations about the ice giant.Read 'Mining Old Data From NASA's Voyager 2 Solves Several Uranus Mysteries'Eyes on the Solar System: UranusEyes of the Solar System uses data and images from NASA missions to give you a simulated view of Uranus.Solar System NewsArticle4 Min ReadNASA Finds Lunar Regolith Limits Meteorites as Source of Earth's WaterBlog4 Min ReadNASA's SunRISE SmallSats Ace Tests, Moving Closer to LaunchBlog1 Min ReadStrong Flare Erupts From Sun25 Min ReadArtemis II: What NASA Learned From Launching Artemis ILearn MoreFor Kids: All About Planet UranusFacts about Planet Uranus for kids.Ice Giant ResourcesA curated collection of resources about ice giants, including activities that can be done at home, as well as videos and animations, images, and posters.NASA Photojournal: Planet UranusImages of planet Uranus from NASA spacecraft.Keep ExploringDiscover More Topics From NASANEptuneSunMercurySolar System Exploration

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38Expedition 39Expedition 40Expedition 42Expedition 43Expedition 44Expedition
45Expedition 46Expedition 47Expedition 48Expedition 49Expedition 50Expedition
51Expedition 52Expedition 53Expedition 54Expedition 55Expedition 56Expedition
57Expedition 59Expedition 61Expedition 65Expedition 67Expedition 69ISS
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InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport)
International Space Station (ISS)
Commercial Crew
Commercial Resupply
SpaceX Commercial Resupply Expedition 36
Expedition 37
Expedition 38
Expedition 39
Expedition 40
Expedition 42
Expedition 43
Expedition 44
Expedition 45
Expedition 46
Expedition 47
Expedition 48
Expedition 49
Expedition 50
Expedition 51
Expedition 52
Expedition 53
Expedition 54
Expedition 55
Expedition 56
Expedition 57
Expedition 59
Expedition 61
Expedition 65
Expedition 67
Expedition 69
ISS Research
Vegetable Production System (VEGGIE)
James Webb Space Telescope (JWST)
Joint Polar Satellite System (JPSS)
Juno
LADEE (Lunar Atmosphere Dust Environment Explorer)
Lucy
Lunar Reconnaissance Orbiter (LRO)
Mariner
Mars 2020
Ingenuity (Helicopter)
Perseverance (Rover)
Mars Climate Orbiter
Mars Exploration Rovers (MER)
Opportunity (Rover)
Spirit (Rover)
Mars Express
Mars Global Surveyor (MGS)
Mars Observer
Mars Odyssey
Mars Orbiter Mission (MOM)
Mars Pathfinder
Mars Phoenix
Mars Polar Lander / Deep Space 2
Mars Reconnaissance Orbiter (MRO)
Mars Sample Return (MSR)
Mars Science Laboratory (MSL)
Curiosity (Rover)
MAVEN (Mars Atmosphere and Volatile Evolution)
N)
MESSENGER (MErcury Surface, Space ENvironment, GEochemistry, and Ranging)
Nancy Grace Roman Space Telescope
New Horizons
OSIRIS-REx (Origins, Spectral Interpretation, Resource Identification, Security-Regolith Explorer)
Parker Solar Probe (PSP)
Quesst (X-59)
Quesst: The Mission
Quesst: The Science
Quesst: The Team
Scientific

BalloonsSmall Satellite MissionsCubeSatsELaNa (Educational Launch of Nanosatellites)MarCO (Mars Cube One)SOFIA (Stratospheric Observatory for Infrared Astronomy) / 747-SPSolar Dynamics Observatory (SDO)Sounding RocketsSpitzer Space TelescopeSTEREO (Solar TERrestrial RELations Observatory)SWOT (Surface Water and Ocean Topography)Technology DemonstrationDeep Space Atomic ClockTeleroboticsTropospheric Emissions: Monitoring of Pollution (TEMPO)VERITAS (Venus Emissivity, Radio Science, InSAR, Topography & Spectroscopy)VikingViking 1Viking 2Voyager ProgramVoyager 1WISE (Wide-field Infrared Survey Explorer)X-57 MaxwellNASA Advisory Council (NAC)NASA Centers & FacilitiesAmes Research CenterAmes Research Center's Science DirectorateArmstrong Flight Research CenterGlenn Research CenterGoddard Institute for Space StudiesGoddard Space Flight CenterJet Propulsion LaboratoryJohnson Space CenterPeople of JohnsonKennedy Space CenterResearch and Technology at Kennedy Space CenterSustainability at Kennedy Space CenterLangley Research CenterMarshall Space Flight CenterMichoud Assembly FacilityNASA HeadquartersStennis Space CenterWallops Flight FacilityNASA DirectoratesAeronautics Research Mission DirectorateIntegrated Aviation Systems ProgramFlight Demos CapabilitiesLow Boom Flight DemonstratorTransformative Aeronautics Concepts ProgramUniversity InnovationUniversity Leadership InitiativeUniversity Student Research ChallengeExploration Systems Development Mission DirectorateScience Mission DirectorateAstrophysics DivisionExoplanet Exploration ProgramEarth Science DivisionHeliophysics DivisionSounding Rockets ProgramMOSAICSPlanetary Science DivisionMars Exploration ProgramOuter Planets & Ocean Worlds ProgramSmallSats ProgramSpace Operations Mission DirectorateSpace Communications & Navigation ProgramSpace Technology Mission DirectorateFlight Opportunities ProgramGame Changing Development ProgramLunar Surface Innovation InitiativeLunar Surface Innovation ConsortiumNASA Innovative Advanced Concepts (NIAC) ProgramPrizes, Challenges, and Crowdsourcing ProgramCentennial ChallengesCenter of Excellence for Collaborative Innovation (CoECI)Prizes, Challenges, & Crowdsourcing NewsSpace Technology Research GrantsEarly Career Faculty (ECF)Technology Demonstration Missions ProgramTechnology TransferSpinoffsNASA en españolCiencia e InvestigaciónNASA HistoryPopular HistoryNOAA (National Oceanic and Atmospheric Administration)Office of International and Interagency Relations (OIIR)Artemis AccordsOrion ProgramPeople of AmesPeople of NASALeadershipPeople of GoddardSpecial ObservancesAsian American and Native Hawaiian Pacific Islander HeritageHispanic Heritage MonthNative American Heritage MonthWomen's History MonthWomen at NASAScience & ResearchAeronautics ResearchAeronautics TechnologyAstrophysicsExoplanet ScienceStudying ExoplanetsBiological & Physical SciencesSpace BiologyAnimal BiologyPlant BiologyEarth ScienceCarbon CycleClimate ScienceSea IceEarth Surface & InteriorWater & Energy

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InstrumentsTechnology for Living in SpaceTechnology for Space TravelTechnology Transfer
& SpinoffsThe Solar SystemAsteroidsApophisBennuDinkineshPotentially Hazardous
Asteroid (PHA)Trojan AsteroidsComets3I/ATLASHalley's CometDwarf
PlanetsCeresErisHaumeaMakemakePlutoEarth's MoonMeteors &
MeteoritesMoonsPlanetsIce GiantsJupiterJupiter MoonsEuropaGanymedeMarsMars
MoonsMercuryNeptuneOcean WorldsSaturnSaturn
MoonsTitanUranusVenusSkywatchingAstronomyEclipsesLunar EclipsesSolar
Eclipses2017 Solar Eclipse2023 Solar Eclipse2024 Solar EclipseNight Sky
NetworkPlanetary TransitsThe SunThe UniverseExoplanetsExoplanet Detection
MethodsExoplanet TransitsExoplanet Discoveries55 Cancri eTRAPPIST-1Gas Giant
ExoplanetsNeptune-Like ExoplanetsSuper-Earth ExoplanetsTerrestrial
ExoplanetsNebulaeSupernova RemnantsThe Search for LifeVideo SeriesVIPER (Volatiles
Investigating Polar Exploration Rover)xEVA & Human Surface MobilityAll Science Org
TermsAir QualityAstronaut PhotographyAstrophysicsCyclonesDust StormsEarth
ObservatoryEarth's AtmosphereAerosolsAir QualityAtmospheric
TemperatureCloudsOzone LayerPrecipitationEarth's Surface and InteriorExEP (Exoplanet
Exploration Program)Heat & RadiationHeliophysicsHistory of ScienceISSLandslidesLife on
EarthHuman DimensionsAgricultureEarth at NightUrban DevelopmentVegetationWater
BloomsWildlifeOceansRemote Sensing TechnologySun-Earth InteractionsVolcanoesWater
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Heritage Month Women's History Month Women at NASA Science & Research Aeronautics Research Aeronautics Technology Astrophysics Exoplanet Science Studying Exoplanets Biological & Physical Sciences Space Biology Animal Biology Plant Biology Earth Science Carbon Cycle Climate Science Sea Ice Earth Surface & Interior Water & Energy Cycle Cryosphere Weather and Atmospheric Dynamics Global Precipitation For Researchers Post Doc Program Research Opportunities in Space and Earth Sciences (ROSES) Heliophysics Auroras Solar Science The Sun & Solar Physics Lunar Science Planetary Science Astrobiology Planetary Environments & Atmospheres Planetary Geosciences & Geophysics Science-enabling Technology Technology Research Tools & Resources Social Media Solar Wind Space Launch System (SLS) System-Wide Safety Technology Communicating and Navigating with Missions Deep Space Network Near Space Network Space Communications Technology High-Tech Computing Manufacturing, Materials, 3-D Printing Radioisotope Power Systems (RPS) Robotics Science Instruments Technology for Living in Space Technology for Space Travel Technology Transfer & Spinoffs The Solar System Asteroids Apophis Bennu Dinkinesh Potentially Hazardous Asteroid (PHA) Trojan Asteroids Comets 3I/ATLAS Halley's Comet Dwarf Planets Ceres Eris Haumea Makemake Pluto Earth's Moon Meteors & Meteorites Moons Planets Ice Giants Jupiter Jupiter Moons Europa Ganymede Mars Mars Moons Mercury Neptune Ocean Worlds Saturn Saturn Moons Titan Uranus Venus Skywatching Astronomy Eclipses Lunar Eclipses Solar Eclipses 2017 Solar Eclipse 2023 Solar Eclipse 2024 Solar Eclipse Night Sky Network Planetary Transits The Sun The Universe Exoplanets Exoplanet Detection Methods Exoplanet Transits Exoplanet Discoveries 55 Cancri e TRAPPIST-1 Gas Giant Exoplanets Neptune-Like Exoplanets Super-Earth Exoplanets Terrestrial Exoplanets Nebulae Supernova Remnants The Search for Life Video Series VIPER (Volatiles Investigating Polar Exploration Rover) xEVA & Human Surface Mobility All Science Org Terms Air Quality Astronaut Photography Astrophysics Cyclones Dust Storms Earth Observatory Earth's Atmosphere Aerosols Air Quality Atmospheric Temperature Clouds Ozone Layer Precipitation Earth's Surface and Interior ExEP (Exoplanet Exploration Program) Heat & Radiation Heliophysics History of Science ISS Landslides Life on Earth Human Dimensions Agriculture Earth at Night Urban Development Vegetation Water Blooms Wildlife Oceans Remote Sensing Technology Sun-Earth Interactions Volcanoes Water on Earth Glaciers & Ice Sheets Ground Water Sea & Lake Ice Snow Surface Water Wildfires Clear Filters

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CareersLife at NASAWhat We DoCommercial Supersonic TechnologyCOVID-19
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TemperatureCloudsHeat & RadiationHistory of Earth ScienceHuman
DimensionsAgricultureEarth at NightUrban DevelopmentLife on EarthVegetationWater
BloomsWildlifeNatural DisastersDust StormsHurricanes &
TyphoonsLandslidesVolcanoesWildfiresOceansRemote Sensing TechnologySun-Earth
InteractionsWater on EarthGround WaterSnowSurface WaterExploration Ground
SystemsGeneralGet InvolvedCitizen ScienceCommunity PartnersData ResourcesGrants &
OpportunitiesSpace Technology GrantsOpportunities For Educators to Get
InvolvedOpportunities For International Participants to Get InvolvedOpportunities For
Researchers to Get InvolvedOpportunities For Students to Get InvolvedOpportunities For
U.S. Citizens to Get InvolvedOpportunities to Contribute to NASA Missions & Get
InvolvedPartner With UsSolar System AmbassadorsHuman Landing System
ProgramHuman Lander ChallengeHuman Research ProgramHumans in
SpaceAstronautsBob HinesFormer AstronautsAlan B. Shepard Jr.Buzz AldrinFrank
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LindgrenLoral O'HaraNichole AyersNicole Aunapu MannStephen G. BowenSunita L.
WilliamsWarren HoburgBecoming an AstronautCommercial SpaceCommercial Space
ProgramsLow Earth Orbit EconomyPrivate Astronaut MissionsThe Future of Commercial
SpaceSpacesuitsWhy Go To SpaceBenefits Back on EarthIn-Situ Resource Utilization
(ISRU)Launch Services ProgramCubeSat Launch InitiativeLearning ResourcesFor Colleges
& UniversitiesMSI ExchangeNASA STEM ProjectsIn-flight Education DownlinksSpace
Samples for LearningFor EducatorsGrades 5 - 8 for EducatorsGrades 9-12 for
EducatorsGrades K - 4 for EducatorsOutside the ClassroomFor Kids and StudentsGrades 5
- 8ExploreGrades 9 - 12Find Your PlaceGrades K - 4Color it!Read it!For
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ObservatoryDART (Double Asteroid Redirection Test)DAVINCI (Deep Atmosphere Venus
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Environmental Satellite)HS3 Hurricane Mission (Hurricane and Severe Storm

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Directorate Astrophysics Division Exoplanet Exploration Program Earth Science
Division Heliophysics Division Sounding Rockets Program MOSAICS Planetary Science
Division Mars Exploration Program Outer Planets & Ocean Worlds Program SmallSats
Program Space Operations Mission Directorate Space Communications & Navigation
Program Space Technology Mission Directorate Flight Opportunities Program Game
Changing Development Program Lunar Surface Innovation Initiative Lunar Surface
Innovation Consortium NASA Innovative Advanced Concepts (NIAC) Program Prizes,
Challenges, and Crowdsourcing Program Centennial Challenges Center of Excellence for
Collaborative Innovation (CoE CI) Prizes, Challenges, & Crowdsourcing News Space
Technology Research Grants Early Career Faculty (ECF) Technology Demonstration Missions
Program Technology Transfer Spinoffs NASA en español Ciencia e Investigación NASA
History Popular History NOAA (National Oceanic and Atmospheric Administration) Office of
International and Interagency Relations (OIIR) Artemis Accords Orion Program People of
Ames People of NASA Leadership People of Goddard Special Observances Asian American
and Native Hawaiian Pacific Islander Heritage Hispanic Heritage Month Native American
Heritage Month Women's History Month Women at NASA Science & Research Aeronautics
Research Aeronautics Technology Astrophysics Exoplanet Science Studying
Exoplanets Biological & Physical Sciences Space Biology Animal Biology Plant Biology Earth
Science Carbon Cycle Climate Science Sea Ice Earth Surface & Interior Water & Energy
Cycle Cryosphere Weather and Atmospheric Dynamics Global Precipitation For
Researchers Post Doc Program Research Opportunities in Space and Earth Sciences
(ROSES) Heliophysics Auroras Solar Science The Sun & Solar Physics Lunar Science Planetary
Science Astrobiology Planetary Environments & Atmospheres Planetary Geosciences &
Geophysics Science-enabling Technology Technology Research Tools & Resources Social
Media Solar Wind Space Launch System (SLS) System-Wide
Safety Technology Communicating and Navigating with Missions Deep Space Network Near
Space Network Space Communications Technology High-Tech Computing Manufacturing,
Materials, 3-D Printing Radioisotope Power Systems (RPS) Robotics Science
Instruments Technology for Living in Space Technology for Space Travel Technology Transfer
& Spinoffs The Solar System Asteroids Apophis Bennu Dinkinesh Potentially Hazardous
Asteroid (PHA) Trojan Asteroids Comets 3I/ATLAS Halley's Comet Dwarf
Planets Ceres Eris Haumea Makemake Pluto Earth's Moon Meteors &
Meteorites Moons Planets Ice Giants Jupiter Jupiter Moons Europa Ganymede Mars Mars
Moons Mercury Neptune Ocean Worlds Saturn Saturn
Moons Titan Uranus Venus Skywatching Astronomy Eclipses Lunar Eclipses Solar
Eclipses 2017 Solar Eclipse 2023 Solar Eclipse 2024 Solar Eclipse Night Sky
Network Planetary Transits The Sun The Universe Exoplanets Exoplanet Detection

MethodsExoplanet TransitsExoplanet Discoveries55 Cancri eTRAPPIST-1Gas Giant
ExoplanetsNeptune-Like ExoplanetsSuper-Earth ExoplanetsTerrestrial
ExoplanetsNebulaeSupernova RemnantsThe Search for LifeVideo SeriesVIPER (Volatiles
Investigating Polar Exploration Rover)xEVA & Human Surface Mobility

3D Resources

Advanced Air Transport Technology

AeroSpace Frontiers

Air Mobility Pathfinders project

Audio

Budget & Annual Reports

Featured Careers

Life at NASA

Commercial Supersonic Technology

COVID-19 Response

Ice & Glaciers

Ozone Layer

Earth Day

Earth Observatory

Earth's Atmosphere

Aerosols

Atmospheric Temperature

Clouds

Heat & Radiation

History of Earth Science

Human Dimensions

Agriculture

Earth at Night

Urban Development

Life on Earth

Vegetation

Water Blooms

Wildlife

Natural Disasters

Dust Storms

Hurricanes & Typhoons

Landslides

Volcanoes

Wildfires

Remote Sensing Technology

Sun-Earth Interactions

Water on Earth

Ground Water

Snow

Surface Water

Exploration Ground Systems

Community Partners

Data Resources

Grants & Opportunities

Space Technology Grants

Opportunities For Educators to Get Involved

Opportunities For International Participants to Get Involved

Opportunities For Researchers to Get Involved

Opportunities For Students to Get Involved

Opportunities For U.S. Citizens to Get Involved

Opportunities to Contribute to NASA Missions & Get Involved

Partner With Us

Solar System Ambassadors

Human Landing System Program

Human Lander Challenge

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Becoming an Astronaut

Commercial Space Programs

The Future of Commercial Space

Spacesuits

Why Go To Space

Benefits Back on Earth

In-Situ Resource Utilization (ISRU)

CubeSat Launch Initiative

For Colleges & Universities

MSI Exchange

In-flight Education Downlinks

Outside the Classroom

Grades 5 - 8

Grades 9 - 12

Find Your Place

Grades K - 4

Color it!

Read it!

Courses & Curriculum for Professionals

Academy of Program / Project & Engineering Leadership

NASA Educator Professional Development Center

NASA Engineering & Safety Center Academy

Project Management & Systems Engineering

Space Samples for Your Classroom

EPSCoR

MUREP

Partner with NASA STEM

Space Grant

STEM Impacts

Advanced Air Mobility

Apollo 11

Apollo 8

Artemis 1

Artemis 3

Gateway Space Station

B200

Huygens

DART (Double Asteroid Redirection Test)

DAVINCI (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging)

Dragonfly

Europa Clipper

ExoMars

Global Precipitation Measurement (GPM)

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IceBridge

Antarctic 2013

ICESat (Ice, Clouds and Land Elevation Satellite)

InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport)

SpaceX Commercial Resupply

Expedition 36

Expedition 37

Expedition 38

Expedition 39

Expedition 40

Expedition 42

Expedition 43

Expedition 44

Expedition 45

Expedition 46

Expedition 47

Expedition 48

Expedition 49

Expedition 50

Expedition 51

Expedition 52

Expedition 53

Expedition 54

Expedition 55

Expedition 56

Expedition 57

Expedition 59

Expedition 61

Expedition 65

Expedition 67

Expedition 69

Vegetable Production System (VEGGIE)

Joint Polar Satellite System (JPSS)

LADEE (Lunar Atmosphere Dust Environment Explorer)

Lucy

Lunar Reconnaissance Orbiter (LRO)

Mariner

Mars 2020

Ingenuity (Helicopter)

Perseverance (Rover)

Mars Climate Orbiter

Mars Exploration Rovers (MER)

Opportunity (Rover)

Spirit (Rover)

Mars Express

Mars Global Surveyor (MGS)

Mars Observer

Mars Odyssey

Mars Orbiter Mission (MOM)

Mars Pathfinder

Mars Phoenix

Mars Polar Lander / Deep Space 2

Mars Reconnaissance Orbiter (MRO)

Mars Sample Return (MSR)

MAVEN (Mars Atmosphere and Volatile Evolution)

MESSENGER (MErcury Surface, Space ENvironment, GEOchemistry, and Ranging)

OSIRIS-REx (Origins, Spectral Interpretation, Resource Identification, Security-Regolith Explorer)

Parker Solar Probe (PSP)

Quesst: The Mission

Quesst: The Team

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Small Satellite Missions

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MarCO (Mars Cube One)

SOFIA (Stratospheric Observatory for Infrared Astronomy) / 747-SP

Solar Dynamics Observatory (SDO)

Sounding Rockets

Spitzer Space Telescope

STEREO (Solar TERrestrial RELations Observatory)

SWOT (Surface Water and Ocean Topography)

Deep Space Atomic Clock

Telerobotics

Tropospheric Emissions: Monitoring of Pollution (TEMPO)

VERITAS (Venus Emissivity, Radio Science, InSAR, Topography & Spectroscopy)

Viking

Viking 1

Viking 2

Voyager Program

Voyager 1

WISE (Wide-field Infrared Survey Explorer)

X-57 Maxwell

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System-Wide Safety

Manufacturing, Materials, 3-D Printing

Technology for Living in Space

Technology for Space Travel

Apophis

Bennu

Dinkinesh

Potentially Hazardous Asteroid (PHA)

3I/ATLAS

Halley's Comet

Makemake

Planets

Ice Giants

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Mars Moons

Ocean Worlds

Saturn Moons

Lunar Eclipses

Solar Eclipses

2017 Solar Eclipse

2023 Solar Eclipse

2024 Solar Eclipse

Night Sky Network

Planetary Transits

Exoplanet Detection Methods

Exoplanet Transits

55 Cancri e

TRAPPIST-1

Gas Giant Exoplanets

Super-Earth Exoplanets

Terrestrial Exoplanets

Nebulae

Supernova Remnants

Video Series

VIPER (Volatiles Investigating Polar Exploration Rover)

xEVA & Human Surface Mobility

All Science Org Terms

Air QualityAstronaut PhotographyAstrophysicsCyclonesDust StormsEarth
ObservatoryEarth's AtmosphereAerosolsAir QualityAtmospheric
TemperatureCloudsOzone LayerPrecipitationEarth's Surface and InteriorExEP (Exoplanet
Exploration Program)Heat & RadiationHeliophysicsHistory of ScienceISSLandslidesLife on
EarthHuman DimensionsAgricultureEarth at NightUrban DevelopmentVegetationWater
BloomsWildlifeOceansRemote Sensing TechnologySun-Earth InteractionsVolcanoesWater
on EarthGlaciers & Ice SheetsGround WaterSea & Lake IceSnowSurface WaterWildfires

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Launch Your Future: NASA OSTEM Internship Webinar Join the NASA internship team for a webinar to explore exciting internship opportunities and better understand eligibility and application requirements for NASA internships. Current NASA interns will discuss their

experiences and share advice for prospective interns. Gain valuable insights to strengthen your application and take the first step toward your NASA journey. Event Date: Feb. 3 [Register Now](#)

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[Register Now](#)

Student Challenge: NASA ORBIT NASA's Opportunities in Research, Business, Innovation, and Technology (ORBIT) challenge is a multi-phase innovation competition designed to empower university and college students to develop next-generation solutions that benefit life on Earth and deep-space exploration. Registration Deadline: Feb. 9 [Register Now](#)

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Student Suborbital Flight Opportunity: RockOn! 2026 Workshop NASA's Wallops Flight Facility in Virginia is hosting a weeklong, hands-on workshop to teach participants how to create a sounding rocket experiment from scratch and launch it into space. Registration Deadline: Feb. 13 [Apply Now](#)

Student Suborbital Flight Opportunity: RockOn! 2026 Workshop

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2026 NASA Revolutionary Aerospace Systems Concepts – Academic Linkage (RASC-AL) Competition The 2026 NASA RASC-AL competition challenges undergraduate and graduate students to push the boundaries of innovation and engineering to develop bold concepts that advance humanity's ability to live and work on the Moon, Mars, and beyond. Entry Deadline: Feb. 23 [Learn More](#)

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NASA Summer 2026 Internship Opportunities Find your place in space with a NASA internship! NASA offers several opportunities for students to undertake meaningful and challenging projects that truly make an impact on humanity. Application Deadline: Feb. 27 [Apply Now](#)

NASA Summer 2026 Internship Opportunities

Find your place in space with a NASA internship! NASA offers several opportunities for students to undertake meaningful and challenging projects that truly make an impact on humanity. Application Deadline: Feb. 27

NASA 2026 Human Lander Challenge Through this challenge, college students contribute to the advancement of HLS technologies, concepts, and approaches. Improvements in these technology areas have the potential to revolutionize NASA's approach to space exploration, and contributions from the academic community are a valuable part of the journey to discovery. Close Date: March 4 [Learn More](#)

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Through this challenge, college students contribute to the advancement of HLS technologies, concepts, and approaches. Improvements in these technology areas have the potential to revolutionize NASA's approach to space exploration, and contributions from the academic community are a valuable part of the journey to discovery. Close Date: March 4

Deep Space Food Challenge: Mars to Table As NASA plans for deep space missions that could last months or years, new ways to feed astronauts that are independent of Earth must be developed. The Deep Space Food Challenge: Mars to Table is a global competition inviting chefs, innovators, culinary experts, higher-education students, and citizen scientists to design a complete, Earth-independent food system for long-duration space missions. Registration Deadline: July 31 [Learn More](#)

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Latest NewsMore NewsArticle2 Min ReadNASA Launches Research Program for Students to Explore Big IdeasArticle3 Min ReadNASA Helps Connect Astronomers and Community Colleges Across the NationArticle3 Min ReadNASA Launches 2026 Lunabotics ChallengeNews Release2 Min ReadNASA, International Astronauts to Address Students from New York3 Min ReadNASA Accepts Proposals for 2026 Student Launch ChallengeArticle2 Min ReadNASA Astronauts to Answer Questions from Students in OhioNews Release2 Min ReadNASA Astronauts to Answer Questions from Students in MinnesotaNews Release3 Min ReadNASA Seeks Proposals for 2026 Human Exploration Rover ChallengeArticle

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3 Min ReadNASA Helps Connect Astronomers and Community Colleges Across the Nation

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Article3 Min ReadNASA Launches 2026 Lunabotics Challenge

3 Min ReadNASA Launches 2026 Lunabotics Challenge

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News Release2 Min ReadNASA, International Astronauts to Address Students from New York

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3 Min ReadNASA Accepts Proposals for 2026 Student Launch ChallengeArticle2 Min ReadNASA Astronauts to Answer Questions from Students in OhioNews Release2 Min ReadNASA Astronauts to Answer Questions from Students in MinnesotaNews Release3 Min ReadNASA Seeks Proposals for 2026 Human Exploration Rover ChallengeArticle

3 Min ReadNASA Accepts Proposals for 2026 Student Launch ChallengeArticle

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2 Min ReadNASA Astronauts to Answer Questions from Students in OhioNews Release

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NASA Student Launch Challenge Student Launch is a research-based, competitive, experiential exploration activity. This project offers multiple challenges reaching a broad audience of middle and high schools, colleges, and universities across the nation. [Learn More](#)

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TITLE: NASA Policy on the Release of Information to News and Information Media

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Encyclopedia

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ContentsScopeApplicabilityPrinciplesResponsibilitiesPublic Information Coordination and ConcurrenceInterviewsPreventing Release of Classified Information to the MediaPreventing Unauthorized Release of Sensitive but Unclassified (SBU) Information and/or Material to the News MediaMultimedia MaterialsNews Releases Concerning International ActivitiesScopeThis directive sets forth policy governing the release of public information, which is defined as information in any form provided to news and information media, especially information that has the potential to generate significant media, or public interest or inquiry. Examples include, but are not limited to, press releases, media advisories, news features, and web postings. Not included under this definition are scientific and technical reports, web postings designed for technical or scientific interchange, and technical information presented at professional meetings or in professional journals.Applicability(a)This policy applies to NASA Headquarters, NASA Centers, and Component Facilities.(b) In the event of any conflict between this policy and any other NASA policy, directive, or regulation, this policy shall govern and supersede any

previous issuance or directive.

Principles(a) NASA, a scientific and technical agency, is committed to a culture of openness with the media and public that values the free exchange of ideas, data, and information as part of scientific and technical inquiry. Scientific and technical information from or about Agency programs and projects will be accurate and unfiltered.(b) Consistent with NASA statutory responsibility, NASA will “provide for the widest practicable and appropriate dissemination of information concerning its activities and the results thereof.” Release of public information concerning NASA activities and the results of NASA activities will be made promptly, factually, and completely.(c) To ensure timely release of information, NASA will endeavor to ensure cooperation and coordination among the Agency’s scientific, engineering, and public affairs communities.(d) In keeping with the desire for a culture of openness, NASA employees may, consistent with this policy, speak to the press and the public about their work.(e) This policy does not authorize or require disclosure of information that is exempt from disclosure under the Freedom of Information Act (5 U.S.C. § 552) or otherwise restricted by statute, regulation, Executive Order, or other Executive Branch policy or NASA policy (e.g., OMB Circulars, NASA Policy Directives). Examples of information not releasable under this policy include, without limitation, information that is, or is marked as, classified information, procurement sensitive information, information subject to the Privacy Act, other sensitive but unclassified information, and information subject to privilege, such as pre-decisional information or attorney-client communications.

Responsibilities(a) The Assistant Administrator for Public Affairs is responsible for developing and administering an integrated Agency-wide communications program, establishing Agency public affairs policies and priorities, and coordinating and reviewing the performance of all Agency public affairs activities. The Assistant Administrator will develop criteria to identify which news releases and other types of public information will be issued nationwide by NASA Headquarters. Decisions to release public information nationwide by NASA Headquarters will be made by the Assistant Administrator for Public Affairs or his/her designee.(b) NASA’s Mission Directorate Associate Administrators and Mission Support Office heads have ultimate responsibility for the technical, scientific, and programmatic accuracy of all information that is related to their respective programs and released by NASA.(c) Under the direction of the Assistant Administrator for Public Affairs, public affairs officers assigned to Mission Directorates are responsible for the timely and efficient coordination of public information covering their respective programs. This coordination includes review by appropriate Mission Directorate officials. It also includes editing by public affairs staff to ensure that public information products are well written and appropriate for the intended audience. However, such editing shall not change scientific or technical data, or the meaning of programmatic content.(d) Center Public Affairs Directors are responsible for implementing their portion of the

Agency's communications program, adhering to Agency policies, procedures, and priorities, and coordinating their activities with Headquarters (and others where appropriate). They are responsible for the quality of public information prepared by Center public affairs officers. They also are responsible for the day-to-day production of public information covering their respective Center activities, which includes obtaining the necessary Center concurrences and coordinating, as necessary, with the appropriate Headquarters public affairs officers.(e) Center Directors have ultimate responsibility for the accuracy of public information that does not require the concurrence of Headquarters. (See "Public information coordination and concurrence," section (d).)(f) All NASA employees are required to coordinate, in a timely manner, with the appropriate public affairs officers prior to releasing information that has the potential to generate significant media, or public interest or inquiry.(g) All NASA public affairs officers are required to notify the appropriate Headquarters public affairs officers in a timely manner about activities or events that have the potential to generate significant media or public interest or inquiry.(h) All NASA public affairs employees are expected to adhere to the following code of conduct: Be honest and accurate in all communications. Honor publication embargoes. Respond promptly to media requests and respect media deadlines. Act promptly to correct mistakes or erroneous information, either internally or externally. Promote the free flow of scientific and technical information. Protect non-public information.(i) All NASA employees are responsible for adhering to plans (including schedules) for activities established by public affairs offices and senior management for the coordinated release of public information.(j) All NASA-funded missions will have a public affairs plan, approved by the Assistant Administrator for Public Affairs, which will be managed by Headquarters and/or a designated NASA Center.(k) Public affairs activities for NASA-funded missions will not be managed by non-NASA institutions, unless authorized by the Assistant Administrator for Public Affairs.(l) The requirements of this directive do not apply to the Office of Inspector General regarding its activities.

Public Information Coordination and Concurrence(a) General. All NASA employees involved in preparing and issuing NASA public information are responsible for proper coordination among Headquarters, Center, and Mission Directorate offices to include review and clearance by appropriate officials prior to issuance. Such coordination will be accomplished through procedures developed and published by the NASA Assistant Administrator for Public Affairs.(b) Coordination. To ensure timely release of public information, Headquarters and Center public affairs officers are required to coordinate to obtain review and clearance by appropriate officials, keep each other informed of changes, delays, or cancellation of releases, and provide advance notification of the actual release.(c) All public information shall be coordinated through the appropriate Headquarters offices, including review by the appropriate Mission Directorate Associate Administrator and mission support office head,

or their designees, to ensure scientific, technical, and programmatic accuracy, and review by the Assistant Administrator of Public Affairs or his/her designee to ensure that public information products are well written and appropriate for the intended audience.(d)

Centers may, however, without the full coordination of Headquarters, issue public information that is institutional in nature, of local interest, or has been deemed not to be a Headquarters release. (The Assistant Administrator for Public Affairs or his/her designee will determine which public information will be issued nationwide by NASA Headquarters.) These releases must be coordinated through the appropriate Center offices and approved by the Center Director and Center Public Affairs Director. The Center Public Affairs Director is required to provide proper notification to the NASA Office of Public Affairs, Headquarters, prior to release. (The Assistant Administrator for Public Affairs shall publish guidelines for the release of public information that may be issued by Centers without clearance from Headquarters' offices.)(e) Dispute Resolution. Any dispute arising from a decision to

proceed or not proceed with the issuance of a news release or other type of public information will be addressed and resolved by the Assistant Administrator for Public Affairs with the appropriate Mission Directorate Associate Administrator, mission support office head, Center Director, and others, such as Center Public Affairs Directors, as necessary. However, the appropriate Mission Directorate Associate Administrator shall be the arbiter of disputes about the accuracy or characterization of programmatic, technical, or scientific information. Additional appeals may be made to the Chief of Strategic Communications and to the Office of the Administrator. When requested by a Center Public Affairs Director, an explanation of the resolution will be provided in writing to all interested Agency parties.

Interviews(a) Only spokespersons designated by the Assistant Administrator for Public Affairs, or his/her designee, are authorized to speak for the Agency in an official capacity regarding NASA policy, programmatic, and budget issues.(b) In response to media interview requests, NASA will offer articulate and knowledgeable spokespersons who can best serve the needs of the media and the American public. However, journalists may have access to the NASA officials they seek to interview, provided those NASA officials agree to be interviewed.(c) NASA employees may speak to the media and the public about their work. When doing so, employees shall notify their immediate supervisor and coordinate with their public affairs office in advance of interviews whenever possible, or immediately thereafter, and are encouraged, to the maximum extent practicable, to have a public affairs officer present during interviews. If public affairs officers are present, their role will be to attest to the content of the interview, support the interviewee, and provide post-interview follow up with the media as necessary.(d) NASA, as an Agency, does not take a position on any scientific conclusions. That is the role of the broad scientific community and the nature of the scientific process. NASA scientists may draw conclusions and may, consistent with this policy, communicate those conclusions to the media. However, NASA

employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee’s official area of expertise or responsibility.(f) Media interviews will be “on-the-record” and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.(g) Audio recordings may be made by NASA with consent of the interviewee.(h) NASA employees are not required to speak to the media.(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.

Preventing Release of Classified Information to the Media(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. See NPR 1600.1, Chapter 5, Section 5.23 for internal NASA guidance on management of classified information. For further guidance that applies to all agencies, see Executive Order 12958, as amended, “Classified National Security Information” and its implementing directive at 32 CFR Parts 2001 and 2004.(b) Any attempt by news media representatives to obtain classified information will be reported through the Headquarters Office of Public Affairs or Installation Public Affairs Office to the Installation Security Office and Office of Security and Program Protection.(c) For classified operations and/or programs managed under the auspices of a DD Form 254, “Contract Security Classification Specification,” all inquiries concerning this activity will be responded to by the appropriate PAO official designated in Item 12 on the DD Form 254.(d) For classified operations and/or information owned by other Government agencies (e.g., DOD, DOE, etc.), all inquiries will be referred to the appropriate Agency public affairs officer as established in written agreements.

Preventing Unauthorized Release of Sensitive but Unclassified (SBU) Information and/or Material to the News Media(a) All NASA SBU information requires accountability and approval for release. Release of SBU information to unauthorized personnel is prohibited. Unauthorized release of SBU information may result in prosecution and/or disciplinary action. Ignorance of NASA policy and procedures regarding SBU information does not release a NASA employee from responsibility for unauthorized release. See NPR 1600.1, Chapter 5, Section

5.24 for guidance on identification, marking, accountability and release of NASA SBU information.(b) Examples of SBU information include: proprietary information of others provided to NASA under nondisclosure or confidentiality agreement; source selection and bid and proposal information; information subject to export control under the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR); information subject to the Privacy Act of 1974; predecisional materials such as national space policy not yet publicly released; pending reorganization plans or sensitive travel itineraries; and information that could constitute an indicator of U.S. government intentions, capabilities, operations, or activities or otherwise threaten operations security.(c) Upon request for access to information/material deemed SBU, coordination must be made with the information/material owner to determine if the information/material may be released. Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.

Multimedia Materials(a) NASA's multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.Audio and/or video files of significant news developments and other events of historic or public interest.Interactive multimedia features that can be incorporated into the Agency's Internet portal for use by internal and external audiences, including the media and the general public.

News Releases Concerning International Activities(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

ContentsScopeApplicabilityPrinciplesResponsibilitiesPublic Information Coordination and ConcurrenceInterviewsPreventing Release of Classified Information to the MediaPreventing Unauthorized Release of Sensitive but Unclassified (SBU) Information

and/or Material to the News MediaMultimedia MaterialsNews Releases Concerning International Activities

Scope

Applicability

Principles

Responsibilities

Public Information Coordination and Concurrence

Interviews

Preventing Release of Classified Information to the Media

Preventing Unauthorized Release of Sensitive but Unclassified (SBU) Information and/or Material to the News Media

Multimedia Materials

News Releases Concerning International Activities

ScopeThis directive sets forth policy governing the release of public information, which is defined as information in any form provided to news and information media, especially information that has the potential to generate significant media, or public interest or inquiry. Examples include, but are not limited to, press releases, media advisories, news features, and web postings. Not included under this definition are scientific and technical reports, web postings designed for technical or scientific interchange, and technical information presented at professional meetings or in professional

journals.Applicability(a)This policy applies to NASA Headquarters, NASA Centers, and Component Facilities.(b) In the event of any conflict between this policy and any other NASA policy, directive, or regulation, this policy shall govern and supersede any previous issuance or directive.Principles(a) NASA, a scientific and technical agency, is committed to a culture of openness with the media and public that values the free exchange of ideas, data, and information as part of scientific and technical inquiry. Scientific and technical information from or about Agency programs and projects will be accurate and unfiltered.(b) Consistent with NASA statutory responsibility, NASA will “provide for the widest practicable and appropriate dissemination of information concerning its activities and the results thereof.” Release of public information concerning NASA activities and the results of NASA activities will be made promptly, factually, and completely.(c) To ensure timely release of information, NASA will endeavor to ensure cooperation and coordination among the Agency’s scientific, engineering, and public affairs communities.(d) In keeping with the

desire for a culture of openness, NASA employees may, consistent with this policy, speak to the press and the public about their work.(e) This policy does not authorize or require disclosure of information that is exempt from disclosure under the Freedom of Information Act (5 U.S.C. § 552) or otherwise restricted by statute, regulation, Executive Order, or other Executive Branch policy or NASA policy (e.g., OMB Circulars, NASA Policy Directives). Examples of information not releasable under this policy include, without limitation, information that is, or is marked as, classified information, procurement sensitive information, information subject to the Privacy Act, other sensitive but unclassified information, and information subject to privilege, such as pre-decisional information or attorney-client communications.

Responsibilities

(a) The Assistant Administrator for Public Affairs is responsible for developing and administering an integrated Agency-wide communications program, establishing Agency public affairs policies and priorities, and coordinating and reviewing the performance of all Agency public affairs activities. The Assistant Administrator will develop criteria to identify which news releases and other types of public information will be issued nationwide by NASA Headquarters. Decisions to release public information nationwide by NASA Headquarters will be made by the Assistant Administrator for Public Affairs or his/her designee.

(b) NASA's Mission Directorate Associate Administrators and Mission Support Office heads have ultimate responsibility for the technical, scientific, and programmatic accuracy of all information that is related to their respective programs and released by NASA.

(c) Under the direction of the Assistant Administrator for Public Affairs, public affairs officers assigned to Mission Directorates are responsible for the timely and efficient coordination of public information covering their respective programs. This coordination includes review by appropriate Mission Directorate officials. It also includes editing by public affairs staff to ensure that public information products are well written and appropriate for the intended audience. However, such editing shall not change scientific or technical data, or the meaning of programmatic content.

(d) Center Public Affairs Directors are responsible for implementing their portion of the Agency's communications program, adhering to Agency policies, procedures, and priorities, and coordinating their activities with Headquarters (and others where appropriate). They are responsible for the quality of public information prepared by Center public affairs officers. They also are responsible for the day-to-day production of public information covering their respective Center activities, which includes obtaining the necessary Center concurrences and coordinating, as necessary, with the appropriate Headquarters public affairs officers.

(e) Center Directors have ultimate responsibility for the accuracy of public information that does not require the concurrence of Headquarters. (See "Public information coordination and concurrence," section (d).)

(f) All NASA employees are required to coordinate, in a timely manner, with the appropriate public affairs officers prior to releasing information that has the potential to generate significant

media, or public interest or inquiry.(g) All NASA public affairs officers are required to notify the appropriate Headquarters public affairs officers in a timely manner about activities or events that have the potential to generate significant media or public interest or inquiry.(h) All NASA public affairs employees are expected to adhere to the following code of conduct:Be honest and accurate in all communications.Honor publication embargoes.Respond promptly to media requests and respect media deadlines.Act promptly to correct mistakes or erroneous information, either internally or externally.Promote the free flow of scientific and technical information.Protect non-public information.(i) All NASA employees are responsible for adhering to plans (including schedules) for activities established by public affairs offices and senior management for the coordinated release of public information.(j) All NASA-funded missions will have a public affairs plan, approved by the Assistant Administrator for Public Affairs, which will be managed by Headquarters and/or a designated NASA Center.(k) Public affairs activities for NASA-funded missions will not be managed by non-NASA institutions, unless authorized by the Assistant Administrator for Public Affairs.(l) The requirements of this directive do not apply to the Office of Inspector General regarding its activities.Public Information Coordination and Concurrence(a) General. All NASA employees involved in preparing and issuing NASA public information are responsible for proper coordination among Headquarters, Center, and Mission Directorate offices to include review and clearance by appropriate officials prior to issuance. Such coordination will be accomplished through procedures developed and published by the NASA Assistant Administrator for Public Affairs.(b) Coordination. To ensure timely release of public information, Headquarters and Center public affairs officers are required to coordinate to obtain review and clearance by appropriate officials, keep each other informed of changes, delays, or cancellation of releases, and provide advance notification of the actual release.(c) All public information shall be coordinated through the appropriate Headquarters offices, including review by the appropriate Mission Directorate Associate Administrator and mission support office head, or their designees, to ensure scientific, technical, and programmatic accuracy, and review by the Assistant Administrator of Public Affairs or his/her designee to ensure that public information products are well written and appropriate for the intended audience.(d) Centers may, however, without the full coordination of Headquarters, issue public information that is institutional in nature, of local interest, or has been deemed not to be a Headquarters release. (The Assistant Administrator for Public Affairs or his/her designee will determine which public information will be issued nationwide by NASA Headquarters.) These releases must be coordinated through the appropriate Center offices and approved by the Center Director and Center Public Affairs Director. The Center Public Affairs Director is required to provide proper notification to the NASA Office of Public Affairs, Headquarters, prior to release. (The Assistant Administrator for Public Affairs shall publish guidelines for

the release of public information that may be issued by Centers without clearance from Headquarters' offices.)(e) Dispute Resolution. Any dispute arising from a decision to proceed or not proceed with the issuance of a news release or other type of public information will be addressed and resolved by the Assistant Administrator for Public Affairs with the appropriate Mission Directorate Associate Administrator, mission support office head, Center Director, and others, such as Center Public Affairs Directors, as necessary. However, the appropriate Mission Directorate Associate Administrator shall be the arbiter of disputes about the accuracy or characterization of programmatic, technical, or scientific information. Additional appeals may be made to the Chief of Strategic Communications and to the Office of the Administrator. When requested by a Center Public Affairs Director, an explanation of the resolution will be provided in writing to all interested Agency parties.

Interviews(a) Only spokespersons designated by the Assistant Administrator for Public Affairs, or his/her designee, are authorized to speak for the Agency in an official capacity regarding NASA policy, programmatic, and budget issues.(b) In response to media interview requests, NASA will offer articulate and knowledgeable spokespersons who can best serve the needs of the media and the American public. However, journalists may have access to the NASA officials they seek to interview, provided those NASA officials agree to be interviewed.(c) NASA employees may speak to the media and the public about their work. When doing so, employees shall notify their immediate supervisor and coordinate with their public affairs office in advance of interviews whenever possible, or immediately thereafter, and are encouraged, to the maximum extent practicable, to have a public affairs officer present during interviews. If public affairs officers are present, their role will be to attest to the content of the interview, support the interviewee, and provide post-interview follow up with the media as necessary.(d) NASA, as an Agency, does not take a position on any scientific conclusions. That is the role of the broad scientific community and the nature of the scientific process. NASA scientists may draw conclusions and may, consistent with this policy, communicate those conclusions to the media. However, NASA employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee's official area of expertise or responsibility.(f) Media interviews will be "on-the-record" and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.(g) Audio recordings may be made by NASA with consent of

the interviewee.(h) NASA employees are not required to speak to the media.(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.

Preventing Release of Classified Information to the Media(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. See NPR 1600.1, Chapter 5, Section 5.23 for internal NASA guidance on management of classified information. For further guidance that applies to all agencies, see Executive Order 12958, as amended, "Classified National Security Information" and its implementing directive at 32 CFR Parts 2001 and 2004.(b) Any attempt by news media representatives to obtain classified information will be reported through the Headquarters Office of Public Affairs or Installation Public Affairs Office to the Installation Security Office and Office of Security and Program Protection.(c) For classified operations and/or programs managed under the auspices of a DD Form 254, "Contract Security Classification Specification," all inquiries concerning this activity will be responded to by the appropriate PAO official designated in Item 12 on the DD Form 254.(d) For classified operations and/or information owned by other Government agencies (e.g., DOD, DOE, etc.), all inquiries will be referred to the appropriate Agency public affairs officer as established in written agreements.

Preventing Unauthorized Release of Sensitive but Unclassified (SBU) Information and/or Material to the News Media(a) All NASA SBU information requires accountability and approval for release. Release of SBU information to unauthorized personnel is prohibited. Unauthorized release of SBU information may result in prosecution and/or disciplinary action. Ignorance of NASA policy and procedures regarding SBU information does not release a NASA employee from responsibility for unauthorized release. See NPR 1600.1, Chapter 5, Section 5.24 for guidance on identification, marking, accountability and release of NASA SBU information.(b) Examples of SBU information include: proprietary information of others provided to NASA under nondisclosure or confidentiality agreement; source selection and bid and proposal information; information subject to export control under the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR); information subject to the Privacy Act of 1974; predecisional materials such as national space policy not yet publicly released; pending reorganization plans or sensitive travel itineraries; and information that could constitute an indicator of U.S. government intentions, capabilities, operations, or activities or otherwise threaten operations security.(c) Upon request for access to information/material deemed SBU, coordination must be made with the information/material owner to determine if the information/material

may be released. Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.Multimedia Materials(a) NASA's multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.Audio and/or video files of significant news developments and other events of historic or public interest.Interactive multimedia features that can be incorporated into the Agency's Internet portal for use by internal and external audiences, including the media and the general public.News Releases Concerning International Activities(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

This directive sets forth policy governing the release of public information, which is defined as information in any form provided to news and information media, especially information that has the potential to generate significant media, or public interest or inquiry. Examples include, but are not limited to, press releases, media advisories, news features, and web postings. Not included under this definition are scientific and technical reports, web postings designed for technical or scientific interchange, and technical information presented at professional meetings or in professional journals.

(a)This policy applies to NASA Headquarters, NASA Centers, and Component Facilities.

(b) In the event of any conflict between this policy and any other NASA policy, directive, or regulation, this policy shall govern and supersede any previous issuance or directive.

(a) NASA, a scientific and technical agency, is committed to a culture of openness with the media and public that values the free exchange of ideas, data, and information as part of scientific and technical inquiry. Scientific and technical information from or about Agency programs and projects will be accurate and unfiltered.

(b) Consistent with NASA statutory responsibility, NASA will “provide for the widest practicable and appropriate dissemination of information concerning its activities and the results thereof.” Release of public information concerning NASA activities and the results of NASA activities will be made promptly, factually, and completely.

(c) To ensure timely release of information, NASA will endeavor to ensure cooperation and coordination among the Agency’s scientific, engineering, and public affairs communities.

(d) In keeping with the desire for a culture of openness, NASA employees may, consistent with this policy, speak to the press and the public about their work.

(e) This policy does not authorize or require disclosure of information that is exempt from disclosure under the Freedom of Information Act (5 U.S.C. § 552) or otherwise restricted by statute, regulation, Executive Order, or other Executive Branch policy or NASA policy (e.g., OMB Circulars, NASA Policy Directives). Examples of information not releasable under this policy include, without limitation, information that is, or is marked as, classified information, procurement sensitive information, information subject to the Privacy Act, other sensitive but unclassified information, and information subject to privilege, such as pre-decisional information or attorney-client communications.

(a) The Assistant Administrator for Public Affairs is responsible for developing and administering an integrated Agency-wide communications program, establishing Agency public affairs policies and priorities, and coordinating and reviewing the performance of all Agency public affairs activities. The Assistant Administrator will develop criteria to identify which news releases and other types of public information will be issued nationwide by NASA Headquarters. Decisions to release public information nationwide by NASA Headquarters will be made by the Assistant Administrator for Public Affairs or his/her designee.

(b) NASA’s Mission Directorate Associate Administrators and Mission Support Office heads have ultimate responsibility for the technical, scientific, and programmatic accuracy of all information that is related to their respective programs and released by NASA.

(c) Under the direction of the Assistant Administrator for Public Affairs, public affairs officers assigned to Mission Directorates are responsible for the timely and efficient coordination of public information covering their respective programs. This coordination includes review by appropriate Mission Directorate officials. It also includes editing by public affairs staff to ensure that public information products are well written and appropriate for the intended audience. However, such editing shall not change scientific or technical data, or the meaning of programmatic content.

(d) Center Public Affairs Directors are responsible for implementing their portion of the Agency's communications program, adhering to Agency policies, procedures, and priorities, and coordinating their activities with Headquarters (and others where appropriate). They are responsible for the quality of public information prepared by Center public affairs officers. They also are responsible for the day-to-day production of public information covering their respective Center activities, which includes obtaining the necessary Center concurrences and coordinating, as necessary, with the appropriate Headquarters public affairs officers.

(e) Center Directors have ultimate responsibility for the accuracy of public information that does not require the concurrence of Headquarters. (See "Public information coordination and concurrence," section (d).)

(f) All NASA employees are required to coordinate, in a timely manner, with the appropriate public affairs officers prior to releasing information that has the potential to generate significant media, or public interest or inquiry.

(g) All NASA public affairs officers are required to notify the appropriate Headquarters public affairs officers in a timely manner about activities or events that have the potential to generate significant media or public interest or inquiry.

(h) All NASA public affairs employees are expected to adhere to the following code of conduct:

Be honest and accurate in all communications.

Honor publication embargoes.

Respond promptly to media requests and respect media deadlines.

Act promptly to correct mistakes or erroneous information, either internally or externally.

Promote the free flow of scientific and technical information.

Protect non-public information.

(i) All NASA employees are responsible for adhering to plans (including schedules) for activities established by public affairs offices and senior management for the coordinated release of public information.

(j) All NASA-funded missions will have a public affairs plan, approved by the Assistant Administrator for Public Affairs, which will be managed by Headquarters and/or a designated NASA Center.

(k) Public affairs activities for NASA-funded missions will not be managed by non-NASA institutions, unless authorized by the Assistant Administrator for Public Affairs.

(l) The requirements of this directive do not apply to the Office of Inspector General regarding its activities.

(a) General. All NASA employees involved in preparing and issuing NASA public information are responsible for proper coordination among Headquarters, Center, and Mission Directorate offices to include review and clearance by appropriate officials prior to issuance. Such coordination will be accomplished through procedures developed and published by the NASA Assistant Administrator for Public Affairs.

(b) Coordination. To ensure timely release of public information, Headquarters and Center public affairs officers are required to coordinate to obtain review and clearance by appropriate officials, keep each other informed of changes, delays, or cancellation of releases, and provide advance notification of the actual release.

(c) All public information shall be coordinated through the appropriate Headquarters offices, including review by the appropriate Mission Directorate Associate Administrator and mission support office head, or their designees, to ensure scientific, technical, and programmatic accuracy, and review by the Assistant Administrator of Public Affairs or his/her designee to ensure that public information products are well written and appropriate for the intended audience.

(d) Centers may, however, without the full coordination of Headquarters, issue public information that is institutional in nature, of local interest, or has been deemed not to be a Headquarters release. (The Assistant Administrator for Public Affairs or his/her designee will determine which public information will be issued nationwide by NASA Headquarters.) These releases must be coordinated through the appropriate Center offices and approved by the Center Director and Center Public Affairs Director. The Center Public Affairs Director is required to provide proper notification to the NASA Office of Public Affairs, Headquarters, prior to release. (The Assistant Administrator for Public Affairs shall publish guidelines for the release of public information that may be issued by Centers without clearance from Headquarters' offices.)

(e) Dispute Resolution. Any dispute arising from a decision to proceed or not proceed with the issuance of a news release or other type of public information will be addressed and resolved by the Assistant Administrator for Public Affairs with the appropriate Mission Directorate Associate Administrator, mission support office head, Center Director, and others, such as Center Public Affairs Directors, as necessary. However, the appropriate Mission Directorate Associate Administrator shall be the arbiter of disputes about the

accuracy or characterization of programmatic, technical, or scientific information. Additional appeals may be made to the Chief of Strategic Communications and to the Office of the Administrator. When requested by a Center Public Affairs Director, an explanation of the resolution will be provided in writing to all interested Agency parties.

(a) Only spokespersons designated by the Assistant Administrator for Public Affairs, or his/her designee, are authorized to speak for the Agency in an official capacity regarding NASA policy, programmatic, and budget issues.

(b) In response to media interview requests, NASA will offer articulate and knowledgeable spokespersons who can best serve the needs of the media and the American public. However, journalists may have access to the NASA officials they seek to interview, provided those NASA officials agree to be interviewed.

(c) NASA employees may speak to the media and the public about their work. When doing so, employees shall notify their immediate supervisor and coordinate with their public affairs office in advance of interviews whenever possible, or immediately thereafter, and are encouraged, to the maximum extent practicable, to have a public affairs officer present during interviews. If public affairs officers are present, their role will be to attest to the content of the interview, support the interviewee, and provide post-interview follow up with the media as necessary.

(d) NASA, as an Agency, does not take a position on any scientific conclusions. That is the role of the broad scientific community and the nature of the scientific process. NASA scientists may draw conclusions and may, consistent with this policy, communicate those conclusions to the media. However, NASA employees who present personal views outside their official area of expertise or responsibility must make clear that they are presenting their individual views – not the views of the Agency – and ask that they be sourced as such.

(e) Appropriated funds may only be used to support Agency missions and objectives consistent with legislative or presidential direction. Government funds shall not be used for media interviews or other communication activities that go beyond the scope of Agency responsibilities and/or an employee's official area of expertise or responsibility.

(f) Media interviews will be "on-the-record" and attributable to the person making the remarks, unless authorized to do otherwise by the Assistant Administrator for Public Affairs or Center Public Affairs Director, or their designees. Any NASA employee providing material to the press will identify himself/herself as the source.

(g) Audio recordings may be made by NASA with consent of the interviewee.

(h) NASA employees are not required to speak to the media.

(i) Public information volunteered by a NASA official will not be considered exclusive to any one media source and will be made available to other sources, if requested.

(a) Release of classified information in any form (e.g., documents, through interviews, audio/visual, etc.) to the news media is prohibited. The disclosure of classified information to unauthorized individuals may be cause for prosecution and/or disciplinary action against the NASA employee involved. Ignorance of NASA policy and procedures regarding classified information does not release a NASA employee from responsibility for preventing any unauthorized release. See NPR 1600.1, Chapter 5, Section 5.23 for internal NASA guidance on management of classified information. For further guidance that applies to all agencies, see Executive Order 12958, as amended, "Classified National Security Information" and its implementing directive at 32 CFR Parts 2001 and 2004.

(b) Any attempt by news media representatives to obtain classified information will be reported through the Headquarters Office of Public Affairs or Installation Public Affairs Office to the Installation Security Office and Office of Security and Program Protection.

(c) For classified operations and/or programs managed under the auspices of a DD Form 254, "Contract Security Classification Specification," all inquiries concerning this activity will be responded to by the appropriate PAO official designated in Item 12 on the DD Form 254.

(d) For classified operations and/or information owned by other Government agencies (e.g., DOD, DOE, etc.), all inquiries will be referred to the appropriate Agency public affairs officer as established in written agreements.

(a) All NASA SBU information requires accountability and approval for release. Release of SBU information to unauthorized personnel is prohibited. Unauthorized release of SBU information may result in prosecution and/or disciplinary action. Ignorance of NASA policy and procedures regarding SBU information does not release a NASA employee from responsibility for unauthorized release. See NPR 1600.1, Chapter 5, Section 5.24 for guidance on identification, marking, accountability and release of NASA SBU information.

(b) Examples of SBU information include: proprietary information of others provided to NASA under nondisclosure or confidentiality agreement; source selection and bid and proposal information; information subject to export control under the International Traffic in Arms Regulations (ITAR) or the Export Administration Regulations (EAR); information subject to the Privacy Act of 1974; predecisional materials such as national space policy not yet publicly released; pending reorganization plans or sensitive travel itineraries; and information that could constitute an indicator of U.S. government intentions, capabilities, operations, or activities or otherwise threaten operations security.

(c) Upon request for access to information/material deemed SBU, coordination must be made with the information/material owner to determine if the information/material may be released. Other organizations that play a part in SBU information identification, accountability and release (e.g., General Counsel, External Relations, Procurement, etc.) must be consulted for assistance and/or concurrence prior to release.

(d) Requests for SBU information from other Government agencies must be referred to the respective Agency public affairs officer.

(a) NASA's multimedia material, from all sources, will be made available to the information media, the public, and to all Agency Centers and contractor installations utilizing contemporary delivery methods and emerging digital technology.

(b) Centers will provide the media, the public, and as necessary, NASA Headquarters with:

Selected prints and original or duplicate files of news-oriented imagery and other digital multimedia material generated within their respective areas.

Selected video material in the highest quality format practical, which, in the opinion of the installations, would be appropriate for use as news feed material or features in pre-produced programs and other presentations.

Audio and/or video files of significant news developments and other events of historic or public interest.

Interactive multimedia features that can be incorporated into the Agency's Internet portal for use by internal and external audiences, including the media and the general public.

(a) Releases of information involving NASA activities, views, programs, or projects involving another country or an international organization require prior coordination and approval by the Headquarters offices of External Relations and Public Affairs.

(b) NASA Centers and Headquarters offices will report all visits proposed by representatives of foreign news media to the public affairs officer for the Office of External Relations for appropriate handling consistent with all NASA policies and procedures.

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TITLE: NASA Kennedy Top 20 Stories of 2025

6 min readNASA Kennedy Top 20 Stories of 2025

Elyna Niles-CarnesDec 22, 2025ArticleNASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025. The spacecraft will carry NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen on a 10-day mission around the Moon and back in early 2026. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to NASA Kennedy's Launch Complex 39B for the wet dress rehearsal.NASA/Kim ShiflettTeams at NASA's Kennedy Space Center in Florida spent 2025 preparing the launch vehicle and its powerhouse SLS (Space Launch System) rocket to launch four astronauts around the Moon for Artemis II in early 2026. The center also celebrated milestones by conducting science experiments at the International Space Station to studying the Sun's solar wind impacts on Earth to traveling to Mars in hopes of one day exploring the Red Planet in person.JANUARYNASA Kennedy Marks New Chapter for Florida Space IndustryKennedy Space Center Director Janet Petro and charter members of the Florida University Space Research Consortium sign a memorandum of understanding in research and development to assist with missions and contribute to NASA's Moon to Mars exploration approach.From left: Jennifer Kunz, Associate Director, Technical, Kennedy Space Center; Kelvin Manning, Deputy Director, Kennedy Space Center; Dr. Kent Fuchs, Interim President, University of Florida; Janet Petro, Director, Kennedy Space Center; Jeanette Nuñez, Florida Lieutenant Governor; Dr. Alexander Cartwright, President, University of Central Florida; Dr. Barry Butler, President, Embry-Riddle Aeronautical University.NASA/Kim ShiflettFirefly Launches Blue Ghost Mission OneFirefly Aerospace launched Blue Ghost Mission One lunar lander with a suite of NASA scientific instruments on January 15, from Launch Complex 39A at NASA Kennedy. The lander and instruments landed March 2 on the Moon.Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Firefly Aerospace's Blue Ghost Mission One lander soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Wednesday, Jan. 15, 2025 as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.NASA/Cory S HustonFEBRUARYIntuitive Machines Launches to the MoonIntuitive Machines' IM-2 Nova C lunar lander launched Feb. 26 from Launch Complex 39A, carrying NASA science and technology demonstrations to the Mons Mouton region of the Moon. IM-2 reached the surface of the Moon on March 6.Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Intuitive Machines' Nova-C lunar lander (IM-2) soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:16 p.m. EST Wednesday, Feb. 26, 2025, as part of NASA's CLPS

(Commercial Lunar Payload Services) initiative. NASA/Frank Michaux MARCH NASA's SpaceX Crew-10 Launch NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov launched March 14 from Launch Complex 39A to the International Space Station for a five-month science mission. Members of NASA's SpaceX Crew-10, from left, Roscosmos cosmonaut Kirill Peskov, mission specialist; NASA astronauts Nichole Ayers, pilot and Anne McClain, commander; and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, mission specialist. SpaceX NASA's SPHEREx, PUNCH Missions Launch A SpaceX Falcon 9 rocket launched on March 11, from Space Launch Complex 4 East at Vandenberg Space Force Base in California carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) and PUNCH (Polarimeter to Unify the Corona and Heliosphere) missions. NASA's Launch Services Program, based at NASA Kennedy managed the launch service for SPHEREx. NASA's SPHEREx observatory is installed in the Titan Thermal Vacuum (TVAC) test Chamber at BAE Systems in Boulder, Colorado, in June 2024. As part of the test setup, the spacecraft and photon shield are covered in multilayer insulation and blankets and surrounded by ground support equipment. Jet Propulsion Laboratory NASA's SpaceX Crew-9 Returns NASA astronauts Nick Hague, Suni Williams, and Butch Wilmore were greeted by dolphins and recovery teams after their SpaceX Dragon spacecraft splashed down on March 18, off the coast of Tallahassee, Florida following their long-duration mission at the International Space Station. Support teams work around a SpaceX Dragon spacecraft shortly after it landed with NASA astronauts Nick Hague, Suni Williams, Butch Wilmore, and Roscosmos cosmonaut Aleksandr Gorbunov aboard in the water off the coast of Tallahassee, Florida, Tuesday, March 18, 2025. Hague, Gorbunov, Williams, and Wilmore are returning from a long-duration science expedition aboard the International Space Station. NASA/Keegan Barber NASA Causeway Bridge Opens The Florida Department of Transportation opened the westbound portion of the NASA Causeway Bridge on March 19, completing construction in both directions spanning the Indian River Lagoon and connecting NASA Kennedy and Cape Canaveral Space Force Station to the mainland. Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023. NASA/Glenn Benson NASA Artemis Teams Complete URT-12 Teams from NASA and the Department of War train during a week-long Underway Recovery Test-12 in March off the coast of California for Artemis II test flight crewmembers and the Orion spacecraft. The series of tests demonstrate and evaluate the processes, procedures, and hardware used in recovery operations for crewed lunar

missions. Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.

NASA/Joel Kowsky
APRIL NASA's SpaceX 32nd Commercial Resupply Mission A SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station. The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's 32nd commercial resupply services mission for the agency to the International Space Station.

SpaceX
JULY Artemis III Begins Processing NASA's Artemis III SLS engine section and boat-tail made the journey from the Space Systems Processing Facility at NASA Kennedy to the spaceport's Vehicle Assembly Building in July to complete integration and check-out testing. Beginning with the Artemis III hardware, NASA moved certain operations to NASA Kennedy to streamline the manufacturing process and enable simultaneous production operations of two core stages. Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one of the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.

NASA/Ronald Beard
AUGUST NASA's SpaceX Crew-11 Launches NASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station. NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.

NASA/Joel Kowsky
NASA's SpaceX Crew-10 Returns NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency's Commercial Crew Program.

Roscosmos
cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station.

NASA/Keegan Barber
NASA's SpaceX 33rd Commercial Resupply Mission A SpaceX Falcon 9 launched the

company's Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A. A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA's SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module's forward port.

NASA Orion Tested, Stacked With Hardware Teams transported NASA's Orion spacecraft from Kennedy's Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS. The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch.

NASA/Cory Huston **SEPTEMBER** NASA Launches IMAP Mission NASA's IMAP (Interstellar Mapping and Acceleration Probe) launched from Launch Complex 39A on Sept. 24, to help researchers better understand the boundary of the heliosphere, a huge bubble created by the Sun surrounding and protecting our solar system. A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.

BAE Systems/Benjamin Fry **NASA's Northrop Grumman Commercial Resupply Mission** A Northrop Grumman Cygnus XL spacecraft atop a SpaceX Falcon 9 rocket lifted off from Launch Complex 39A to the International Space Station delivering NASA science investigations, supplies, and equipment as part of the agency's partnership to resupply the orbiting laboratory. Northrop Grumman's Cygnus XL

cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume.

NASA OCTOBER Orion Integrated With SLS Rocket
Teams stacked NASA's Orion spacecraft with its launch abort system on the agency's SLS rocket in High Bay 3 of the Vehicle Assembly Building at NASA Kennedy on Oct. 20 for the agency's Artemis II mission. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to Launch Complex 39B for the wet dress rehearsal.

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025.

NASA/Kim Shiflett NOVEMBER NASA's ESCAPEDE Begins Journey to Mars
NASA's ESCAPEDE (Escape and Plasma Acceleration and Dynamics Explorers) twin spacecraft launched aboard Blue Origin's New Glenn rocket on Nov. 13 from Launch Complex 36 at Cape Canaveral Space Force Station. Its twin orbiters will take simultaneous observations from different locations around Mars to reveal how the solar wind interacts with Mars' magnetic environment and how this interaction drives the planet's atmospheric escape.

Near Cape Canaveral Lighthouse, Blue Origin's New Glenn rocket carrying NASA's twin ESCAPEDE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPEDE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.

Blue Origin NASA, European Partners Launch Sea Satellite
A SpaceX Falcon 9 rocket carrying the U.S.-European Sentinel-6B satellite launched at Nov. 16 from Space Launch Complex 4 East at Vandenberg Space Force Base in California. Sentinel-6B will observe Earth's ocean, measuring sea levels to improve weather forecasts and flood predictions, safeguard public safety, benefit commercial industry, and protect coastal infrastructure.

A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.

SpaceX DECEMBER NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA astronaut Jeremy Hansen participated in a dry dress rehearsal at NASA Kennedy on Dec. 20 to mimic launch day operations for the Artemis II launch. The crew donned their

spacesuits, exited the Neil A. Operations and Checkout Building, and took the journey to the Vehicle Assembly Building, up the mobile launcher to the crew access arm, and entered the Orion spacecraft that will take them around the Moon and back to Earth. From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all. NASA/Aubrey Gemignani

Elyna Niles-Carnes Dec 22, 2025 Article

Elyna Niles-Carnes

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025. The spacecraft will carry NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen on a 10-day mission around the Moon and back in early 2026. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to NASA Kennedy's Launch Complex 39B for the wet dress rehearsal. NASA/Kim Shiflett Teams at NASA's Kennedy Space Center in Florida spent 2025 preparing the launch vehicle and its powerhouse SLS (Space Launch System) rocket to launch four astronauts around the Moon for Artemis II in early 2026. The center also celebrated milestones by conducting science experiments at the International Space Station to studying the Sun's solar wind impacts on Earth to traveling to Mars in hopes of one day exploring the Red Planet in person. JANUARY NASA Kennedy Marks New Chapter for Florida Space Industry Kennedy Space Center Director Janet Petro and charter members of the Florida University Space Research Consortium sign a memorandum of understanding in research and development to assist with missions and contribute to NASA's Moon to Mars exploration approach. From left: Jennifer Kunz, Associate Director, Technical, Kennedy Space Center; Kelvin Manning, Deputy Director, Kennedy Space Center; Dr. Kent Fuchs, Interim President, University of Florida; Janet Petro, Director, Kennedy Space Center; Jeanette Nuñez, Florida Lieutenant Governor; Dr. Alexander Cartwright, President,

University of Central Florida; Dr. Barry Butler, President, Embry-Riddle Aeronautical University.

NASA/Kim Shifflet **Firefly Launches Blue Ghost Mission One** Firefly Aerospace launched Blue Ghost Mission One lunar lander with a suite of NASA scientific instruments on January 15, from Launch Complex 39A at NASA Kennedy. The lander and instruments landed March 2 on the Moon.

Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Firefly Aerospace's Blue Ghost Mission One lander soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Wednesday, Jan. 15, 2025 as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.

NASA/Cory S Houston **FEBRUARY** **Intuitive Machines Launches to the Moon** Intuitive Machines' IM-2 Nova C lunar lander launched Feb. 26 from Launch Complex 39A, carrying NASA science and technology demonstrations to the Mons Mouton region of the Moon. IM-2 reached the surface of the Moon on March 6.

Creating a golden streak in the night sky, a SpaceX Falcon 9 rocket carrying Intuitive Machines' Nova-C lunar lander (IM-2) soars upward after liftoff from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:16 p.m. EST Wednesday, Feb. 26, 2025, as part of NASA's CLPS (Commercial Lunar Payload Services) initiative.

NASA/Frank Michaux **MARCH** **NASA's SpaceX Crew-10 Launch** NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov launched March 14 from Launch Complex 39A to the International Space Station for a five-month science mission.

Members of NASA's SpaceX Crew-10, from left, Roscosmos cosmonaut Kirill Peskov, mission specialist; NASA astronauts Nichole Ayers, pilot and Anne McClain, commander; and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, mission specialist.

SpaceX **NASA's SPHEREx, PUNCH Missions Launch** A SpaceX Falcon 9 rocket launched on March 11, from Space Launch Complex 4 East at Vandenberg Space Force Base in California carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) and PUNCH (Polarimeter to Unify the Corona and Heliosphere) missions. NASA's Launch Services Program, based at NASA Kennedy managed the launch service for SPHEREx.

NASA's SPHEREx observatory is installed in the Titan Thermal Vacuum (TVAC) test Chamber at BAE Systems in Boulder, Colorado, in June 2024. As part of the test setup, the spacecraft and photon shield are covered in multilayer insulation and blankets and surrounded by ground support equipment.

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Support teams work around a SpaceX Dragon spacecraft shortly after it landed with NASA astronauts Nick Hague, Suni Williams, Butch Wilmore, and Roscosmos cosmonaut Aleksandr Gorbunov aboard in the water off the coast of Tallahassee, Florida, Tuesday, March 18, 2025. Hague, Gorbunov,

Williams, and Wilmore are returning from a long-duration science expedition aboard the International Space Station.

NASA/Keegan Barber
NASA Causeway Bridge Opens
The Florida Department of Transportation opened the westbound portion of the NASA Causeway Bridge on March 19, completing construction in both directions spanning the Indian River Lagoon and connecting NASA Kennedy and Cape Canaveral Space Force Station to the mainland. Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023.

NASA/Glenn Benson
NASA Artemis Teams Complete URT-12
Teams from NASA and the Department of War train during a week-long Underway Recovery Test-12 in March off the coast of California for Artemis II test flight crewmembers and the Orion spacecraft. The series of tests demonstrate and evaluate the processes, procedures, and hardware used in recovery operations for crewed lunar missions.

Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.

NASA/Joel Kowsky
APRIL
NASA's SpaceX 32nd Commercial Resupply Mission
A SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station. The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's 32nd commercial resupply services mission for the agency to the International Space Station.

SpaceX
JULY
Artemis III Begins Processing
NASA's Artemis III SLS engine section and boat-tail made the journey from the Space Systems Processing Facility at NASA Kennedy to the spaceport's Vehicle Assembly Building in July to complete integration and check-out testing. Beginning with the Artemis III hardware, NASA moved certain operations to NASA Kennedy to streamline the manufacturing process and enable simultaneous production operations of two core stages.

Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.

NASA/Ronald Beard
AUGUST
NASA's SpaceX Crew-11 Launches
NASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan

Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station. NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program. NASA/Joel Kowsky NASA's SpaceX Crew-10 Returns NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency's Commercial Crew Program. Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station. NASA/Keegan Barber NASA's SpaceX 33rd Commercial Resupply Mission A SpaceX Falcon 9 launched the company's Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A. A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA's SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module's forward port. NASA Orion Tested, Stacked With Hardware Teams transported NASA's Orion spacecraft from Kennedy's Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS. The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch. NASA/Cory Huston SEPTEMBER NASA Launches IMAP Mission NASA's IMAP (Interstellar Mapping and Acceleration Probe) launched from Launch

Complex 39A on Sept. 24, to help researchers better understand the boundary of the heliosphere, a huge bubble created by the Sun surrounding and protecting our solar system. A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On-Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.

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JANUARY NASA Kennedy Marks New Chapter for Florida Space Industry

Kennedy Space Center Director Janet Petro and charter members of the Florida University Space Research Consortium sign a memorandum of understanding in research and development to assist with missions and contribute to NASA's Moon to Mars exploration approach.

From left: Jennifer Kunz, Associate Director, Technical, Kennedy Space Center; Kelvin Manning, Deputy Director, Kennedy Space Center; Dr. Kent Fuchs, Interim President, University of Florida; Janet Petro, Director, Kennedy Space Center; Jeanette Nuñez, Florida Lieutenant Governor; Dr. Alexander Cartwright, President, University of Central Florida; Dr. Barry Butler, President, Embry-Riddle Aeronautical University. NASA/Kim Shifflet

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The Florida Department of Transportation opened the westbound portion of the NASA Causeway Bridge on March 19, completing construction in both directions spanning the Indian River Lagoon and connecting NASA Kennedy and Cape Canaveral Space Force Station to the mainland.

Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023. NASA/Glenn Benson

Cars drive over the newly completed westbound portion (right side of photo) of the NASA Causeway Bridge leading away from NASA's Kennedy Space Center in Florida on Wednesday, March 19, 2025. The Florida Department of Transportation (FDOT) opened the span on Tuesday, March 18, 2025, alongside its twin on the eastbound side, which has accommodated traffic in both directions since FDOT opened it on June 9, 2023.

NASA/Glenn Benson

NASA Artemis Teams Complete URT-12

Teams from NASA and the Department of War train during a week-long Underway Recovery Test-12 in March off the coast of California for Artemis II test flight crewmembers and the Orion spacecraft. The series of tests demonstrate and evaluate the processes, procedures, and hardware used in recovery operations for crewed lunar missions.

Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean. NASA/Joel Kowsky

Waves break inside USS Somerset as the Crew Module Test Article, a full scale mockup of the Orion spacecraft, is tethered during Underway Recovery Test-12 off the coast of California, Wednesday, March 26, 2025. During the test, NASA and Department of Defense teams are practicing to ensure recovery procedures are validated as NASA plans to send Artemis II astronauts around the Moon and splashdown in the Pacific Ocean.

APRIL NASA's SpaceX 32nd Commercial Resupply Mission

A SpaceX Falcon 9 rocket and a Dragon spacecraft carrying nearly 6,700 pounds of scientific investigations, food, supplies, and equipment launched on April 21 from Launch Complex 39A to the International Space Station.

The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's

32nd commercial resupply services mission for the agency to the International Space Station.SpaceX

The SpaceX Falcon 9 rocket carrying the Dragon spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, April 21, on the company's 32nd commercial resupply services mission for the agency to the International Space Station.

JULYArtemis III Begins Processing

NASA's Artemis III SLS engine section and boat-tail made the journey from the Space Systems Processing Facility at NASA Kennedy to the spaceport's Vehicle Assembly Building in July to complete integration and check-out testing. Beginning with the Artemis III hardware, NASA moved certain operations to NASA Kennedy to streamline the manufacturing process and enable simultaneous production operations of two core stages.

Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one of the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.NASA/Ronald Beard

Teams from NASA's Kennedy Space Center in Florida integrate NASA's Artemis III SLS (Space Launch System) core stage engine section with its boat-tail inside the spaceport's Vehicle Assembly Building on Wednesday, July 30, 2025. The boat-tail is a fairing-like structure that protects the bottom end of the core stage, while the engine section is one of the most complex and intricate parts of the rocket stage that will help power the Artemis missions to the Moon.

NASA/Ronald Beard

AUGUSTNASA's SpaceX Crew-11 Launches

NASA astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov launched aboard a SpaceX Dragon spacecraft and its Falcon 9 rocket on Aug. 1 from Launch Complex 39A bound for a long-duration mission to the International Space Station.

NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.NASA/Joel Kowsky

NASA's SpaceX Crew-11 mission is the eleventh crew rotation mission of the SpaceX Dragon spacecraft and Falcon 9 rocket to the International Space Station as part of the agency's Commercial Crew Program.

NASA's SpaceX Crew-10 Returns

NASA astronauts Anne McClain and Nicole Ayers, JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi, and Roscosmos cosmonaut Kirill Peskov became the first Commercial Crew to splash down in the Pacific Ocean off the coast of California on Aug. 9, completing their nearly five-month mission at the orbiting outpost as part of the agency's Commercial Crew Program.

Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station. NASA/Keegan Barber

Roscosmos cosmonaut Kirill Peskov, left, NASA astronauts Nichole Ayers, Anne McClain, and JAXA (Japan Aerospace Exploration Agency) astronaut Takuya Onishi returned after 147 days in space as part of Expedition 73 aboard the International Space Station.

NASA's SpaceX 33rd Commercial Resupply Mission

A SpaceX Falcon 9 launched the company's Dragon spacecraft carrying more than 5,000 pounds of food, crew supplies, science investigations, spacewalk equipment, and more to the space station on Aug. 24 from Launch Complex 39A.

A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA's SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module's forward port. NASA

A SpaceX Dragon cargo spacecraft with its nosecone open and carrying over 5,000 pounds of science, supplies, and hardware for NASA's SpaceX CRS-33 mission approaches the International Space Station for an automated docking to the Harmony module's forward port.

Orion Tested, Stacked With Hardware

Teams transported NASA's Orion spacecraft from Kennedy's Multi-Payload Processing Facility to the Launch Abort System Facility in August where crews integrated the 44-foot-tall launch abort system. The Orion spacecraft will send NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA (Canadian Space Agency) astronaut Jeremy Hansen

around the Moon for the Artemis II mission in early 2026. The launch abort system is designed to carry the crew to safety in the event of an emergency atop the SLS.

The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch. NASA/Cory Huston

The launch abort tower on NASA's Artemis II Orion spacecraft is pictured inside the Launch Abort System Facility at the agency's Kennedy Space Center in Florida on Wednesday, Aug. 27, 2025, after teams with NASA's Exploration Ground Systems Program installed the tower on Wednesday, Aug. 20, 2025. Positioned at the top of Orion, the 44-foot-tall launch abort system is designed to carry the crew to safety in the event of an emergency during launch or ascent, with its three solid rocket motors working together to propel Orion – and astronauts inside – away from the rocket for a safe landing in the ocean, or detach from the spacecraft when it is no longer needed. The final step to complete integration will be the installation of the ogive fairings, which are four protective panels that will shield the crew module from the severe vibrations and sounds it will experience during launch.

NASA/Cory Huston

SEPTEMBER NASA Launches IMAP Mission

NASA's IMAP (Interstellar Mapping and Acceleration Probe) launched from Launch Complex 39A on Sept. 24, to help researchers better understand the boundary of the heliosphere, a huge bubble created by the Sun surrounding and protecting our solar system.

A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space

weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system. BAE Systems/Benjamin Fry

A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On-Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.

BAE Systems/Benjamin Fry

NASA's Northrop Grumman Commercial Resupply Mission

A Northrop Grumman Cygnus XL spacecraft atop a SpaceX Falcon 9 rocket lifted off from Launch Complex 39A to the International Space Station delivering NASA science investigations, supplies, and equipment as part of the agency's partnership to resupply the orbiting laboratory.

Northrop Grumman's Cygnus XL cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume. NASA

Northrop Grumman's Cygnus XL cargo craft, carrying over 11,000 pounds of new science and supplies for the Expedition 73 crew, is pictured in the grips of the International Space Station's Canadarm2 robotic arm following its capture. Both spacecraft were orbiting 257 miles above Tanzania. Cygnus XL is Northrop Grumman's expanded version of its previous Cygnus cargo craft increasing its payload capacity and pressurized cargo volume.

OCTOBER Orion Integrated With SLS Rocket

Teams stacked NASA's Orion spacecraft with its launch abort system on the agency's SLS rocket in High Bay 3 of the Vehicle Assembly Building at NASA Kennedy on Oct. 20 for the agency's Artemis II mission. Teams will begin conducting a series of verification tests ahead of rolling out the integrated SLS rocket to Launch Complex 39B for the wet dress rehearsal.

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025. NASA/Kim Shiflett

NASA's Artemis II Orion spacecraft with its launch abort system is stacked atop the agency's SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Monday, Oct. 20, 2025.

NOVEMBER NASA's ESCAPE Begins Journey to Mars

NASA's ESCAPE (Escape and Plasma Acceleration and Dynamics Explorers) twin spacecraft launched aboard Blue Origin's New Glenn rocket on Nov. 13 from Launch Complex 36 at Cape Canaveral Space Force Station. Its twin orbiters will take simultaneous observations from different locations around Mars to reveal how the solar wind interacts with Mars' magnetic environment and how this interaction drives the planet's atmospheric escape.

Near Cape Canaveral Lighthouse, Blue Origin's New Glenn rocket carrying NASA's twin ESCAPE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars. Blue Origin

Near Cape Canaveral Lighthouse, Blue Origin's New Glenn rocket carrying NASA's twin ESCAPE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.

Blue Origin

NASA, European Partners Launch Sea Satellite

A SpaceX Falcon 9 rocket carrying the U.S.-European Sentinel-6B satellite launched at Nov. 16 from Space Launch Complex 4 East at Vandenberg Space Force Base in California. Sentinel-6B will observe Earth's ocean, measuring sea levels to improve weather forecasts and flood predictions, safeguard public safety, benefit commercial industry, and protect coastal infrastructure.

A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.SpaceX

A SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.

DECEMBER

NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and CSA astronaut Jeremy Hansen participated in a dry dress rehearsal at NASA Kennedy on Dec. 20 to mimic launch day operations for the Artemis II launch. The crew donned their spacesuits, exited the Neil A. Operations and Checkout Building, and took the journey to the Vehicle Assembly Building, up the mobile launcher to the crew access arm, and entered the Orion spacecraft that will take them around the Moon and back to Earth.

From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.NASA/Aubrey Gemignani

From right to left, NASA astronauts Christina Koch, mission specialist; Reid Wiseman, commander; Victor Glover, pilot; and CSA (Canadian Space Agency) astronaut Jeremy Hansen, mission specialist are seen as they depart the Neil A. Armstrong Operations and Checkout Building to board their Orion spacecraft atop NASA's Space Launch System rocket inside the Vehicle Assembly Building as part of the Artemis II countdown

demonstration test, Saturday, Dec. 20, 2025, at NASA's Kennedy Space Center in Florida. For this operation, the Artemis II crew and launch teams are simulating the launch day timeline including suit-up, walkout, and spacecraft ingress and egress. Through the Artemis campaign, NASA will send astronauts to explore the Moon for scientific discovery, economic benefits, and to build the foundation for the first crewed missions to Mars, for the benefit of all.

NASA/Aubrey Gemignani

URL: <https://www.nasa.gov/kennedy/launch-services-program/>

TITLE: Launch Services Program

Launch Services ProgramLaunch Services ProgramNASA's Launch Services Program (LSP) is responsible for launching rockets delivering spacecraft that observe the Earth, visit other planets and explore the universe – from weather satellites to telescopes to Mars rovers and more.LSP functions as a broker, matching spacecraft with the best-suited rockets, managing the launch process, providing support from pre-mission planning to post-launch. LSP helps implement NASA's policy of a mixed-fleet launch strategy, which uses both existing and emerging domestic launch capabilities to assure access to space.Learn Moreabout Launch Services ProgramLatest NewsMore LSP NewsBlog2 Min ReadNASA's Pandora Satellite Acquires SignalBlog1 Min ReadNASA's Exoplanet Observing Satellite Separated From RocketBlog1 Min ReadLiftoff of NASA's Newest Planet-Observing SatelliteBlog3 Min ReadNASA's Pandora Mission, CubeSats Ready for Flight3 Min ReadWelcome to Launch Day for NASA's Pandora Mission, CubeSatsBlog6 Min ReadNASA Kennedy Top 20 Stories of 2025Article4 Min ReadNASA Cost-Saving Technology Demo is Ready for LaunchBlog5 Min ReadNASA, SpaceX Launch US-European Satellite to Monitor Earth's OceansNews ReleaseRocketsFirefly Aerospace AlphaUnited Launch Alliance Atlas VRocket Lab ElectronSpaceX Falcon 9SpaceX Falcon HeavyBlue Origin New GlennNorthrop Grumman PegasusXLUnited Launch Alliance VulcanFirefly Aerospace AlphaFirefly's small-lift rocket, Alpha, is equipped to launch more than 1,000 kg to low Earth orbit. Alpha can be launched at Firefly's launch facilities at the Vandenberg Space Force Base in California and new launch capabilities coming soon at the Mid-Atlantic Regional Spaceport (MARS) on Wallops Island, Virginia as early as 2025.Read MoreFirefly Aerospace's "Noise of Summer" launched eight CubeSats for NASA on its Venture-Class Launch Services Demonstration 2 contract in July of 2024.Firefly Aerospace/Trevor MahlmannUnited Launch Alliance Atlas VAtlas V uses a standard common core booster, up to five solid rocket boosters (SRBs), a Centaur upper stage in a single- or dual-engine configuration, and one of several sizes of payload fairings. Atlas V is on NASA's Launch Services II contract.Read MoreA United Launch Alliance Atlas V 541 rocket lifts off from

Space Launch Complex 41 at Cape Canaveral Air Force Station in Florida on July 30, 2020, at 7:50 a.m. EDT, carrying NASA's Mars Perseverance rover and Ingenuity helicopter. NASA/Tony Gray and Tim Powers

Rocket Lab Electron Rocket Lab's Electron is a reusable orbital-class small rocket. Capturing and refllying Electron's first stage enables higher launch frequency without expanding production and lowers launch costs. Rocket Lab has three launch pads at two launch sites, including two launch pads at a private orbital launch site located in New Zealand and a third pad at NASA's Wallops Flight Facility in Virginia. [Read More](#)

A Rocket Lab Electron rocket lifts off Launch Complex 1, Pad B, in Māhia, New Zealand on May 8 at 1 p.m. New Zealand time (May 7 at 9 p.m. EDT), carrying two NASA CubeSats designed to study tropical cyclones, including hurricanes and typhoons. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats will provide data on temperature, precipitation, water vapor, and clouds by measuring microwave frequencies, providing insight into storm formation and intensification. [Read More](#)

Rocket Lab SpaceX Falcon 9 Falcon 9 is a reusable, two-stage rocket designed and manufactured by SpaceX for the reliable and safe transport of people and payloads into Earth orbit and beyond. Falcon 9 is on NASA's Launch Services II contract. [Read More](#)

NASA's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield. [Read More](#)

SpaceX Falcon Heavy Falcon Heavy is composed of three reusable Falcon 9 nine-engine cores whose 27 Merlin engines together generate more than 5 million pounds of thrust at liftoff, equal to approximately eighteen 747 aircraft. As one of the world's most powerful operational rockets, Falcon Heavy can lift nearly 64 metric tons (141,000 lbs.) to orbit. Falcon Heavy is on NASA's Launch Services II contract. [Read More](#)

A SpaceX Falcon Heavy rocket carrying NASA's Europa Clipper spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 12:06 p.m. EDT on Monday, Oct. 14, 2024. After launch, the spacecraft plans to fly by Mars in February 2025, then back by Earth in December 2026, using the gravity of each planet to increase its momentum. With help of these "gravity assists," Europa Clipper will achieve the velocity needed to reach Jupiter in April 2030. [Read More](#)

SpaceX Blue Origin New Glenn Named after John Glenn, the first American astronaut to orbit Earth, New Glenn is a single-configuration, heavy-lift orbital launch vehicle with a first stage designed for a minimum of 25 flights. Powered by seven Blue Engine 4 (BE-4) engines, it can carry more than 13 metric tons to geostationary transfer orbit and 45 metric tons to low Earth orbit. New Glenn is on NASA's Launch Services II

contract.[Read More](#)Blue Origin's New Glenn rocket carrying NASA's twin ESCAPEDE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPEDE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.[Blue Origin](#)Northrop Grumman PegasusXLThe three-stage Pegasus rocket is used to deploy small satellites weighing up to 1,000 pounds (453.59 kg) into low-Earth orbit. Pegasus is carried aloft by the Stargazer L-1011 aircraft to approximately 40,000 feet over open ocean, where it is released and free-falls five seconds before igniting its first stage rocket motor. With its unique delta-shaped wing, Pegasus typically delivers satellites into orbit in a little over 10 minutes. Pegasus is on NASA's Launch Services II contract.[Read More](#)The Northrop Grumman Pegasus XL rocket, carrying NASA's Ionospheric Connection Explorer (ICON) at the Skid Strip at Cape Canaveral Air Force Station in Florida as it prepared for launch in October 2019. The rocket is attached beneath the company's L-1011 Stargazer aircraft which launches the rocket once in the air.[NASA](#)United Launch Alliance VulcanVulcan is United Launch Alliance's next generation rocket available in four standard offering configurations including zero, two, four and six solid rocket booster variants. ULA will launch Vulcan from both Cape Canaveral Space Force Station in Florida and Vandenberg Space Force Base in California. Vulcan is on NASA's Launch Services II contract.[Read More](#)United Launch Alliance's Vulcan rocket is rolled out of the Vertical Integration Facility at Space Launch Complex 41 on Cape Canaveral Space Force Station in Florida in preparation for a commercial launch in January 2024.[NASA/Cory S Huston](#)Launch SitesKennedy Space CenterCape CanaveralVandenbergWallopsKwajalein AtollAdditional SitesKennedy Space CenterOne of two primary launch sites for NASA's Launch Vehicles. Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit.[View Site](#)A SpaceX Falcon 9 rocket with NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On-Lagrange 1 (SWFO-L1) spacecraft atop stands vertical at Launch Complex 39A as the sun sets on Monday, Sept. 22, 2025, at the agency's Kennedy Space Center in Florida. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.[BAE Systems/Benjamin Fry](#)Cape Canaveral Space Force StationOne of two primary launch sites for NASA's Launch Vehicles. Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is

ideal for spacecraft requiring a west-east or equatorial orbit. [View Site](#) A United Launch Alliance Atlas V rocket with the Lucy spacecraft aboard is seen in this 2 minute and 30 second exposure photograph as it launches from Space Launch Complex 41, Saturday, Oct. 16, 2021, at Cape Canaveral Space Force Station in Florida. Lucy will be the first spacecraft to study Jupiter's Trojan Asteroids. Like the mission's namesake – the fossilized human ancestor, “Lucy,” whose skeleton provided unique insight into humanity's evolution – Lucy will revolutionize our knowledge of planetary origins and the formation of the solar system.

[NASA/Bill Ingalls](#) Vandenberg Space Force Base One of two primary launch sites for NASA's Launch Vehicles. Located along California's central coast between Los Angeles and San Francisco, Vandenberg is preferred for spacecraft requiring a north-south, or polar, orbit. [Visit Site](#) A SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission lifts off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 10:44 a.m. PST Friday, Nov. 28, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.

[SpaceX](#) Wallops Flight Facility Located along the Atlantic Coast on Wallops Island, Virginia. Operated by Goddard Space Flight Center in Greenbelt, Maryland, Wallops is NASA's principal facility for suborbital research programs. [View Site](#) A Northrop Grumman Antares rocket carrying a Cygnus resupply spacecraft is seen on the Mid-Atlantic Regional Spaceport's Pad-0A, Thursday, October 1, 2020, at NASA's Wallops Flight Facility in Virginia. Northrop Grumman's 14th contracted cargo resupply mission with NASA to the International Space Station will deliver nearly 8,000 pounds of science and research, crew supplies and vehicle hardware to the orbital laboratory and its crew. The CRS-14 Cygnus spacecraft is named after the first female astronaut of Indian descent, Kalpana Chawla.

[NASA/Patrick Black](#) Reagan Test Site, Kwajalein Atoll An additional Expendable Launch Vehicle launch location in the Republic of the Marshall Islands in the North Pacific. The Kwajalein site is located between Hawaii and Australia and is used for missions requiring equatorial orbits and low inclinations. [Visit Site](#) Orbital Science Corp.'s L-1011 aircraft “Stargazer” flies over the runway on Kwajalein Atoll with the company's Pegasus rocket slung underneath. Kwajalein is part of the Marshall Islands chain in the Pacific Ocean. It's also part of the Reagan Test Site and used for launches of NASA, commercial and military missions.

[NASA](#) Additional Launch Sites Additional launch sites specific to LSP's VADR launches include Rocket Lab's Launch Complex 1 in Mahia, New Zealand; and SpaceX's Starbase in Boca Chica, Texas. Kodiak Island, located in Alaska, is considered one of the best locations in the world for polar launch operations, providing a wide launch azimuth

and unobstructed downrange flight path.[Read More](#)A wet dress rehearsal is underway for Rocket Lab's Electron rocket at Launch Complex 1 in Mahia, New Zealand on April 28, 2023. NASA's Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) CubeSats are secured in the payload fairing atop the rocket.[Rocket Lab](#)[Roman](#)The Nancy Grace Roman Space Telescope will be able to block starlight to directly see exoplanets and planet-forming disks, complete a statistical census of planetary systems in our galaxy, and settle essential questions in the areas of dark energy, exoplanets, and infrared astrophysics.[Learn More](#)[about Roman](#)[Upcoming Missions](#)[Learn more about future missions managed by NASA's Launch Services Program.](#)[More LSP Missions](#)[Pandora](#)Pandora is a small satellite designed to characterize exoplanet atmospheres and their host stars. It is slated to observe at least 20 different planets during its one year of science operations.[Read More](#)[Gateway](#)Astronauts will explore the scientific mysteries of deep space with Gateway, humanity's first space station around the Moon, and chart a path for the first human missions to Mars and beyond.[Read More](#)[Dragonfly](#)Dragonfly, the first-of-its-kind rotorcraft to explore another world, will fly to various locations on Saturn's moon Titan. The Dragonfly rotorcraft will break the barriers for exploration of other planetary bodies.[Read More](#)[Learn More About Launch Services Program](#)[Launch Services Program Awards](#)View the latest contract award news from Launch Services Program.[Venture-Class Acquisition of Dedicated and Rideshare](#)Find out how NASA is enabling greater access to space for science and technology missions through the agency's VADR (Venture-Class Acquisition of Dedicated and Rideshare) contract.[LSP Launch Archive](#)Check out detailed information about the spacecraft and launch vehicles that have lofted important scientific missions into Earth's orbit and to the outer reaches of our solar system.[LSP YouTube](#)Watch videos of past Launch Services Program missions.[Media Resources](#)[Learn more about Launch Services Program.](#)[About CubeSat Launch Initiative](#)Since its inception, NASA's CubeSat Launch Initiative has launched over 150 CubeSats on more than 40 Educational Launch of Nanosatellites (ELaNa) missions.[NASA's CubeSat Launch initiative \(CSLI\)](#) provides low-cost access to space for U.S. educational institutions, informal educational institutions such as museums and science centers, non-profits with an education/outreach component, and NASA centers for early career workforce development. The initiative's intent is to inspire and develop the next generation of scientists, engineers, and technologists by offering a unique opportunity to conduct scientific research and develop/demonstrate novel technologies in space.[Learn More](#)[about CubeSat Launch Initiative](#)Once Launch Services Program pairs CubeSats to a launch, an Educational Launch of Nanosatellites (ELaNa) mission number is assigned. Pictured is ELaNa 19 / Venture Class CubeSats (CHOMPTT) inside Rocket Lab's facility, located at Huntington Beach California.[Image credit: NASA](#) A SpaceX Falcon 9 rocket carrying NASA's R5-S7

(Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission stands vertical on the launch pad of Space Launch Complex 4 East at Vandenberg Space Force Base in California on Thursday, Nov. 27, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users.

SpaceX long exposure photo shows two streaks – the SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifting off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025, and the rocket's first stage returning minutes later to land at Vandenberg's Landing Zone 4 East. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans.

SpaceX Blue Origin's New Glenn rocket carrying NASA's twin ESCAPEDE (Escape and Plasma Acceleration and Dynamics Explorers) spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPEDE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.

Blue Origin A SpaceX Falcon 9 rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On–Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.

SpaceX NASA's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield.

SpaceX A SpaceX Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch

Complex 4 East at Vandenberg Space Force Base in California on Tuesday, March 11, 2025. SPHEREx will use its telescope to provide an all-sky spectral survey, creating a 3D map of the entire sky to help scientists investigate the origins of our universe. PUNCH will study origins of the Sun's outflow of material, or the solar wind, capturing continuous 3D images of the Sun's corona and the solar wind's journey into the solar system. SpaceX A reflection in the water shows NASA's Europa Clipper spacecraft atop SpaceX's Falcon Heavy rocket at Launch Pad 39A on Sunday, Oct. 13, 2024, at the agency's Kennedy Space Center in Florida ahead of launch to Jupiter's icy moon, Europa. The spacecraft will complete nearly 50 flybys of Europa to determine if there are conditions suitable for life beyond Earth. Launch is targeting 12:06 p.m. EDT on Monday, Oct. 14, from Launch Complex 39A at Kennedy Space Center in Florida. SpaceX A United Launch Alliance Atlas V 541 rocket, carrying the National Oceanic and Atmospheric Administration's (NOAA) Geostationary Operational Environmental Satellite-T (GOES-T), lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station in Florida on March 1, 2022. Liftoff was at 4:38 p.m. EST. GOES-T is the third satellite in the GOES-R series that will continue to help meteorologists observe and predict local weather events that affect public safety. GOES-T will be renamed GOES-18 once it reaches geostationary orbit. GOES-18 will go into operational service as GOES West to provide critical data for the U.S. West Coast, Alaska, Hawaii, Mexico, Central America, and the Pacific Ocean. The launch was managed by NASA's Launch Services Program based at Kennedy Space Center in Florida, America's multi-user spaceport. NASA/ Kevin O'Connell and Kevin Davis A Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab's Launch Complex 1 on the Mahia Peninsula of New Zealand. A SpaceX Falcon Heavy rocket with the Psyche spacecraft onboard is seen as it is rolled to the launch pad at Launch Complex 39A as preparations continue for the Psyche mission, Tuesday, Oct. 10, 2023, at NASA's Kennedy Space Center in Florida. NASA's Psyche spacecraft will travel to a metal-rich asteroid by the same name orbiting the Sun between Mars and Jupiter to study its composition. The spacecraft also carries the agency's Deep Space Optical Communications technology demonstration, which will test laser communications beyond the Moon. Photo Credit: (NASA/ Aubrey Gemignani) NASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. PACE is NASA's newest earth-observing satellite that will help increase our understanding of Earth's oceans, atmosphere, and climate by delivering hyperspectral observations of microscopic marine organisms called phytoplankton, as well new data on clouds and aerosols. SpaceX Meet Our Team More Biographies Albert Sierra Albert Sierra is the program manager of the Launch Services Program at Kennedy. Read More Jennifer W.

Lyons Jenny Lyons is the deputy program manager of the Launch Services Program at Kennedy. [Read More](#) Tim Dunn Tim Dunn is senior launch director of the Launch Services Program at Kennedy. [Read More](#) [Facebook](#)

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Launch Services Program NASA's Launch Services Program (LSP) is responsible for launching rockets delivering spacecraft that observe the Earth, visit other planets and explore the universe – from weather satellites to telescopes to Mars rovers and more. LSP functions as a broker, matching spacecraft with the best-suited rockets, managing the launch process, providing support from pre-mission planning to post-launch. LSP helps implement NASA's policy of a mixed-fleet launch strategy, which uses both existing and emerging domestic launch capabilities to assure access to space. [Learn More about Launch Services Program](#)

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Rockets

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Launch Sites

Kennedy Space Center Cape Canaveral Vandenberg Wallops Kwajalein Atoll Additional Sites Kennedy Space Center One of two primary launch sites for NASA's Launch Vehicles. Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit. View Site A SpaceX Falcon 9 rocket with NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow On-Lagrange 1 (SWFO-L1) spacecraft atop stands vertical at Launch Complex 39A as the sun sets on Monday, Sept. 22, 2025, at the agency's Kennedy Space Center in Florida. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system. BAE Systems/Benjamin Fry Cape Canaveral Space Force Station One of two primary launch sites for NASA's Launch Vehicles. Located along Florida's central Atlantic coast between Jacksonville and Miami, our nation's premiere spaceport is ideal for spacecraft requiring a west-east or equatorial orbit. View Site A United Launch Alliance Atlas V rocket with the Lucy spacecraft aboard is seen in this 2 minute and 30 second exposure photograph as it launches from Space Launch Complex 41, Saturday, Oct. 16, 2021, at Cape Canaveral Space Force Station in Florida. Lucy will be the first spacecraft to study Jupiter's Trojan Asteroids. Like the mission's namesake – the fossilized human ancestor, "Lucy," whose skeleton provided unique insight into humanity's evolution – Lucy will revolutionize our knowledge of planetary origins and the formation of the solar system. NASA/Bill Ingalls Vandenberg Space Force Base One of two primary launch sites for NASA's Launch Vehicles. Located along California's central coast between Los Angeles and San Francisco, Vandenberg is preferred for spacecraft requiring a north-south, or polar, orbit. Visit Site A SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-

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About CubeSat Launch Initiative Since its inception, NASA's CubeSat Launch Initiative has launched over 150 CubeSats on more than 40 Educational Launch of Nanosatellites (ELaNa) missions. NASA's CubeSat Launch initiative (CSLI) provides low-cost access to space for U.S. educational institutions, informal educational institutions such as museums and science centers, non-profits with an education/outreach component, and NASA centers for early career workforce development. The initiative's intent is to inspire and develop the next generation of scientists, engineers, and technologists by offering a unique opportunity to conduct scientific research and develop/demonstrate novel technologies in space. **Learn More** about CubeSat Launch Initiative Once Launch Services Program pairs CubeSats to a launch, an Educational Launch of Nanosatellites (ELaNa) mission number is assigned. Pictured is ELaNa 19 / Venture Class CubeSats (CHOMPTT) inside Rocket Lab's facility, located at Huntington Beach California. Image credit: NASA

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A SpaceX Falcon 9 rocket carrying NASA's R5-S7 (Realizing Rapid, Reduced-cost high-Risk Research project Spacecraft 7) CubeSat along with several other satellites as part of the company's Transporter-15 mission stands vertical on the launch pad of Space Launch Complex 4 East at Vandenberg Space Force Base in California on Thursday, Nov. 27, 2025. The latest in a series of spacecraft, R5-S7 will explore ways to get multiple technology prototypes into low Earth orbit rapidly and at a low cost, accelerating the demonstration of these technologies in orbit and allowing engineers and scientists to more quickly prove them and make them available to NASA missions and other users. Space X's long exposure photo shows two streaks – the SpaceX Falcon 9 rocket carrying the international Sentinel-6B spacecraft lifting off from Space Launch Complex 4 East at Vandenberg Space Force Base in California at 9:21 p.m. PST Sunday, Nov. 16, 2025, and the rocket's first stage returning minutes later to land at Vandenberg's Landing Zone 4 East. A collaboration between NASA, ESA (European Space Agency), EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites), and the National Oceanic and Atmospheric Administration (NOAA), Sentinel-6B is designed to measure sea levels down to roughly an inch for about 90% of the world's oceans. Space X's Blue Origin's New Glenn rocket carrying NASA's twin ESCAPE (Escape and Plasma Acceleration and Dynamics Explorers)

spacecraft launches at 3:55 p.m. EST, Thursday, Nov. 13, 2025, from Launch Complex 36 at Cape Canaveral Space Force Station in Florida. The ESCAPE mission, built by Rocket Lab, will study how solar wind and plasma interact with Mars' magnetosphere and how this interaction drives the planet's atmospheric escape to prepare for future human missions on Mars.

Blue Origin's New Glenn rocket carrying NASA's IMAP (Interstellar Mapping and Acceleration Probe), the agency's Carruthers Geocorona Observatory, and National Oceanic and Atmospheric Administration's (NOAA) Space Weather Follow-On-Lagrange 1 (SWFO-L1) spacecraft lifts off from Launch Complex 39A at NASA's Kennedy Space Center in Florida at 7:30 a.m. EDT Wednesday, Sept. 24, 2025. The missions will each focus on different effects of the solar wind — the continuous stream of particles emitted by the Sun — and space weather — the changing conditions in space driven by the Sun — from their origins at the Sun to their farthest reaches billions of miles away at the edge of our solar system.

SpaceX's TRACERS (Tandem Reconnection and Cusp Electrodynamics Reconnaissance Satellites) mission launches at 11:13 a.m. PDT (2:13 p.m. EDT) on Wednesday, July 23, 2025, atop a SpaceX Falcon 9 rocket at Space Launch Complex 4 East at Vandenberg Space Force Base in California. The TRACERS mission will study magnetic reconnection around Earth — a process in which electrically charged plasmas exchange energy in the atmosphere — to understand how the Sun's solar wind interacts with the magnetosphere, Earth's protective magnetic shield.

SpaceX's Falcon 9 rocket, carrying NASA's SPHEREx (Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer) observatory and PUNCH (Polarimeter to Unify the Corona and Heliosphere) satellites, launches from Space Launch Complex 4 East at Vandenberg Space Force Base in California on Tuesday, March 11, 2025. SPHEREx will use its telescope to provide an all-sky spectral survey, creating a 3D map of the entire sky to help scientists investigate the origins of our universe. PUNCH will study origins of the Sun's outflow of material, or the solar wind, capturing continuous 3D images of the Sun's corona and the solar wind's journey into the solar system.

SpaceX's Falcon Heavy rocket, carrying NASA's Europa Clipper spacecraft, launches from Launch Pad 39A on Sunday, Oct. 13, 2024, at the agency's Kennedy Space Center in Florida ahead of launch to Jupiter's icy moon, Europa. The spacecraft will complete nearly 50 flybys of Europa to determine if there are conditions suitable for life beyond Earth. Launch is targeting 12:06 p.m. EDT on Monday, Oct. 14, from Launch Complex 39A at Kennedy Space Center in Florida.

United Launch Alliance's Atlas V 541 rocket, carrying the National Oceanic and Atmospheric Administration's (NOAA) Geostationary Operational Environmental Satellite-T (GOES-T), lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station in Florida on March 1, 2022. Liftoff was at 4:38 p.m. EST. GOES-T is the third satellite in the GOES-R series that will continue to help meteorologists observe and predict local weather events that affect public safety. GOES-T will be renamed GOES-18 once it

reaches geostationary orbit. GOES-18 will go into operational service as GOES West to provide critical data for the U.S. West Coast, Alaska, Hawaii, Mexico, Central America, and the Pacific Ocean. The launch was managed by NASA's Launch Services Program based at Kennedy Space Center in Florida, America's multi-user spaceport. NASA/Kevin O'Connell and Kevin Davis A Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab's Launch Complex 1 on the Mahia Peninsula of New Zealand. A SpaceX Falcon Heavy rocket with the Psyche spacecraft onboard is seen as it is rolled to the launch pad at Launch Complex 39A as preparations continue for the Psyche mission, Tuesday, Oct. 10, 2023, at NASA's Kennedy Space Center in Florida. NASA's Psyche spacecraft will travel to a metal-rich asteroid by the same name orbiting the Sun between Mars and Jupiter to study its composition. The spacecraft also carries the agency's Deep Space Optical Communications technology demonstration, which will test laser communications beyond the Moon. Photo Credit: (NASA/Aubrey Gemignani) NASA's PACE (Plankton, Aerosol, Cloud, ocean Ecosystem) spacecraft, atop a SpaceX Falcon 9 rocket, successfully lifts off from Space Launch Complex 40 at Cape Canaveral Space Force Station in Florida at 1:33 a.m. EST Thursday, Feb. 8. PACE is NASA's newest earth-observing satellite that will help increase our understanding of Earth's oceans, atmosphere, and climate by delivering hyperspectral observations of microscopic marine organisms called phytoplankton, as well new data on clouds and aerosols. SpaceX

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A Rocket Lab Electron rocket that will carry the Cislunar Autonomous Positioning System Technology Operations and Navigation Experiment (CAPSTONE) spacecraft is seen at Rocket Lab's Launch Complex 1 on the Mahia Peninsula of New Zealand.

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TITLE: Earth — A Photo-Essay

Earth ObservatoryEarth — A Photo-EssayOf all celestial bodies within reach or view, as far as we can see, out to the edge, the most wonderful and marvelous and mysterious is turning out to be our own planet earth. There is nothing to match it anywhere, not yet anyway. —Lewis ThomasA photo-essay from NASA's Earth Science

DivisionDownloadEarthinPDF,MOBI(Kindle), orePubformats.ForewordSixty years ago, with the launch of Explorer 1, NASA made its first observations of Earth from space. Fifty years ago, astronauts left Earth orbit for the first time and looked back at our “blue marble.” All of these years later, as we send spacecraft and point our telescopes past the outer edges of the solar system, as we study our planetary neighbors and our Sun in exquisite detail, there remains much to see and explore at home.We are still just getting to know Earth through the tools of science. For centuries, painters, poets, philosophers, and photographers have sought to teach us something about our home through their art.This book stands at an intersection of science and art. From its origins, NASA has studied our planet in novel ways, using ingenious tools to study physical processes at work—from beneath the crust to the edge of the atmosphere. We look at it in macrocosm and microcosm, from the flow of one mountain stream to the flow of jet streams. Most of all, we look at Earth as a system, examining the cycles and processes—the water cycle, the carbon cycle, ocean circulation, the movement of heat—that interact and influence each other in a complex, dynamic dance across seasons and decades.We measure particles, gases, energy, and fluids moving in, on, and around Earth. And like artists, we study the light—how it bounces, reflects, refracts, and gets absorbed and changed. Understanding the light and the pictures it composes is no small feat, given the rivers of air and gas moving between our satellite eyes and the planet below.For all of the dynamism and detail we can observe from orbit, sometimes it is worth stepping back and simply admiring Earth. It is a beautiful, awe-inspiring place, and it is the only world most of us will ever know.NASA has a unique vantage point for observing the beauty and wonder of Earth and for making sense of it. Looking back from space, astronaut Edgar Mitchell once called Earth “a sparkling blue and white jewel,” and it does dazzle the eye. The planet's palette of colors and textures and shapes—far more than just blues and whites—are spread across the pages of this book.We chose these images because they inspire. They tell a story of a 4.5-billion-year-old planet

where there is always something new to see. They tell a story of land, wind, water, ice, and air as they can only be viewed from above. They show us that no matter what the human mind can imagine, no matter what the artist can conceive, there are few things more fantastic and inspiring than the world as it already is. The truth of our planet is just as compelling as any fiction. We hope you enjoy this satellite view of Earth. It is your planet. It is NASA's mission.

Michael Carlowicz Earth Observatory Managing Editor

atmosphere The astonishing thing about the Earth... is that it is alive.... Aloft, floating free beneath the moist, gleaming membrane of bright blue sky, is the rising Earth, the only exuberant thing in this part of the cosmos.... It has the organized, self-contained look of a live creature, full of information, marvelously skilled in handling the Sun.— Lewis Thomas, *The Lives of a Cell*

Explore water We shall not cease from exploration, and the end of all our exploring will be to arrive where we started and know the place for the first time.—T.S. Eliot, “Little Gidding”

Explore land Earth and sky, woods and fields, lakes and rivers, the mountain and the sea, are excellent schoolmasters, and teach some of us more than we can ever learn from books.—John Lubbock, *The Use of Life*

Explore ice and snow It seems to me that the natural world is the greatest source of excitement; the greatest source of visual beauty; the greatest source of intellectual interest. It is the greatest source of so much in life that makes life worth living.—David Attenborough

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Imagery and data courtesy of: NASA Earth Observatory U.S. Geological Survey (USGS) and NASA Landsat Program International Space Station (ISS) Crew Earth Observations Facility LANCE/EOSDIS MODIS Rapid Response Team MABEL Science Team Level-1 and Atmosphere Archive & Distribution System Distributed Active Archive Center (LAADS DAAC) EO-1 Science Team Suomi National Polar-orbiting Partnership (Suomi NPP) NASA Ocean Biology Processing Group NASA/METI/ERSDAC/JAROS/Japan ASTER Science Team

Adapted for the web by Paul Przyborski

About the Authors Michael Carlowicz is managing editor of the NASA Earth Observatory. He has written about Earth science and geophysics since 1991 for several NASA divisions, the American Geophysical Union, the Woods Hole Oceanographic Institution, and in three popular science books. He is a baseball player and fan, a longtime singer and guitarist, and the proud father of three science and engineering majors.

Lawrence Friedl directs the Applied Sciences Program in the Earth Science Division of NASA's Science Mission Directorate. He works to enable innovative and practical uses of data from Earth-observing satellites. He has worked at the U.S. Environmental Protection Agency and as a Space Shuttle flight controller in NASA's Mission Control Center. He and his wife have three children, and he enjoys ultimate frisbee and hiking.

Kevin Ward manages NASA's Earth Observatory Group, including the Earth Observatory, Visible Earth, NASA Earth Observations (NEO), and EONET. He holds a master's degree in library and information science and has spent more than 20 years developing Web-accessible resources in support of NASA Earth science communications. He and his wife have a son

and a deep love of music. Kathy Carroll supports the Earth Science Division in the Science Mission Directorate at NASA Headquarters. She previously worked as a manager and organizer at for-profit and non-profit organizations and on political campaigns. She is a diehard baseball and hockey fan, and she volunteers with animal rescue organizations. Stephen Schaeberle is a graphic designer with the Communications Support Services Center at NASA Headquarters. He holds a bachelor of fine arts from the Pratt Institute, and he has received numerous awards and honors for his work and designs. He enjoys boating and fishing on the Chesapeake Bay.

Acknowledgments Just a few names end up on the title page of a book, but it takes an entire cast of people to bring it from idea to draft to finished product. The cast for Earth begins with Maxine Aldred, Andrew Cooke, Tun Hla, and Lisa Jirousek, who shepherded the words and images through design and layout. Thanks are also due to Kathryn Hansen, Pola Lem, Rebecca Lindsey, Holli Riebeek, Michon Scott, and Adam Voiland, whose reporting and writing contributions gave this book its depth. Joshua Stevens, Robert Simmon, Jesse Allen, Jeff Schmaltz, Michael Taylor, and Norman Kuring applied their strong visual sense and processing skills to make each image pop with color and texture while remaining scientifically accurate. We owe a debt to our scientific and outreach colleagues, who keep the satellites running, the sensors sensing, and the data and imagery flowing. Every one of the images in this book is publicly available through the Internet, truly making science accessible to every citizen. The Landsat teams at the U.S. Geological Survey and NASA, the LANCE/EOSDIS MODIS Rapid Response Team, and the NASA Earth Observatory deserve extra gratitude for making our planet visible to the scientist and the layman every day.

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Earth — A Photo-Essay Of all celestial bodies within reach or view, as far as we can see, out to the edge, the most wonderful and marvelous and mysterious is turning out to be our own planet earth. There is nothing to match it anywhere, not yet anyway. —Lewis Thomas

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Foreword

Sixty years ago, with the launch of Explorer 1, NASA made its first observations of Earth from space. Fifty years ago, astronauts left Earth orbit for the first time and looked back at our “blue marble.” All of these years later, as we send spacecraft and point our telescopes past the outer edges of the solar system, as we study our planetary neighbors and our Sun in exquisite detail, there remains much to see and explore at home. We are still just getting to know Earth through the tools of science. For centuries, painters, poets, philosophers, and photographers have sought to teach us something about our home through their art. This book stands at an intersection of science and art. From its origins, NASA has studied our planet in novel ways, using ingenious tools to study physical processes at work—from beneath the crust to the edge of the atmosphere. We look at it in macrocosm and microcosm, from the flow of one mountain stream to the flow of jet streams. Most of all, we look at Earth as a system, examining the cycles and processes—the water cycle, the carbon cycle, ocean circulation, the movement of heat—that interact and influence each other in a complex, dynamic dance across seasons and decades.

We measure particles, gases, energy, and fluids moving in, on, and around Earth. And like artists, we study the light—how it bounces, reflects, refracts, and gets absorbed and changed. Understanding the light and the pictures it composes is no small feat, given the rivers of air and gas moving between our satellite eyes and the planet below. For all of the dynamism and detail we can observe from orbit, sometimes it is worth stepping back and simply admiring Earth. It is a beautiful, awe-inspiring place, and it is the only world most of us will ever know. NASA has a unique vantage point for observing the beauty and wonder of Earth and for making sense of it. Looking back from space, astronaut Edgar Mitchell once called Earth “a sparkling blue and white jewel,” and it does dazzle the eye. The planet’s palette of colors and textures and shapes—far more than just blues and whites—are spread across the pages of this book.

We chose these images because they inspire. They tell a story of a 4.5-billion-year-old planet where there is always something new to see. They tell a story of land, wind, water, ice, and air as they can only be viewed from above. They show us that no matter what the human mind can imagine, no matter what the artist can conceive, there are few things more fantastic and inspiring than the world as it already is. The truth of our planet is just as compelling as any fiction. We hope you enjoy this satellite view of Earth. It is your planet. It is NASA’s mission. Michael Carlowicz Earth Observatory Managing Editor

atmosphereThe astonishing thing about the Earth... is that it is alive.... Aloft, floating free beneath the moist, gleaming membrane of bright blue sky, is the rising Earth, the only exuberant thing in this part of the cosmos.... It has the organized, self-contained look of a live creature, full of information, marvelously skilled in handling the Sun.— Lewis

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TITLE: Earth Science at Work

Earth (ESD)Earth Science at WorkNASA Earth Sciencehelps Americans respond to challenges and societal needs— such as wildland fires, hurricanes, and water supplies for farming — by putting actionable satellite information into the hands of decisionmakers.

NASA conceives, builds, and flies cutting-edge satellites and instruments; harvests data from those tools and from commercial partners; combines it with expert insights and the world's most advanced models — all to advance solutions for monitoring, prediction, and management.

Why it matters Using the power of space-based observations, innovative technologies, engineering, science, and partnerships, NASA can offer a comprehensive, digital view of Earth as a system. NASA works across commercial and industrial sectors and federal and international agencies to:

- generate the scientific data necessary to inform decisions from the local to state to federal to international level
- leverage innovation from academia and the private sector to develop novel applications of Earth science information
- create products and analyses that directly address societal needs
- link technology, research & development, and actionable science to ensure rapid transfer from innovation to impact.

advance America's space exploration goals, using our Earth tools and insights to better prepare us for a sustainable presence on the Moon and Mars.

Featured Video NASA Earth: Powering America's Future Did you know the very same cutting-edge technology NASA uses to study other planets is exactly what we use to study our own?

From the ground, air, and skies, NASA gathers information crucial to life here on Earth, and we build instruments and technologies and programs that equip the US to meet critical, real-world challenges.

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By 2030, the potential economic value of EO (Earth observations) could exceed \$700 billion globally...better data can inform decision-making, optimize solutions and enable innovation.

Amplifying the Global Value of Earth Observation World Economic Forum report

What We Do NASA's research and data provide a competitive advantage to American businesses and support the U.S. economy.

Mapping rare Earth minerals for use in domestic, technology-dominant industries such as telecommunications, energy generation, and aerospace.

Aiding U.S. agriculture with continuous measurements of water resources, crop health, and global production that inform business decisions, improve farm and ranch efficiency, and reduce input costs, resulting in a positive impact from the commodity-trading floor to the grocery store.

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NASA, BLM, USGS/Yolanda Fong-Sam NASA Earth supports national security with valuable information and tools.

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Satellite observations at night indicate that oil, gas, and mining have expanded across Earth's northern latitudes.

Data is the key. If we can measure

it, we can quantify it, we can solve it. Dwane Roth Farmer and OpenET Data User, Finney County, Kansas NASA is improving the nation's ability to forecast and respond to severe storms, droughts, wildland fires, and other natural hazards. Offering near real-time data products, early warning tools, and disaster mapping portals. Developing an advanced landslide forecast model. Improving our understanding of hurricanes and sharing those insights and tools with NOAA's National Hurricane Center and the Joint Typhoon Warning Center. Working with the U.S. Forest Service and other agencies to provide key observations of conditions before, during, and after wildland fires- such as estimating fuel loads and soil moisture. A selection of four images showing a volcano erupting, flooded streets in Vermont, a hurricane seen from space, and fires burning a hillside. NASA Earth Science Division Our data supports decision-making that affects human health and quality of life. Satellite measurements of temperature, humidity, and precipitation help predict outbreaks of mosquito-borne illnesses. Satellite observations and models help assess freshwater resources from the surface through the root zone to the deep aquifer. Observations of the amount of freshwater stored on land and underground (left) contribute to our understanding of how water availability puts a strain on farmers and communities, potentially leading to famine, conflicts, poverty, and an increased risk of disease, including those transmitted by mosquitoes (right). NASA Earth Observatory, CDC NASA's consistent sea level rise data is the best way and the best tool for us to make sure that we spend our money carefully and wisely so that the future generations get to enjoy our beaches and bays and rivers just as much as I did. Casi Callaway Chief Resilience Officer, City of Mobile, Alabama NASA is uniquely positioned to take scientific and societal questions from conception to public utility. We develop and fund technological leaps, build and launch the satellites and instruments, gather and disseminate the data, analyze the findings, and work with commercial, industry, academic, and government partners to serve societal needs. NASA Earth Science Division Studying Earth from Space Since the 1960s, we have been observing our own planet because it is the one we need to know best. NASA applies ingenuity and expertise gained from decades of planetary and deep-space exploration to the study of our home planet. From Headquarters and multiple NASA centers and labs, the Earth Science Division operates more than 20 satellites in orbit, sponsors hundreds of research programs and studies, and funds opportunities to put data to use for societal needs. We develop new ways to observe the oceans, land, ice, atmosphere, and life, and we measure how changes in one drive changes in others over the short and long term. While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world. NASA Research and Data in Action Article 5 min read OpenET Study Helps Water Managers and Farmers Put NASA Data to Work Article 4 min read NASA and Forest Service

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NASA's consistent sea level rise data is the best way and the best tool for us to make sure that we spend our money carefully and wisely so that the future generations get to enjoy our beaches and bays and rivers just as much as I did. Casi Callaway Chief Resilience Officer, City of Mobile, Alabama

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Studying Earth from Space

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We develop new ways to observe the oceans, land, ice, atmosphere, and life, and we measure how changes in one drive changes in others over the short and long term.

While listening to and collaborating with industry leaders, international partners, academic institutions, and other users of our data, we deepen knowledge of our planet, drive innovations, and deliver science to help inform decisions that benefit the nation and the world.

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Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration) This illustration represents half the disk of gas and dust surrounding the protostar EC 53. Stellar outbursts periodically form crystalline silicates, which are launched up and out to the edges of the system, where comets and other icy rocky bodies may eventually form. **Illustration: NASA, ESA, CSA, Elizabeth Wheatley (STScI)**

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Goddard Space Flight Center
Jan 21, 2026
Article Contents
Image: Protostar EC 53 in the Serpens Nebula (NIRCam Image)
Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration)
Look ahead
Downloads & Related Information
Related Links

Contents
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Image: Silicate Crystallization and Movement Near Protostar EC 53 (Illustration)
Look ahead
Downloads & Related Information
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Look ahead

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Why go to space? Human space exploration addresses fundamental questions about our place in the universe and the history of our solar system. NASA's exploration vision is anchored in providing value for humanity by answering some of the most fundamental questions: Why are we here? How did it all begin? Are we alone? What comes next? [Learn More about Why go to space?](#)

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URL: <https://www.nasa.gov/mission/expedition-72/>

TITLE: Expedition 72

ISSExpedition 72Mission TypeHumans in Space
meet the crew
Sunni Williams, Butch Wilmore, Nick Hague, Don Pettit, Alexey Ovchinin, Ivan Vagner, Aleksandr Gorbunov
startSep. 23, 2024LandingApril 19, 2025
The official portrait of the International Space Station's Expedition 72 crew. At the top (from left) are, Roscosmos cosmonaut and

Flight Engineer Alexey Ovchinin, NASA astronaut and space station Commander Suni Williams, and NASA astronaut and Flight Engineer Butch Wilmore. In the middle row are, Roscosmos cosmonaut and Flight Engineer Ivan Vagner and NASA astronaut and Flight Engineer Don Pettit. In the bottom row are, Roscosmos cosmonaut and Flight Engineer Aleksandr Gorbunov and NASA astronaut and Flight Engineer Nick Hague.

NASA/Bill Stafford/Robert Markowitz

Crew Members Suni Williams, Commander Butch Wilmore, Flight Engineer Nick Hague, Flight Engineer Don Pettit, Flight Engineer Alexey Ovchinin, Flight Engineer Ivan Vagner, Flight Engineer Aleksandr Gorbunov, Flight Engineer Crew and Cargo

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Spacewalks Date: Jan. 30, 2025 Duration: 5 hours, 26 minutes

Spacewalkers: Suni Williams, Butch Wilmore

Date: Jan. 16, 2025 Duration: 6 hours

Spacewalkers: Nick Hague, Suni Williams

Date: Dec. 19, 2024 Duration: 7 hours, 17 minutes

Spacewalkers: Alexey Ovchinin, Ivan Vagner

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SpaceX Crew-9 Science The SpaceX Crew-9 members will investigate blood cell development in space, X-rays emitted by neutron stars, taking a vitamin supplement to improve vision, space agriculture, and more.

SpaceX Crew-10 Science The SpaceX Crew-10 members are studying how microgravity affects production of protein by microalgae, testing a microscope to capture microbial activity, and studying genetic activity in biofilms.

Mission Imagery The Expedition 72 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 72 gallery.

Space Station Imagery View imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.

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Crew Members

Suni Williams, Commander Butch Wilmore, Flight Engineer Nick Hague, Flight Engineer Don Pettit, Flight Engineer Alexey Ovchinin, Flight Engineer Ivan Vagner, Flight Engineer Aleksandr Gorbunov, Flight Engineer

Crew and Cargo Missions

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SpaceX Crew-10 ScienceThe SpaceX Crew-10 members are studying how microgravity affects production of protein by microalgae, testing a microscope to capture microbial activity, and studying genetic activity in biofilms.

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Mission ImageryThe Expedition 72 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 72 gallery.

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Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.

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URL: <https://www.nasa.gov/mission/expedition-73/>

TITLE: Expedition 73

ISSExpedition 73Mission TypeHumans in Spacemeet the crewAlexey Zubritsky, Sergey Ryzhikov, Jonny Kim, Kimiya Yui, Oleg Platonov, Zena Cardman, Mike FinckestartApril 19, 2025LandingNov. 8, 2025Crew MembersAlexey Zubritsky, Flight EngineerSergey Ryzhikov, CommanderJonny Kim, Flight EngineerKimiya Yui, Flight EngineerOleg Platonov, Flight EngineerZena Cardman, Flight EngineerMike Fincke, Flight EngineerCrew and Cargo Missions12/8/2025 – Soyuz MS-27 Undock12/1/2025 – Cygnus NG-23

Reinstallation11/27/2025 – Soyuz MS-28 Dock11/27/2025 – Soyuz MS-28
 Launch11/24/2025 – Cygnus NG-23 Uninstallation10/29/2025 – HTV-X1
 Capture10/25/2025 – HTV-X1 Launch9/18/2025 – Cygnus NG-23
 Capture/Installation9/14/2025 – Cygnus NG-23 Launch9/13/2025 – Progress 93
 Dock9/11/2025 – Progress 93 Launch9/9/2025 – Progress 91 Undock8/25/2025 – SpaceX
 CRS-33 Dock8/24/2025 – SpaceX CRS-33 Launch8/9/2025 – SpaceX Crew-10
 Splashdown8/8/2025 – SpaceX Crew-10 Undock8/2/2025 – SpaceX Crew-11 Dock8/1/2025
 – SpaceX Crew-11 Launch7/15/2025 – Axiom Mission 4 Splashdown7/14/2025 – Axiom
 Mission 4 Undock7/5/2025 – Progress 92 Dock7/3/2025 – Progress 92 Launch7/1/2025 –
 Progress 90 Undock6/26/2025 – Axiom Mission 4 Dock6/25/2025 – Axiom Mission 4
 Launch5/23/2025 – SpaceX CRS-32 Undock4/22/2025 – SpaceX CRS-32 Dock4/21/2025 –
 SpaceX CRS-32 Launch4/19/2025 – Soyuz MS-26 Land4/19/2025 – Soyuz MS-26
 UndockSpacewalksDate:Oct. 28, 2025Duration:6 hours, 54 minutesSpacewalkers:Sergey
 Ryzhikov, Alexey ZubritskyDate:Oct. 16, 2025Duration:6 hours, 9
 minutesSpacewalkers:Sergey Ryzhikov, Alexey ZubritskyDate:May 1, 2025Duration:6 hours,
 35 minutesSpacewalkers:Anne McClain,Nichole AyersExpedition 73 Image Gallery727
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Alexey Zubritsky, Sergey Ryzhikov, Jonny Kim, Kimiya Yui, Oleg Platonov, Zena Cardman, Mike Finck

start April 19, 2025 Landing Nov. 8, 2025

start April 19, 2025

Landing Nov. 8, 2025

Nov. 8, 2025

Alexey Zubritsky, Flight Engineer Sergey Ryzhikov, Commander Jonny Kim, Flight Engineer Kimiya Yui, Flight Engineer Oleg Platonov, Flight Engineer Zena Cardman, Flight Engineer Mike Finck, Flight Engineer

12/8/2025 – Soyuz MS-27 Undock 12/1/2025 – Cygnus NG-23 Reinstallation 11/27/2025 – Soyuz MS-28 Dock 11/27/2025 – Soyuz MS-28 Launch 11/24/2025 – Cygnus NG-23 Uninstallation 10/29/2025 – HTV-X1 Capture 10/25/2025 – HTV-X1 Launch 9/18/2025 – Cygnus NG-23 Capture/Installation 9/14/2025 – Cygnus NG-23 Launch 9/13/2025 – Progress 93 Dock 9/11/2025 – Progress 93 Launch 9/9/2025 – Progress 91 Undock 8/25/2025 – SpaceX CRS-33 Dock 8/24/2025 – SpaceX CRS-33 Launch 8/9/2025 – SpaceX Crew-10 Splashdown 8/8/2025 – SpaceX Crew-10 Undock 8/2/2025 – SpaceX Crew-11 Dock 8/1/2025 – SpaceX Crew-11 Launch 7/15/2025 – Axiom Mission 4 Splashdown 7/14/2025 – Axiom Mission 4 Undock 7/5/2025 – Progress 92 Dock 7/3/2025 – Progress 92 Launch 7/1/2025 – Progress 90 Undock 6/26/2025 – Axiom Mission 4 Dock 6/25/2025 – Axiom Mission 4 Launch 5/23/2025 – SpaceX CRS-32 Undock 4/22/2025 – SpaceX CRS-32 Dock 4/21/2025 – SpaceX CRS-32 Launch 4/19/2025 – Soyuz MS-26 Land 4/19/2025 – Soyuz MS-26 Undock

Date: Oct. 28, 2025 Duration: 6 hours, 54 minutes Spacewalkers: Sergey Ryzhikov, Alexey Zubritsky Date: Oct. 16, 2025 Duration: 6 hours, 9 minutes Spacewalkers: Sergey Ryzhikov,

Alexey Zubritsky Date: May 1, 2025 Duration: 6 hours, 35 minutes Spacewalkers: Anne McClain, Nichole Ayers

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URL: <https://www.nasa.gov/mission/expedition-70/>

TITLE: Expedition 70

ISS Expedition 70 Expedition 70 researched heart health, cancer treatments, space manufacturing techniques, and more during their long-duration stay in Earth orbit.

Mission Type ISS Expedition

Meet the crew Andreas Mogensen, Jasmin Moghbeli, Satoshi Furukawa, Loral O'Hara, Konstantin Borisov, Oleg Kononenko, Nikolai Chub

start Sept. 27, 2023

END April 5, 2024

iss 070-s-002 (March 6, 2023) — The official Expedition 70 crew portrait with (top row from left) Roscosmos cosmonauts Nikolai Chub, Konstantin Borisov, and Oleg Kononenko; JAXA (Japan Aerospace Exploration Agency) astronaut Satoshi Furukawa; and NASA astronaut Loral O'Hara. In the front row are, ESA (European Space Agency) astronaut and Expedition 70 Commander Andreas Mogensen and NASA astronaut Jasmin Moghbeli.

NASA The official patch of the International Space Station's Expedition 70 crew. The Expedition 70 patch is designed around the central yin-yang symbol representing balance; first and foremost, the balance of our beautiful planet Earth that is encircled by the yin-yang symbol and which forms part of the Expedition number. In our exploration of space, we are reminded of the uniqueness of Earth; the further we push the boundaries of human existence, the stronger our longing for our home planet grows. As our understanding of the cosmos expands, so does our understanding of Earth, and although we live in an ever-changing world, we recognize the need for a planet in balance to ensure our future. Space exploration is also about creating the future of our dreams. The tentative first steps we take today will hopefully become a well-trodden path in the future. This is represented stylistically by the "retro-futuristic" design of the patch, which mimics the design of the posters depicting the future from the early days of the space age. It is also emphasized by the yellow, orange, and red colors suggesting a sunrise. Lastly, the dynamism in the depiction of the number 7 suggests not only the physical launch into space, but also humanity's progress towards the future.

Crew Members Andreas Mogensen, Commander Jasmin Moghbeli, Flight Engineer Satoshi Furukawa, Flight Engineer Loral O'Hara, Flight Engineer Konstantin Borisov, Flight Engineer Oleg Kononenko, Flight Engineer Nikolai Chub, Flight Engineer

Crew and Cargo Missions

4/6/2024	- Soyuz MS-24 Land
4/5/2024	- Soyuz MS-24 Undock
3/25/24	- Soyuz MS-25 Dock
3/23/24	- Soyuz MS-25 Launch
3/23/24	- SpaceX CRS-30 Dock
3/21/2024	- SpaceX CRS-30 Launch
3/12/2024	- SpaceX Crew-7 Splashdown
3/11/2024	- SpaceX Crew-7 Undock
3/5/2024	- SpaceX Crew-8 Dock
3/3/2024	- SpaceX Crew-8 Launch
2/17/24	- ISS Progress 87 Dock
2/14/24	- ISS Progress 87 Launch
2/7/2024	- Axiom Mission 3 Undock
2/1/2024	- Cygnus CRS-20 Capture
1/30/2024	- Cygnus CRS-20 Launch
1/20/2024	- Axiom Mission 3 Dock
1/18/2024	- Axiom Mission 3 Launch
12/21/2023	- SpaceX CRS-29 Undock
12/1/23	- ISS Progress 86 Launch/Dock
11/11/23	- SpaceX CRS-29 Dock
11/9/23	- SpaceX CRS-29 Launch
9/27/23	- Soyuz MS-23 Undock/Land
9/15/23	- Soyuz MS-24 Launch/Dock
8/27/23	- SpaceX Crew-7 Dock
8/26/23	- SpaceX Crew-7 Launch

Spacewalks

Date:	Nov. 1, 2023	Duration:	6 hours, 42 minutes
Spacewalkers:	Jasmin Moghbeli, Loral O'Hara	Date:	Oct. 25, 2023
		Duration:	7 hours,

41 minutesSpacewalkers:Oleg Kononenko, Nikolai ChubExpedition 70 Image Gallery493 ImagesGo To GalleryGo To GalleryExpedition 70 ResourcesMission BlogsThe daily blog updates for Expedition 70 highlighting crew and cargo missions, spacewalks, and advanced space research.Mission ScienceLearn more about the range of microgravity science encompassing human research, space physics, and more the Expedition 70 crew has conducted.SpaceX Crew-7 ScienceThe SpaceX Crew-7 members compared Earth and space sleep patterns, monitored numerous astronaut health parameters, examined disinfectants to clean spacecraft surfaces, and more.SpaceX Crew-8 ScienceThe SpaceX Crew-8 members explored the causes of neurodegenerative disorders, protecting plants from space radiation, space-caused changes in eye structure and vision, and more.Mission ImageryThe Expedition 70 crew is taking images of the Earth, the orbiting station, their daily lives aboard, and the experiments they conduct during their stay, plus so much more. Check out some of the images in the Expedition 70 gallery.Space Station ImageryView imagery taken from the International Space Station since 1999 of daily crew activities, spacewalks, visiting vehicles, and the Earth below.Expedition 70 NewsMore Expedition 70 NewsBlog1 Min ReadNASA, SpaceX Adjust Crew-8 Undocking DateArticle3 Min ReadWelcome Back to Planet Earth, Expedition 70 Crew!Blog2 Min ReadSoyuz Lands Returning O'Hara, Two Crewmates Back to EarthBlog1 Min ReadLoral O'Hara, Two Crewmates Returning to Earth on NASA TV2 Min ReadSoyuz Spacecraft Undocks to Return Three Crewmates to EarthBlog2 Min ReadThree Crew Members Departing Station Live on NASA TVBlog2 Min ReadTrio Boards Soyuz Crew Ship and Closes HatchBlog3 Min ReadTrio Finalizes Packing, Science Activities Before Friday DepartureBlogKeep ExploringKeep ExploringCommercial Crew Program OverviewSpace Station Research and TechnologyStation Benefits for HumanityHumans In Space

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Undock2/1/2024 – Cygnus CRS-20 Capture1/30/2024 – Cygnus CRS-20 Launch1/20/2024
– Axiom Mission 3 Dock1/18/2024 – Axiom Mission 3 Launch12/21/2023 – SpaceX CRS-29
Undock12/1/23 – ISS Progress 86 Launch/Dock11/11/23 – SpaceX CRS-29 Dock11/9/23 –
SpaceX CRS-29 Launch9/27/23 – Soyuz MS-23 Undock/Land9/15/23 – Soyuz MS-24
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ChamberElectromagnetic SpectrumELFIN (Electron Losses & Fields Investigation)Ellen
OchoaEllen S. BakerEllington FieldElliot M. See Jr.Elliptical GalaxiesEllison S.
OnizukaEmergency & Pandemic ResponseEMIT (Earth Surface Mineral Dust Source
Investigation)EnceladusEnvironmental Management Division (EMD)Environmental Science
Services Administration (ESSA)EnVisionEquator-SER-2ERBS (Earth Radiation Budget
Satellite)Eric A. BoeErisErosEscaPADE (Escape and Plasma Acceleration and Dynamics
Explorers)EspañolEstablished Program to Stimulate Competitive Research (EPSCoR)Eta
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Exploration ProgramExoplanet ScienceExoplanet Travel
BureauExoplanetsExosphereExpedition 1Expedition 2Expedition 3Expedition 4Expedition
5Expedition 6Expedition 7Expedition 8Expedition 9Expedition 10Expedition 11Expedition
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30Expedition 31Expedition 32Expedition 33Expedition 34Expedition 35Expedition
36Expedition 37Expedition 38Expedition 39Expedition 40Expedition 41Expedition

42Expedition 43Expedition 44Expedition 45Expedition 46Expedition 47Expedition
48Expedition 49Expedition 50Expedition 51Expedition 52Expedition 53Expedition
54Expedition 55Expedition 56Expedition 57Expedition 58Expedition 59Expedition
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66Expedition 67Expedition 68Expedition 69Expedition 70ExpeditionsExploration
Capabilities DivisionEFT 1 (Exploration Flight Test 1)Exploration Ground
SystemsExploration Operations DivisionExploration Systems Development Mission
DirectorateExplorerExplorer 33Explorer 35Explorer 49Explorers ProgramExtraterrestrial
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Weather EventsEZIE (Electrojet Zeeman Imaging Explorer)FF / A-18F. Drew GaffneyF-15BF-
15DF-18Faces of NASAFAST (Fast Auroral SnapshoT Explorer)FASTSat (Fast, Affordable,
Science and Technology Satellite)FBCE (Flow Boiling and Condensation
Experiment)Featured CareersFermi Gamma-Ray Space TelescopeFernando CaldeiroFilm &
Documentary GuidelinesFind Your PlaceFirst Woman Graphic NovelFission Surface Power
(FSP)FLARE (Flammability Limits At Reduced-g Experiment)Flight Demos CapabilitiesFlight
InnovationFlight Opportunities ProgramFlight ProgramFlight ProvidersFlight
SimulationFluid PhysicsFor Colleges & UniversitiesFor EducatorsFor Kids and StudentsFor
ProfessionalsFormer AstronautsFrancis R. ScobeeFranco MalerbaFrank BormanFrank L.
Culbertson Jr.Frank RubioFranklin R. Chang-DíazFred W. Haise Jr.Fred W. LeslieFrederick D.
GregoryFrederick H. HauckFrederick W. SturckowFruit Fly Lab (FFL)Fundamental
PhysicsFUSE (Far Ultraviolet Spectroscopic Explorer)FutureFlight CentralGG. David LowG.
Reid WisemanGaiaGalaxiesGalaxies, Stars, & Black Holes ResearchGalaxy of
HorrorsGALEX (Galaxy Evolution Explorer)GalileoGalileo Jupiter Atmospheric ProbeGame
Changing Development ProgramGamma RaysGamma-Ray BurstsGanymedeGarrett E.
ReismanGary E. PaytonGas Giant ExoplanetsGas GiantsGateway ProgramGateway Space
StationGDC (Geospace Dynamics Constellation)GEDI (Global Ecosystem Dynamics
Investigation)GeminiGemini IGemini IIGemini IIIGemini IVGemini VGemini VIGemini
VIIGemini VIIIGemini IXGemini XGemini XIIGemini XIIGeminidsGenesisGRIP (Genesis and
Rapid Intensification Processes)Gente de la NASAGeneral George D. NelsonGeorge D.
ZamkaGeosat (U.S. Navy GEOdetic SATellite)Geospace Dynamics Constellation
(GDC)GeotailGerald CarrGet InvolvedGet SocialGII-STAGiottoGlaciersGlenn Research
CenterGLIMR (Geosynchronous Littoral Imaging and Monitoring Radiometer)Global
HawkGloryGoddard Institute for Space StudiesGoddard Space Flight CenterGOES
(Geostationary Operational Environmental Satellite)GOES-NGOES-RGOLD (Global-scale
Observations of the Limb and Disk)GPIM (Green Propellant Infusion Mission)GPM (Global
Precipitation Global Precipitation Measurement)GRACE (Gravity Recovery And Climate
Experiment)GRACE-FO (Gravity Recovery and Climate Experiment Follow-on)Grades 5 –
8Grades 5 – 8 for EducatorsGrades 9 – 12Grades 9-12 for EducatorsGrades K – 4Grades K –

4 for Educators
 GRAIL (Gravity Recovery And Interior Laboratory)
 Grants & Opportunities
 Gravity Probe B (GP-B)
 Green Aviation Tech
 Greenhouse Effect
 Greenhouse Gasses
 Gregory C. Johnson
 Gregory E. Chamitoff
 Gregory H. Johnson
 Gregory J. Harbaugh
 Gregory Jarvis
 Gregory T. Linteris
 Griffin (lander)
 Ground Facilities for ISS
 Guion S. Bluford Jr.
 Gulfstream C-20A
 Gulfstream G-III
 Gulfstream G-V
 Gulfstream IV
 Guy S. Gardner
 HH-135
 HALCA (Highly Advanced Laboratory for Communications and Astronomy)
 Hale-Bopp
 Halley's Comet
 HaloSat
 Harrison H. Schmitt
 Haumea
 Hayabusa
 Hayabusa 2
 Heidemarie M. Stefanyshyn-Piper
 Heliophysics
 Heliophysics Division
 Heliophysics Research Program
 Helios 1
 Helios 2
 Heliosphere
 HelioSwarm
 Helix Nebula
 Henry W. Hartsfield Jr.
 HERA (Human Exploration Research Analog)
 HERMES (Heliophysics Environmental and Radiation Measurement Experiment Suite)
 Herschel Space Observatory
 HETE-1 (High Energy Transient Explorer 1)
 HETE-2 (High Energy Transient Explorer 2)
 High-Tech Computing
 Hinode (Solar B)
 Hipparcos
 Hi-Rate Composite Aircraft Manufacturing
 Hispanic Heritage Month
 History: NASA
 Hiten / Hagomoro
 Hitomi (ASTRO-H)
 Horsehead Nebula
 HS3 Hurricane Mission (Hurricane and Severe Storm Sentinel)
 HU-25
 Hubble Space Telescope
 Human Dimensions
 Human Exploration of Space: Why We Go
 Human Health and Performance
 Human Landing System Program
 Human Research Program
 Human Space Travel Research
 Human Spaceflight Capabilities Division
 Humans in Space
 Hurricanes
 Hybrid Thermally Efficient Core
 Hyperion
 Hypersonic Technology
 HypsIRI (Hyperspectral Infrared Imager)
 IIBEX (Interstellar Boundary Explorer)
 Ice Giants
 IceBridge
 ICESat (Ice, Clouds and Land Elevation Satellite)
 ICESat-2 (Ice, Cloud and land Elevation Satellite-2)
 ICON (Ionospheric Connection Explorer)
 IdaleH-3 (International Extreme Ultraviolet Hitchhiker)
 Ilan Ramon
 IMAGE (Imager for Magnetopause-to-Aurora Global Exploration)
 Images
 Images & Media Guidelines
 IMAP (Interstellar Mapping and Acceleration Probe)
 IMP-8 (Interplanetary Monitoring Platform 8)
 In-flight Education Downlinks
 INCUS (Investigation of Convective Updrafts)
 Infrared Space Observatory (ISO)
 Ingenuity (Helicopter)
 InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport)
 Institutional Funding
 Integrated Aviation Systems Program
 Interactives
 International Gamma-ray Astrophysics Laboratory (INTEGRAL)
 International Space Station (ISS)
 International Space Station Division
 Internships
 ISRU (In-Situ Resource Utilization)
 Io
 Ionosphere
 IRIS (Interface Region Imaging Spectrograph)
 Irregular Galaxies
 ISEE-3/ICE (International Sun-Earth Explorer-3 / International Cometary Explorer)
 ISERV (ISS SERVIR Environmental Research and Visualization System)
 ISS-RapidScat (Rapid Scatterometer)
 ISS Research
 IUE (International Ultraviolet Explorer)
 IXPE (Imaging X-ray Polarimetry Explorer)
 Jack D. Fischer
 Jack Hathaway
 Jack R. Lousma
 Jake Garn
 James A. Lovell Jr.
 James A. McDivitt
 James A. Pawelczyk
 James B. Irwin
 James C. Adamson
 James D. A. Van Hoften
 James D. Halsell

Jr. James D. Wetherbee James F. Buchli James F. Reilly James H. Newman James M. Kelly James P. Bagian James P. Dutton James S. Voss James Webb Space Telescope (JWST) Janet L. Kavand Janice E. Voss Janus Jasmin Moghbeli Jason-1 Jason-3 Jason-CS (Continuity of Service) / Sentinel-6 Jay Clark Buckey Jeanette J. Epps Jean-Jacques Favier Jeffrey A. Hoffman Jeffrey N. Williams Jeffrey S. Ashby Jerome Apt Jerry L. Ross Jerry M. Linenger Jessica U. Meir Jessica Watkins Jessica Wittner Jet Propulsion Laboratory Joan E. Higginbotham Joe F. Edwards Jr. Joe H. Engle John A. Llewellyn John B. Herrington John D. Olivas John E. Blaha John H. Casper John H. Glenn Jr. John L. Phillips John L. Swigert Jr. John M. Fabian John M. Grunsfeld John M. Lounge John O. Creighton John S. Bull John W. Young Johnson Flight Operations Johnson Space Center Johnson's Mission Control Center Joint Agency Satellite Division Jon A. McBride Jonny Kim José M. Hernández Joseph M. Acaba Joseph P. Allen Joseph P. Kerwin Joseph R. Tanner Josh A. Cassada JPSS (Joint Polar Satellite System) JPSS-3 JPSS-4 Judith A. Resnik JUICE (Jupiter Icy Moons Explorer) Juno Jupiter Jupiter Moons JWST (James Webb Space Telescope) KK. Megan McArthur Kaguya Kalpana Chawla Karen L. Nyberg Karl G. Henize Karol J. Bobko Katherine Johnson IV & V Facility Kathleen Rubins Kathryn C. Thornton Kathryn D. Sullivan Kathryn P. Hire Kayla Barron Keck Interferometer (KI) Kennedy Space Center Kenneth D. Bowersox Kenneth D. Cameron Kenneth D. Cockrell Kenneth S. Reightler Jr. Kenneth T. Ham Kent V. Rominger Kepler / K2 Kepler 22B Kevin A. Ford Kevin P. Chilton Kevin R. Kregel Kjell N. Lindgren Knowledge Inventory For Professionals Kuiper Airborne Observatory (KAO) LL. Blaine Hammond Jr. L. Gordon Cooper Jr. La Tierra y calentamiento global LADEE (Lunar Atmosphere Dust Environment Explorer) LAGEOS (LAsER GEodynamics Satellite) Landsat Landsat 7 Landsat 8 / LDCM (Landsat Data Continuity Mission) Landsat 9 Langley Research Center Laser Communications Relay Launch Services Office Launch Services Program Laurel B. Clark Lawrence J. Delucas LBTI (Large Binocular Telescope Interferometer) LCROSS (Lunar Crater Observation and Sensing Satellite) Learning Resources Lee J. Archambault Lee M. Morin LEIA (Lunar Explorer Instrument for space biology Applications) Leland D. Melvin Lenticular Galaxies Leonid K. Kadenyuk Leonid MAC (Multi-instrument Aircraft Campaign) Leonids Leroy Chiao Lessons Learned for Professionals LGBTQ Pride Libera Life at NASA LIS (Lightning Imaging Sensor) Linda M. Godwin LISA (Laser Interferometer Space Antenna) Lisa M. Nowak Lodewijk Van Den Berg LOFTID (Low-Earth Orbit Flight Test of an Inflatable Decelerator) Logistics Management Division (LMD) Logo & Media Usage Loral O'Hara Loren J. Shriver Loren W. Acton Low Boom Flight Demonstrator Low-Density Supersonic Decelerator Low-Earth Orbit Economy LRO (Lunar Reconnaissance Orbiter) LSII Consortium Lucy Luke Delany Luna H-Map (Lunar Polar Hydrogen Mapper) Lunar Discovery & Exploration Program Lunar Eclipses Lunar Lander Simulation Lunar Orbiter Lunar Orbiter 1 Lunar Orbiter 2 Lunar Orbiter 3 Lunar Orbiter 4 Lunar Orbiter 5 Lunar Prospector Lunar Science Lunar Surface Innovation Consortium Lunar

Surface Innovation InitiativeLunar Surface Technology ResearchLunar TrailblazerLyridsMM.
Scott CarpenterMae C. JemisonMagellanMagnetarsMagnetosphereMAIA (Multi-Angle
Imager for Aerosols)Management AstronautsManley Lanier Carter Jr.Manufacturing,
Materials, 3-D PrintingMarCO (Mars Cube One)Marcos BerriosMargaret Rhea
SeddonMarinerMariner 1Mariner 2Mariner 3Mariner 4Mariner 5Mariner 6Mariner 7Mariner
8Mariner 9Mariner 10Mario Runco Jr.Mark C. LeeMark E. KellyMark L. PolanskyMark N.
BrownMark T. Vande HeiMarsMars 2020Mars Campaign Development DivisionMars
Climate OrbiterMars Exploration ProgramMars Exploration Rovers (MER)Mars ExpressMars
Global Surveyor (MGS)Mars MoonsMars ObserverMars OdysseyMars Orbiter Mission
(MOM)Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE)Mars PathfinderMars
PhoenixMars Polar Lander / Deep Space 2Mars Reconnaissance Orbiter (MRO)Mars
Sample Return (MSR)Mars Science Laboratory (MSL)Marsha IvinsMarshall Space Flight
CenterMarshall Test Facility and Support InfrastructureMartian Moon Exploration
(MMX)Martin J. FettmanMary E. WeberMary L. CleaveMaterials ScienceMaterials Science
Research Rack (MSRR)Matthew DominickMAVEN (Mars Atmosphere and Volatile
Evolution)Media ResourcesMegan McArthurMerchandise
ApprovalsMercuryMesosphereMESSENGER (MErcury Surface, Space ENvironment,
GEochemistry, and Ranging)Messier CatalogMeteor ShowersMeteors & MeteoritesMetOp
(Meteorological Operational satellite)Michael A. BakerMichael CollinsMichael
FinckeMichael E. FossumMichael E. Lopez-AlegriaMichael J. BloomfieldMichael J.
ForemanMichael J. MassiminoMichael J. McCulleyMichael J. SmithMichael L.
CoatsMichael L. GernhardtMichael P. AndersonMichael R. BarrattMichael R.
CliffordMichael S. HopkinsMichael T. GoodMichel F. CurtisMichoud Assembly
FacilityMicro-10Microbial Tracking (MT)MicrobiologyMillie Hughes-FulfordMini-RF
InstrumentMinXSS (Miniature X-ray Solar Spectrometer)MirandaMission EquityMission
Support DirectorateMissionsMMS (Magnetospheric Multiscale)Modeling, Analysis and
Prediction (MAP) ProgramModerate Resolution Imaging Spectroradiometer (MODIS)Modern
FiguresModular Optoelectric Multispectral Scanner (MOMS)MoonsMOS (Marine
Observation Satellite)MSI ExchangeMSL InstrumentationMulti-Spectral Fluorescence
Imaging System (Spectrum)MultimediaMUREPMUSE (Multi-slit Solar Explorer)Museum
AllianceNN. Jan DavisNAAMES (North Atlantic Aerosols and Marine Ecosystems
Study)Nancy Grace Roman Space TelescopeNancy J. Currie-GreggNanoSail-DNASA
Advisory Council (NAC)NASA Aeronautics CommitteeNASA AircraftNASA AppsNASA
Centers & FacilitiesNASA DirectoratesNASA e-BooksNASA Educator Professional
Development CenterNASA en españolNASA Engineering and Safety CenterNASA
Engineering & Safety Center AcademyNASA HeadquartersNASA HistoryNASA History
OfficeNASA Home & CityNASA ImagesNASA Innovative Advanced Concepts (NIAC)
ProgramNASA InteractivesNASA iTech ProgramNASA Knowledge ProgramNASA Open

Source SoftwareNASA PlusNASA PodcastsNASA Safety CenterNASA Safety Center
Professional DevelopmentNASA Shared Services CenterNASA Social Media ContactsNASA
Socials ProgramNASA STEMNASA STEM ProjectsNASA TVNASA WorldwindNational
Aeronautics Research InstituteNational Space Council Users' Advisory Group (NSpC
UAG)Native American Heritage MonthNatural DisastersNEA Scout (Near Earth Asteroid
Scout)NEAR ShoemakerNear Space NetworkNear-Earth Object (NEO)Nebulae
GameNEEMO (NASA Extreme Environment Mission Operations)Neil A. ArmstrongNeil
Gehrels Swift ObservatoryNeil W. Woodward IIINEK/SIRIUS (Nezemnyy Eksperimental'nyy
Kompleks/Scientific International Research In a Unique terrestrial Station)NEO Surveyor
(Near-Earth Object Surveyor Space Telescope)NEOWISENeptuneNeptune MoonsNeptune-
Like ExoplanetsNeutral Buoyancy LabNeutron StarsNew Frontiers ProgramNew
HorizonsNewslettersNEXIS (NASA's Exploration & In-space Services)Next Gen STEMNIAC
StudiesNIAC SymposiumNICER (Neutron star Interior Composition Explorer)Nicholas J. M.
PatrickNichole AyersNicole A. MannNicole P. StottNight Sky NetworkNimbusNISAR (NASA-
ISRO Synthetic Aperture Radar)NOAA (National Oceanic and Atmospheric
Administration)NOAA-20 (JPSS-1)NOAA-21 (JPSS-2)NOAA-N PrimeNorman E.
ThagardNorthrop Grumman Commercial ResupplyNova-C (Lander)NozomiNuSTAR
(Nuclear Spectroscopic Telescope Array)OO / OREOS (Organism / Organic Exposure to
Orbital Stresses)Ocean Surface Topography Mission (OSTM) / Jason-2Ocean
WorldsOceanographyOceansOceans Melting Greenland (OMG)OCO (Orbiting Carbon
Observatory)OCO-2 (Orbiting Carbon Observatory 2)OCO-3 (Orbiting Carbon Observatory
3)Office of International and Interagency Relations (OIIR)Office of Legislative and
Intergovernmental Affairs (OLIA)Office of Small Business Programs (OSBP)Office of
Strategic Infrastructure (OSI)Office of Technology, Policy and Strategy (OTPS)Office of the
Chief Engineer (OCE)Office of the Chief Financial Officer (OCFO)Office of the Chief Health
and Medical Officer (OCHMO)Office of the Chief Information Officer (OCIO)Office of the
Chief Scientist (OCS)Office of the General Counsel (OGC)One Year CrewOpportunities For
Educators to Get InvolvedOpportunities For International Participants to Get
InvolvedOpportunities For Researchers to Get InvolvedOpportunities For Students to Get
InvolvedOpportunities For U.S. Citizens to Get InvolvedOpportunities to Contribute to
NASA Missions & Get InvolvedOpportunity (Rover)ORACLES (ObseRvations of Aerosols
above CLOUDS and their IntERactionS)Orbiting Astronomical Observatory (OAO)ORFEUS
(Orbiting and Retrievable Far and Extreme Ultraviolet Spectrometer)-SPAS
IIOrganizationsOrigin & Evolution of the UniverseOrion Multi-Purpose Crew VehicleOrion
NebulaOrion SpacecraftOrionids OSAM-1 (On-Orbit Servicing, Assembly, and
Manufacturing 1)OSIRIS-REx (Origins, Spectral Interpretation, Resource Identification,
Security-Regolith Explorer)OumuamuaOuter Planets & Ocean Worlds ProgramOutside the
ClassroomOV-10Owen K. GarriottOzone LayerOzone Mapping and Profiler Suite (OMPS)PP-

3PACE (Plankton, Aerosol, Cloud, Ocean Ecosystem)Pamela A. MelroyPandoraParker Solar Probe (PSP)Partner AstronautsPartner with NASA STEMPartner With UsPatricia C. RobertsonPatrick G. ForresterPaul D. Scully-PowerPaul J. WeitzPaul S. LockhartPaul W. RichardsPeggy A. WhitsonPeople of NASAPeregrine Lunar LanderPerseidsPerseverance (Rover)Peter J. K. WisoffPFMI (Pore Formation and Mobility Investigation)PharmaSatPhilae (Lander)Philip K. ChapmanPhobosPhoebePhysical SciencesPhysical Sciences ProgramPI ResourcesPierre J. ThuotPiers J. SellersPioneerPioneer 0 / Able 1Pioneer 1 / Able 2Pioneer 2Pioneer 3Pioneer 4Pioneer 5Pioneer 6Pioneer 7Pioneer 8Pioneer 9Pioneer 10Pioneer 11Pioneer P-3 / Able 4BPioneer P-30 / Able 5APioneer P-31 / Able 5BPioneer VenusPioneer Venus 1Pioneer Venus 2Pioneer-EPlanckPlanet XPlanetary DefensePlanetary Defense Coordination OfficePlanetary Environments & AtmospheresPlanetary Geosciences & GeophysicsPlanetary SciencePlanetary Science DivisionPlanetary TransitsPlanetsPlant BiologyPlasma Kristall 4 (PK-4)PlayPlay it!Plume-Surface Interaction (PSI) ProjectPlutoPluto MoonsPodcastsPolarPopular HistoryPost Doc ProgramPotentially Hazardous Asteroid (PHA)PredatorPREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment)PRESat (PharmaSat Risk Evaluation Satellite)Private Astronaut MissionsPrizes, Challenges & CrowdsourcingPrizes, Challenges & Crowdsourcing NewsProject Management & Systems EngineeringProject MercuryProtostarsProxima Centauri bPsyche AsteroidPsyche MissionPulsarsPUNCH (Polarimeter to Unify the Corona and Heliosphere)QQ-PACE (CubeSat Particle Aggregation and Collision Experiment)QuadrantidsQuasarsQuesst (X-59)Quesst: The FlightsQuesst: The MissionQuesst: The ScienceQuesst: The TeamQuesst: The VehicleQuikSCAT (Quick Scatterometer)RRADARSAT-1 (Radar Satellite 1)RAD-SEEDRaja ChariRandolph J. BresnikRangerRanger 1Ranger 2Ranger 3Ranger 4Ranger 5Ranger 6Ranger 7Ranger 8Ranger 9Read it!Red DwarfsRed GiantsRedstone 3 (Freedom 7)Redstone 4 (Liberty Bell 7)Research & Analytics ProgramResearch ArchiveResearch Opportunities in Space and Earth Sciences (ROSES)Revolutionary Vertical Lift TechnologyRex J. WalheimRheaRHESSI (Reuven Ramaty High Energy Solar Spectroscopic Imager)Richard A. MastracchioRichard A. SearfossRichard F. Gordon Jr.Richard H. TrulyRichard J. HiebRichard M. LinnehanRichard M. MullaneRichard N. RichardsRichard O. CoveyRichard R. ArnoldRick D. HusbandRings of JupiterRings of NeptuneRings of SaturnRing-Sheared Drop (RSD)RLL (Robotic Lunar Lander)Robb KulinRobert A. R. ParkerRobert C. SpringerRobert D. CabanaRobert F. OvermyerRobert J. CenkerRobert L. BehnkenRobert L. CrippenRobert L. Curbeam Jr.Robert L. GibsonRobert L. Satcher Jr.Robert L. StewartRobert Shane KimbroughRobotic Lunar Exploration ProgramRoboticsRodent Research (RR)Roger B. ChaffeeRoger K. CrouchRonald A. PariseRonald E. EvansRonald E. McNairRonald J. Garan Jr.Ronald J. GrabeRonald M. SegaRosalind Franklin (Rover)ROSAT (ROentgen SATellite)RosettaRoy D. Bridges Jr.RS-25Russell L. SchweickartRXTE (Rossi X-ray Timing Explorer)RyuguSS. Christa McAuliffeS.

David GriggsS3VISAC-B (Satélite de Aplicaciones Científicas-B)SAGE III (Stratospheric Aerosol and Gas Experiment)SAGE-III Meteor-3M (Stratospheric Aerosol and Gas Experiment III on Meteor-3M)SakigakeSally K. RideSAMPEX (Solar Anomalous and Magnetospheric Particle Explorer)Samuel T. DurranceSandra H. MagnusSaturnSaturn MoonsSBIRSBIR / STTRScience & ResearchScience ActivationScience for EveryoneScience in the AirScience InstrumentsScience LeadershipScience Mission DirectorateScience of Space ExplorationScience SolicitationsScience-enabling TechnologyScientific BalloonsScott D. AltmanScott D. TingleScott E. ParazynskiScott J. HorowitzScott J. KellySea IceSEAC4RS (Studies of Emissions, Atmospheric Composition, Clouds and Climate Coupling by Regional Surveys)SeasatSeaWiFS (Sea-viewing Wide Field-of-view Sensor)SeaWindsSentinel-6 Michael Freilich SatelliteSentinel-6BSerena M. Auñón-ChancellorSeres Humanos en el espacioSERVIR (Regional Visualization and Monitoring System)Seyfert GalaxiesShannon W. LucidShannon WalkerSherwood C. SpringShoemaker-Levy 9Shuttle-MirSidney M. GutierrezSierra UAS #2SimLabsSIR-C/X-SAR (Shuttle Imaging Radar-C / X-Band Synthetic Aperture Radar)Sistema solarSkylabSkywatchingSkywatching TipsSmall Bodies of the Solar SystemSmall Business Innovation Research / Small BusinessSmall Satellite MissionsSmall Spacecraft Technology ProgramSmallSats ProgramSMAP (Soil Moisture Active Passive)SMART-1 (Small Missions for Advanced Research in Technology 1)SNOE (Student Nitric Oxide Explorer)Social MediaSOFIA (Stratospheric Observatory for Infrared Astronomy) / 747-SPSoft Matter Dynamics (SMD) / FOAMSoftware to LicenseSOHO (Solar and Heliospheric Observatory)Solar CruiserSolar Dynamics Observatory (SDO)Solar EclipsesSolar Electric Propulsion (SEP)Solar FlaresSolar OrbiterSolar Sail – Beyond Plum BrookSolar System AmbassadorsSolar System Exploration Research Virtual InstituteSolar Terrestrial Probes ProgramSolar WindSolid Fuel Ignition and Extinction (SoFIE)Solidification Using a Baffle in Sealed Ampoules (SUBSA)SORCE (Solar Radiation and Climate Experiment)SORTIE (Scintillation Observations & Response of the Ionosphere to Electrodynamics)Sounding RocketsSounding Rockets ProgramSounds & RingtonesSouthern Delta AquariidsSpace BiologySpace Biology ProgramSpace Biosciences at AmesSpace Communications & Navigation Program (SCaN)Space Communications TechnologySpace Environment Testbeds (SET-1)Space Environments Testing Management Office (SETMO)Space Flight Awareness (SFA)Space GrantSLS (Space Launch System)Space Life & Physical Sciences Research & Applications DivisionSpace Nuclear Propulsion (SNP)Space Operations Directorate Resource Management OfficeSpace Operations Mission DirectorateSpace Samples for Your ClassroomSpace ShuttleSpace Station Research OpportunitiesSpace Tech Graduate ResearchSpace Technology 5Space Technology 6Space Technology 7 / Disturbance Reduction System (DRS)Space Technology GrantsSpace Technology Mission DirectorateSpace Technology Research GrantsSpace Vehicle Mockup FacilitySpace

WeatherSpaceships and RocketsSpacesuitsSpaceX Commercial
ResupplySpartanSpectrum ManagementSPHEREx (Spectro-Photometer for the History of
the Universe and Ices Explorer)SpinoffsSpiral GalaxiesSpirit (Rover)Spitzer Space
TelescopeSpot the International Space StationSRTM (Shuttle Radar Topography
Mission)ST5Stanley G. LoveStardust / Stardust NEXtStarlinerStarsSTEM Engagement at
NASASTennis Space CenterStennis Test Facility and Support InfrastructureStephanie D.
WilsonStephen D. ThorneStephen G. BowenStephen K. RobinsonStephen N. FrickStephen
S. OswaldSTEREO (Solar TERrestrial RELations Observatory)Steven A. HawleySteven L.
SmithSteven R. NagelSteven R. SwansonSteven W. LindseyStory MusgraveSTPStrange New
WorldsStrategic Integration and Management DivisionStrategic Partnerships
GuidelinesStratosphereSTS- 1STS- 2STS- 3STS- 4STS- 5STS- 6STS- 7STS- 8STS- 9STS-
26STS- 27STS- 28STS- 29STS- 30STS- 31STS- 32STS- 33STS- 34STS- 35STS- 36STS- 37STS-
38STS- 39STS- 40STS- 41STS- 41BSTS- 41CSTS- 41DSTS- 41GSTS- 42STS- 43STS- 44STS-
45STS- 46STS- 47STS- 48STS- 49STS- 50STS- 51STS- 51ASTS- 51BSTS- 51CSTS- 51DSTS-
51FSTS- 51GSTS- 51ISTS- 51JSTS- 51LSTS- 52STS- 53STS- 54STS- 55STS- 56STS- 57STS-
58STS- 59STS- 60STS- 61STS- 61ASTS- 61BSTS- 61CSTS- 62STS- 63STS- 64STS- 65STS-
66STS- 67STS- 68STS- 69STS- 70STS- 71STS- 72STS- 73STS- 74STS- 75STS- 76STS- 77STS-
78STS- 79STS- 80STS- 81STS- 82STS- 83STS- 84STS- 85STS-86STS- 87STS- 88STS- 89STS-
90STS- 91STS- 92STS- 93STS- 94STS- 95STS- 96STS- 97STS- 98STS- 99STS- 100STS-
101STS- 102STS- 103STS- 104STS- 105STS- 106STS- 107STS- 108STS- 109STS- 110STS-
111STS- 112STS- 113STS- 114STS- 115STS- 116STS- 117STS- 118STS- 119STS- 120STS-
121STS- 122STS- 123STS- 124STS- 125STS- 126STS- 127STS- 128STS- 129STS- 130STS-
131STS- 132STS- 133STS- 134STS- 135Stuart A. RoosaStudying ExoplanetsSubmillimeter
Wave Astronomy Satellite (SWAS)Sunita L. WilliamsSunRISE (Sun Radio Interferometer
Space Experiment)SunspotsSuomi NPP (Suomi National Polar-orbiting Partnership)Super
King AirSuper-Earth ExoplanetsSupermoonsSupernovaeSupersonic
FlightSurveyorSurveyor 1Surveyor 2Surveyor 3Surveyor 4Surveyor 5Surveyor 6Surveyor
7Surveyor Model 1Surveyor Model 2Surveyor Model SD-1Susan J. HelmsSusan L.
KilrainSustainability at Kennedy Space CenterSustainable Flight DemonstratorSustainable
Flight National PartnershipSuzakuSWOT (Surface Water and Ocean
Topography)Synchronous Meteorological Satellite (SMS)System-Wide SafetyTT-34CT-38
Astronaut TrainerTamara E. JerniganTaylor WangTech Demo MissionsTech Dev ProjectsTech
PortfolioTechnologyTechnology DemonstrationTechnology Demonstration Missions
ProgramTechnology for Living in SpaceTechnology for Space TravelTechnology
HighlightsTechnology ResearchTechnology Transfer &
SpinoffsTecnologíaTeleroboticsTempel 1Terence T. HenricksTerraTerrence W.
WilcuttTerrestrial ExoplanetsTerrestrial PlanetsTerriers (Tomographic Experiment Using
Radiative Recombinative Ionospheric EUV and Radio Sources)Terry J. HartTerry W. Virts

Jr.TESS (Transiting Exoplanet Survey Satellite)Tethered Satellite System (TSS)TG-14The Big BangThe Future of Commercial SpaceThe Great Red SpotThe Habitable ZoneThe Human Body in SpaceThe Kuiper BeltThe Milky WayThe North StarThe Oort CloudThe Search for LifeThe Solar SystemThe SunThe UniverseTHEMIS (Time History of Events and Macroscale Interactions During Substorms)THEMIS-ARTEMIS (Time History of Events and Macroscale Interactions During Substorms – Acceleration, Reconnection, Turbulence and Electrodynamics of Moon's Interaction with the Sun)Theodore C. FreemanThermosphereThomas D. AkersThomas D. JonesThomas H. MarshburnThomas J. HennenThomas K. Mattingly IIThomas P. StaffordTIMED (Thermosphere Ionosphere Mesosphere Energetics and Dynamics Mission)Timothy J. CreamerTimothy L. KopratiTIROS (Television Infrared Observation Satellite Program)TitanTOMS-EP (Total Ozone Mapping Spectrometer – Earth Probe)TOPEX / Poseidon (ocean TOPography Experiment)TornadoesTotal and Spectral Solar Irradiance Sensor – 1 (TSIS-1)Total and Spectral Solar Irradiance Sensor – 2 (TSIS-2)TP / ACOTRACE (Transition Region and Coronal Explorer)Trace Gas Orbiter (TGO)Tracking Aerosol Convection Interactions Experiment (TRACER)Tracking and Data Relay Satellite (TDRS)Tracy Caldwell DysonTransformational ToolsTechnologiesTransformative Aeronautics Concepts ProgramTritonTrojan AsteroidsTropical Rainfall Measuring Mission (TRMM)TROPICS (Time-Resolved Observations of Precipitation Structure and Storm Intensity with a Constellation of Smallsats)TroposphereTropospheric Emissions: Monitoring of Pollution (TEMPO)TsunamisTWINS (Two Wide-Angle Imaging Neutral-Atom Spectrometers)Twins StudyTyler N. HagueUUAS in the NASUUAS Traffic ManagementUH-1HUH-IIUlyssesUnderway Recovery Test 10Universe of MonstersUniversity InnovationUniversity Leadership InitiativeUniversity Student Research ChallengeUniversoUpper Atmosphere Research Satellite (UARS)UranusUranus MoonsUrban Air Mobility SimulationUSGS (United States Geological Survey)VValles MarinerisVan Allen ProbesVance D. BrandVegetable Production System (VEGGIE)VenusVenus ExpressVERITAS (Venus Emissivity, Radio Science, InSAR, Topography & Spectroscopy)Vertical Motion SimulatorVestaVictor J. GloverVideoVikingViking 1Viking 2Viking 400VIPER (Volatiles Investigating Polar Exploration Rover)Virgil I. GrissomVirtual Guest ProgramVisible Infrared Imaging Radiometer Suite (VIIRS)VolcanoesVoyager 1Voyager 2Voyager ProgramWWallops Flight FacilityWalter CunninghamWalter M. Schirra Jr.Warren HoburgWater & Energy CycleWater on EarthWB-57Weather and Atmospheric DynamicsWendy B. LawrenceWetLab-2What We DoWhite DwarfsWhite Sands Test FacilityWhite Sands Test Facility and Support InfrastructureWhy Go To SpaceWild 2WildfiresWilkinson Microwave Anisotropy Probe (WMAP)William A. AndersWilliam A. OefeleinWilliam A. PaillesWilliam B. LenoirWilliam C. McCoolWilliam E. ThorntonWilliam F. FisherWilliam F. ReaddyWilliam G. GregoryWilliam M. ShepherdWilliam R. PogueWilliam S. McArthur Jr.Wind MissionWinston E. ScottWIRE (Wide-Field Infrared

Explorer)WISE (Wide-field Infrared Survey Explorer)Women at NASAWomen's History
MonthXXelene Lunar Lander (XL-1)xEVA & Human Surface MobilityXMM-Newton (X-ray
Multi-Mirror Newton)XRISM (X-Ray Imaging and Spectroscopy Mission)YYO-
3AYohkohYvonne Darlene CagleZZena CardmanZero Boil-Off Tank (ZBOT-NC)

A to Z Topics ListingUse this A to Z Topics listing to discover all things
NASA!ABCDEFGHIJKLMN**OP**QRSTUVWXYZ

A to Z Topics ListingUse this A to Z Topics listing to discover all things NASA!

A to Z Topics Listing

Use this A to Z Topics listing to discover all things NASA!

ABCDEFGHIJKLMN**OP**QRSTUVWXYZ

A

B

C

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Q

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T

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V

W

X

Y

Z

2023 Annular Eclipse

3D Tissue Chips

67P/Churyumov-Gerasimenko

AAM Drones and You

Accessibility at NASA

Accessing Flight Tests

ACE (Advanced Composition Explorer)

Acerca de NASA

ACME (Advanced Combustion via Microgravity Experiments)

AcrimSat (Active Cavity Irradiance Monitor Satellite)

ACT (Atmospheric Carbon and Transport) – America

ACTE (Adaptive Compliant Trailing Edge)

Active Galaxies

ADEOS (Advanced Earth Observing Satellite) / MIDORI

Advanced Air Vehicles Program

Advanced Composites

Advanced Studies & Concepts (PACE)

Advanced Spacesuits

Aerospace Safety Advisory Panel

AIM (Aeronomy of Ice in the Mesosphere)

Air Mobility Pathfinders Project

Air Traffic Control Labs

AirMOSS (Airborne Microwave Observatory of Subcanopy and Subsurface)

AirSTAR

AJAX (Alpha Jet Atmospheric eXperiment)

Akatsuki/Venus Climate Orbiter mission (PLANET-C)

Alan G. Poindexter

Alan L. Bean

Albert Sacco Jr.

Alfred M. Worden

Analog Missions

Andre Douglas

Andrew J. Feustel

Andrew M. Allen

Andrew R. Morgan

Andrew S. W. Thomas

Anil Menon

Anna L. Fisher

Anne C. McClain

Antarctic Stations

Antarctic Station NSF

Anthony W. England

APH (Advanced Plant Habitat)

API

Apollo Program

Apollo 1

Apollo 7

Apollo 9

Apollo 10

Apollo 12

Apollo 13

Apollo 14

Apollo 15

Apollo 15 Subsatellite

Apollo 16

Apollo 16 Subsatellite

Apollo 17

Applied Sciences Program

Apply for Funding

Apps

Aqua

Aquarius

ARCTAS (Arctic Research of the Composition of the Troposphere from Aircraft and Satellites)

ARIEL (Atmospheric Remote-sensing Infrared Exoplanet Large-survey)

ARISE 2014

Armstrong Test Facility (Plum Brook)

Artemis 4

Artemis Campaign Development Division

[ASCA \(Advanced Satellite for Cosmology and Astrophysics\)](#)

[Asteroid Initiative](#)

[ASTHROS \(Astrophysics Stratospheric Telescope for High Spectral Resolution Observations at Submillimeter-wavelengths\)](#)

[Astro Observatory 1](#)

[Astro Observatory 2](#)

[Astrogram](#)

[Astronaut Candidates](#)

[Astrophysics Explorers Program](#)

[Astrophysics Pioneers](#)

[Astrophysics Research Program](#)

[Atmospheric Composition](#)

[ATLAS \(Atmospheric Laboratory of Applications and Science\)](#)

[Atmospheric Tomography Mission \(ATom\)](#)

[ATS \(Applications Technology Satellite Program\)](#)

[ATTREX \(Airborne Tropical Tropopause Experiment\)](#)

[Aura](#)

[Awarded Abstracts](#)

[AWE \(Atmospheric Waves Experiment\)](#)

[B-377 Super Guppy](#)

[Barbara R. Morgan](#)

[BARREL \(Balloon Array for Radiation-belt Relativistic Electron Losses\)](#)

[Barry E. Wilmore](#)

[BBXRT \(Broad Band X-ray Telescope\)](#)

[BECCAL \(Bose Einstein Condensates & Cold Atoms Lab\)](#)

[Become an Astronaut](#)

Benefits to Humanity

Benefits to Science

Benjamin Alvin Drew

BepiColombo

Beresheet

Bernard A. Harris Jr.

Betelgeuse

Bill Nelson

Binary Stars

Biological Experiment-01 (BioExpt-01)

BION / Biocosmos

Black History Month

Blazars

Blue Ghost

Boeing Starliner

Bonnie J. Dunbar

Borrelly

Brand Guidelines Center

Brent W. Jett

Brewster H. Shaw Jr.

Brian Duffy

Brian T. O'Leary

BRIC (Biological Research in Canisters)

Brown Dwarfs

Bruce E. Melnick

Bruce McCandless

Bryan D. O'Connor

Byron K. Lichtenberg

C-130

C-20A

C-9

Caldwell Catalog

CALIPSO (Cloud-Aerosol Lidar and Infrared Pathfinder Satellite)

Callie First Graphic Novel

Callisto

Candidate Astronauts

Carl E. Walz

Carl J. Meade

Carlos I. Noriega

Carruthers Geocorona Observatory (GLIDE)

CARVE (Carbon in Arctic Reservoirs Vulnerability Experiment)

Cassini-Huygens

Catherine G. Coleman

CATS (Cloud-Aerosol Transport System)

Centennial Challenges News

Center Innovation Fund Center of Excellence for Collaborative Innovation (CoECI)

CERES (Clouds and the Earth's Radiant Energy System)

CeREs (Compact Radiation Belt Explorer)

Ceres Dwarf Planet

Cessna C206

CGRO (Compton Gamma Ray Observatory)

CHAMP (Challenging Mini-satellite Payload)

Chandra X-ray Observatory

Chandrayaan-1

Chandrayaan-2

Chandrayaan-3

Charles A. Bassett II

Charles Camarda

Charles Conrad Jr.

Charles D. Gemar

Charles D. Walker

Charles E. Brady Jr.

Charles F. Bolden Jr.

Charles G. Fullerton

Charles J. Precourt

Charles Lacy Veach

Charles M. Duke Jr.

Charles O. Hobaugh

Charon

Chiaki Mukai

CHIPS (Cosmic Hot Interstellar Plasma Spectrometer)

Christina Birch

Christina H. Koch

Christopher J. Ferguson

Christopher J. Loria

Christopher J. Cassidy

Christopher L. Williams

CINDI (Coupled Ion Neutral Dynamic Investigation)

CIPHER (Complement of Integrated Protocols for Human Exploration Research)

Cirrus SR-22

Clayton C. Anderson

Clementine

Clifton C. Williams Jr.

Cloudsat

CLPS (Commercial Lunar Payload Services)

Cluster II

COBE (Cosmic Background Explorer)

Columbia 300

Commercial Crew Rockets

Commercial Crew Spacecraft

Commercial Lunar Payload Services (CLPS)

Community of Practice Webinar

Complex Fluids

Conjunction Assessment Risk Analysis Team (CARA)

Constellations

CONTOUR (COmet Nucleus TOUR)

Convergent Aeronautics Solutions

CORAL (COral Reef Airborne Observatory)

Coronal Mass Ejections

COSI (Compton Spectrometer and Imager)

Cosmic Origins Program

Crab Nebula

Crew Health and Performance Exploration Analog (CHAPEA)

Crew Vehicle Systems Research Facility

Cross Program Integration

CRRES (Combined Release and Radiation Effects Satellite)

Cryogenic Fluid Management (CFM)

Cryogenic Propellant Handling

CubeSat Or Microsat Probabilistic and Analogies Cost Tool (COMPACT)

CuPID (Cusp Plasma Imaging Detector)

Curtis L. Brown Jr.

CUSP (CubeSat Mission to Study Solar Particles)

Cycles & Focus Areas

CYGNSS (Cyclone Global Navigation Satellite System)

Dale A. Gardner

Daniel C. Brandenstein

Daniel C. Burbank

Daniel M. Tani

Daniel T. Barry

Daniel W. Bursch

David A. Wolf

David C. Hilmer

David C. Leestma

David M. Brown

David M. Walker

David R. Scott

Dawn

DC-8

Deep Impact / EPOXI (Extrasolar Planet Observation and Deep Impact Extended Investigation)

Deep Space 1

Deep Space Network (DSN)

Deep Space Optical Communications (DSOC)

Deimos

Deniz Burnham

Desert RATS (Research and Technology Studies)

(DECLIC) DEvice for the study of Critical Liquids and Crystallization

DHC

Didymos & Dimorphos

Diffuse X-Ray Spectrometer (DXS)

DISCOVER AQ (Deriving Information on Surface Conditions from COlumn and VERTically Resolved Observations Relevant to Air Quality)

Discovery Program

Dominic A. Antonelli

Dominic L. Gorie

Don L. Lind

Donald A. Thomas

Donald E. Williams

Donald H. Peterson

Donald K. Slayton

Donald L. Holmquest

Donald R. McMonagle

Donald R. Pettit

Donn F. Eisele

Dorothy M. Metcalf-Lindenburger

Douglas G. Hurley

Douglas H. Wheelock

Dragon Crew Spacecraft

Draper SERIES-2

Droughts

DSCOVR (Deep Space Climate Observatory)

Duane G. Carey

Duane Graveline

Dynamics Explorer (DE)

:envihab

Early Career Faculty

Earth Information Center

Earth Observing-1 (EO-1)

Earth Science Data Systems Program

Earth Science Technology Office

Earth System Observatory (ESO)

Earthquakes

Earth's Atmosphere

Earth's Magnetic Field

Earth's Vital Signs

Ecostress (ECOSystem Spaceborne Thermal Radiometer Experiment on Space Station)

Edgar D. Mitchell

Edward G. Gibson

Edward G. Givens Jr.

Edward H. White II

Edward M. Fincke

Edward T. Lu

Eileen M. Collins

Electrified Powertrain Flight Demo

ElectroHydroDynamic (EHD) Test Chamber

Electromagnetic Spectrum

ELFIN (Electron Losses & Fields Investigation)

Ellen Ochoa

Ellen S. Baker

Ellington Field

Elliot M. See Jr.

Elliptical Galaxies

Ellison S. Onizuka

Emergency & Pandemic Response

EMIT (Earth Surface Mineral Dust Source Investigation)

Enceladus

Environmental Management Division (EMD)

Environmental Science Services Administration (ESSA)

EnVision

Equator-S

ERBS (Earth Radiation Budget Satellite)

Eric A. Boe

Eros

EscaPADE (Escape and Plasma Acceleration and Dynamics Explorers)

Established Program to Stimulate Competitive Research (EPSCoR)

Eta Aquarids

E-TBEx (Enhanced Tandem Beacon Experiment)

Euclid

Eugene A. Cernan

Eugene H. Trinh

EUVE (Extreme-Ultraviolet Explorer)

Everyday Extraordinary Stories

Exosphere

Expedition 1

Expedition 2

Expedition 3

Expedition 4

Expedition 5

Expedition 6

Expedition 7

Expedition 8

Expedition 9

Expedition 10

Expedition 11

Expedition 12

Expedition 13

Expedition 14

Expedition 15

Expedition 16

Expedition 17

Expedition 18

Expedition 19

Expedition 20

Expedition 21

Expedition 22

Expedition 23

Expedition 24

Expedition 25

Expedition 26

Expedition 27

Expedition 28

Expedition 29

Expedition 30

Expedition 31

Expedition 32

Expedition 33

Expedition 34

Expedition 35

Expedition 41

Expedition 58

Expedition 60

Expedition 62

Expedition 63

Expedition 66

Expeditions

Exploration Capabilities Division

EFT 1 (Exploration Flight Test 1)

Exploration Operations Division

Explorer

Explorer 33

Explorer 35

Explorer 49

Explorers Program

Extraterrestrial Life

Extreme Ultraviolet High-Throughput Spectroscopic Telescope (EUVST) Epsilon

Extreme Weather Events

EZIE (Electrojet Zeeman Imaging Explorer)

F / A-18

F. Drew Gaffney

F-18

FAST (Fast Auroral SnapshoT Explorer)

FASTSat (Fast, Affordable, Science and Technology Satellite)

FBCE (Flow Boiling and Condensation Experiment)

Fermi Gamma-Ray Space Telescope

Fernando Caldeiro

Film & Documentary Guidelines

First Woman Graphic Novel

Fission Surface Power (FSP)

FLARE (Flammability Limits At Reduced-g Experiment)

Flight Program

Flight Providers

Flight Simulation

Francis R. Scobee

Franco Malerba

Frank Borman

Frank L. Culbertson Jr.

Franklin R. Chang-Díaz

Fred W. Haise Jr.

Fred W. Leslie

Frederick D. Gregory

Frederick H. Hauck

Frederick W. Sturckow

Fruit Fly Lab (FFL)

FUSE (Far Ultraviolet Spectroscopic Explorer)

FutureFlight Central

G. David Low

G. Reid Wiseman

Gaia

Galaxies, Stars, & Black Holes Research

GALEX (Galaxy Evolution Explorer)

Galileo

Galileo Jupiter Atmospheric Probe

Gamma Rays

Gamma-Ray Bursts

Garrett E. Reisman

Gary E. Payton

Gas Giants

Gateway Program

GDC (Geospace Dynamics Constellation)

GEDI (Global Ecosystem Dynamics Investigation)

Gemini I

Gemini II

Gemini III

Gemini IV

Gemini V

Gemini VI

Gemini VII

Gemini VIII

Gemini IX

Gemini X

Gemini XI

Gemini XII

Geminids

Genesis

GRIP (Genesis and Rapid Intensification Processes)

General George D. Nelson

George D. Zamka

Geosat (U.S. Navy GEOdetic SATellite)

Geospace Dynamics Constellation (GDC)

Geotail

Gerald Carr

Get Social

GII-STA

Giotto

Glaciers

GLIMR (Geosynchronous Littoral Imaging and Monitoring Radiometer)

Glory

GOES-N

GOES-R

GOLD (Global-scale Observations of the Limb and Disk)

GPIM (Green Propellant Infusion Mission)

GPM (Global Precipitation Global Precipitation Measurement)

GRACE (Gravity Recovery And Climate Experiment)

GRACE-FO (Gravity Recovery and Climate Experiment Follow-on)

Grades 5 – 8

Grades 5 – 8 for Educators

Grades 9 – 12

Grades K – 4

Grades K – 4 for Educators

GRAIL (Gravity Recovery And Interior Laboratory)

Gravity Probe B (GP-B)

Greenhouse Effect

Greenhouse Gasses

Gregory C. Johnson

Gregory E. Chamitoff

Gregory H. Johnson

Gregory J. Harbaugh

Gregory Jarvis

Gregory T. Linteris

Griffin (lander)

Ground Facilities for ISS

Guion S. Bluford Jr.

Gulfstream C-20A

Gulfstream G-III

Gulfstream G-V

Gulfstream IV

Guy S. Gardner

H-135

HALCA (Highly Advanced Laboratory for Communications and Astronomy)

Hale-Bopp

Halley's Comet

HaloSat

Harrison H. Schmitt

Hayabusa

Hayabusa 2

Heidemarie M. Stefanyshyn-Piper

Heliophysics Research Program

Helios 1

Helios 2

Heliosphere

HelioSwarm

Helix Nebula

Henry W. Hartsfield Jr.

HERA (Human Exploration Research Analog)

HERMES (Heliophysics Environmental and Radiation Measurement Experiment Suite)

Herschel Space Observatory

HETE-1 (High Energy Transient Explorer 1)

HETE-2 (High Energy Transient Explorer 2)

Hinode (Solar B)

Hipparcos

Hi-Rate Composite Aircraft Manufacturing

History: NASA

Hiten / Hagomoro

Hitomi (ASTRO-H)

Horsehead Nebula

HU-25

Human Exploration of Space: Why We Go

Human Health and Performance

Human Spaceflight Capabilities Division

Hurricanes

Hybrid Thermally Efficient Core

Hyperion

Hypersonic Technology

HyspIRI (Hyperspectral Infrared Imager)

IBEX (Interstellar Boundary Explorer)

ICESat-2 (Ice, Cloud and land Elevation Satellite-2)

ICON (Ionospheric Connection Explorer)

Ida

IEH-3 (International Extreme Ultraviolet Hitchhiker)

Ilan Ramon

IMAGE (Imager for Magnetopause-to-Aurora Global Exploration)

Images & Media Guidelines

IMAP (Interstellar Mapping and Acceleration Probe)

IMP-8 (Interplanetary Monitoring Platform 8)

INCUS (Investigation of Convective Updrafts)

Infrared Space Observatory (ISO)

Institutional Funding

International Gamma-ray Astrophysics Laboratory (INTEGRAL)

International Space Station Division

ISRU (In-Situ Resource Utilization)

Io

Ionosphere

IRIS (Interface Region Imaging Spectrograph)

Irregular Galaxies

ISEE-3/ICE (International Sun-Earth Explorer-3 / International Cometary Explorer)

ISERV (ISS SERVIR Environmental Research and Visualization System)

ISS-RapidScat (Rapid Scatterometer)

IUE (International Ultraviolet Explorer)

IXPE (Imaging X-ray Polarimetry Explorer)

Jack D. Fischer

Jack Hathaway

Jack R. Lousma

Jake Garn

James A. Lovell Jr.

James A. McDivitt

James A. Pawelczyk

James B. Irwin

James C. Adamson

James D. A. Van Hoften

James D. Halsell Jr.

James D. Wetherbee

James F. Buchli

James F. Reilly

James H. Newman

James M. Kelly

James P. Bagian

James P. Dutton

James S. Voss

Janet L. Kavandi

Janice E. Voss

Janus

Jason-1

Jason-CS (Continuity of Service) / Sentinel-6

Jay Clark Buckey

Jean-Jacques Favier

Jeffrey A. Hoffman

Jeffrey N. Williams

Jeffrey S. Ashby

Jerome Apt

Jerry L. Ross

Jerry M. Linenger

Jessica U. Meir

Jessica Wittner

Joan E. Higginbotham

Joe F. Edwards Jr.

Joe H. Engle

John A. Llewellyn

John B. Herrington

John D. Olivas

John E. Blaha

John H. Casper

John H. Glenn Jr.

John L. Phillips

John L. Swigert Jr.

John M. Fabian

John M. Grunsfeld

John M. Lounge

John O. Creighton

John S. Bull

John W. Young

Johnson Flight Operations

Johnson's Mission Control Center

Joint Agency Satellite Division

Jon A. McBride

Jonny Kim

José M. Hernández

Joseph M. Acaba

Joseph P. Allen

Joseph P. Kerwin

Joseph R. Tanner

Josh A. Cassada

JPSS (Joint Polar Satellite System)

JPSS-3

JPSS-4

Judith A. Resnik

JUICE (Jupiter Icy Moons Explorer)

JWST (James Webb Space Telescope)

Kaguya

Kalpana Chawla

Karen L. Nyberg

Karl G. Henize

Karol J. Bobko

Katherine Johnson IV & V Facility

Kathleen Rubins

Kathryn C. Thornton

Kathryn D. Sullivan

Kathryn P. Hire

Kayla Barron

Keck Interferometer (KI)

Kenneth D. Bowersox

Kenneth D. Cameron

Kenneth D. Cockrell

Kenneth S. Reightler Jr.

Kenneth T. Ham

Kent V. Rominger

Kepler / K2

Kepler 22B

Kevin A. Ford

Kevin P. Chilton

Kevin R. Kregel

Knowledge Inventory For Professionals

Kuiper Airborne Observatory (KAO)

L. Blaine Hammond Jr.

L. Gordon Cooper Jr.

La Tierra y calentamiento global

LAGEOS (LAsEr GEOdynamics Satellite)

Landsat 7

Landsat 8 / LDCM (Landsat Data Continuity Mission)

Landsat 9

Laser Communications Relay

Launch Services Office

Laurel B. Clark

Lawrence J. Delucas

LBTI (Large Binocular Telescope Interferometer)

LCROSS (Lunar Crater Observation and Sensing Satellite)

Lee J. Archambault

Lee M. Morin

LEIA (Lunar Explorer Instrument for space biology Applications)

Leland D. Melvin

Lenticular Galaxies

Leonid K. Kadenyuk

Leonid MAC (Multi-instrument Aircraft Campaign)

Leonids

Leroy Chiao

Lessons Learned for Professionals

LGBTQ Pride

Libera

LIS (Lightning Imaging Sensor)

Linda M. Godwin

LISA (Laser Interferometer Space Antenna)

Lisa M. Nowak

Lodewijk Van Den Berg

LOFTID (Low-Earth Orbit Flight Test of an Inflatable Decelerator)

Logistics Management Division (LMD)

Logo & Media Usage

Loral O'Hara

Loren J. Shriver

Loren W. Acton

Low-Density Supersonic Decelerator

Low-Earth Orbit Economy

LRO (Lunar Reconnaissance Orbiter)

LSII Consortium

Luke Delany

LunaH-Map (Lunar Polar Hydrogen Mapper)

Lunar Discovery & Exploration Program

Lunar Lander Simulation

Lunar Orbiter

Lunar Orbiter 1

Lunar Orbiter 2

Lunar Orbiter 3

Lunar Orbiter 4

Lunar Orbiter 5

Lunar Prospector

Lunar Surface Technology Research

Lunar Trailblazer

Lyrids

M. Scott Carpenter

Mae C. Jemison

Magellan

Magnetars

Magnetosphere

MAIA (Multi-Angle Imager for Aerosols)

Management Astronauts

Manley Lanier Carter Jr.

Marcos Berrios

Margaret Rhea Seddon

Mariner 1

Mariner 2

Mariner 3

Mariner 4

Mariner 5

Mariner 6

Mariner 7

Mariner 8

Mariner 9

Mariner 10

Mario Runco Jr.

Mark C. Lee

Mark E. Kelly

Mark L. Polansky

Mark N. Brown

Mark T. Vande Hei

Mars Campaign Development Division

Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE)

Marsha Ivins

Marshall Test Facility and Support Infrastructure

Martian Moon Exploration (MMX)

Martin J. Fettman

Mary E. Weber

Mary L. Cleave

Materials Science Research Rack (MSRR)

Matthew Dominick

Megan McArthur

Merchandise Approvals

Mesosphere

Messier Catalog

Meteor Showers

MetOp (Meteorological Operational satellite)

Michael A. Baker

Michael Collins

Michael Fincke

Michael E. Fossum

Michael E. Lopez-Alegria

Michael J. Bloomfield

Michael J. Foreman

Michael J. Massimino

Michael J. McCulley

Michael J. Smith

Michael L. Coats

Michael L. Gernhardt

Michael P. Anderson

Michael R. Barratt

Michael R. Clifford

Michael S. Hopkins

Michael T. Good

Michel F. Curtis

Micro-10

Microbial Tracking (MT)

Millie Hughes-Fulford

Mini-RF Instrument

MinXSS (Miniature X-ray Solar Spectrometer)

Miranda

Mission Equity

Mission Support Directorate

MMS (Magnetospheric Multiscale)

Modeling, Analysis and Prediction (MAP) Program

Moderate Resolution Imaging Spectroradiometer (MODIS)

Modern Figures

Modular Optoelectric Multispectral Scanner (MOMS)

MOS (Marine Observation Satellite)

MSL Instrumentation

Multi-Spectral Fluorescence Imaging System (Spectrum)

MUSE (Multi-slit Solar Explorer)

Museum Alliance

N. Jan Davis

NAAMES (North Atlantic Aerosols and Marine Ecosystems Study)

Nancy J. Currie-Gregg

NanoSail-D

NASA Aeronautics Committee

NASA Engineering and Safety Center

NASA History Office

NASA Home & City

NASA Interactives

NASA iTech Program

NASA Knowledge Program

NASA Open Source Software

NASA Plus

NASA Safety Center

NASA Safety Center Professional Development

NASA Shared Services Center

NASA Social Media Contacts

NASA Socials Program

NASA STEM

NASA Worldwind

National Aeronautics Research Institute

National Space Council Users' Advisory Group (NSpC UAG)

NEA Scout (Near Earth Asteroid Scout)

NEAR Shoemaker

Near-Earth Object (NEO)

Nebulae Game

NEEMO (NASA Extreme Environment Mission Operations)

Neil A. Armstrong

Neil Gehrels Swift Observatory

Neil W. Woodward III

NEK/SIRIUS (Nezemnyy Eksperimental'nyy Kompleks/Scientific International Research In a Unique terrestrial Station)

NEO Surveyor (Near-Earth Object Surveyor Space Telescope)

NEOWISE

Neptune Moons

Neutral Buoyancy Lab

Neutron Stars

New Frontiers Program

NExIS (NASA's Exploration & In-space Services)

NIAC Studies

NIAC Symposium

NICER (Neutron star Interior Composition Explorer)

Nicholas J. M. Patrick

Nicole A. Mann

Nicole P. Stott

Nimbus

NISAR (NASA-ISRO Synthetic Aperture Radar)

NOAA-20 (JPSS-1)

NOAA-21 (JPSS-2)

NOAA-N Prime

Norman E. Thagard

Northrop Grumman Commercial Resupply

Nova-C (Lander)

Nozomi

NuSTAR (Nuclear Spectroscopic Telescope Array)

O / OREOS (Organism / Organic Exposure to Orbital Stresses)

Ocean Surface Topography Mission (OSTM) / Jason-2

Oceanography

Oceans Melting Greenland (OMG)

OCO (Orbiting Carbon Observatory)

OCO-2 (Orbiting Carbon Observatory 2)

OCO-3 (Orbiting Carbon Observatory 3)

Office of Legislative and Intergovernmental Affairs (OLIA)

Office of Strategic Infrastructure (OSI)

Office of Technology, Policy and Strategy (OTPS)

Office of the Chief Engineer (OCE)

Office of the Chief Financial Officer (OCFO)

Office of the Chief Health and Medical Officer (OCHMO)

Office of the Chief Information Officer (OCIO)

Office of the Chief Scientist (OCS)

Office of the General Counsel (OGC)

One Year Crew

ORACLES (ObseRvations of Aerosols above CLouds and their IntEractionS)

Orbiting Astronomical Observatory (OAO)

ORFEUS (Orbiting and Retrievable Far and Extreme Ultraviolet Spectrometer)-SPAS II

Orion Nebula

Orion Spacecraft

Orionids OSAM-1 (On-Orbit Servicing, Assembly, and Manufacturing 1)

Oumuamua

OV-10

Owen K. Garriott

Ozone Mapping and Profiler Suite (OMPS)

P-3

PACE (Plankton, Aerosol, Cloud, Ocean Ecosystem)

Pamela A. Melroy

Partner Astronauts

Patricia C. Robertson

Patrick G. Forrester

Paul D. Scully-Power

Paul J. Weitz

Paul S. Lockhart

Paul W. Richards

Peggy A. Whitson

Peregrine Lunar Lander

Perseids

Peter J. K. Wisoff

PFMI (Pore Formation and Mobility Investigation)

PharmaSat

Philae (Lander)

Philip K. Chapman

Phobos

Phoebe

Physical Sciences

PI Resources

Pierre J. Thuot

Piers J. Sellers

Pioneer

Pioneer 0 / Able 1

Pioneer 1 / Able 2

Pioneer 2

Pioneer 3

Pioneer 4

Pioneer 5

Pioneer 6

Pioneer 7

Pioneer 8

Pioneer 9

Pioneer 10

Pioneer 11

Pioneer P-3 / Able 4B

Pioneer P-30 / Able 5A

Pioneer P-31 / Able 5B

Pioneer Venus

Pioneer Venus 1

Pioneer Venus 2

Pioneer-E

Planck

Planetary Defense Coordination Office

Plasma Kristall 4 (PK-4)

Play

Play it!

Plume-Surface Interaction (PSI) Project

Pluto Moons

Polar

Predator

PREFIRE (Polar Radiant Energy in the Far-InfraRed Experiment)

PRESat (PharmaSat Risk Evaluation Satellite)

Prizes, Challenges & Crowdsourcing

Prizes, Challenges & Crowdsourcing News

Proxima Centauri b

Psyche Asteroid

Psyche Mission

Pulsars

PUNCH (Polarimeter to Unify the Corona and Heliosphere)

Q-PACE (CubeSat Particle Aggregation and Collision Experiment)

Quadrantids

Quasars

Quesst: The Flights

Quesst: The Vehicle

QuikSCAT (Quick Scatterometer)

RADARSAT-1 (Radar Satellite 1)

RAD-SEED

Raja Chari

Randolph J. Bresnik

Ranger

Ranger 1

Ranger 2

Ranger 3

Ranger 4

Ranger 5

Ranger 6

Ranger 7

Ranger 8

Ranger 9

Red Dwarfs

Red Giants

Redstone 3 (Freedom 7)

Redstone 4 (Liberty Bell 7)

Research & Analytics Program

Research Archive

Revolutionary Vertical Lift Technology

Rex J. Walheim

Rhea

RHESSI (Reuven Ramaty High Energy Solar Spectroscopic Imager)

Richard A. Mastracchio

Richard A. Searfoss

Richard F. Gordon Jr.

Richard H. Truly

Richard J. Hieb

Richard M. Linnehan

Richard M. Mullane

Richard N. Richards

Richard O. Covey

Richard R. Arnold

Rick D. Husband

Rings of Jupiter

Rings of Neptune

Rings of Saturn

Ring-Sheared Drop (RSD)

RLL (Robotic Lunar Lander)

Robb Kulin

Robert A. R. Parker

Robert C. Springer

Robert D. Cabana

Robert F. Overmyer

Robert J. Cenker

Robert L. Behnken

Robert L. Crippen

Robert L. Curbeam Jr.

Robert L. Gibson

Robert L. Satcher Jr.

Robert L. Stewart

Robert Shane Kimbrough

Robotic Lunar Exploration Program

Rodent Research (RR)

Roger B. Chaffee

Roger K. Crouch

Ronald A. Parise

Ronald E. Evans

Ronald E. McNair

Ronald J. Garan Jr.

Ronald J. Grabe

Ronald M. Sega

Rosalind Franklin (Rover)

ROSAT (ROentgen SATellite)

Rosetta

Roy D. Bridges Jr.

RS-25

Russell L. Schweickart

RXTE (Rossi X-ray Timing Explorer)

Ryugu

S. Christa McAuliffe

S. David Griggs

S3VI

SAC-B (Satélite de Aplicaciones Científicas-B)

SAGE III (Stratospheric Aerosol and Gas Experiment)

SAGE-III Meteor-3M (Stratospheric Aerosol and Gas Experiment III on Meteor-3M)

Sakigake

Sally K. Ride

SAMPEX (Solar Anomalous and Magnetospheric Particle Explorer)

Samuel T. Durrance

Sandra H. Magnus

SBIR

SBIR / STTR

Science Leadership

Science of Space Exploration

Science Solicitations

Scott D. Altman

Scott D. Tingle

Scott E. Parazynski

Scott J. Horowitz

Scott J. Kelly

SEAC4RS (Studies of Emissions, Atmospheric Composition, Clouds and Climate Coupling by Regional Surveys)

Seasat

SeaWiFS (Sea-viewing Wide Field-of-view Sensor)

SeaWinds

Sentinel-6 Michael Freilich Satellite

Sentinel-6B

Serena M. Auñón-Chancellor

Seres Humanos en el espacio

SERVIR (Regional Visualization and Monitoring System)

Seyfert Galaxies

Shannon W. Lucid

Shannon Walker

Sherwood C. Spring

Shoemaker-Levy 9

Shuttle-Mir

Sidney M. Gutierrez

Sierra UAS #2

SimLabs

SIR-C/X-SAR (Shuttle Imaging Radar-C / X-Band Synthetic Aperture Radar)

Skywatching Tips

Small Bodies of the Solar System

Small Business Innovation Research / Small Business

Small Spacecraft Technology Program

SMAP (Soil Moisture Active Passive)

SMART-1 (Small Missions for Advanced Research in Technology 1)

SNOE (Student Nitric Oxide Explorer)

Soft Matter Dynamics (SMD) / FOAM

Software to License

SOHO (Solar and Heliospheric Observatory)

Solar Cruiser

Solar Electric Propulsion (SEP)

Solar Flares

Solar Orbiter

Solar Sail – Beyond Plum Brook

Solar System Exploration Research Virtual Institute

Solar Terrestrial Probes Program

Solid Fuel Ignition and Extinction (SoFIE)

Solidification Using a Baffle in Sealed Ampoules (SUBSA)

SORCE (Solar Radiation and Climate Experiment)

SORTIE (Scintillation Observations & Response of the Ionosphere to Electrodynamics)

Sounds & Ringtones

Southern Delta Aquariids

Space Biosciences at Ames

Space Communications & Navigation Program (SCaN)

Space Environment Testbeds (SET-1)

Space Environments Testing Management Office (SETMO)

Space Flight Awareness (SFA)

SLS (Space Launch System)

Space Life & Physical Sciences Research & Applications Division

Space Nuclear Propulsion (SNP)

Space Operations Directorate Resource Management Office

Space Station Research Opportunities

Space Tech Graduate Research

Space Technology 5

Space Technology 6

Space Technology 7 / Disturbance Reduction System (DRS)

Space Vehicle Mockup Facility

Space Weather

Spartan

Spectrum Management

Spiral Galaxies

Spot the International Space Station

SRTM (Shuttle Radar Topography Mission)

ST5

Stanley G. Love

Stardust / Stardust NExT

Starliner

Stennis Test Facility and Support Infrastructure

Stephanie D. Wilson

Stephen D. Thorne

Stephen K. Robinson

Stephen N. Frick

Stephen S. Oswald

Steven A. Hawley

Steven L. Smith

Steven R. Nagel

Steven R. Swanson

Steven W. Lindsey

Story Musgrave

STP

Strategic Integration and Management Division

Strategic Partnerships Guidelines

Stratosphere

STS- 1

STS- 2

STS- 3

STS- 4

STS- 5

STS- 6

STS- 7

STS- 8

STS- 9

STS- 26

STS- 27

STS- 28

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STS- 30

STS- 31

STS- 32

STS- 33

STS- 34

STS- 35

STS- 36

STS- 37

STS- 38

STS- 39

STS- 40

STS- 41

STS- 41B

STS- 41C

STS- 41D

STS- 41G

STS- 42

STS- 43

STS- 44

STS- 45

STS- 46

STS- 47

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STS- 50

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STS- 133

STS- 134

STS- 135

Stuart A. Roosa

Submillimeter Wave Astronomy Satellite (SWAS)

SunRISE (Sun Radio Interferometer Space Experiment)

Sunspots

Suomi NPP (Suomi National Polar-orbiting Partnership)

Supermoons

Supernovae

Surveyor 1

Surveyor 2

Surveyor 3

Surveyor 4

Surveyor 5

Surveyor 6

Surveyor 7

Surveyor Model 1

Surveyor Model 2

Surveyor Model SD-1

Susan J. Helms

Susan L. Kilrain

Sustainable Flight National Partnership

Suzaku

Synchronous Meteorological Satellite (SMS)

T-38 Astronaut Trainer

Tamara E. Jernigan

Taylor Wang

Tech Demo Missions

Tech Dev Projects

Tech Portfolio

Technology Highlights

Tempel 1

Terence T. Henricks

Terra

Terrence W. Wilcutt

Terrestrial Planets

Terriers (Tomographic Experiment Using Radiative Recombinative Ionospheric EUV and Radio Sources)

Terry J. Hart

Terry W. Virts Jr.

TESS (Transiting Exoplanet Survey Satellite)

Tethered Satellite System (TSS)

TG-14

The Great Red Spot

The Habitable Zone

The Human Body in Space

The Milky Way

The North Star

THEMIS (Time History of Events and Macroscale Interactions During Substorms)

THEMIS-ARTEMIS (Time History of Events and Macroscale Interactions During Substorms – Acceleration, Reconnection, Turbulence and Electrodynamics of Moon's Interaction with the Sun)

Theodore C. Freeman

Thermosphere

Thomas D. Akers

Thomas D. Jones

Thomas H. Marshburn

Thomas J. Hennen

Thomas K. Mattingly II

Thomas P. Stafford

TIMED (Thermosphere Ionosphere Mesosphere Energetics and Dynamics Mission)

Timothy J. Creamer

Timothy L. Kopra

TIROS (Television Infrared Observation Satellite Program)

TOMS-EP (Total Ozone Mapping Spectrometer – Earth Probe)

TOPEX / Poseidon (ocean TOPography EXperiment)

Tornadoes

Total and Spectral Solar Irradiance Sensor – 1 (TSIS-1)

Total and Spectral Solar Irradiance Sensor – 2 (TSIS-2)

TP / ACO

TRACE (Transition Region and Coronal Explorer)

Trace Gas Orbiter (TGO)

Tracking Aerosol Convection Interactions Experiment (TRACER)

Tracking and Data Relay Satellite (TDRS)

Tracy Caldwell Dyson

Transformational Tools

Technologies

Triton

Tropical Rainfall Measuring Mission (TRMM)

TROPICS (Time-Resolved Observations of Precipitation Structure and Storm Intensity with a Constellation of Smallsats)

Troposphere

Tsunamis

TWINS (Two Wide-Angle Imaging Neutral-Atom Spectrometers)

Twins Study

Tyler N. Hague

UAS in the NAS

UAS Traffic Management

UH-1H

UH-II

Ulysses

Underway Recovery Test 10

Universe of Monsters

Upper Atmosphere Research Satellite (UARS)

Uranus Moons

Urban Air Mobility Simulation

USGS (United States Geological Survey)

Valles Marineris

Van Allen Probes

Vance D. Brand

Venus Express

Vertical Motion Simulator

Victor J. Glover

Viking 400

Virgil I. Grissom

Visible Infrared Imaging Radiometer Suite (VIIRS)

Walter Cunningham

Walter M. Schirra Jr.

Wendy B. Lawrence

WetLab-2

White Dwarfs

White Sands Test Facility and Support Infrastructure

Wild 2

Wilkinson Microwave Anisotropy Probe (WMAP)

William A. Anders

William A. Oefelein

William A. Pailes

William B. Lenoir

William C. McCool

William E. Thornton

William F. Fisher

William F. Readdy

William G. Gregory

William M. Shepherd

William R. Pogue

William S. McArthur Jr.

Wind Mission

Winston E. Scott

WIRE (Wide-Field Infrared Explorer)

Women's History Month

Xelene Lunar Lander (XL-1)

XMM-Newton (X-ray Multi-Mirror Newton)

XRISM (X-Ray Imaging and Spectroscopy Mission)

YO-3A

Yohkoh

Yvonne Darlene Cagle

Zena Cardman

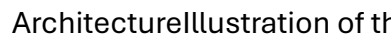
Zero Boil-Off Tank (ZBOT-NC)

URL: <https://www.nasa.gov/topics/moon-to-mars/>

TITLE: Destinations

Humans in SpaceDestinationsNASA is taking a steppingstone approach to human exploration in space. Building on NASA's 60 years of exploration experience and more than 20 years of continuous human presence on the International Space Station in low Earth orbit, we will extend humanity farther into space than ever before. The International Space Station has built the foundation to conduct complex operations in space, perform research in a microgravity environment, foster a growing space economy, and forge international partnerships toward a common goal. Artemis missions will establish our long-term presence at the Moon as astronauts explore more of the lunar surface than ever before to learn about the origins of the solar system and prepare for humanity's next giant leap: human missions to Mars.Quick FactsThe space station has been continuously occupied since November 2000.Missions to the Moon are about 1,000 times farther from Earth than missions to the International Space Station.Thanks to nearly 20 years of continuous human habitation on the space station, future life support systems for Mars can be designed with a 36% reduction in mass.An international partnership of five space agencies from 15 countries operates the International Space Station.ExploreLearn more about NASA's destinations for human exploration from the orbiting laboratory in low-Earth orbit, to Artemis missions at the Moon, and leading to the boldest mission yet: sending humans to Mars.All Humans in Space TopicsInternational Space StationAstronauts on the orbiting laboratory are helping us prove many of the technologies and communications systems needed for human missions in deep space, including to the Moon and Mars. The space station also advances our understanding of how the body reacts and changes in space and how to protect astronaut health.Read MoreEarth's MoonArtemis missions will collaborate with commercial and international partners to demonstrate innovative technologies on the lunar surface and around the Moon for scientific discovery and to prepare for human missions to Mars.Read MoreMarsEngineers and scientists around the country are developing the technologies astronauts will use to one day live and work on Mars and safely return home from humanity's next giant leap. NASA is working to address the challenges of living farther from Earth, such as using existing resources, options for disposing of trash, and more.Read MoreCommercial Destinations in Low Earth OrbitNASA's Commercial Low Earth Orbit Development Program is supporting the development of commercially owned and operated low Earth orbit destinations from which NASA, along with other customers, can purchase services and stimulate the growth of

commercial activities close to Earth.[Read More](#)[Architecture](#)[Moon to Mars](#)

[Architecture](#)NASA's Moon to Mars architecture represents the hardware and operations needed for human missions to the Moon and Mars, and how they function together as a system.The architecture is not a mission, a manifest, or a set of requirements, but it does define the elements — rockets, spacecraft, rovers, spacesuits, communications relays, and more — that will be incrementally developed and delivered to the Moon and Mars for long-term, human-led scientific discovery in deep space.[Learn More](#)[about Moon to Mars Architecture](#)NASA That's one small step for (a) man; one giant leap for mankind.Neil ArmstrongApollo astronaut[Latest News](#)[News Release](#)[5 Min Read](#)[NASA Sets Briefings for SpaceX Crew-12 Mission to Space Station](#)[Article](#)[4 Min Read](#)[NASA Selects Participants to Track Artemis II Mission](#)[5 Min Read](#)[NASA Astronaut Suni Williams Retires](#)[News Release](#)[3 Min Read](#)[NASA Unlocks Golden Age of Innovation, Exploration in Trump's First Year](#)[News Release](#)[25 Min Read](#)[Artemis II: What NASA Learned From Launching Artemis I](#)[Keep Exploring](#)[Discover More Topics From NASA](#)[International Space Station](#)[Artemis](#)[Humans to Mars](#)Like the Moon, Mars is a rich destination for scientific discovery and a driver of technologies that will enable humans...[Humans In Space](#)

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Explore Learn more about NASA's destinations for human exploration from the orbiting laboratory in low-Earth orbit, to Artemis missions at the Moon, and leading to the boldest mission yet: sending humans to Mars. All Humans in Space Topics International Space Station Astronauts on the orbiting laboratory are helping us prove many of the technologies and communications systems needed for human missions in deep space, including to the Moon and Mars. The space station also advances our understanding of how the body reacts and changes in space and how to protect astronaut health. Read More Earth's Moon Artemis missions will collaborate with commercial and international partners to demonstrate innovative technologies on the lunar surface and around the Moon for scientific discovery and to prepare for human missions to Mars. Read More Mars Engineers and scientists around the country are developing the technologies astronauts will use to one day live and work on Mars and safely return home from humanity's next giant leap. NASA is working to address the challenges of living farther from Earth, such as using existing resources, options for disposing of trash, and more. Read More Commercial Destinations in Low Earth Orbit NASA's Commercial Low Earth Orbit Development Program is supporting the development of commercially owned and operated low Earth orbit destinations from which NASA, along with other customers, can purchase services and stimulate the growth of commercial activities close to Earth. Read More

Explore Learn more about NASA's destinations for human exploration from the orbiting laboratory in low-Earth orbit, to Artemis missions at the Moon, and leading to the boldest mission yet: sending humans to Mars. All Humans in Space Topics

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All Humans in Space Topics

International Space Station Astronauts on the orbiting laboratory are helping us prove many of the technologies and communications systems needed for human missions in deep space, including to the Moon and Mars. The space station also advances our understanding of how the body reacts and changes in space and how to protect astronaut health. Read More Earth's Moon Artemis missions will collaborate with commercial and international partners to demonstrate innovative technologies on the lunar surface and around the Moon for scientific discovery and to prepare for human missions to Mars. Read

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Architecture
Moon to Mars Architecture
NASA's Moon to Mars architecture represents the hardware and operations needed for human missions to the Moon and Mars, and how they function together as a system. The architecture is not a mission, a manifest, or a set of requirements, but it does define the elements — rockets, spacecraft, rovers, spacesuits, communications relays, and more — that will be incrementally developed and delivered to the Moon and Mars for long-term, human-led scientific discovery in deep space.[Learn More](#)
about Moon to Mars Architecture
Illustration of three astronaut silhouettes
NASA

Architecture
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Illustration of three astronaut silhouettes
NASA

Illustration of three astronaut silhouettes

That's one small step for (a) man; one giant leap for mankind.
Neil Armstrong
Apollo astronaut

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Neil ArmstrongApollo astronaut

Neil Armstrong

Apollo astronaut

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Humans to MarsLike the Moon, Mars is a rich destination for scientific discovery and a driver of technologies that will enable humans...

Like the Moon, Mars is a rich destination for scientific discovery and a driver of technologies that will enable humans...

URL: <https://www.nasa.gov/humans-in-space/commercial-space/low-earth-orbit-economy/commercial-destinations-in-low-earth-orbit/>

TITLE: CommercialSpace Stations

Humans in SpaceCommercialSpace StationsNASA is supporting the development of commercially owned and operated space stations in low Earth orbit from which the agency, along with other customers, can purchase services and stimulate the growth of commercial activities in microgravity. NASA is using a phased approach to support the design, development, and demonstration of commercial space stations followed by a phase to finalize certification and procure services.[Read More](#)[about CommercialSpace Stations](#)[Featured Video](#)[America's Future in Low Earth Orbit](#)NASA plans to transition to a new low Earth orbit model featuring commercial space stations to continue leveraging the unique environment and microgravity research benefits.[Watch Now](#)[Commercial Space Station News](#)[Article3 min read](#)[NASA Seeks Industry Input on Next Phase of Commercial Space Stations](#)[Article2 min read](#)[NASA Sees Key Progress on Starlab Commercial Space Station](#)[Article2 min read](#)[NASA Helps with Progress on Vast's Haven-1 Commercial Space Station](#)[Article3 min read](#)[NASA Sees Progress on Blue Origin's Orbital Reef Design Development](#)[Article3 min read](#)[NASA, Axiom Space Change Assembly Order of Commercial Space Station](#)[Article3 min read](#)[NASA Sees Progress on Starlab Commercial Space Station Development](#)[Article6 min read](#)[NASA Commercial Partners Make Progress on Low Earth Orbit Projects](#)[Article3 min read](#)[NASA Quiet Space Fan Research to Benefit Commercial Space Stations](#)[Article3 min read](#)[NASA Supports Burst Test for Orbital Reef Commercial Space Station](#)[Article2 min read](#)[NASA Shares Use Requirements with Commercial Destination Partners](#)[Article3 min read](#)[NASA Shares Lessons of Human Systems Integration with Industry](#)[Article5 min read](#)[NASA Shares Medical Expertise with New Space Station Partners](#)[Article3 min read](#)[NASA Sees Progress on Blue Origin's Orbital Reef Life Support System](#)[Article4 min read](#)[NASA Adjusts Agreements to Benefit Commercial Station Development](#)[Article4 min read](#)[NASA's Commercial Partners Continue Progress on New Space Stations](#)[Article3 min read](#)[NASA Partners Combine Efforts for Low Earth Orbit Commercial Station](#)[Article4 min read](#)[NASA's Commercial Partners Pass Milestones for New Space Stations](#)[Article5 min read](#)[How NASA's Work Led to Commercial Spaceflight Revolution](#)[Article3 min read](#)[NASA's Commercial Partners Move Needle for Space Station Destinations](#)[Article4 min read](#)[NASA Seeks Input for Commercial Low Earth Orbit Destinations](#)[Article2 min read](#)[NASA Seeks Feedback on Commercial Destination Certification](#)[Keep Exploring](#)[Discover More Topics From NASA](#)[Low Earth Orbit Economy](#)[Humans In Space](#)[Private Astronaut Missions](#)[Commercial Space](#)

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Featured VideoAmerica's Future in Low Earth OrbitNASA plans to transition to a new low Earth orbit model featuring commercial space stations to continue leveraging the unique environment and microgravity research benefits.[Watch Now](#)

America's Future in Low Earth Orbit

NASA plans to transition to a new low Earth orbit model featuring commercial space stations to continue leveraging the unique environment and microgravity research benefits.

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Exhibition view of NASA's Earth Information CenterNASA/Dan Goods

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Destiny Doran International Space Station Research Communications Team Jan 15, 2026 [Article Contents](#) [Bolstering bone resilience](#) [Observing Earth and beyond](#) [Space catch](#) [Tracking internal temperature](#) [A new cargo vehicle](#) [Making nutrients on demand](#) [Celebrating a historic milestone](#) NASA's SpaceX Crew-11 mission with agency astronauts Zena Cardman and Mike Fincke, JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui, and Roscosmos cosmonaut Oleg Platonov returned to Earth after a long-duration mission aboard the International Space Station. During their stay, Cardman, Fincke, and Yui contributed more than 850 hours of research to help prepare humanity for the return to the Moon and future missions to Mars, while improving life back on Earth. Here's a glimpse into the science completed during the Crew-11 mission: [Bolstering bone resilience](#) NASA astronaut Zena Cardman works with bone stem cells aboard the International Space Station to improve our understanding of how bone loss occurs during spaceflight. Studying bone cell activity in microgravity could help researchers learn how to control bone loss to protect astronauts' bone density during future long-duration space missions and inform treatments for diseases like osteoporosis on Earth. [Learn more about MABL-B](#). [Observing Earth and beyond](#) JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui photographs the Earth from the International Space Station's

cupola. For more than 40 years, astronauts have used hand-held cameras to capture millions of images documenting Earth's geographic features, weather patterns, urban growth, changes to its surface, and the impacts of natural disasters such as hurricanes and floods. Astronauts also use the cupola and other viewports aboard the space station to gaze into the cosmos without Earth's atmospheric interference. Just as viewing Earth from 250 miles above provides a new perspective on our home planet, looking out into the stars from the orbiting laboratory offers a clearer view of our universe.

Space catch NASA astronaut Mike Fincke poses aboard the International Space Station with a new device designed to test an inflatable capture bag's ability to open, close, and stay airtight in microgravity. This technology could be used to remove space debris from orbit, protecting future spacecraft and crew members. It also may enable trapping samples during exploration missions and support the capture and mining of small asteroids. [Learn more about Capture Bag Demo.](#)

Tracking internal temperature NASA astronaut Mike Fincke wears a temperature-monitoring headband that tracks how the human body regulates its core temperature during spaceflight. Adjusting to living and working aboard the International Space Station can influence human temperature regulation. This headband provides an easy, non-invasive way to collect temperature data while astronauts conduct their daily activities. The sensor is also being tested on Earth and may help prevent hyperthermia in people working in high-temperature environments. [Learn more about T-Mini.](#)

A new cargo vehicle JAXA's (Japan Aerospace Exploration Agency) new cargo resupply spacecraft, HTV-X1, is shown after being captured by the International Space Station's Canadarm2 robotic arm during the Crew-11 mission. The spacecraft launched from Tanegashima Space Center on Oct. 26, 2025, delivering approximately 12,800 pounds of science, supplies, and hardware to the orbital complex. New cargo spacecraft expand the station's capability to support more research and receive critical supplies. [Making nutrients on demand](#)

JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui holds yogurt bags produced aboard the International Space Station that could provide important nutrients during missions far from Earth. Certain nutrients degrade when stored for long periods of time, and deficiency in even one can lead to illness. Researchers are building on previous experiments to develop a method for producing on-demand vitamins and nutrients in space using microorganisms. [Learn more about BioNutrients-3.](#)

Celebrating a historic milestone The Expedition 73 crew poses for a portrait to commemorate 25 years of continuous human presence aboard the International Space Station. In the front row from left, NASA astronaut Jonny Kim, Roscosmos cosmonaut Sergey Ryzhikov, and Roscosmos cosmonaut Alexey Zubritsky. In the back row, Roscosmos cosmonaut Oleg Platonov, NASA astronauts Mike Fincke and Zena Cardman, and JAXA (Japan Aerospace Exploration Agency) astronaut Kimiya Yui. A truly global endeavor, the space station has been visited by more than 290 people from 26 countries, along with a variety of international and

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ContentsBolstering bone resilienceObserving Earth and beyondSpace catchTracking internal temperatureA new cargo vehicleMaking nutrients on demandCelebrating a historic milestone

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Artemis II astronauts, from left, NASA astronaut Victor Glover (left), CSA (Canadian Space Agency) astronaut Jeremy Hansen, NASA astronauts Christina Koch and Reid Wiseman stand on the crew access arm of the mobile launcher at Launch Pad 39B as part of an integrated ground systems test at Kennedy Space Center in Florida on Wednesday, Sept. 20. The test ensures the ground systems team is ready to support the crew timeline on launch day. Photo Credit: NASA/Frank Michaux [Subscribe to Updates](#) [Domestic Engagement](#) [International Engagement](#) [Hear from us!](#) Fill out the form linked below if you'd like to be added to the mailing list for updates on NASA's Moon to Mars

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Moon to Mars Architecture – Deep Dive

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Subtopics

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Space TravelThe path to the Moon, Mars, and beyond requires technologies to get us where we need to go quickly, safely and efficiently. Space travel includes launch and in-space propulsion systems, cryogenic fluid management, and thermal management, as well as navigation and landing systems to get our supplies, equipment, and robotic or human explorers to diverse surface destinations.

Living in SpaceTechnologies for living in space enable our robotic and human explorers to achieve their mission goals. This means everything from having power to do their work, to the habitats or spacecraft they'll call home. Our astronauts also need technologies to make food and eat, exercise, sleep and maintain their overall wellbeing.

Manufacturing and MaterialsAdvanced manufacturing technologies for both terrestrial and in-space purposes will make commercial and exploration missions more efficient and affordable. We're developing new materials with improved or combined properties, learning how to build infrastructure and innovating on manufacturing processes.

RoboticsNASA uses robotic systems to explore other planets and objects in our solar system as precursors to crewed missions, assist astronauts on the International Space Station, study the universe, and much more. This also includes autonomous systems that enable robotics, spacecraft and aircraft to operate in a dynamic environment independent of external control.

Science InstrumentsScience instruments are state-of-the-art tools for acquiring information about planetary geology, atmosphere, and environmental conditions. This includes technologies for remote sensing, mapping and observation capabilities, analyzing light from the universe, and even searching for signs of life.

High-Tech ComputingSpace-qualified computing technology is designed to address the most computationally-intensive part of a mission. But for the future, NASA is going to need more power and more processing capabilities. We also work on modeling and simulation tech, software, and unique electronics and computing hardware for flight systems, whether in space or the atmosphere.

Space CommunicationsReliable space communication systems are critical to every NASA mission. Spacecraft commands, never-before-seen images, and scientific data are sent and received daily by NASA's giant antennas on Earth. From the Voyager mission

exploring beyond our solar system, to astronauts onboard the International Space Station, space communications provide the crucial connection to our home planet. Starling NASA's Starling mission is advancing the readiness of various technologies for cooperative groups of spacecraft – also known as distributed missions, clusters, or swarms. Starling will demonstrate technologies to enable multipoint science data collection by several small spacecraft flying in swarms. The six-month mission will use four CubeSats in low-Earth orbit to test four technologies that let spacecraft operate in a synchronized manner without resources from the ground. Visit the Starling Mission Page about Starling NASA Technology 19 Images Go To Gallery Go To Gallery Podcasts Explore NASA's podcasts to learn more about how NASA Technology is ever evolving. Learn More about Podcasts Your Orbiting Laboratory Station Technology Demonstration Since the first crew's arrival aboard more than twenty years ago, the International Space Station has evolved into a state-of-the-art scientific lab. Learn More iss069e018537 (June 9, 2023) — NASA astronaut and Expedition 68 Flight Engineer Woody Hoburg rides the Canadarm2 robotic arm while maneuvering a roll-out solar array toward the International Space Station's truss structure 257 miles above the Pacific Ocean. In the rear, is the SpaceX Dragon crew vehicle that docked to the Harmony module's forward port on March 3 carrying four SpaceX Crew-6 crew members. faces of technology videos Go to Faces of Tech Video Playlist Lindsay Kaldon Fission Surface Power, Project Manager at NASA's Glenn Research Center Jason Frieman Research Electrical Engineer at NASA's Glenn Research Center Beau Compton Aerospace Engineer at NASA's Glenn Research Center Wanessa Priesmeyer Flight Test Campaign Manager for NASA's Flight Opportunities program Lauren Best Ameen Deputy Manager for the Cryogenic Fluid Management Portfolio Project Office at NASA Paromita Mitra Avionics Systems Manager at NASA Women of Technology Celebrating the Women who Bring NASA Technology to Life Angela Herblet Manager for NASA's Prize Competitions Kellin Slater Project Manager at Colorado State University Amber Soja Program Manager at NASA's Langley Research Center Emily Gargulinski NASA Research Engineer at the National Institute of Aerospace Ali Guarneros Luna Aerospace Engineer at NASA's Ames Research Center Anjie Emmett Mechanical Engineer at NASA's Langley Research Center Prital Johnson Mechanical Engineer at NASA's Kennedy Space Center Kevin Grossman Materials Scientist at NASA's Kennedy Space Center Bob Carter Materials Engineer at NASA's Glenn Research Center Social Media @NASA_Technology @NASATechnology NASA Space Tech Channel Keep Exploring Explore Technology Areas Space Technology Mission Directorate Artemis Space Station Technology Demonstration Missions

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Why the Moon's Icy South Pole is a Hot Target for NASA

Ranked Civil Space Shortfalls STMD is refining its strategy for prioritizing technology investments, with the goal of evolving into a stronger and more resilient national tech base for civil space. [Learn More](#)

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NASA's Next-Generation Solar Sail Mission

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NASA Bolsters Golden Age of Exploration with Technology Priorities

Explore Technology Subtopics

Technology Transfer and Spinoffs The NASA Technology Transfer program ensures that innovations developed for exploration and discovery are broadly available to the public, maximizing the benefit to the nation. NASA ensures inventions for space find practical uses closer to home. They often become solutions to different challenges – ones we didn't set out to solve. Called NASA spinoffs, these technologies benefit life on Earth in the form of commercial products.

Space Travel The path to the Moon, Mars, and beyond requires technologies to get us where we need to go quickly, safely and efficiently. Space travel includes launch and in-space propulsion systems, cryogenic fluid management, and thermal management, as well as navigation and landing systems to get our supplies, equipment, and robotic or human explorers to diverse surface destinations.

Living in Space Technologies for living in space enable our robotic and human explorers to achieve their mission goals. This means everything from having power to do their work, to the habitats or spacecraft they'll call home. Our astronauts also need technologies to make food and eat, exercise, sleep and maintain their overall wellbeing.

Manufacturing and Materials Advanced manufacturing technologies for both terrestrial and in-space purposes will make commercial and exploration missions more efficient and affordable. We're developing new materials with improved or combined properties, learning how to build infrastructure and innovating on manufacturing processes.

Robotics NASA uses robotic systems to explore other planets and objects in our solar system as precursors to crewed missions, assist astronauts on the International Space Station, study the universe, and much more. This also includes autonomous systems that enable robotics, spacecraft and aircraft to operate in a dynamic environment independent of external control.

Science Instruments Science instruments are state-of-the-art tools for acquiring information about planetary geology, atmosphere, and environmental conditions. This includes technologies for remote sensing, mapping and observation capabilities, analyzing light from the universe, and even searching for signs of life.

High-Tech Computing Space-qualified computing technology is designed to address the most computationally-intensive part of a mission. But for the future, NASA is going to need more power and more processing capabilities. We also work on modeling and simulation tech, software, and unique electronics and computing hardware for flight systems, whether in space or the atmosphere.

Space Communications Reliable space communication systems are critical to every NASA mission. Spacecraft commands, never-before-seen images, and scientific data are sent and received daily by NASA's giant antennas on Earth. From the Voyager mission exploring beyond our solar system, to astronauts onboard the International Space Station, space communications provide the crucial connection to our home planet.

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Space Communications

Starling NASA's Starling mission is advancing the readiness of various technologies for cooperative groups of spacecraft – also known as distributed missions, clusters, or swarms. Starling will demonstrate technologies to enable multipoint science data collection by several small spacecraft flying in swarms. The six-month mission will use four CubeSats in low-Earth orbit to test four technologies that let spacecraft operate in a synchronized manner without resources from the ground. [Visit the Starling Mission Page](#) [about Starling](#)

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Station Technology Demonstration

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[iss069e018537](#) (June 9, 2023) — NASA astronaut and Expedition 68 Flight Engineer Woody Hoburg rides the Canadarm2 robotic arm while maneuvering a roll-out solar array toward the International Space Station's truss structure 257 miles above the Pacific Ocean. In the rear, is the SpaceX Dragon crew vehicle that docked to the Harmony module's forward port on March 3 carrying four SpaceX Crew-6 crew members.

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Loura HalJan 12, 2026ArticleNASAAs NASA prepares for long-duration missions to the Moon that will pave the way for human exploration on Mars, the agency is tapping into America's expanding space economy to help guide its strategic technology investments. This initiative, led by NASA's Space Technology Mission Directorate invites collaboration from U.S. industry leaders, academic institutions, and other government agencies to help prioritize critical technology development needs – known as shortfalls – identified for future science and exploration missions. "NASA wants to hear directly from the nation's brightest minds to drive solutions for our greatest technology needs as we lead America's exploration through the solar system," said Greg Stover, acting associate administrator for NASA's Space Technology Mission Directorate (STMD) at NASA Headquarters in Washington. "As we pursue collaboration with industry to support our most ambitious missions and increase agility, prioritizing NASA's technology efforts ensures the most efficient and impactful progress for the agency and its stakeholders." Until Friday, Feb. 20, NASA will collect input from the aerospace community on consolidated technology shortfalls, such as developing infrastructure and capabilities for long-term operations in the lunar and Martian environments. Technology stakeholders will participate in virtual meetings, provide feedback, and submit their shortfall ranking to the agency. This effort builds on NASA's first shortfall ranking exercise in 2024 which asked participants to rank 187 civil space shortfalls, resulting in an integrated list of technology priorities. Based on the invaluable feedback provided by stakeholders in the first exercise, NASA has streamlined the process by consolidating the shortfalls into 32 broader, integrated categories, each addressing specific needs to provide further definition and context. This restructuring maintains the original content's depth while creating a more efficient and accessible feedback mechanism for participants. NASA will analyze and aggregate the rankings to produce priority lists for each stakeholder group, which will be made publicly available for continued collaboration. This prioritization framework will guide NASA's evaluation of current technology development efforts to identify necessary adjustments within its existing portfolios. The shortfall prioritization process may inspire new investments within NASA or spark innovative partnerships with external stakeholders. This initiative also has the potential to unlock emerging commercial opportunities and accelerate growth in the U.S. space economy. As NASA nears its next mission to the Moon, prioritizing the most important and impactful efforts helps NASA appropriately direct available resources to best support mission needs for the agency and the nation. To maintain this collaborative approach, STMD plans to conduct feedback sessions and workshops every three years with industry, academia, and other government agencies,

creating a dynamic process that continuously incorporates stakeholder insights and end-user perspectives. The agency remains committed to refining this engagement framework, ensuring it delivers maximum value to all participants while advancing America's leadership in space exploration and technology development. To review the list of technology shortfalls and add input to NASA Space Technology's prioritization effort, visit: <https://www.nasa.gov/spacetechpriorities> By: Jasmine Hopkins Facebook logo@NASATechnology@NASA_Technology

Laura Hall Jan 12, 2026 Article

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Space Technology Mission DirectorateSpace Technology Mission DirectorateTechnology drives exploration and the space economy. NASA's Space Technology Mission Directorate (STMD) aims to transform future missions while ensuring American leadership in aerospace. STMD develops, demonstrates, and transfers new space technologies that benefit NASA, commercial, and other government missions.Learn More About STMDabout Space Technology Mission DirectorateRanked Civil Space ShortfallsNASA invites you to contribute to the prioritization of technology shortfalls critical to civil space by February 20. Your feedback will help shape the future of space technology.Submit FeedbackLatest NewsMore Space Tech NewsArticle1 Min ReadNIH Supplements, Facts First ChallengeArticle4 Min ReadNASA Back for Seconds with New Food System Design Challenge1 Min Read2026 Civil Space Shortfall RankingArticle2 Min ReadComplement-ARIE NAMs Reduction to Practice ChallengeArticle3 Min ReadNASA Bolsters Golden Age of Exploration with Technology PrioritiesArticleStarlingNASA's Starling mission is advancing the readiness of various technologies for cooperative groups of spacecraft – also known as distributed missions, clusters, or swarms. Starling will demonstrate technologies to enable multipoint science data collection by several small spacecraft flying in swarms. The six-month mission will use four CubeSats in low-Earth orbit to test four technologies that let spacecraft operate in a synchronized manner without resources from the ground.Visit the Starling Mission Pageabout StarlingProgramsCenter Innovation FundThe purpose of the Center Innovation Fund is to stimulate and encourage creativity and innovation within the NASA Centers in addressing the technology needs of NASA and the nation.Go to Center Innovation FundFlight OpportunitiesFlight Opportunities rapidly demonstrates promising technologies for space exploration, discovery, and the expansion of space commerce through suborbital testing with industry flight providers. The program matures capabilities needed for NASA missions and commercial applications while strategically investing in the growth of the U.S. commercial spaceflight industry.Go to Flight OpportunitiesGame Changing DevelopmentGCD seeks to identify and rapidly mature high-impact capabilities

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Rachel Barry Jan 23, 2026 Article A close-up view of a portion of a "relatively fresh" crater, looking southeast, as photographed during the third Apollo 15 lunar surface moonwalk. Credit: NASA A new NASA study of its Apollo lunar soils clarifies the Moon's record of meteorite impacts and timing of water delivery. These findings place upper bounds on how much water meteorites could have supplied later in Earth's history. Research has previously shown that meteorites may have been a significant source of Earth's water as they bombarded our planet early in the solar system's development. In a paper published Tuesday in the Proceedings of the National Academy of Sciences, researchers led by Tony Gargano, a postdoctoral fellow at NASA's Johnson Space Center and the Lunar and Planetary Institute (LPI), both in Houston, used a novel method for analyzing the dusty debris that covers the Moon's surface called regolith. They learned that even under generous assumptions, meteorite delivery since about four billion years ago could only have supplied a small fraction of Earth's water. The Moon serves as an ancient archive of the impact history the Earth-Moon system has experienced over billions of years. Where Earth's dynamic crust and weather erase such records, lunar samples preserve them. The records don't come without challenge, though. Traditional methods of studying regolith have relied on analyzing metal-loving elements. These elements can get muddled by repeated impacts on the Moon, making it harder to untangle and reconstruct what the original meteoroids contained. Enter triple oxygen isotopes, high precision "fingerprints" that take advantage of the fact that oxygen, the dominant element by mass in rocks, is unaffected by impact or other external forces. The isotopes offer a clearer understanding of the composition of meteorites that impacted the Earth-Moon system. The oxygen-isotope measurements revealed that at least ~1% by mass of the regolith contained material from carbon-rich meteorites that were partially vaporized when they hit the Moon. Using the known properties of such meteorites allowed the team to calculate the amount of water that would have been carried within. "The lunar regolith is one of the rare places we can still interpret a time-integrated record of what was hitting Earth's neighborhood for billions of years," said Gargano. "The oxygen-isotope fingerprint lets us pull an impactor signal out of a mixture that's been melted, vaporized, and reworked countless times." The findings have implications for our understanding of water sources on Earth and the Moon. When scaled up by roughly 20 times to account for the substantially higher rate of impacts on Earth, the cumulative water shown in the model made up only a small percent of the water in Earth's

oceans. That makes it difficult to reconcile the hypothesis that late delivery of water-rich meteorites was the dominant source of Earth's water. "Our results don't say meteorites delivered no water," added co-author Justin Simon, a planetary scientist at NASA Johnson's Astromaterials Research and Exploration Science Division. "They say the Moon's long-term record makes it very hard for late meteorite delivery to be the dominant source of Earth's oceans." For the Moon, the implied delivery since about 4 billion years ago is tiny on an Earth-ocean scale but is not insignificant for the Moon. The Moon's accessible water inventory is concentrated in small, permanently shadowed regions at the North and South Poles. These are some of the coldest spots in the solar system and introduce unique opportunities for scientific discovery and potential resources for lunar exploration when NASA lands astronauts on the Moon through Artemis III and beyond. The samples analyzed for this study came from parts of the Moon near the equator on the side of the Moon facing Earth, where all six Apollo missions landed. The rocks and dust collected more than 50 years ago continue to reveal new insights but are constrained to a small portion of the Moon. Samples delivered through Artemis will open the door for a new generation of discoveries for decades to come. "I'm part of the next generation of Apollo scientists — people who didn't fly the missions, but who were trained on the samples and the questions Apollo made possible," said Gargano. "The value of the Moon is that it gives us ground truth: real, physical material we can measure in the lab and use to anchor what we infer from orbital data and telescopes. I can't wait to see what the Artemis samples have to teach us and the next generation about our place in the solar system." For more information on NASA's Astromaterials Research and Exploration Science Division, visit: <https://science.nasa.gov/astromaterials> Karen Fox / Molly Wasser Headquarters, Washington 240-285-5155 / 240-419-1732 karen.c.fox@nasa.gov / molly.l.wasser@nasa.gov Victoria Segovia NASA's Johnson Space Center 281-483-5111 victoria.segovia@nasa.gov Read More Facebook logo@NASASolarSystem @NASASolarSystem Instagram logo@NASASolarSystem

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A close-up view of a portion of a "relatively fresh" crater, looking southeast, as photographed during the third Apollo 15 lunar surface moonwalk. Credit: NASA A new NASA study of its Apollo lunar soils clarifies the Moon's record of meteorite impacts and timing of water delivery. These findings place upper bounds on how much water meteorites could have supplied later in Earth's history. Research has previously shown that meteorites may have been a significant source of Earth's water as they bombarded our planet early in the solar system's development. In a paper published Tuesday in the Proceedings to the

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Pioneering spaceflight and innovation

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TITLE: NASA's Day of Remembrance Honors Fallen Heroes of Exploration

3 min readNASA's Day of Remembrance Honors Fallen Heroes of Exploration

Jennifer M. DoorenJan 20, 2026RELEASE26-008NASA HeadquartersThe Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA's Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration.Credit: NASA/Bill IngallsNASA will observe its annual Day of Remembrance on Thursday, Jan. 22, which includes commemorating the crews of Apollo 1 and the space shuttles Challenger and Columbia. The event is traditionally held every year on the fourth Thursday of January, as all three astronaut accidents happened around the end of the month.“On NASA's Day of Remembrance, we pause to honor the members of the NASA family who lost their lives while pushing the boundaries of exploration and discovery,” said NASA Administrator Jared Isaacman. “We remember them not to retreat from risk, but to respect it — to learn, to improve, and continue onward. Their sacrifice and the strength of their families will forever inspire us as we continue to reach for the stars and pursue the secrets of the universe.”Isaacman will lead an observance at 1 p.m. EST at Arlington National Cemetery in Virginia, which will begin with a wreath-laying ceremony at the Tomb of the Unknown Soldier, followed by observances for the Apollo 1, Challenger, and Columbia crews.Several agency centers also will hold observances:Johnson Space Center in HoustonNASA Johnson will hold a commemoration at 10 a.m. CST at the Astronaut Memorial Grove with remarks by Center Director Vanessa Wyche, NASA astronaut Jasmin Moghbeli, and Cheryl McNair, widow of Challenger astronaut Ronald McNair. The event will have a moment of silence, a NASA T-38 flyover, taps performed by the Texas A&M Squadron 17, and a procession placing flowers at Apollo I, Challenger, and Columbia memorial trees.Kennedy Space Center in FloridaNASA Kennedy and the

Astronauts Memorial Foundation will host a ceremony at the Space Shuttle Atlantis building at Kennedy's Visitor Complex at 11 a.m. EST. The event will include musical guests, a bell ringing commemoration, a moment of silence, and wreath-laying. Kelvin Manning, deputy director at NASA Kennedy, and Bob Cabana, former NASA associate administrator and Kennedy center director, will provide remarks during the ceremony, which will livestream on the center's Facebookpage.Ames Research Center in California's Silicon ValleyNASA Ames will hold a remembrance ceremony at 1 p.m. PST that includes remarks from Center Director Eugene Tu, a moment of silence, and bell ringing commemoration for each astronaut lost in service.Langley Research Center in Hampton, VirginiaNASA Langley will hold a remembrance ceremony at 1 p.m. EST with acting Center Director Trina Dyal, followed by placing flags at the Langley Workers Memorial.Marshall Space Flight Center in Huntsville, AlabamaNASA Marshall will hold a candle-lighting ceremony and wreath placement at 9:30 a.m. CST and include remarks from Rae Ann Meyer, Marshall's acting center director, and Bill Hill, director of Safety and Mission Assurance at Marshall.Stennis Space Flight Center in Bay St. Louis, MississippiNASA Stennis and the NASA Shared Services Center will hold a wreath-laying ceremony and moment of silence at 10:30 a.m. CST with remarks from Center Director John Bailey and Anita Harrell, NASA Shared Services Center executive director.The agency also is paying tribute to its fallen astronauts with special online content, updated on NASA's Day of Remembrance, at:[https://www.nasa.gov/dor-end-Bethany Stevens / Elizabeth Shaw](https://www.nasa.gov/dor-end-Bethany-Stevens-Elizabeth-Shaw)Headquarters, Washington202-358-1600bethany.c.stevens@nasa.gov/elizabeth.a.shaw@nasa.gov

Jennifer M. DoorenJan 20, 2026RELEASE26-008NASA Headquarters

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The Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA's Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration.Credit: NASA/Bill IngallsNASA will observe its annual Day of Remembrance on Thursday, Jan. 22, which includes commemorating the crews of Apollo 1 and the space shuttles Challenger and Columbia. The event is traditionally held every year on the fourth Thursday of January, as all three astronaut accidents happened around the end of the month.“On NASA's Day of Remembrance, we pause to honor the members of the NASA family who lost their lives while pushing the boundaries of exploration and discovery,” said NASA Administrator Jared Isaacman. “We remember them not to retreat from risk, but to respect it — to learn, to improve, and

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The Space Shuttle Columbia Memorial is seen during a wreath laying ceremony that was part of NASA’s Day of Remembrance, Thursday, Jan. 23, 2025, at Arlington National Cemetery in Virginia. Wreaths were laid in memory of those men and women who lost their lives in the quest for space exploration. Credit: NASA/Bill Ingalls

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Langley Research Center in Hampton, Virginia

NASA Langley will hold a remembrance ceremony at 1 p.m. EST with acting Center Director Trina Dyal, followed by placing flags at the Langley Workers Memorial.

Marshall Space Flight Center in Huntsville, Alabama

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Stennis Space Flight Center in Bay St. Louis, Mississippi

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inviting the public to join the agency's Artemis II test flight as four astronauts venture around the Moon and back to test systems and hardware needed for deep space exploration. As part of the agency's "Send Your Name with Artemis II" effort, anyone can claim their spot by signing up today. Participants can download a boarding pass with their name on it as a collectable. Send Your Name Around the Moon about Launch Your Name Around Moon in 2026 on NASA's Artemis II Mission Boarding passes will carry participants' names on NASA's Artemis II mission in 2026. Credit: NASA Dive Into NASA STEM Search STEM Resources Use the STEM Search tool to explore hands-on activities, interactive features, videos, facilitated learning guides, and more! Explore Opportunities For Students Multiple STEM challenges and opportunities reaching a broad audience of middle and high schools, technical colleges, and universities across the nation. Science for Everyone With so many ways to connect with and experience NASA's mission of scientific exploration and discovery, there's something here to awaken the curiosity in everyone. Discover STEM Careers at NASA The Surprisingly STEM video series provides a behind-the-scenes look at just a few of the unexpected roles that help NASA accomplish its goals in space and on Earth. Explore the series and learn about other NASA STEM careers in the NASA Workforce and Careers. Sign-up for the NASA EXPRESS Newsletter Keep up with the latest NASA STEM happenings by subscribing to the NASA EXPRESS newsletter. Find crafting ideas, science experiments with household items and videos to watch as a family. Updates on workshops, internships, contests and student challenges are also included. Explore Resources for International Audiences NASA fosters international collaboration to support joint missions and research efforts, while also offering engaging STEM activities—such as challenges and online resources—available globally within the existing framework of the STEM Engagement program. get social with nasa stem engagement NASA Kahoot! Learn With NASA YouTube Learn With NASA Pinterest Learn With NASA X Learn With NASA Facebook Keep Exploring Discover More Topics From NASA e-Books Requests for Exhibits, Artifacts, or Speakers Podcasts Games and Interactives

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2026 on NASA's Artemis II MissionBoarding passes will carry participants' names on
NASA's Artemis II mission in 2026.Credit: NASA

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Challenge: Build a Space Biology Knowledge Engine
Country/Territory: Germany
Team SpaceGenes+ created an interactive dashboard designed to help researchers uncover how radiation and microgravity together impact astronaut health at the molecular level. It gives researchers and mission planners an easy way to identify important molecular changes, supporting more effective protection strategies for long-duration spaceflight.
[Learn more about SpaceGenes+’ project](#)

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[Learn more about Resonant Exoplanets’ project](#)

Best Use of Technology Award: Twisters
Team Members: Fernando A., Marcelo T., Mariana D., Regina R., Regina F.
Challenge: Will It Rain on My Parade?
Country/Territory: Mexico
Team Twisters developed SkySense, a web-app platform that uses NASA Earth observation data and AI analysis to provide ultra-local, personalized weather predictions and to analyze weather variables such as rain, wind, temperature, humidity, and visibility, generating real-time risk assessments and suggesting the safest time windows for activities.
[Learn more about Twisters’ project](#)

Galactic Impact Award: Astro Sweepers: We Catch What Space Leaves Behind
Team Members: Harshiv T., Pragathy S., Pratik J., Sherlin D., Yousra H., Zienab E.
Challenge: Commercializing Low Earth Orbit (LEO)
Country/Territory: Universal
Team Astro Sweepers developed an end-to-end orbital debris compliance and risk intelligence platform that automatically ingests public orbital data to generate Debris Assessment Software reports and compute the Astro Sweepers Risk Index for every resident space object. This project considers the operational, regulatory, and environmental challenges of commercialized space travel.
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Most Inspirational Award: Photonics Odyssey
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Country/Territory: India
 Photonics Odyssey reimagined satellite internet as a sovereign national infrastructure rather than a private service, proposing a phased-array antenna approach that reduces ground dependency and expands broadband access to remote regions of India. The concept aims to help connect more than 700 million people who lack access to broadband internet. Learn more about Photonics Odysseys' project

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Art & Technology Award: Zumorroda-X
Team Members: Alaa A., Esraa A., Malak S., Mennatulla E.
Challenge: NASA Farm Navigators: Using NASA Data Exploration in Agriculture
Country/Territory: Egypt
 Team Zumorroda-X created mini games that allow players to step into the shoes of a farmer who sets off on an epic journey around the world. Through this game, players can learn how farmers globally adapt to heat waves, flooding, and other environmental challenges. Learn more about Zumorroda-Xs' project

Local Impact Award: QUEÑARIS
Team Members: Borax Q., Carlos Y., Marcelo S., Máximo S., Oscar M., Pamela P.
Challenge: BloomWatch: An Earth Observation Application for Global Flowering Phenology
Country/Territory: Peru
 Team QUEÑARIS' project addresses critical water scarcity in Peru's second-largest city, Arequipa, caused by the degradation of queñua forests, which are vital for water retention. Their platform combines native microorganisms, NASA satellite data, drones, and artificial intelligence to accelerate tree growth, identify the best areas for reforestation, and monitor ecosystem health. Learn more about QUEÑARIS' project

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More than 114,000 participants came together to address challenges created by NASA subject matter experts. These challenges ranged in complexity and topic, tasking participants with everything from leveraging artificial intelligence, to improving access to NASA research, and developing tools to evaluate air quality.

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scientists, regional decision-makers and members of the public, this challenge demonstrates the excitement of discovery and the real-world applications of agency data. Space apps also fosters a global community of creative and innovative ideas.”

The winners were determined from more than 11,500 project submissions and judged by subject matter experts from NASA and agency partners:

Best Use of Science Award: SpaceGenes+Team Members: Saloni T.Challenge: Build a Space Biology Knowledge EngineCountry/Territory: Germany

Team SpaceGenes+ created an interactive dashboard designed to help researchers uncover how radiation and microgravity together impact astronaut health at the molecular level. It gives researchers and mission planners an easy way to identify important molecular changes, supporting more effective protection strategies for long-duration spaceflight.Learn more about SpaceGenes+’ project

Best Use of Data Award: Resonant ExoplanetsTeam Members: Adhvaidh S., Gabriel S., Jack A., Sahil S.Challenge: A World Away: Hunting for Exoplanets with AICountry/Territory: United StatesTeam Resonant Exoplanets developed an AI-powered system that ingests large sets of telescope and satellite data, including spectra from missions like the James Webb Space Telescope. This tool automatically analyzes data for exoplanets and detects possible biosignatures, rather than identifying them manually.Learn more about Resonant Exoplanets’ project

Best Use of Technology Award: TwistersTeam Members: Fernando A., Marcelo T., Mariana D., Regina R., Regina F.Challenge: Will It Rain on My Parade?Country/Territory: MexicoTeam Twisters developed SkySense, a web-app platform that uses NASA Earth observation data and AI analysis to provide ultra-local, personalized weather predictions and to analyze weather variables such as rain, wind, temperature, humidity, and visibility, generating real-time risk assessments and suggesting the safest time windows for activities.Learn more about Twisters’ project

Galactic Impact Award: Astro Sweepers: We Catch What Space Leaves BehindTeam Members: Harshiv T., Pragathy S., Pratik J., Sherlin D., Yousra H., Zienab E.Challenge: Commercializing Low Earth Orbit (LEO)Country/Territory: Universal EventTeam Astro Sweepers developed an end-to-end orbital debris compliance and risk intelligence platform that automatically ingests public orbital data to generate Debris Assessment Software reports and compute the Astro Sweepers Risk Index for every resident space object. This project considers the operational, regulatory, and environmental challenges of commercialized space travel.Learn more about Astro Sweepers’ project

Best Mission Concept Award: PureFlowTeam Members: Esthefany M., João F., Laiza L., Lara D., Pedro H., Thayane D.**Challenge:** Your Home in Space: The Habitat Layout
CreatorCountry/Territory: BrazilPureFlow developed an interactive systems engineering platform that allows users to design, model in 3D, and validate space habitats, and then test the design against real space-weather threats, such as solar storms. This system considers the critical functions required for living in space, including waste management, power, life support, communications, and more.Learn more about PureFlows' project

Most Inspirational Award: Photonics OdysseyTeam Members: Manish D., Deeraj K., Prasanth G., Rajalingam N., Rashi M., Sakthi R.**Challenge:** Commercializing Low Earth Orbit (LEO)**Country/Territory:** IndiaPhotonics Odyssey reimagined satellite internet as a sovereign national infrastructure rather than a private service, proposing a phased-array antenna approach that reduces ground dependency and expands broadband access to remote regions of India. The concept aims to help connect more than 700 million people who lack access to broadband internet.Learn more about Photonics Odysseys' project

Best Use of Storytelling Award: HerCode SpaceTeam Members: Alice R., Joselyn R., Paula C., Pierina J.**Challenge:** Stellar Stories: Space Weather Through the Eyes of Earthlings**Country/Territory:** Universal EventHerCode Space combined NASA data and heliophysics concepts with powerful storytelling and vibrant illustrations to teach kids how space weather affects daily life and why it matters. HerCode Science hopes their story, "A Solar Tale," can bridge science and imagination, and bring heliophysics to life in classrooms, libraries, and outreach programs.Learn more about HerCode Spaces' project

Global Connection Award: Gaia+LEOTeam Members: Adam H., Katia L., Prajwal S., Upendra K.**Challenge:** Commercializing Low Earth Orbit (LEO)**Country/Territory:** United StatesTeam Gaia+LEO developed a mixed-integer optimization framework that co-designs orbital and terrestrial data-center networks to support large-scale AI training and climate modeling in orbit. Their goal is to reduce the power, and water demands of Earth-based systems and help accelerate the shift toward space-based, green computing within the emerging orbital economy.Learn more about Gaia+LEOs' project

Art & Technology Award: Zumorroda-XTeam Members: Alaa A., Esraa A., Malak S., Mennatulla E.**Challenge:** NASA Farm Navigators: Using NASA Data Exploration in Agriculture**Country/Territory:** EgyptTeam Zumorroda-X created mini games that allow players to step into the shoes of a farmer who sets off on an epic journey around the world. Through this game, players can learn how farmers globally adapt to heat waves, flooding, and other environmental challenges.Learn more about Zumorroda-Xs' project

Local Impact Award: QUEÑARIS Team Members: Borax Q., Carlos Y., Marcelo S., Máximo S., Oscar M., Pamela P. Challenge: BloomWatch: An Earth Observation Application for Global Flowering Phenology Country/Territory: Peru Team QUEÑARIS' project addresses critical water scarcity in Peru's second-largest city, Arequipa, caused by the degradation of queñua forests, which are vital for water retention. Their platform combines native microorganisms, NASA satellite data, drones, and artificial intelligence to accelerate tree growth, identify the best areas for reforestation, and monitor ecosystem health. Learn more about QUEÑARIS' project

Stay up to date with #SpaceApps by following these accounts: X: @SpaceApps Instagram: @nasa_spaceapps Facebook: @spaceappschallenge YouTube: @NASASpaceAppsChallenge

NASA Space Apps is funded by NASA's Earth Science Division through a contract with Booz Allen Hamilton, Mindgrub, and SecondMuse.

To learn more about what inspired these winning projects, visit:

<https://www.spaceappschallenge.org>

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TITLE: NASA Launches Research Program for Students to Explore Big Ideas

NASA STEM TeamDec 16, 2025ArticleNASA is now accepting concepts for a new research challenge. The Opportunities in Research, Business, Innovation, and Technology (ORBIT) challenge is a multi-phase innovation competition designed to empower university and college students to develop next-generation solutions that benefit life on Earth and deep-space exploration. With up to \$380,000 in total prize funding, NASA's ORBIT challenges student teams to bring their most forward-thinking concepts to the table, either utilizing NASA intellectual property or conceptualizing their own. Teams are tasked with conducting targeted research, designing early mockups or models, and performing feasibility analyses to refine their ideas. Finalists then advance to a live showcase where they present their work to a panel of expert judges, who evaluate the proposals and select winners based on the teams' final pitches and responses to questions. The ORBIT has two challenge tracks for teams to choose from. The ORBIT Earth track requires teams to select a NASA-owned patent and develop novel commercial or nonprofit applications addressing real-world problems. From adapting aerospace materials for disaster response and preparedness, to repurposing space-based sensors for healthcare, students must demonstrate clear pathways to public benefit. The ORBIT Space track asks teams to design new system concepts aligned with NASA's current and future missions, particularly supporting the Artemis program's goal of establishing a sustainable human presence on the Moon and preparing for eventual missions to Mars and beyond. Students will create technically feasible designs for everything from lunar habitats that could house future Artemis astronauts to deep space robotics that open more pathways to in-situ resource utilization. Teams that successfully integrate objectives from both tracks may qualify for an optional integration bonus. This challenge accelerates innovation in areas critical to NASA's future goals while cultivating a pipeline of interdisciplinary talent. By engaging the next generation in NASA's dual mission to explore space and improve life on Earth, ORBIT inspires students to join the agency's talent network while delivering tangible benefits to American communities and industries. Beyond monetary awards, participants stand to gain mentorship from NASA experts, access to agency facilities, and hands-on experience in systems design, entrepreneurship, and commercialization. For complete competition

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URL: <https://www.nasa.gov/mission/artemis-ii/>

TITLE: Artemis II

Artemis Artemis II The first crewed Artemis flight marks a key step toward long-term return to the Moon and future missions to Mars. Future Mission WHY we go Scientific Discovery LUNAR DESTINATION Earth's Moon MEET THE CREW Our Artemis Astronauts Artemis II builds on the success of the uncrewed Artemis I in 2022, and will demonstrate a broad range of capabilities needed on deep space missions. The Artemis II test flight will be NASA's first mission with crew aboard the SLS (Space Launch System) rocket and Orion spacecraft. Mission Type Crewed Lunar Flyby CREW SIZE 4 Launch No Earlier Than Feb. 6, 2026 mission duration 10 Days Artemis II News More Mission News Article 4 Min Read NASA Selects Participants to Track Artemis II Mission Article 3 Min Read I Am Artemis: Dustin Gohmert 38 Min Read Artemis II: Recovery Article 5 Min Read NASA's Artemis II Mission to Fly Legacy Keepsakes with Astronaut Crew 2 Min Read Launch Pad Preparations Progress Ahead of Artemis II Wet Dress Rehearsal Blog 1 Min Read NASA's Artemis II Rocket and Spacecraft Make Their Way to Launch Pad Image Article 3 Min Read NASA Unlocks Golden Age of Innovation, Exploration in Trump's First Year News Release 25 Min Read Artemis II: What NASA Learned From Launching Artemis I Our Artemis II Crew Meet the astronauts who will venture around the Moon on Artemis II, the first crewed flight aboard NASA's human deep space capabilities, paving the way for future lunar surface missions. Forging New Frontiers about Our Artemis II Crew The Spacecraft The Rocket The Ground Systems The Science NASA's Newest Spacecraft Orion is developed to be capable of sending astronauts to the Moon and is a crucial step toward eventually sending crews on to Mars. The Orion spacecraft will serve as the exploration vehicle that will carry and sustain the crew on Artemis missions to the Moon and return them safely to Earth. Orion will launch on NASA's new heavy-lift rocket, the SLS (Space Launch System). About Orion Technicians with NASA's Exploration Ground Systems team use a crane to lift and secure NASA's Orion spacecraft on top of the SLS (Space Launch System) rocket in High Bay 3 of the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on

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Artemis II Flight PathThis visualization depicts the Artemis II mission's 10-day "figure-eight" trajectory, featuring a high-altitude lunar flyby that will send four astronauts further into deep space than any humans in history by 2026.

Visualizing Artemis IIabout Artemis II Flight PathArtemis II Astronauts35 ImagesGo To GalleryGo To GalleryArtemis II Boarding PassSend Your Name Around the MoonArtemis II will test NASA's deep space capabilities as humans fly on the Space Launch System rocket and Orion spacecraft for the first time. Join the mission by launching your name around the moon alongside NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and Canadian Space Agency astronaut Jeremy Hansen.

Claim Your Passabout Send Your Name Around the MoonArtemis II crew posterNASADiscover MoreArtemis BlogArtemis II Crew PatchJoin ArtemisNASA en españolNASA's EyesMedia ResourcesArtemis II Media ContactsKeep ExploringDiscover MoreArtemisSpaceships and RocketsHumans In SpaceDestinations

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Our Artemis II CrewMeet the astronauts who will venture around the Moon on Artemis II, the first crewed flight aboard NASA's human deep space capabilities, paving the way for future lunar surface missions.Forging New Frontiersabout Our Artemis II Crew

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About the Space Launch System

Teams with NASA's Exploration Ground Systems and primary contractor Amentum use a massive crane to lift to vertical the SLS (Space Launch System) Moon rocket for Artemis II on Saturday, March 22, 2025, inside the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida. NASA/Frank Michaux

Teams with NASA's Exploration Ground Systems and primary contractor Amentum use a massive crane to lift to vertical the SLS (Space Launch System) Moon rocket for Artemis II on Saturday, March 22, 2025, inside the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida.

The Infrastructure Preparing the infrastructure to support NASA's Space Launch System (SLS) rocket and Orion spacecraft. Exploration Ground Systems, based at Kennedy Space Center in Florida, develops and operates the systems and facilities needed to process, launch, and recover rockets and spacecraft for NASA's Artemis missions. About Exploration Ground Systems Engineers and technicians with NASA's Exploration Ground Systems Program prepare to lift the left center booster segment shown with the iconic NASA "worm" insignia for the agency's SLS (Space Launch System) rocket to the Vehicle Assembly Building at NASA's Kennedy Space Center in Florida on Friday, Jan. 24, 2025. NASA/Frank Michaux

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What We'll Learn Artemis II science operations will lay the foundation for safe and efficient human exploration of the Moon and Mars. The Artemis II mission will carry astronauts farther from Earth and closer to the Moon than any human has been in over half a century. From this unique vantage point and environment, the Artemis II crew will work with scientists on Earth to facilitate science investigations to inform future human spaceflight missions. About the Science The AVATAR (A Virtual Astronaut Tissue Analog Response) investigation will use organ-on-a-chip devices, or organ chips, to study the effects of increased radiation and microgravity on human health.

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Artemis II Flight PathThis visualization depicts the Artemis II mission's 10-day "figure-eight" trajectory, featuring a high-altitude lunar flyby that will send four astronauts further into deep space than any humans in history by 2026.
Visualizing Artemis IIabout Artemis II Flight Path

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Artemis II Boarding PassSend Your Name Around the MoonArtemis II will test NASA's deep space capabilities as humans fly on the Space Launch System rocket and Orion spacecraft for the first time. Join the mission by launching your name around the moon alongside NASA astronauts Reid Wiseman, Victor Glover, Christina Koch, and Canadian Space Agency astronaut Jeremy Hansen.Claim Your Passabout Send Your Name Around the MoonArtemis II crew posterNASA

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Meet the Astronauts Aboard the Space Station

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Info for Parents, Teachers, and CaregiversNASA provides a safe place for children to play games as they learn about NASA and its missions. NASA encourages you to visit NASA Kids' Club and use its games and activities to inspire the next generation. On this site, you will find games of various skill levels for children pre-K through grade 4. These games support national education standards in science, technology, engineering and mathematics.Learn More

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TITLE: Air Traffic Solutions

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Demonstration 2 (ATD-2) tools that air traffic managers began testing in 2017. [NASA / Jim Banke](#) Air traffic controllers test NASA-developed concepts in FutureFlight Central, an ultra-realistic, 360-degree, simulated control tower at NASA Ames Research Center. [NASA / Dominic Hart](#) Air Traffic Solutions Gallery [7 Images](#) [Go To Gallery](#) [Go To Gallery](#) Here is Some Other Cool NASA Aeronautics Stuff to Check Out [Aeronautics @ Home](#) A collection of activities and lessons to learn more about NASA Aeronautics and the science of flight that you can do at home. [STEM Learning Resources](#) Explore hands-on activities, interactive lesson plans, educator guides, and other downloadable content related to aeronautics. [Aeronautics Backgrounds](#) Take video calls from your home or office in the cockpit of the X-59 or from inside a wind tunnel! And other aviation themes. [Free E-Books to Download](#) Add these books covering NASA Aeronautics and other aviation and space topics to your electronic library. [NASA X Video Files](#) This binge worthy page collects aeronautics-themed topics from the popular NASA X video series. [Facebook logo](#) [@NASA](#) [@NASAaero](#) [@NASAes](#) [@NASA](#) [@NASAaero](#) [@NASA_es](#) [Instagram logo](#) [@NASA](#) [@NASAaero](#) [@NASA_es](#) [LinkedIn logo](#) [@NASA](#) Other Aeronautics topics [Even More Aeronautics](#) [Supersonic Flight](#) Research about flying faster than the speed of sound. [Flight Innovation](#) Stories about pioneering the frontiers of 21st century flight. [Drones & You](#) All about new ways to get from here to there in the air. [Ultra-Efficient Aviation](#) Learn about research to make aviation more sustainable. [Air Traffic Solutions](#) Read about how NASA is opening up the sky for all. [Keep Exploring](#) [Discover More Topics From NASA](#) [Missions](#) [Humans In Space](#) [Solar System Exploration](#) [Eyes on the Solar System](#) [Explore NASA's History](#)

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The air traffic solutions NASA is working on today build from a foundation of research that goes back decades. Three recently concluded projects in particular are helping to develop new ways of doing things in the 21st century. The two features presented here help tell those project's stories.

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MultimediaNASA ImagesNASA has always told its story through its images, a few of which have become icons of human history.TodayImage Of The DayRed, Green Light ShowA green and red aurora streams across Earth's horizon above the city lights of Europe in this Jan. 19, 2026, photograph, which looks north across Italy toward Germany. The International Space Station was orbiting 262 miles above the Mediterranean Sea at approximately 10:02 p.m. local time when the image was captured.Browse Image ArchiveTodayImage Of The DayView imageNASA HistoryFlickr Commons has more than 3,000 photos from NASA's history. All photographs on the NASAcommons Flickr account originate from the compilation of photos from NASA's image and video library and other archived sources.Viewabout NASA HistoryNASA Image ResourcesImage Usage GuidelinesImage and Video LibraryNASA HQ Photo OfficeEarth ObservatoryFaces of NASAHubble Space TelescopeJames Webb Space TelescopeNASA Spacewalkers24 ImagesGo To GalleryGo To GalleryDiscover MoreGames and InteractivesBlogsNewslettersWays to Watch

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TITLE: NASA's Chandra Releases Deep Cut From Catalog of Cosmic Recordings

Lee MohonJan 23, 2026ArticleContentsVisual DescriptionNews Media ContactLike a recording artist who has had a long career,NASA's Chandra X-ray Observatoryhas a "back catalog" of cosmic recordings that is impossible to replicate. To access these X-ray tracks, or observations, the ultimate compendium has been developed: theChandra Source Catalog(CSC).The CSC contains theX-raydata detected up to the end of 2020 by Chandra, the world's premier X-ray telescope and one of NASA's "Great Observatories." The latest version of the CSC, known as CSC 2.1, contains over 400,000 unique compact and extended sources and over 1.3 million individual detections in X-ray light.beforeafterThis image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkThis image is the sum of 86 observations added together, representing over three million seconds of Chandra observing time. It spans just about 60 light-years across, which is a veritable pinprick on the entire sky. The underlying image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively. The annotations on the image show where Chandra has detected over 3,300 individual sources in this field of view over a 22-year timeframe.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkbeforeafterThis image contains lower-, medium-, and higher-energy X-rays in red, green, and blue respectively.NASA/CXC/SAO; Image Processing: NASA/CXC/SAO/N. WolkThis image is the sum of 86 observations added together, representing over three

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Because many X-ray sources have been observed multiple times over the life of the Chandra mission, this sonification represents those repeat X-ray sightings over time through different notes. To view this video please enable JavaScript, and consider upgrading to a web browser that supports HTML5 videoChandra Source Catalog Sonification. NASA/CXC/SAO/K. Arcand, SYSTEM Sounds (M. Russo, A. Santaguida)

In the view of the sky, projected in a similar way to how the Earth is often depicted in world maps, the core of the Milky Way is in the center and the Galactic plane is horizontal across the middle of the image. A circle appears at the position of each detection and the size of the circle is determined by the number of detections in that location over time. A year counter appears at the top of the frame. Since Chandra continues to be fully operational, the text changes to “... and beyond” after 2021 as the telescope continues to collect observations. During the video, a collage of images produced by Chandra fades in as a background. In the final frames of the video, thumbnail images representing the thousands of Chandra

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beforeafter

before

after

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Chelsea Gohd Jan 05, 2026 Article Contents Jupiter beams bright, Saturn and the Moon cozy up, and the Beehive Cluster appears Skywatching Highlights Transcript Jupiter beams bright, Saturn and the Moon cozy up, and the Beehive Cluster appears Jupiter is at its biggest and brightest all year, the Moon and Saturn pair up, and the Beehive Cluster buzzes into view. Skywatching Highlights Jan. 10: Jupiter at opposition Jan. 23: Saturn and Moon conjunction Jan. (throughout): Beehive Cluster Transcript Jupiter is at its biggest and brightest The Moon and Saturn share the sky And the beehive cluster makes an appearance That's what's up, this January January 10, Jupiter will be at its most brilliant of the entire year! This night, Jupiter will be at what's called "opposition," meaning that Earth will be directly between Jupiter and the Sun. NASA/JPL-Caltech In this alignment, Jupiter will appear bigger and brighter in the night sky than it will all year - talk about starting off the new year bright! To see Jupiter at its best this year, look to the east and all evening long, you'll be able to see the planet in the constellation Gemini. It will be one of the brightest

objects in the night sky (only the moon and Venus will be brighter) Saturn and the Moon will share the sky on January 23rd as part of a conjunction! NASA/JPL-Caltech A conjunction is when objects in the sky look close together even though they're actually far apart. To spot the pair, look to the west and you'll see Saturn just below the moon, sparkling in the night sky. The beehive cluster will be visible in the night sky throughout January! The beehive cluster, more formally known as Messier 44, or M44, is made of at least 1,000 stars! It's an open star cluster, meaning it's a loosely-bound group of stars. There are thousands of open star clusters like the beehive in the Milky Way Galaxy! NASA/JPL-Caltech To see the beehive cluster, look to the eastern night sky after sunset and before midnight throughout the month - especially great nights to spot the cluster are around the middle of January when the cluster isn't too high or low in the sky to see. With dark skies you might be able to spot the beehive with just your eyes, but binoculars or a small telescope will help. Here are the phases of the Moon for January. NASA/JPL-Caltech You can stay up to date on all of NASA's missions exploring the solar system and beyond at science.nasa.gov. I'm Chelsea Gohd from NASA's Jet Propulsion Laboratory, and that's What's Up for this month.

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What's Up

URL: <https://www.nasa.gov/directorates/stmd/nasa-back-for-seconds-with-new-food-system-design-challenge/>

TITLE: NASA Back for Seconds with New Food System Design Challenge

4 min readNASA Back for Seconds with New Food System Design Challenge

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capabilities and elements. NASA is getting ready to send four astronauts around the Moon with Artemis II, laying the foundation for sustainable missions to the lunar surface and paving the way for human exploration on Mars. As the agency considers deep space endeavors that could last months or years, it must develop ways to feed astronauts beyond sending supplies from Earth. That is why NASA is launching the Deep Space Food Challenge: Mars to Table, a new global competition inviting chefs, innovators, culinary experts, higher-education students, and citizen scientists to design a complete, Earth-independent food system for long-duration space missions. "In the future, exploration missions will grow in both duration and distance from Earth. This will make the critical question of feeding our astronauts more complex, requiring innovative solutions to allow for long-term human exploration of space," said Greg Stover, acting associate administrator of NASA's Space Technology Missions Directorate at NASA Headquarters in Washington. "Opening the door to ideas from beyond the agency strengthens NASA's ability to operate farther from Earth with greater independence." Mars to Table builds on NASA's first Deep Space Food Challenge by seeking to integrate multiple food production and preparation methods into a holistic, self-sustaining system designed for use on Mars. This new challenge is open now until July 31 to the global public and carries a prize purse of up to \$750,000. "Future crews on the Moon and Mars will need food systems that are nutritious, sustainable, and fully independent from Earth," said Jarah Meador, program executive for NASA's Prizes, Challenges, and Crowdsourcing Program at NASA Headquarters. "Food will play a pivotal role in the overall health and happiness of future deep space explorers. The Mars to Table Challenge is about bringing all those pieces together into one comprehensive design." Solvers are tasked with creating a complete meal plan suitable for astronauts living on Mars, using a NASA-created mission scenario as their guide. Each team will design a full food system concept, including a detailed operations plan and system design layout that supports a surface mission. Teams must consider every detail – from nutritional balance and taste to safety, usability, and integration with NASA's Environmental Control and Life Support Systems. Participants in the Mars to Table Challenge are also encouraged to address food security on Earth. Innovative growth systems designed for space could make fresh food production possible in harsh, remote, or resource-limited areas, such as research stations located at Earth's poles or in rural areas with limited access to traditional supply chains. "This challenge isn't just about feeding astronauts; it's about feeding people anywhere," said Jennifer Edmunson, acting program manager for NASA's Centennial Challenges at NASA's Marshall Spaceflight Center in Huntsville, Alabama. "Novel meals that are compact, shelf-stable, and nutrient-rich could expand culinary options for groups like military personnel or disaster relief responders. By solving for Mars and future planetary expeditions, we can also find solutions for Earth." NASA's Centennial Challenges have a 20-year legacy of engaging the public to solve complex problems that benefit

NASA's broader initiatives. Past challenges have spurred advances in robotics, additive manufacturing, power and energy, textiles, chemistry, and biology. Mars to Table is a collaborative, cross-program Centennial Challenge with support from NASA's Division of Biological and Physical Sciences, Heliophysics Division, Planetary Science Program, Human Research Program, Mars Campaign Office, and Earth Science Division. Subject matter experts at the agency's Johnson Space Center in Houston and Kennedy Space Center in Florida support the challenge. This challenge is part of the Prizes, Challenges and Crowdsourcing program within NASA's Space Technology Mission Directorate. NASA has partnered with the Methuselah Foundation and contracted Floor23 Digital to support the administration and management of this challenge. To learn more about the challenge, including timelines, submission requirements, and future webinar dates, visit: <https://www.deepspacefood.org/marstotable> By Savannah Bullard Facebook logo@NASATechnology@NASA_Technology

Laura Hall Jan 13, 2026 Article

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By Savannah Bullard

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DetailsLast UpdatedJan 16, 2026EditorLoura Hall

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URL: <https://www.nasa.gov/learning-resources/for-colleges-universities/nasa-launches-2026-lunabotics-challenge/>

TITLE: NASA Launches 2026 Lunabotics Challenge

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Jason Costa Sep 08, 2025 Article Students prepare their robots to enter Artemis Arena during NASA's Lunabotics competition on May 20, 2025, at the Center for Space Education near the Kennedy Space Center Visitor Complex in Florida. NASA/Isaac Watson As college students across the country embark upon the academic year, NASA is giving them something else to look forward to – the agency's 2026 Lunabotics Challenge. Teams interested in participating can submit their applications and supporting materials through NASA's Stem Gateway portal beginning Monday, Sept. 8. Key dates and challenge details are available in the 2026 Lunabotics Challenge Guidebook. Once all applications and supporting materials are received and evaluated, NASA will notify the selected teams to begin the challenge. Student teams participating in this year's challenge will create robots capable of building berms out of lunar regolith – the loose, fragmental material on the Moon's surface. Structures like these will be important during lunar missions as blast protection during lunar landings and launches, shading for cryogenic propellant tank farms, radiation shielding around nuclear power plants, and other uses critical to future Moon missions. "We are excited to continue the Lunabotics competition for universities as NASA develops new Moon to Mars technologies for the Artemis program," said Robert Mueller, senior technologist at NASA, as well as co-founder and chief judge of the Lunabotics competition. "Excavating and moving regolith is a fundamental need to build infrastructure on the Moon and Mars and this competition creates 21st century skills in the future workforce." An in-person qualifying event will be held May 12-17, 2026, at the University of Central Florida's Space Institute's Exolith Lab in Orlando. From this round, the top 10 teams will be invited to bring their robots to the final competition on May 19-21, at

the Kennedy Space Center Visitor Complex's Artemis Arena in Florida, which has an area filled with a lunar regolith simulant. The team scoring the most points will receive the Lunabotics Grand Prize and participate in an exhibition-style event at NASA Kennedy. By encouraging innovative construction techniques and assessing student designs and data the same way it does its own prototypes, NASA casts a wider net to find innovative solutions to challenges inherent in future Artemis missions, like developing future lunar excavators, in-situ resource utilization capabilities, and living on the Moon or Mars. With its multidisciplinary approach, Lunabotics also serves as a workforce pipeline, with teams gaining valuable hands-on experience in computer coding, engineering, manufacturing, fabricating, and other crucial skills, while also receiving technical expertise in space technology development. NASA's Lunabotics Challenge, held annually since 2010, is one of several Artemis Student Challenges. The two-semester competition provides U.S. college and technical school teams an opportunity to design, build, and operate a prototype lunar robot using NASA systems engineering processes. Competitions help NASA get innovative design and operational data, reduce risks, and cultivate new ideas needed to return to the Moon under the Artemis campaign to prepare for human exploration of Mars. To learn more about Lunabotics, visit: <https://www.nasa.gov/learning-resources/lunabotics-challenge/>

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URL: <https://www.nasa.gov/centers-and-facilities/marshall/nasa-seeks-proposals-for-2026-human-exploration-rover-challenge/>

TITLE: NASA Seeks Proposals for 2026 Human Exploration Rover Challenge

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Beth Ridgeway Aug 15, 2025 Article NASA now is accepting proposals from student teams for a contest to design, build, and test rovers for Moon and Mars exploration through Sept. 15. Known as the Human Exploration Rover Challenge, student rovers should be capable of traversing a course while completing mission tasks. The challenge handbook has guidelines for remote-controlled and human-powered divisions. The cover of the HERC 2026 handbook, which is now available online. “Last year, we saw a lot of success with the debut of our remote-controlled division and the addition of middle school teams,” said Vemitra Alexander, the activity lead for the challenge at NASA’s Marshall Space Flight Center in Huntsville, Alabama. “We’re looking forward to building on both our remote-controlled and human-powered divisions with new challenges for the students, including rover automation.” This year’s mission mimics future Artemis missions to the lunar surface. Teams are challenged to test samples of soil, water, and air from sites along a half-mile course that includes a simulated field of asteroid debris, boulders, erosion ruts, crevasses, and an ancient streambed. Human-powered rover teams will play the role of two astronauts in a lunar terrain vehicle and must use a custom-built task tool to manually collect samples needed for testing. Remote-controlled rover teams will act as a pressurized rover, and the rover itself will contain the tools necessary to collect and test samples onboard. “NASA’s Human Exploration Rover Challenge creates opportunities for students to develop the skills they need to be successful STEM professionals,” said Alexander. “This challenge will help students see themselves in the mission and give them the hands-on experience needed to advance technology and become the workforce of tomorrow.” Seventy-five teams comprised of more than 500 students participated in the agency’s 31st rover challenge in 2025. Participants represented 35 colleges and universities, 38 high schools, and two middle schools, across 20 states, Puerto Rico, and 16 nations around the world. The 32nd annual competition will culminate with an in-person event April 9-11, 2026, at the U.S. Space & Rocket Center near NASA Marshall. The rover challenge is one of NASA’s Artemis Student Challenges, reflecting the goals of the Artemis campaign, which seeks to explore the Moon for scientific discovery, technology advancement, and to learn how to live and work on another world as we prepare for human missions to Mars. NASA uses such challenges to encourage students to pursue degrees and careers in the fields of science, technology, engineering, and mathematics. Since its inception in 1994, more than 15,000 students have participated in the rover challenge – with many former students now working at NASA or within the aerospace industry. To learn more about HERC, visit: <https://www.nasa.gov/roverchallenge/>

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Beth Ridgeway

Aug 15, 2025

NASA now is accepting proposals from student teams for a contest to design, build, and test rovers for Moon and Mars exploration through Sept. 15. Known as the Human Exploration Rover Challenge, student rovers should be capable of traversing a course while completing mission tasks. The challenge handbook has guidelines for remote-controlled and human-powered divisions. The cover of the HERC 2026 handbook, which is now available online. “Last year, we saw a lot of success with the debut of our remote-controlled division and the addition of middle school teams,” said Vemitra Alexander, the activity lead for the challenge at NASA’s Marshall Space Flight Center in Huntsville, Alabama. “We’re looking forward to building on both our remote-controlled and human-powered divisions with new challenges for the students, including rover automation.” This year’s mission mimics future Artemis missions to the lunar surface. Teams are challenged to test samples of soil, water, and air from sites along a half-mile course that includes a simulated field of asteroid debris, boulders, erosion ruts, crevasses, and an ancient streambed. Human-powered rover teams will play the role of two astronauts in a lunar terrain vehicle and must use a custom-built task tool to manually collect samples needed for testing. Remote-controlled rover teams will act as a pressurized rover, and the rover itself will contain the tools necessary to collect and test samples onboard. “NASA’s Human Exploration Rover Challenge creates opportunities for students to develop the skills they need to be successful STEM professionals,” said Alexander. “This challenge will help students see themselves in the mission and give them the hands-on experience needed to advance technology and become the workforce of tomorrow.” Seventy-five teams comprised of more than 500 students participated in the agency’s 31st rover challenge in 2025. Participants represented 35 colleges and universities, 38 high schools, and two middle schools, across 20 states, Puerto Rico, and 16 nations around the world. The 32nd annual competition will culminate with an in-person event April 9-11, 2026, at the U.S. Space & Rocket Center near NASA Marshall. The rover challenge is one of NASA’s Artemis Student Challenges, reflecting the goals of the Artemis campaign, which seeks to explore the Moon for scientific discovery, technology advancement, and to learn how to live and work on another world as we prepare for human missions to Mars. NASA uses such challenges to encourage students to pursue degrees and careers in the fields of science, technology, engineering, and mathematics. Since its inception in 1994, more than 15,000 students have participated in the rover challenge – with many former students now working

at NASA or within the aerospace industry. To learn more about HERC, visit: <https://www.nasa.gov/roverchallenge/>

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NASA STEM TeamDec 01, 2025ArticleContentsWhat is AI?How is NASA using AI?Advice From NASA AI ExpertsCareer CornerExplore MoreThis article is for students grades 5-8.

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NASA’s Perseverance Mars rover alongside the rock nicknamed “Cheyava Falls,” in this July 23, 2024, selfie made up of 62 individual images. “Cheyava Falls,” which has features that may bear on the question of whether the Red Planet was long ago home to microscopic life, is to the left of the rover near the center of the image. The small hole visible in the rock is where Perseverance collected the “Sapphire Canyon” core sample.

NASA/JPL-Caltech/MSSS

How is NASA using AI?NASA has found uses for AI in many of its missions and programs. For missions to the Moon, AI can use satellite imagery to create detailed 3D maps of dark craters. This data could help scientists plan missions, spot hazards, and even identify where future crews might find water ice. On Mars, the Perseverance rover uses AI to drive itselfautonomously. It takes pictures of the ground, sees obstacles, and chooses the safest path.AI also helps NASA search for planets outside our solar system. For example, AI has helpedcitizen scientistsfind over 10,000 pairs of binary stars. These pairs orbit each other and block each other’s light. This information could help scientists search for new planets and learn more about how stars form.

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Contents
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TITLE: Helio Highlight: Comets, the Sun, and You

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Nicholas Oakes
Chantil Hunt Estevez
Dec 02, 2025
Article Contents
The Sun’s Smallest Companions
Messengers from Afar
The New Kid on the Block
How Heliophysics Can Help
What We Can Learn Together
Resources for Educators
↓
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There's a lot orbiting the Sun. Aside from the eight planets, there are billions of smaller objects, too. One type is the asteroids, hunks of rock left over from the formation of

the solar system. They mostly orbit in two belts: one between the orbits of Mars and Jupiter, and one out beyond Neptune. Comets are icy bodies which swing through the solar system on long, highly elliptical orbits. Most of them orbit far beyond Pluto, in the Oort Cloud. When comets swing in close to the Sun, its heat causes their outer shells to melt. Then the solar wind pushes against this loosened material and causes it to trail away from the Sun in a great, glowing tail. These tails are one of the most dynamic images that we associate with comets, as they can bloom up to many times larger than the size of planets as observed from Earth. When the comet departs the inner solar system, the tail fades as the comet freezes up again. Comet C/2013 A1 as seen by the Hubble Space Telescope, showcasing the dusty tail flowing behind the comet. NASA's Messengers from Afar. Most comets, including the famous Halley's Comet, come from within our solar system. That is to say that they were born in the same protoplanetary disk that birthed the Sun and the other bodies around it. We know this mostly by studying their orbits, since the paths they carve through space could only occur if they began on a closed orbit around the Sun. But these visitors can come from farther away, too. Much farther. On October 19, 2017, the University of Hawaii's Pan-STARRS1 telescope, which is funded by NASA's Near-Earth Object Observations (NEOO) Program, tracked a possible asteroid in Earth's neighborhood. Continued observations showed that it was travelling at over 196,000 miles per hour (87.3 kilometers per second) and was accelerating into the inner solar system. That's comet behavior. Initially dubbed 1I/2017 U1, the object was given the name 'Oumuamua by the International Astronomical Union (IAU). This means "a messenger arriving from afar and arriving first" in Hawaiian, and recognizes that it is the first interstellar object known to have visited our solar system. It went beyond Saturn's orbit in early 2019 and is now heading back into deep space. An artist's concept of the first detected interstellar object, 1I/2017, later named 'Oumuamua by the International Astronomical Union. NASA/ESA/STScI. The New Kid on the Block 'Oumuamua was first, but not last. Scientists now estimate that about one such object flies through our solar system every year – and now we know how to look for them. On July 1, 2025, the NASA-funded ATLAS (Asteroid Terrestrial-impact Last Alert System) survey telescope in Rio Hurtado, Chile, reported a new interstellar visitor. It was given the name 3I/ATLAS. 3I/ATLAS has been extensively observed since then. The Hubble Space Telescope observed it on July 21, 2025, when it was about 277 million miles (446 million kilometers) away from Earth. These observations revealed a teardrop-shaped tail flying off of the nucleus, which scientists estimate to be anywhere up to 3.5 miles (5.6 kilometers) in diameter. Other observations have come in from assets as diverse as the James Webb Space Telescope, the Perseverance rover on Mars, and ESA/NASA's Solar and Heliospheric Observatory (SOHO). How Heliophysics Can Help. The Sun is so bright that its light would burn out many of the telescopes which would otherwise observe 3I/ATLAS as it approaches perihelion, or closest approach to the Sun. But NASA's heliophysics mission

fleet is designed to observe the Sun closely, meaning that these spacecraft can keep an eye on 3I/ATLAS as it nears the Sun. This will allow scientists to gather vital data about how comets interact with the Sun, the solar wind, and more, as well as how interstellar objects like 3I/ATLAS differ from objects native to our own solar system. A faint image of comet 3I/ATLAS as observed by ESA/NASA's SOHO mission between Oct. 15-26, 2025. The comet appears as a slight brightening in the center of the image. This mission, launched in December 2025, has discovered many comets in its years of observing the Sun. Lowell Observatory/Qicheng Zhang One instrument observing 3I/ATLAS is SOHO's Large Angle and Spectrometric Coronagraph (LASCO). A coronagraph is a telescope which blocks light from the solar disk in order to see the Sun's faint atmosphere, or corona. LASCO has three coronagraphs that image the corona from 1.1 to 32 solar radii (one solar radius is about 420,000 miles, or 700,000 kilometers). LASCO was able to see 3I/ATLAS when it made its closest approach to the Sun on October 30, 2025. At this closest approach, the object was about 1.36 astronomical units (AU) from the Sun, or about 126 millions miles (203 million kilometers). It won't make its closest approach to Earth until sometime around December 19, when it will be about 1.8 AU from our planet. From there, it will proceed back out of the solar system on a long, dark journey through interstellar space.

What We Can Learn Together

While 3I/ATLAS makes its close approach to the Sun, LASCO and other such instruments will help gather as much data as possible. After the images they return are processed, this data should help scientists to better understand where this strange visitor comes from and what it is made of. Using data gathered on objects like 3I/ATLAS, scientists can work backwards to make more educated inferences about the origins of planetary systems, including our own. Heliophysics can also use data from comets to make discoveries about how the Sun and other stars interact with objects in their orbit. For example, scientists used observations of the tails of comets to help prove the existence of the solar wind. Studying comets has also shown how the Sun's roughly 11-year sunspot cycle contributes to different levels of radiation output. Scientists associated with the Stardust sample return mission to comet Wild 2 celebrate the successful return of cometary material. NASA Sample return missions are another opportunity for scientists to learn about how the Sun affects the rest of the solar system. In 2006, the Stardust mission returned samples of Comet Wild 2, and in 2023, OSIRIS-Rex brought back samples from asteroid Bennu. Studies of these samples have offered clues about how the early components of present day planetary bodies were affected by solar radiation, how the solar wind has changed over time, and more. These cooperative interactions go far beyond 3I/ATLAS. Learning about the Sun helps us to better understand other stars, just as our observations of other stars helps us to better understand our Sun. By focusing on areas where disciplines like heliophysics, planetary science, and astrophysics intersect and overlap, we gain even more insight on our place in the universe.

Resources for Educators ↓

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neighborhood. Continued observations showed that it was travelling at over 196,000 miles per hour (87.3 kilometers per second) and was accelerating into the inner solar system. That's comet behavior. Initially dubbed 1I/2017 U1, the object was given the name 'Oumuamua by the International Astronomical Union (IAU). This means "a messenger arriving from afar and arriving first" in Hawaiian, and recognizes that it is the first interstellar object known to have visited our solar system. It went beyond Saturn's orbit in early 2019 and is now heading back into deep space. An artist's concept of the first detected interstellar object, 1I/2017, later named 'Oumuamua by the International Astronomical Union. NASA/ESA/STScI The New Kid on the Block 'Oumuamua was first, but not last. Scientists now estimate that about one such object flies through our solar system every year – and now we know how to look for them. On July 1, 2025, the NASA-funded ATLAS (Asteroid Terrestrial-impact Last Alert System) survey telescope in Rio Hurtado, Chile, reported a new interstellar visitor. It was given the name 3I/ATLAS. 3I/ATLAS has been extensively observed since then. The Hubble Space Telescope observed it on July 21, 2025, when it was about 277 million miles (446 million kilometers) away from Earth. These observations revealed a teardrop-shaped tail flying off of the nucleus, which scientists estimate to be anywhere up to 3.5 miles (5.6 kilometers) in diameter. Other observations have come in from assets as diverse as the James Webb Space Telescope, the Perseverance rover on Mars, and ESA/NASA's Solar and Heliospheric Observatory (SOHO). How Heliophysics Can Help The Sun is so bright that its light would burn out many of the telescopes which would otherwise observe 3I/ATLAS as it approaches perihelion, or closest approach to the Sun. But NASA's heliophysics mission fleet is designed to observe the Sun closely, meaning that these spacecraft can keep an eye on 3I/ATLAS as it nears the Sun. This will allow scientists to gather vital data about how comets interact with the Sun, the solar wind, and more, as well as how interstellar objects like 3I/ATLAS differ from objects native to our own solar system. A faint image of comet 3I/ATLAS as observed by ESA/NASA's SOHO mission between Oct. 15-26, 2025. The comet appears as a slight brightening in the center of the image. This mission, launched in December 2025, has discovered many comets in its years of observing the Sun. Lowell Observatory/Qicheng Zhang One instrument observing 3I/ATLAS is SOHO's Large Angle and Spectrometric Coronagraph (LASCO). A coronagraph is a telescope which blocks light from the solar disk in order to see the Sun's faint atmosphere, or corona. LASCO has three coronagraphs that image the corona from 1.1 to 32 solar radii (one solar radius is about 420,000 miles, or 700,000 kilometers). LASCO was able to see 3I/ATLAS when it made its closest approach to the Sun on October 30, 2025. At this closest approach, the object was about 1.36 astronomical units (AU) from the Sun, or about 126 millions miles (203 million kilometers). It won't make its closest approach to Earth until sometime around December 19, when it will be about 1.8 AU from our planet. From there, it will proceed back out of the

solar system on a long, dark journey through interstellar space. What We Can Learn Together While 3I/ATLAS makes its close approach to the Sun, LASCO and other such instruments will help gather as much data as possible. After the images they return are processed, this data should help scientists to better understand where this strange visitor comes from and what it is made of. Using data gathered on objects like 3I/ATLAS, scientists can work backwards to make more educated inferences about the origins of planetary systems, including our own. Heliophysics can also use data from comets to make discoveries about how the Sun and other stars interact with objects in their orbit. For example, scientists used observations of the tails of comets to help prove the existence of the solar wind. Studying comets has also shown how the Sun's roughly 11-year sunspot cycle contributes to different levels of radiation output. Scientists associated with the Stardust sample return mission to comet Wild 2 celebrate the successful return of cometary material. NASA sample return missions are another opportunity for scientists to learn about how the Sun affects the rest of the solar system. In 2006, the Stardust mission returned samples of Comet Wild 2, and in 2023, OSIRIS-Rex brought back samples from asteroid Bennu. Studies of these samples have offered clues about how the early components of present day planetary bodies were affected by solar radiation, how the solar wind has changed over time, and more. These cooperative interactions go far beyond 3I/ATLAS. Learning about the Sun helps us to better understand other stars, just as our observations of other stars helps us to better understand our Sun. By focusing on areas where disciplines like heliophysics, planetary science, and astrophysics intersect and overlap, we gain even more insight on our place in the universe. Resources for Educators ↓

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Comparing Comets Teacher Guide (Grades 6-12+) For this activity, your students will play the role of cometary scientist, using images to observe and compare the surfaces of two comet nuclei from close range.

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Goddard Digital Team Jan 16, 2025 Article NASA Space Apps has named 10 global winners, recognizing teams from around the world for their exceptional innovation and collaboration during the 2024 NASA Space Apps Challenge. As the largest annual global hackathon, this event invites participants to leverage open data from NASA and its space agency partners to tackle real-world challenges on Earth and in space. Last year's hackathon welcomed 93,520 registered participants, including space, science, technology, and storytelling enthusiasts of all ages. Participants gathered at local events in 163 countries and territories, forming teams to address the challenges authored by NASA subject matter experts. These challenges included subjects/themes/questions in ocean ecosystems, exoplanet exploration, Earth observation, planetary seismology, and more. The 2024 Global Winners were determined out of 9,996 project submissions and judged by subject matter experts from NASA and space agency partners. "These 10 exceptional teams created projects that reflect our commitment to understanding our planet and exploring beyond, with the potential to transform Earth and space science for the benefit of all," said Dr. Keith Gaddis, NASA Space Apps Challenge program scientist at NASA Headquarters in Washington. "The NASA Space Apps Challenge showcases the potential of every idea and individual. I am excited to see how these innovators will shape and inspire the future of science and exploration." You can watch the Global Winners Announcement here to meet these winning teams and learn about the inspiration behind their projects.

2024 NASA Space Apps Challenge Global Winners

Best Use of Science Award: WMPGang Team
Members: Dakota C., Ian C., Maximilian V., Simon S.
Challenge: Create an Orrery Web App that Displays Near-Earth Objects
Country/Territory: Waterloo, Canada
Using their skills in programming, data analysis, and visualization, WMPGang created a web app that identifies satellite risk zones using real-time data on Near-Earth Objects and meteor streams. Learn more about WMPGang's SkyShield: Protecting Earth and Satellites from Space Hazards project.

Best Use of Data Award: GaamaRamma Team
Members: Aakash H., Arun G., Arthur A., Gabriel A., May K.
Challenge: Leveraging Earth Observation Data for Informed Agricultural Decision-Making
Country/Territory: Universal Event, United States
GaamaRamma's team of tech enthusiasts aimed to create a sustainable way to help farmers efficiently manage water availability in the face of drought, pests, and disease. Learn more about GaamaRamma's Waterwise project.

Best Use of Technology Award: 42 QuakeHeroes Team
Members: Alailton A., Ana B., Gabriel C., Gustavo M., Gustavo T., Larissa M.
Challenge: Seismic Detection Across the Solar System
Country/Territory: Maceió, Brazil
Team 42 QuakeHeroes employed a deep neural

network model to identify the precise locations of seismic events within time-series data. They used advanced signal processing techniques to isolate and analyze unique components of non-stationary signals. Learn more about 42 QuakeHeroes' project

Galactic Impact Award: NVS-knot Team Members: Oksana M., Oleksandra M., Prokipchyn Y., Val K. Challenge: Leveraging Earth Observation Data for Informed Agricultural Decision-Making Country/Territory: Kyiv, Ukraine The NVS-knot team assessed planting conditions using surface soil moisture and evapotranspiration data, then created an app that empowers farmers to manage planting risks. Learn more about NVS-knot's 2plant | ! 2plant project

Best Mission Concept Award: AsturExplorers Team Members: Coral M., Daniel C., Daniel V., Juan B., Samuel G., Vladimir C. Challenge: Landsat Reflectance Data: On the Fly and at Your Fingertips Country/Territory: Gijón, Spain AsturExplorers created Landsat Connect, a web app that provides a simple, intuitive way to track Landsat satellites and access Landsat surface reflectance data. The app also allows users to set a target location and receive notifications when Landsat satellites pass over their area. Learn more about AsturExplorers' Landsat Connect project

Most Inspirational Award: Innovisionaries Team Members: Rikzah K., Samira K., Shafeeqa J., Umamah A. Challenge: SDGs in the Classroom Country/Territory: Sharjah, United Arab Emirates Innovisionaries developed Eco-Metropolis to inspire sustainability through gameplay. This city-building game engages players in making critical urban planning and resource management decisions based on real-world environmental data. Learn more about Innovisionaries' Eco-Metropolis: Sustainable City Simulation project

Best Storytelling Award: TerraTales Team Members: Ahmed R., Fatma E., Habiba A., Judy A., Maya M. Challenge: Tell Us a Climate Story! Country/Territory: Cairo, Egypt TerraTales shared stories of how Earth's changing climate affects three unique regions: Egypt, Brazil, and Germany. The web app also features an artificial intelligence (AI) model for climate forecasting and an interactive game to encourage users to make eco-friendly choices. Learn more about TerraTale's project

Global Connection Award: Asteroid Destroyer Team Members: Kapeesh K., Khoi N., Sathyajit L., Satyam S. Challenge: Navigator for the Habitable Worlds Observatory (HWO): Mapping the Characterizable Exoplanets in our Galaxy Country/Territory: Saskatoon, Canada Team Asteroid Destroyer honed in on exoplanets, utilizing data processing and machine learning techniques to map exoplanets based on size, temperature, and distance. Learn more about Asteroid Destroyer's project

Art & Technology Award: Connected Earth Museum Team Members: Gabriel M., Luc R., Lucas R., Mattheus L., Pedro C., Riccardo S. Challenge: Imagine our Connected Earth Country/Territory: Campinas, Brazil Team Connected Earth Museum created an immersive virtual museum experience to raise awareness of Earth's changing climate. An AI host guides users through an interactive gallery featuring 3D and 2D visualizations, including a time series on Earth and ocean temperatures, population density, wildfires, and more. Learn more about Connected Earth

Museums' project
Local Impact Award:
Team I.O.
Team Members: Frank R., Jan K., Raphael R., Ryan Z., Victoria M.
Challenge: Community Mapping
Country/Territory: Florianópolis, Brazil
Team I.O. bridges the gap between complex Geographic Information Systems data and user-friendly communication, making critical environmental information accessible to everyone, regardless of technical expertise.
Learn more about Team I.O.'s G.R.O.W. (Global Recovery and Observation of Wildfires) project
Want to take part in the 2025 NASA Space Apps Challenge?
Mark your calendars for October 4 and 5! Registration will open in July. At that time, participants will be able to register for a local event hosted by NASA Space Apps leads from around the world. You can stay connected with NASA Space Apps on Facebook, Instagram, and X. Space Apps is funded by NASA's Earth Science Division through a contract with Booz Allen Hamilton, Mindgrub, and Second Muse.

Goddard Digital Team
Jan 16, 2025
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Team 42 Quake Heroes employed a deep neural network model to identify the precise locations of seismic events within time-series data. They used advanced signal processing techniques to isolate and analyze unique components of non-stationary signals. [Learn more about 42 Quake Heroes' project](#)

Galactic Impact Award: NVS-knot Team Members: Oksana M., Oleksandra M., Prokipchyn Y., Val K. **Challenge:** Leveraging Earth Observation Data for Informed Agricultural Decision-Making **Country/Territory:** Kyiv, Ukraine

The NVS-knot team assessed planting conditions using surface soil moisture and evapotranspiration data, then created an app that empowers farmers to manage planting risks. [Learn more about NVS-knot's 2plant | ! 2plant project](#)

Best Mission Concept Award: Astur Explorers Team Members: Coral M., Daniel C., Daniel V., Juan B., Samuel G., Vladimir C. **Challenge:** Landsat Reflectance Data: On the Fly and at Your Fingertips **Country/Territory:** Gijón, Spain

Astur Explorers created Landsat Connect, a web app that provides a simple, intuitive way to track Landsat satellites and access Landsat surface reflectance data. The app also allows users to set a target location and receive notifications when Landsat satellites pass over their area. [Learn more about Astur Explorers' Landsat Connect project](#)

Most Inspirational Award: Innovisionaries Team Members: Rikzah K., Samira K., Shafeeqa J., Umamah A. **Challenge:** SDGs in the Classroom **Country/Territory:** Sharjah, United Arab Emirates

Innovisionaries developed Eco-Metropolis to inspire sustainability through gameplay. This city-building game engages players in making critical urban planning and resource management decisions based on real-world environmental data. [Learn more about Innovisionaries' Eco-Metropolis: Sustainable City Simulation project](#)

Best Storytelling Award: Terra Tales Team Members: Ahmed R., Fatma E., Habiba A., Judy A., Maya M. **Challenge:** Tell Us a Climate Story! **Country/Territory:** Cairo, Egypt

Terra Tales shared stories of how Earth's changing climate affects three unique regions: Egypt, Brazil, and Germany. The web app also features an artificial intelligence (AI) model for climate forecasting and an interactive game to encourage users to make eco-friendly choices. [Learn more about Terra Tale's](#)

projectGlobal Connection Award:Asteroid DestroyerTeam Members:Kapeesh K., Khoi N., Sathyajit L., Satyam S.Challenge:Navigator for the Habitable Worlds Observatory (HWO): Mapping the Characterizable Exoplanets in our GalaxyCountry/Territory:Saskatoon, CanadaTeam Asteroid Destroyer honed in on exoplanets, utilizing data processing and machine learning techniques to map exoplanets based on size, temperature, and distance.Learn more about Asteroid Destroyer's projectArt & Technology Award:Connected Earth MuseumTeam Members:Gabriel M., Luc R., Lucas R., Mattheus L., Pedro C., Riccardo S.Challenge:Imagine our Connected EarthCountry/Territory:Campinas, BrazilTeam Connected Earth Museum created an immersive virtual museum experience to raise awareness of Earth's changing climate. An AI host guides users through an interactive gallery featuring 3D and 2D visualizations, including a time series on Earth and ocean temperatures, population density, wildfires, and more.Learn more about Connected Earth Museums' projectLocal Impact Award:Team I.O.Team Members:Frank R., Jan K., Raphael R., Ryan Z., Victoria M.Challenge:Community MappingCountry/Territory:Florianópolis, BrazilTeam I.O. bridges the gap between complex Geographic Information Systems data and user-friendly communication, making critical environmental information accessible to everyone, regardless of technical expertise.Learn more about Team I.O.'sG.R.O.W. (Global Recovery and Observation of Wildfires)projectWant to take part in the 2025 NASA Space Apps Challenge?Mark your calendars for October 4 and 5! Registration will open in July. At that time, participants will be able to register for a local event hosted by NASA Space Apps leads from around the world. You can stay connected with NASA Space Apps onFacebook,Instagram, andX.Space Apps is funded by NASA'sEarth Science Divisionthrough a contract withBooz Allen Hamilton,Mindgrub, andSecondMuse.

NASA Space Apps has named 10 global winners, recognizing teams from around the world for their exceptional innovation and collaboration during the 2024 NASA Space Apps Challenge. As the largest annual global hackathon, this event invites participants to leverage open data from NASA and its space agency partners to tackle real-world challenges on Earth and in space.

Last year's hackathon welcomed 93,520 registered participants, including space, science, technology, and storytelling enthusiasts of all ages. Participants gathered at local events in 163 countries and territories, forming teams to address the challenges authored by NASA subject matter experts. These challenges included subjects/themes/questions in ocean ecosystems, exoplanet exploration, Earth observation, planetary seismology, and more.

The2024 Global Winnerswere determined out of 9,996 project submissions and judged by subject matter experts from NASA and space agency partners.

“These 10 exceptional teams created projects that reflect our commitment to understanding our planet and exploring beyond, with the potential to transform Earth and space science for the benefit of all,” said Dr. Keith Gaddis, NASA Space Apps Challenge program scientist at NASA Headquarters in Washington. “The NASA Space Apps Challenge showcases the potential of every idea and individual. I am excited to see how these innovators will shape and inspire the future of science and exploration.”

You can watch the Global Winners Announcement [here](#) to meet these winning teams and learn about the inspiration behind their projects.

2024 NASA Space Apps Challenge Global Winners

Best Use of Science Award: WMPGang

Team Members: Dakota C., Ian C., Maximilian V., Simon S.

Challenge: Create an Orrery Web App that Displays Near-Earth Objects

Country/Territory: Waterloo, Canada

Using their skills in programming, data analysis, and visualization, WMPGang created a web app that identifies satellite risk zones using real-time data on Near-Earth Objects and meteor streams.

Learn more about WMPGang’s [SkyShield: Protecting Earth and Satellites from Space Hazards](#) project

Best Use of Data Award: GaamaRamma

Team Members: Aakash H., Arun G., Arthur A., Gabriel A., May K.

Challenge: Leveraging Earth Observation Data for Informed Agricultural Decision-Making

Country/Territory: Universal Event, United States

GaamaRamma’s team of tech enthusiasts aimed to create a sustainable way to help farmers efficiently manage water availability in the face of drought, pests, and disease.

Learn more about GaamaRamma’s [Waterwise](#) project

Best Use of Technology Award: 42 QuakeHeroes

Team Members: Alailton A., Ana B., Gabriel C., Gustavo M., Gustavo T., Larissa M.

Challenge: Seismic Detection Across the Solar System

Country/Territory: Maceió, Brazil

Team 42 QuakeHeroes employed a deep neural network model to identify the precise locations of seismic events within time-series data. They used advanced signal processing techniques to isolate and analyze unique components of non-stationary signals.

[Learn more about 42 QuakeHeroes' project](#)

Galactic Impact Award:NVS-knot

Team Members:Oksana M., Oleksandra M., Prokipchyn Y., Val K.

Country/Territory:Kyiv, Ukraine

The NVS-knot team assessed planting conditions using surface soil moisture and evapotranspiration data, then created an app that empowers farmers to manage planting risks.

[Learn more about NVS-knot's 2plant | ! 2plantproject](#)

Best Mission Concept Award:AsturExplorers

Team Members:Coral M., Daniel C., Daniel V., Juan B., Samuel G., Vladimir C.

Challenge:Landsat Reflectance Data: On the Fly and at Your Fingertips

Country/Territory:Gijón, Spain

AsturExplorers created Landsat Connect, a web app that provides a simple, intuitive way to track Landast satellites and access Landsat surface reflectance data. The app also allows users to set a target location and receive notifications when Landsat satellites pass over their area.

[Learn more about AsturExplorers' Landsat Connectproject](#)

Most Inspirational Award:Innovisionaries

Team Members:Rikzah K., Samira K., Shafeeqa J., Umamah A.

Challenge:SDGs in the Classroom

Country/Territory:Sharjah, United Arab Emirates

Innovisionaries developed Eco-Metropolis to inspire sustainability through gameplay. This city-building game engages players in making critical urban planning and resource management decisions based on real-world environmental data.

[Learn more about Innovisionaries'Eco-Metropolis: Sustainable City Simulationproject](#)

Best Storytelling Award:TerraTales

Team Members:Ahmed R., Fatma E., Habiba A., Judy A., Maya M.

Challenge:Tell Us a Climate Story!

Country/Territory:Cairo, Egypt

TerraTales shared stories of how Earth's changing climate affects three unique regions: Egypt, Brazil, and Germany. The web app also features an artificial intelligence (AI) model for climate forecasting and an interactive game to encourage users to make eco-friendly choices.

Learn more about TerraTale's project

Global Connection Award:Asteroid Destroyer

Team Members:Kapeesh K., Khoi N., Sathyajit L., Satyam S.

Challenge:Navigator for the Habitable Worlds Observatory (HWO): Mapping the Characterizable Exoplanets in our Galaxy

Country/Territory:Saskatoon, Canada

Team Asteroid Destroyer honed in on exoplanets, utilizing data processing and machine learning techniques to map exoplanets based on size, temperature, and distance.

Learn more about Asteroid Destroyer's project

Art & Technology Award:Connected Earth Museum

Team Members:Gabriel M., Luc R., Lucas R., Mattheus L., Pedro C., Riccardo S.

Challenge:Imagine our Connected Earth

Country/Territory:Campinas, Brazil

Team Connected Earth Museum created an immersive virtual museum experience to raise awareness of Earth's changing climate. An AI host guides users through an interactive gallery featuring 3D and 2D visualizations, including a time series on Earth and ocean temperatures, population density, wildfires, and more.

Learn more about Connected Earth Museums' project

Local Impact Award:Team I.O.

Team Members:Frank R., Jan K., Raphael R., Ryan Z., Victoria M.

Challenge:Community Mapping

Country/Territory: Florianópolis, Brazil

Team I.O. bridges the gap between complex Geographic Information Systems data and user-friendly communication, making critical environmental information accessible to everyone, regardless of technical expertise.

Learn more about Team I.O.'s G.R.O.W. (Global Recovery and Observation of Wildfires) project

Want to take part in the 2025 NASA Space Apps Challenge? Mark your calendars for October 4 and 5! Registration will open in July. At that time, participants will be able to register for a local event hosted by NASA Space Apps leads from around the world. You can stay connected with NASA Space Apps on Facebook, Instagram, and X.

Space Apps is funded by NASA's Earth Science Division through a contract with Booz Allen Hamilton, Mindgrub, and SecondMuse.

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